

Practical 1

Practical 1-A:

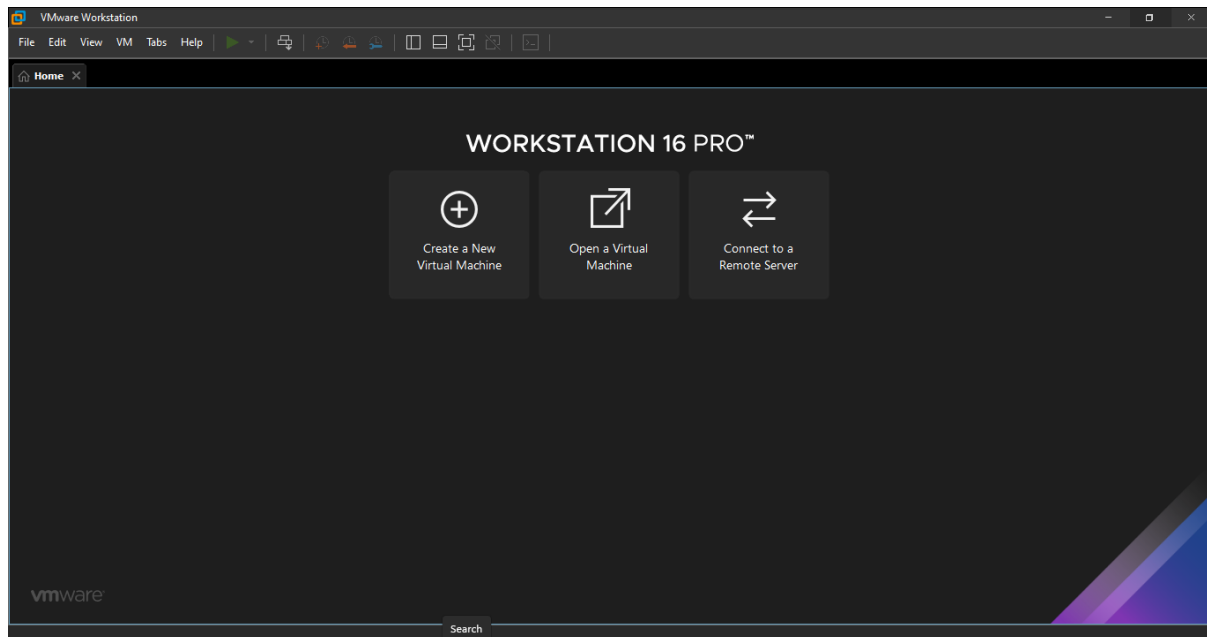
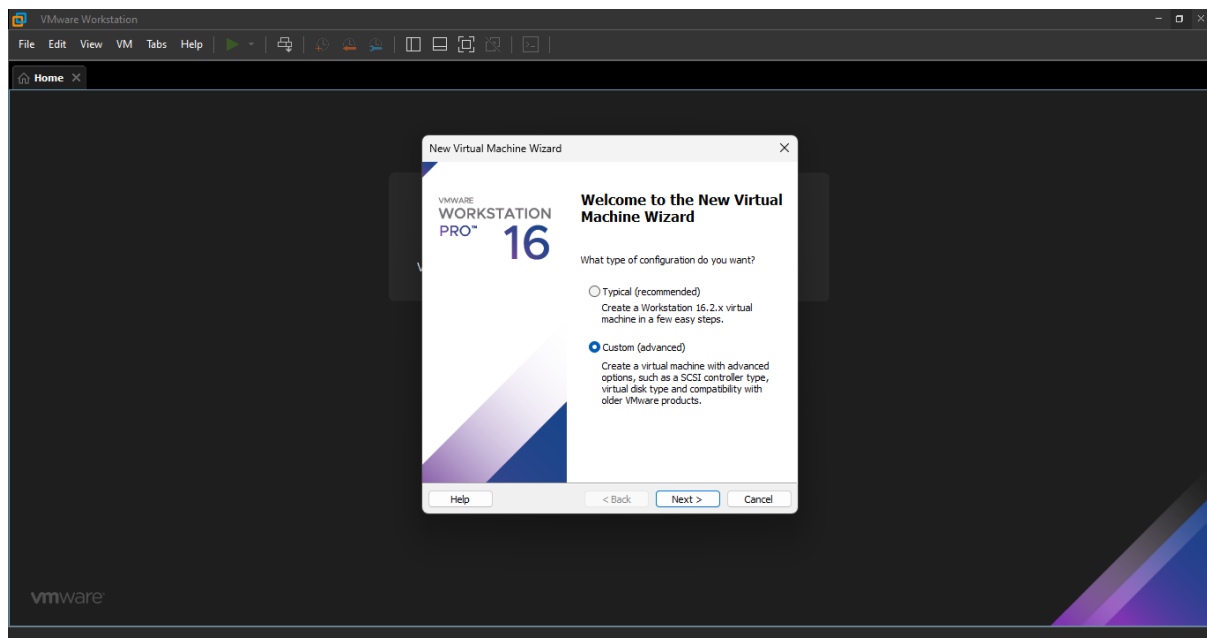
Aim: Create and Manage Cloud using SCVMM 2019

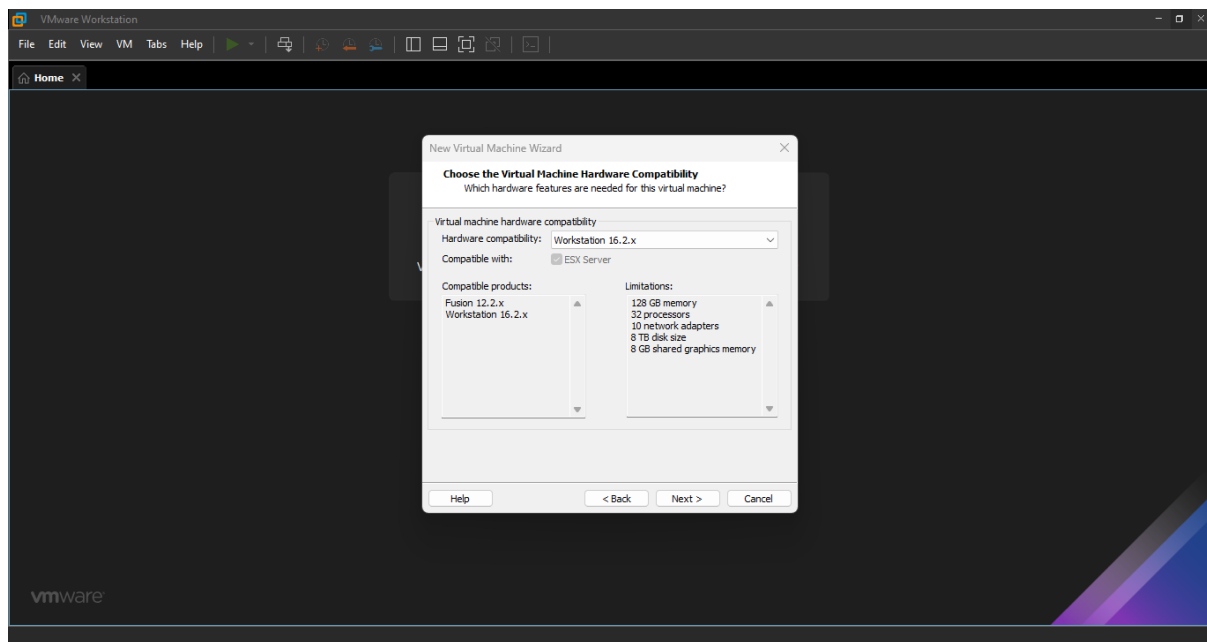
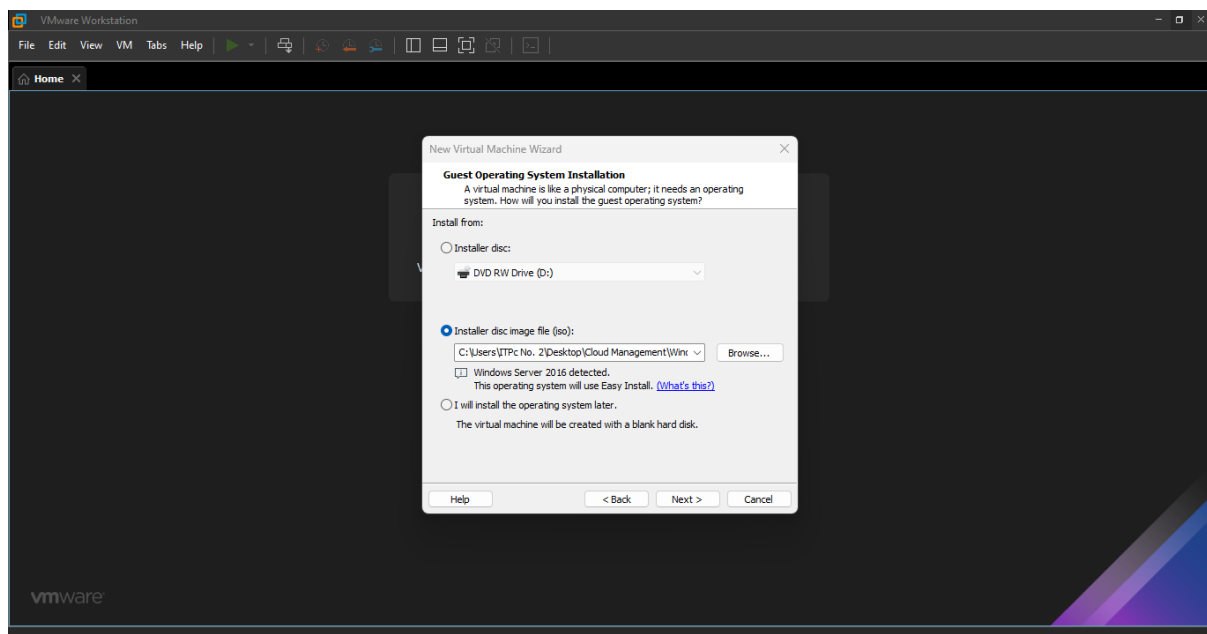
Writeup:

This image shows a single page of white paper with horizontal blue lines. The lines are evenly spaced and run across the width of the page, typical of notebook or ledger paper. There are no margins, text, or other markings on the page.

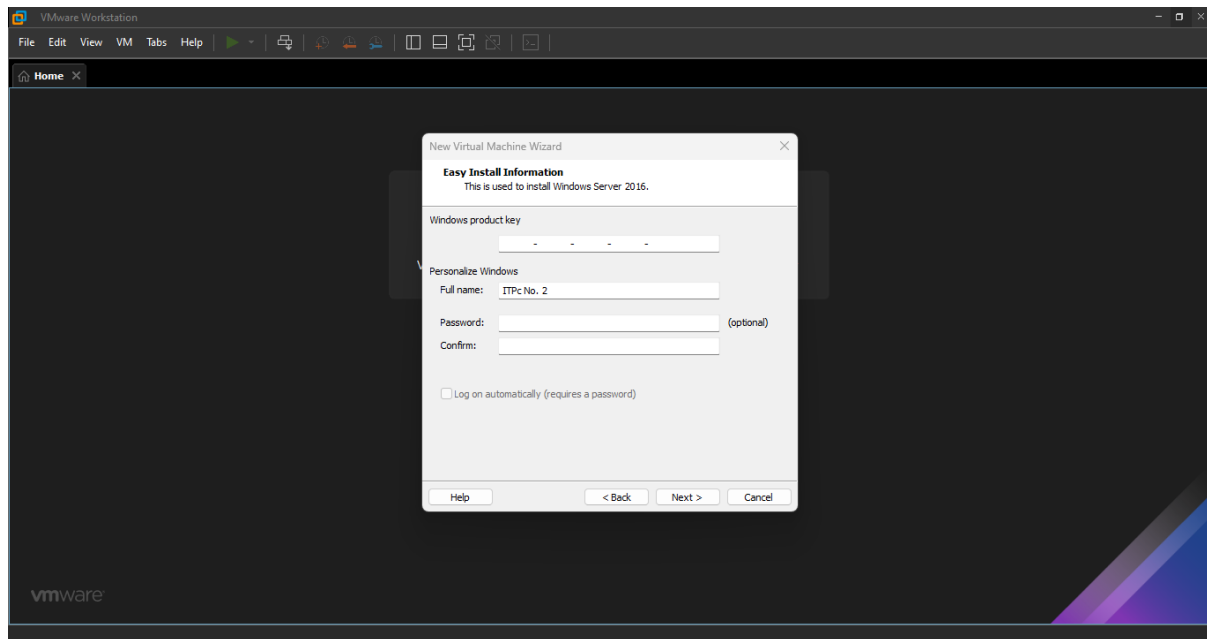
Step 1: Creation of Virtual Machine (Windows Server 2016)

Open VMware Workstation Pro and Click on **Create a New Virtual Machine**.

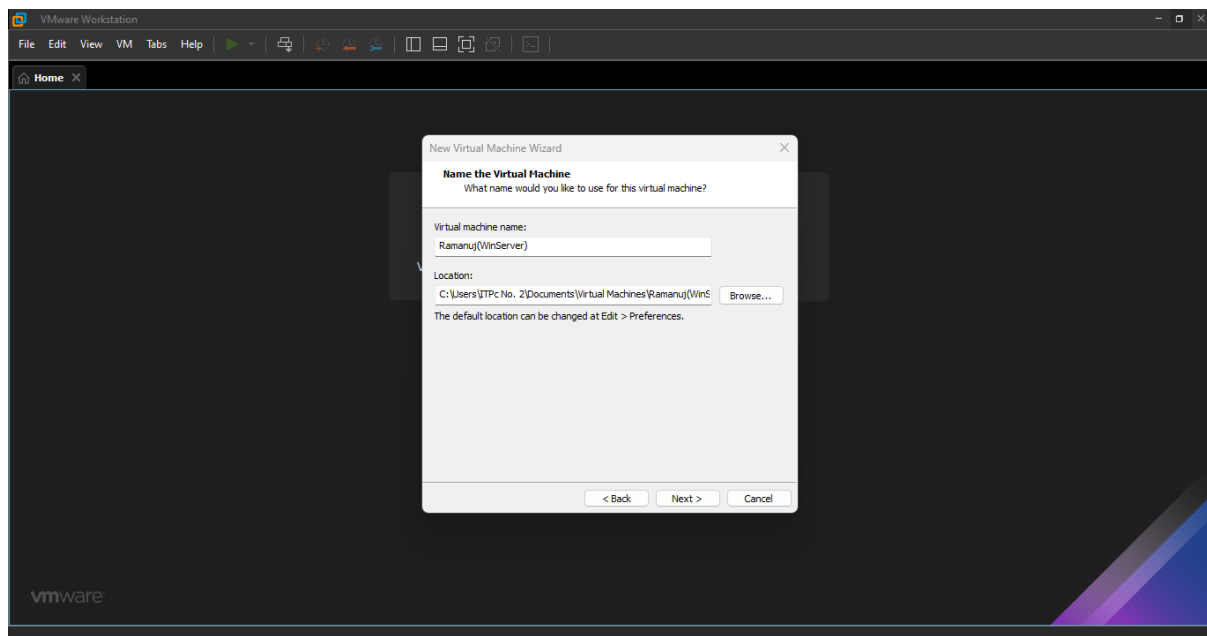
**Step 2: Select Custom.**

Step 3: Keep the Hardware Compatibility as default.**Step 4: Click on **Installer Disc Image File** and choose the Windows Server iso file from the directory.**

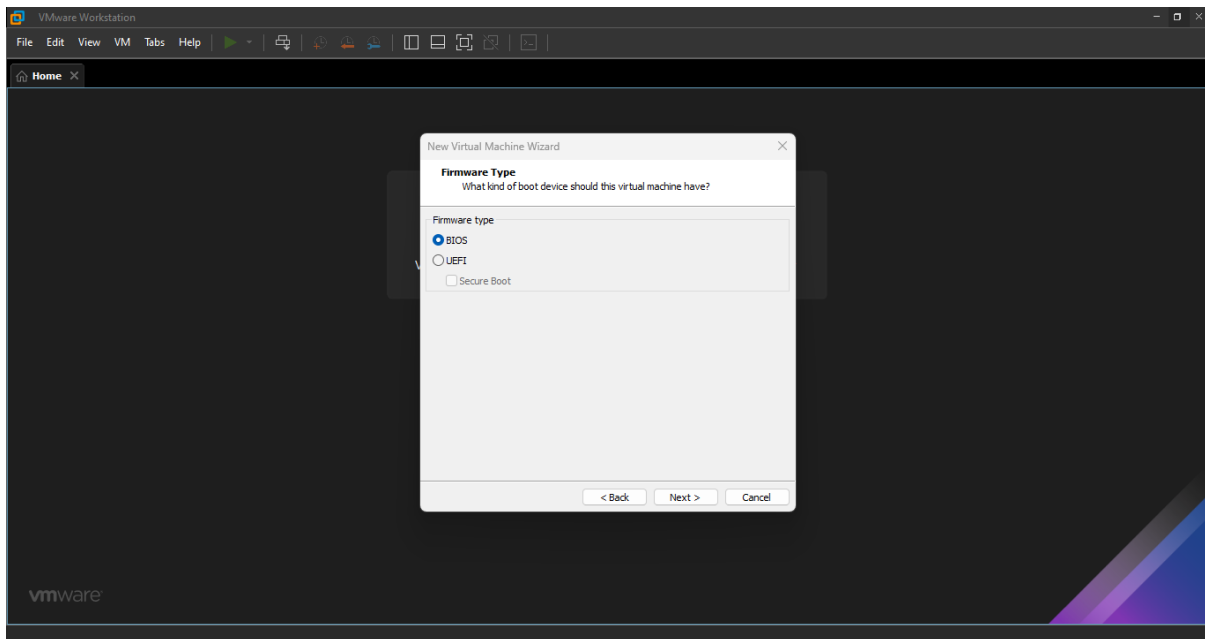
Step 5: Leave this Windows as default and Click **Next**. There will be a pop-up Click **Ok**.



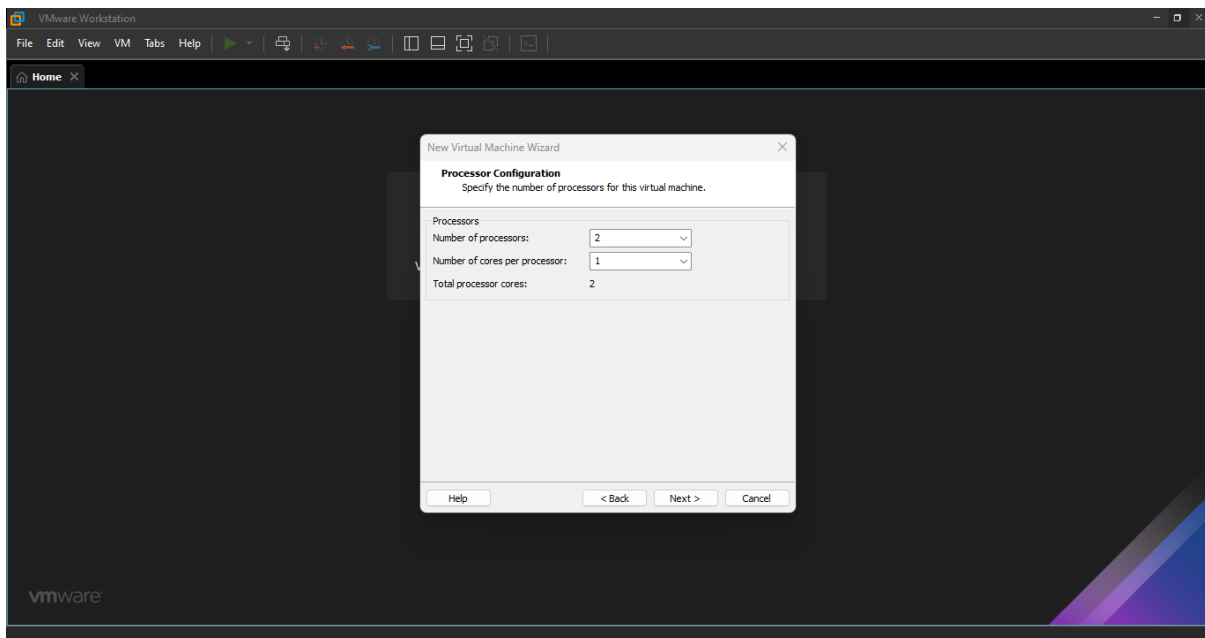
Step 6: Name your Virtual Machine, in this case its **Ramanuj(WinServer)**.

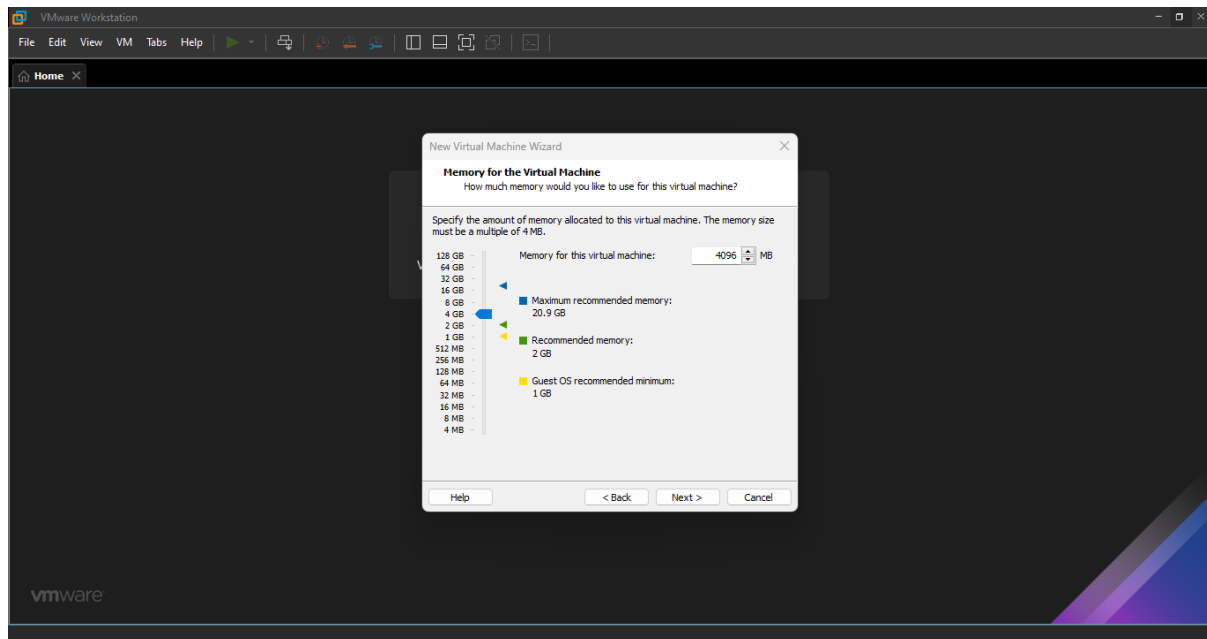
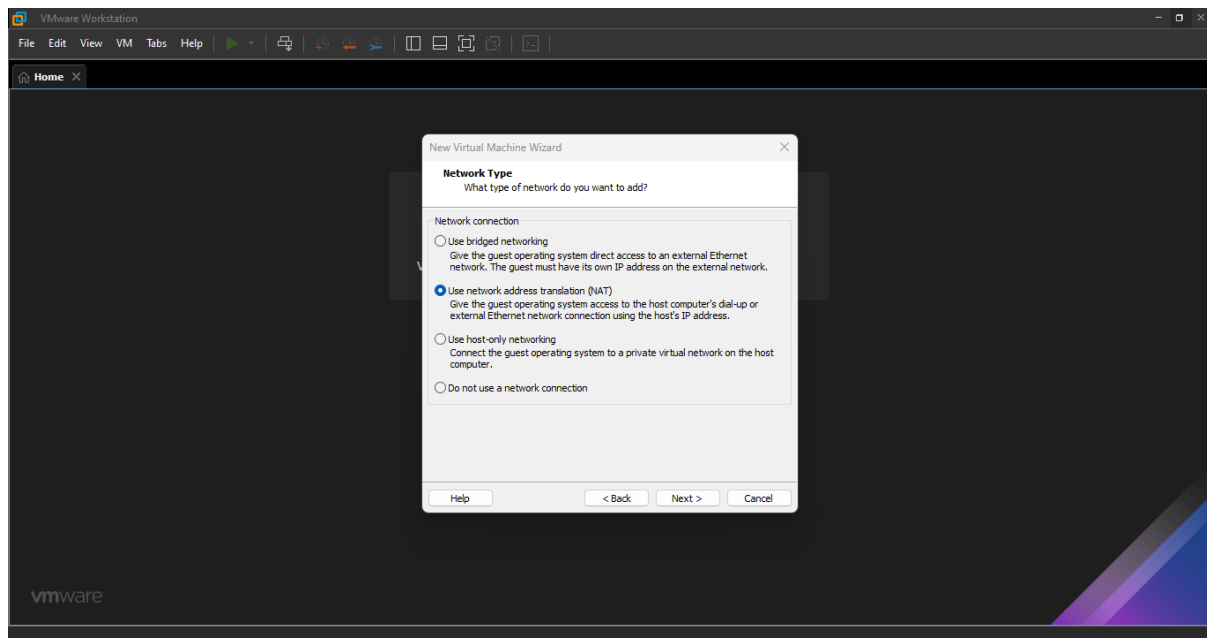


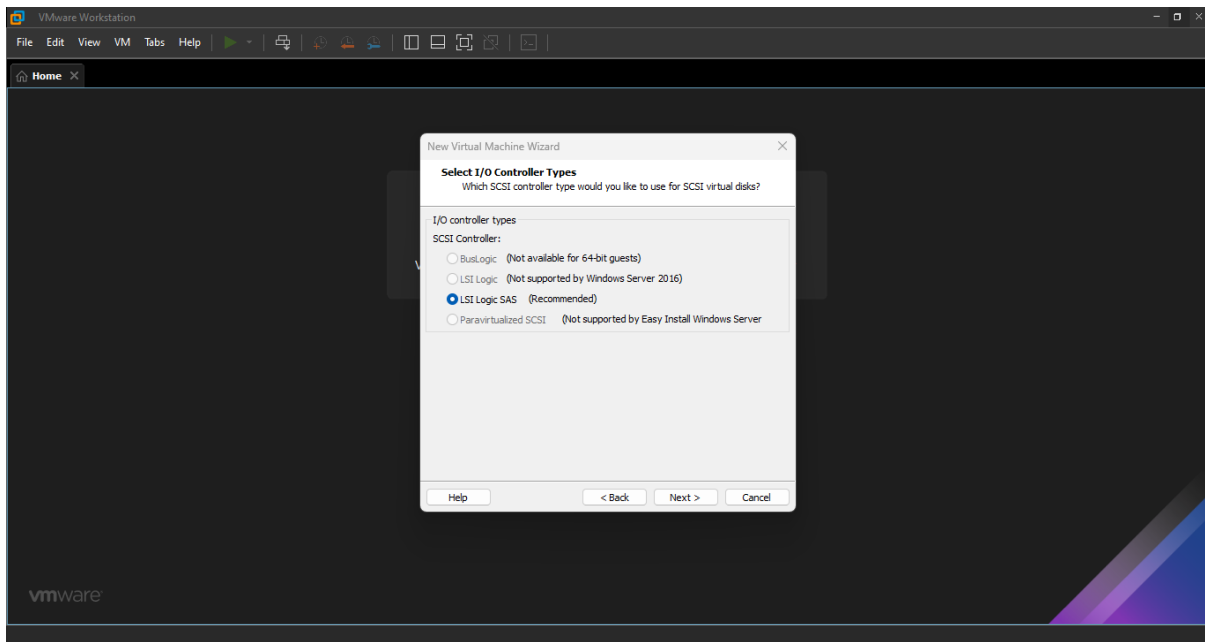
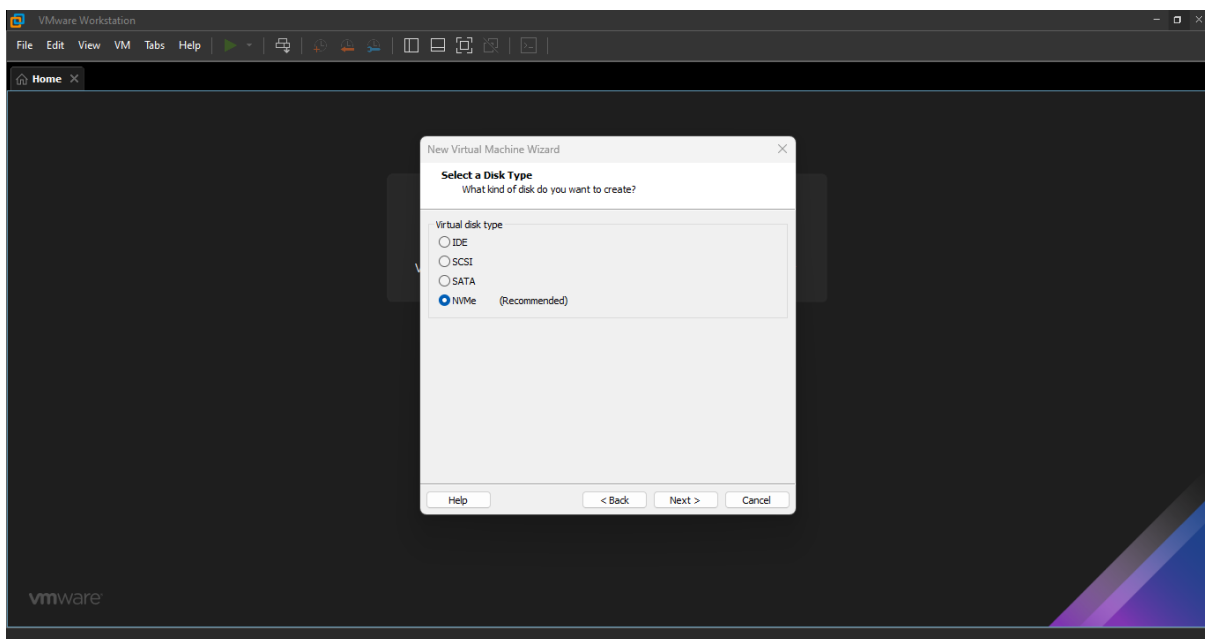
Step 7: Click on **BIOS** and Click **Next**.

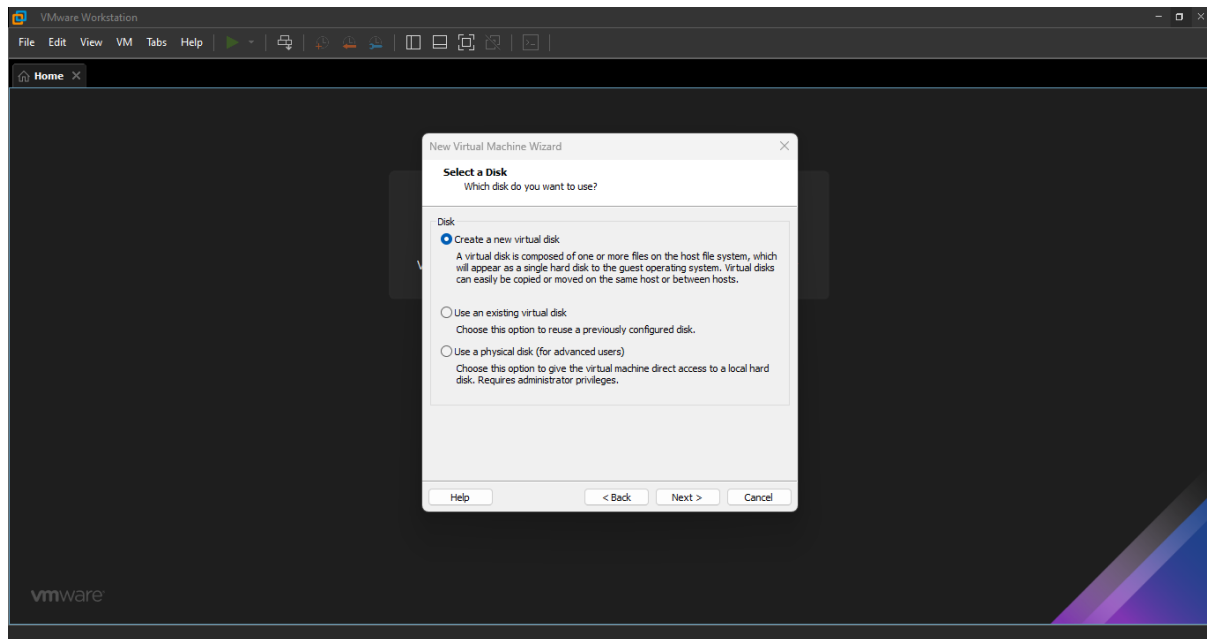
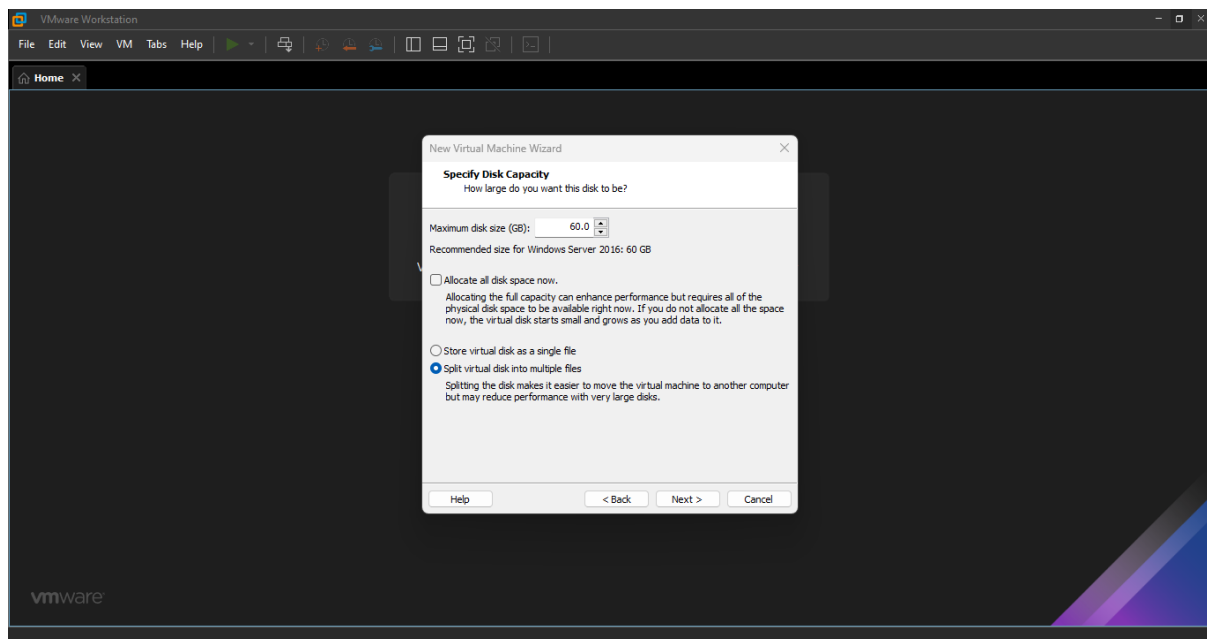


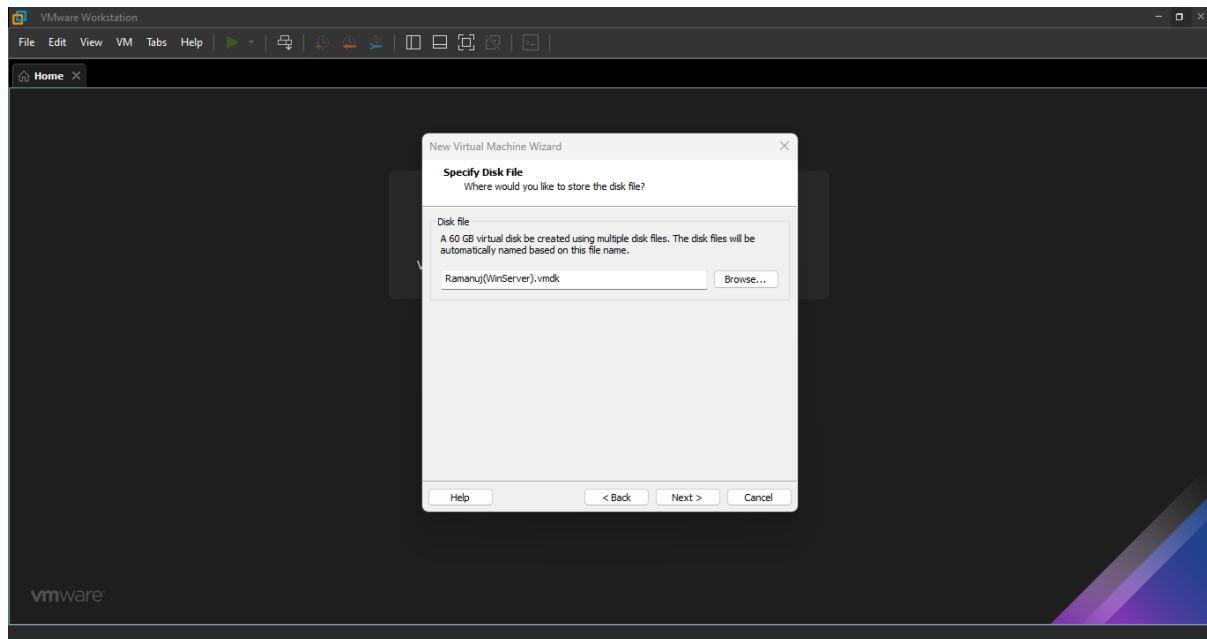
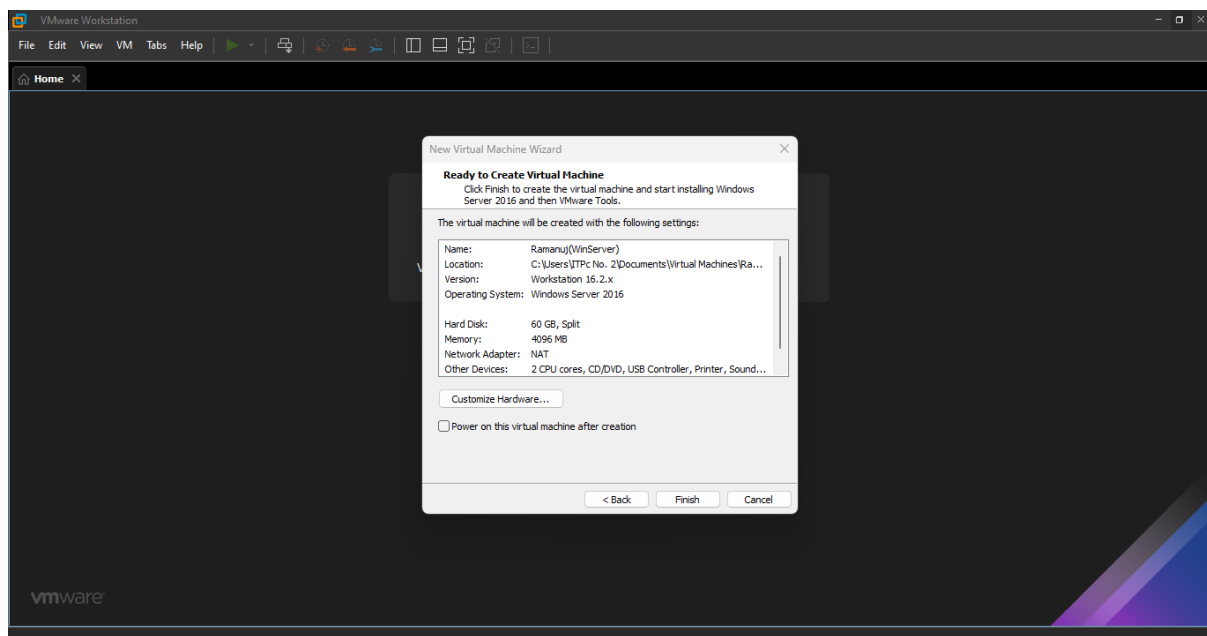
Step 8: Keep Default Values and Click **Next**.



Step 9: Change the RAM to 4096 MB and Click Next.**Step 10: Select Network Address Translation (NAT) and Click Next.**

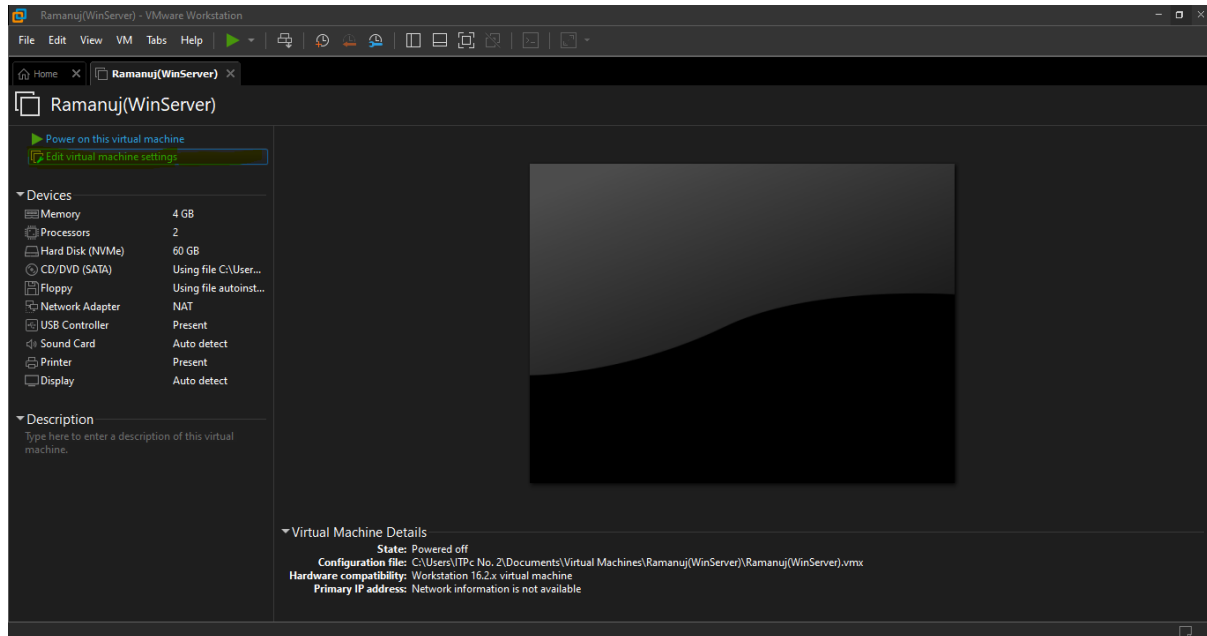
Step 11: Select LSI Logic SAS and Click Next.**Step 12: Keep the Default NVMe and Click Next.**

Step 13: Click on Create new virtual disk and Click Next**Step 14: Keep the default value of 60GB and Select Split Virtual Disk into Multiple Files and Click Next.**

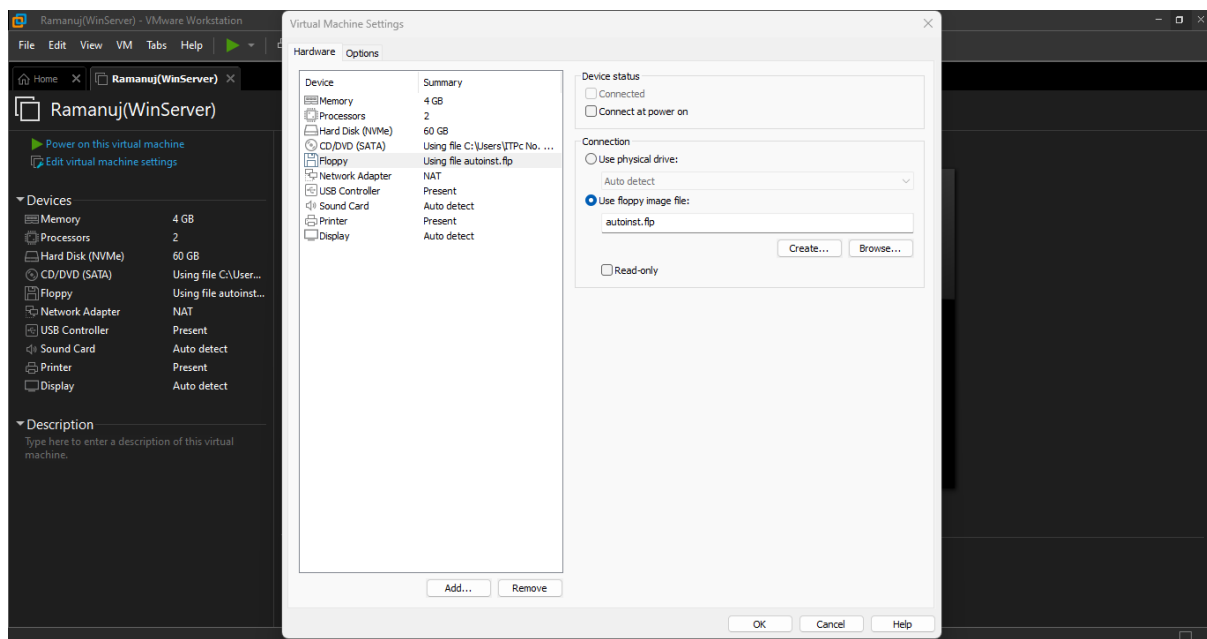
Step 15: Leave it as Default and Click Next**Step 16: Click on Finish**

Step 17: Before Powering ON the Virtual Machine we have to complete a few pre-requisites:-

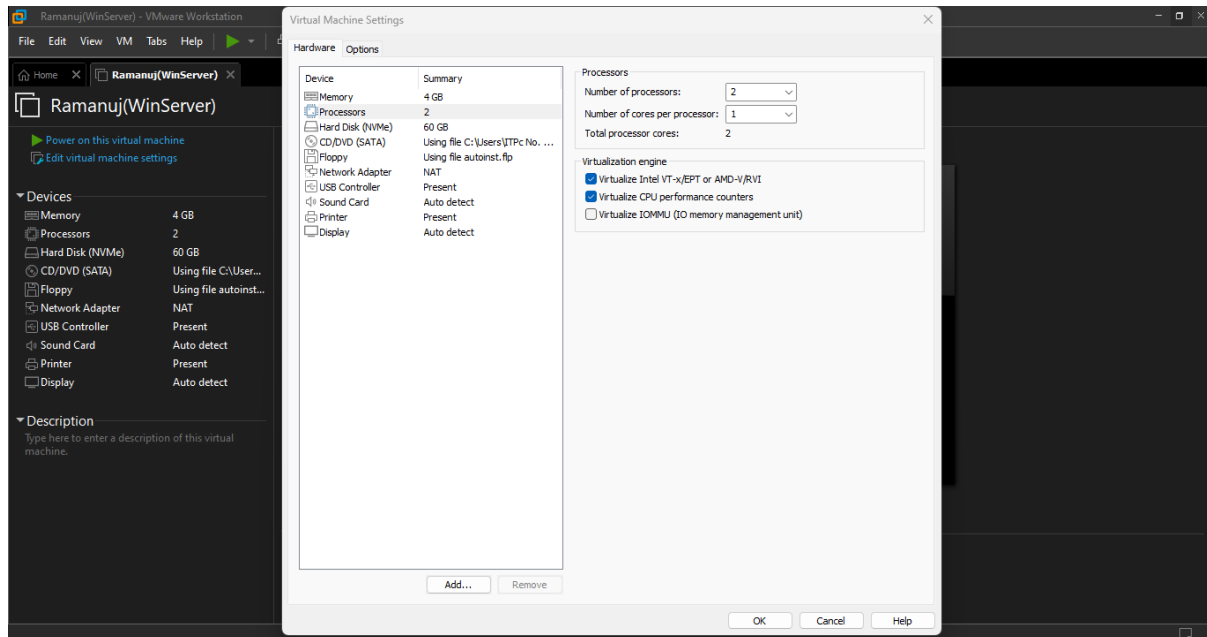
- Click on **Edit Virtual Machine Settings**



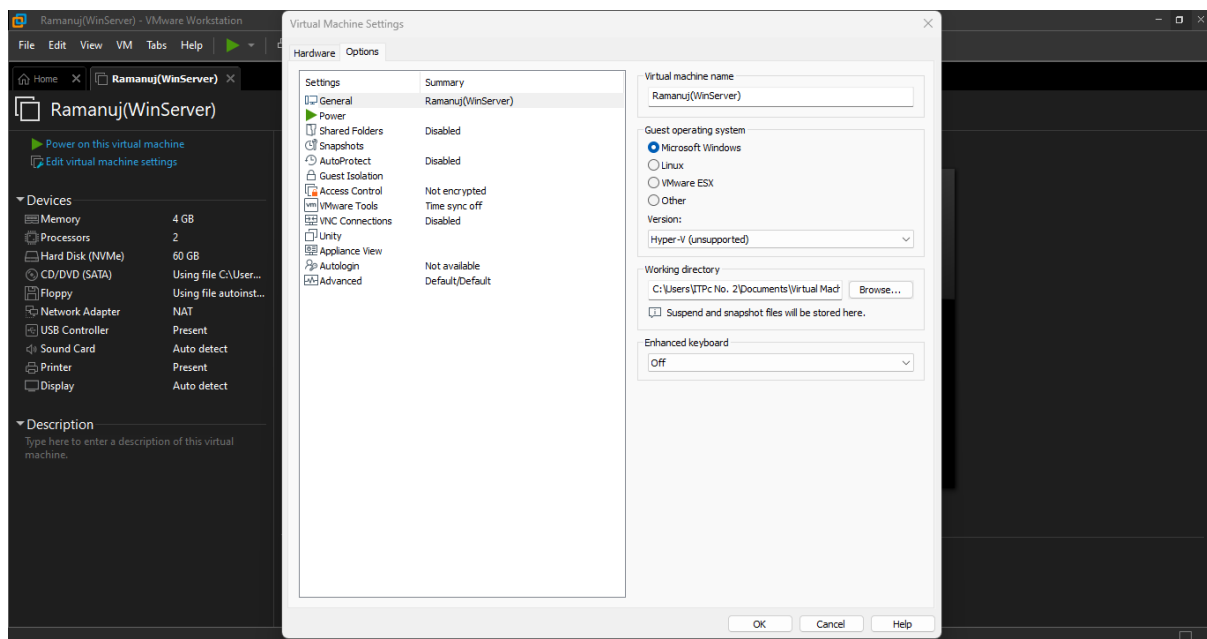
- In Virtual Machine Settings, in Hardware Click on **Floppy** and UnClick the **Connect at Power On** option.



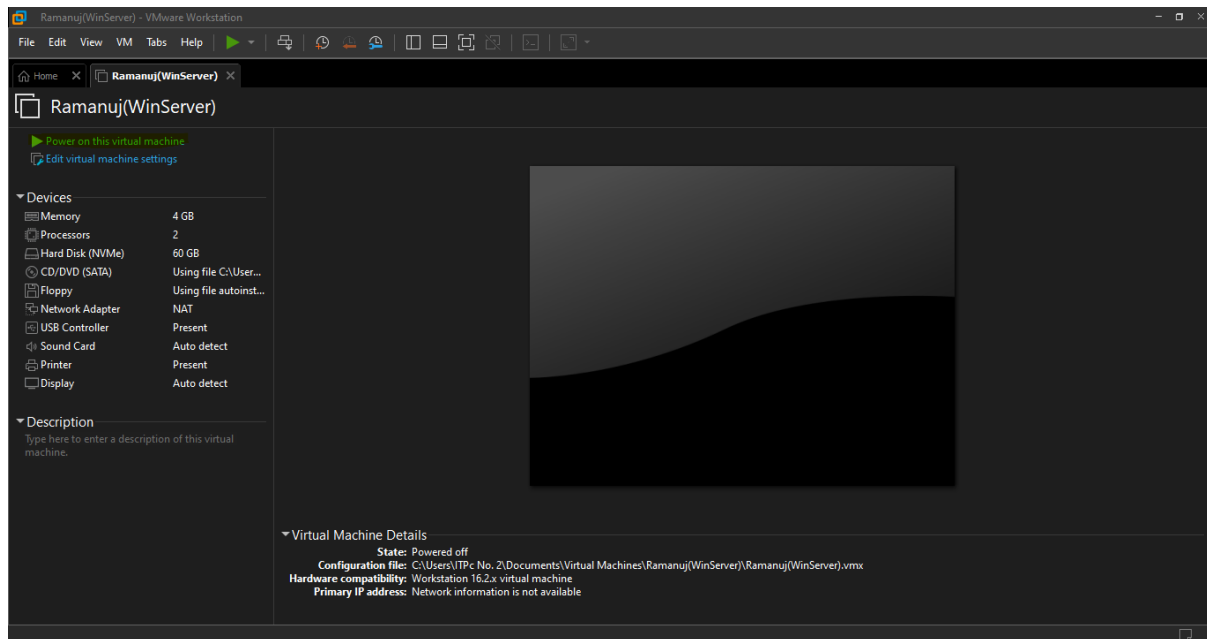
- In Hardware now Click on **Processors**
- Within Processors enable **Virtualize Intel VT-x/EPT or AMD-V/RVI**
- Also enable **Virtualize CPU Performance Counters**



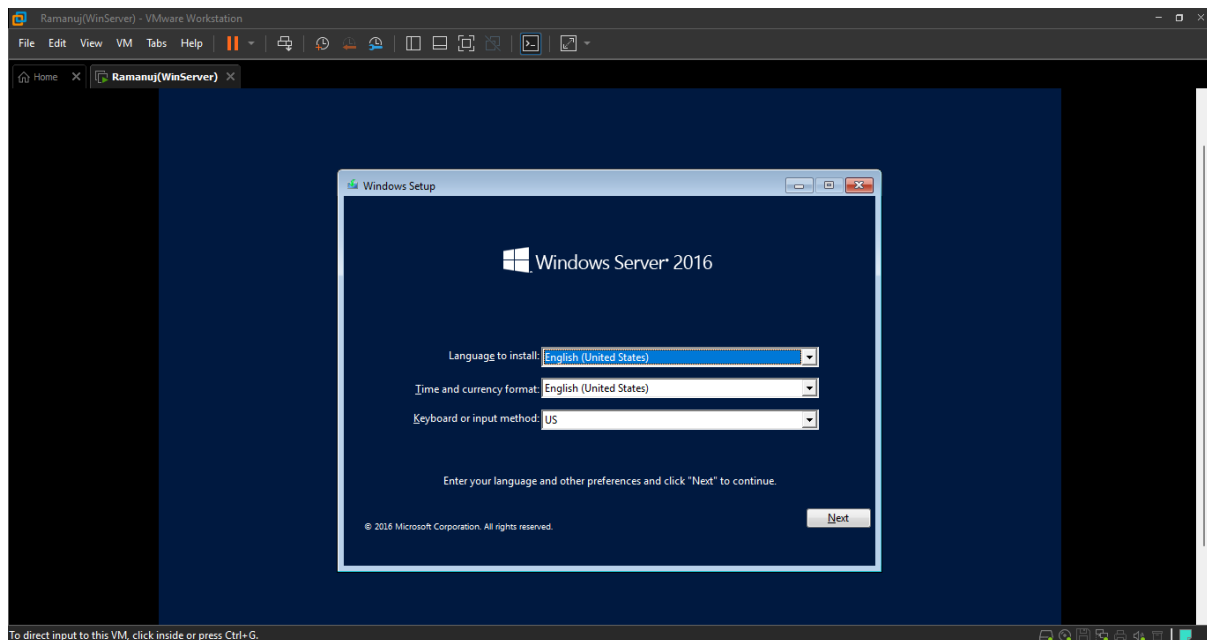
- In Virtual Machine Settings Click on **Options**
- Within Options Click on **General**
- In General Click on **Version** and change it to **Hyper-V Unsupported**



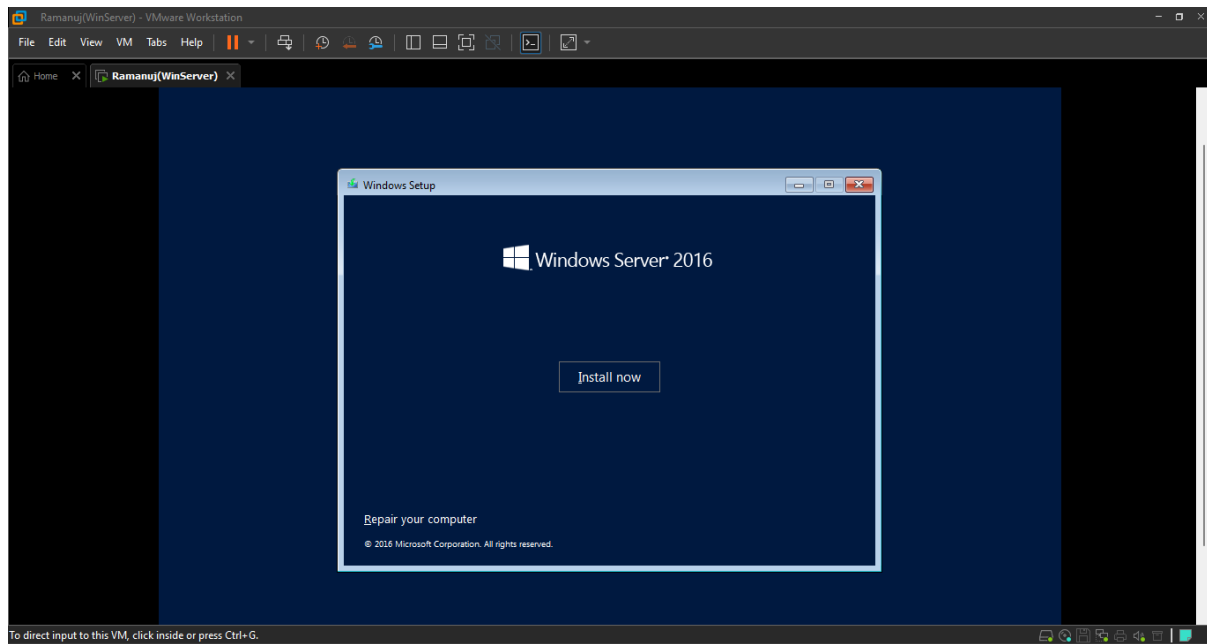
Step 18: Once all the pre-requisites are met, we can turn on the virtual machine.



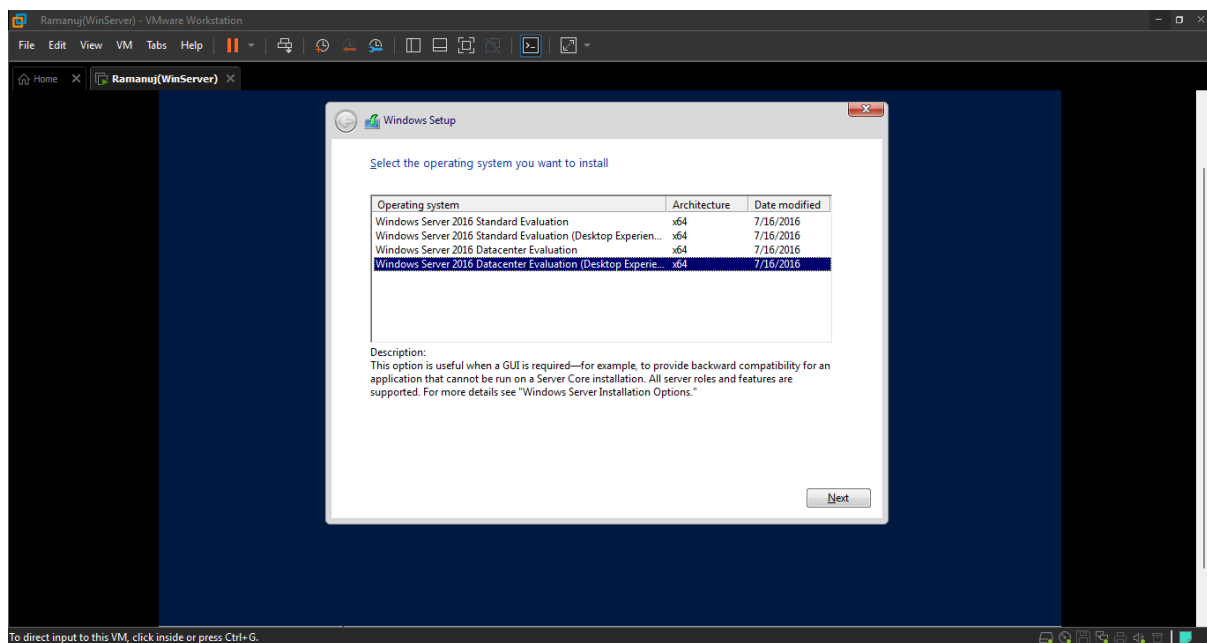
Step 19: VM Powers On and Windows starts to boot up, Click on **Next**.

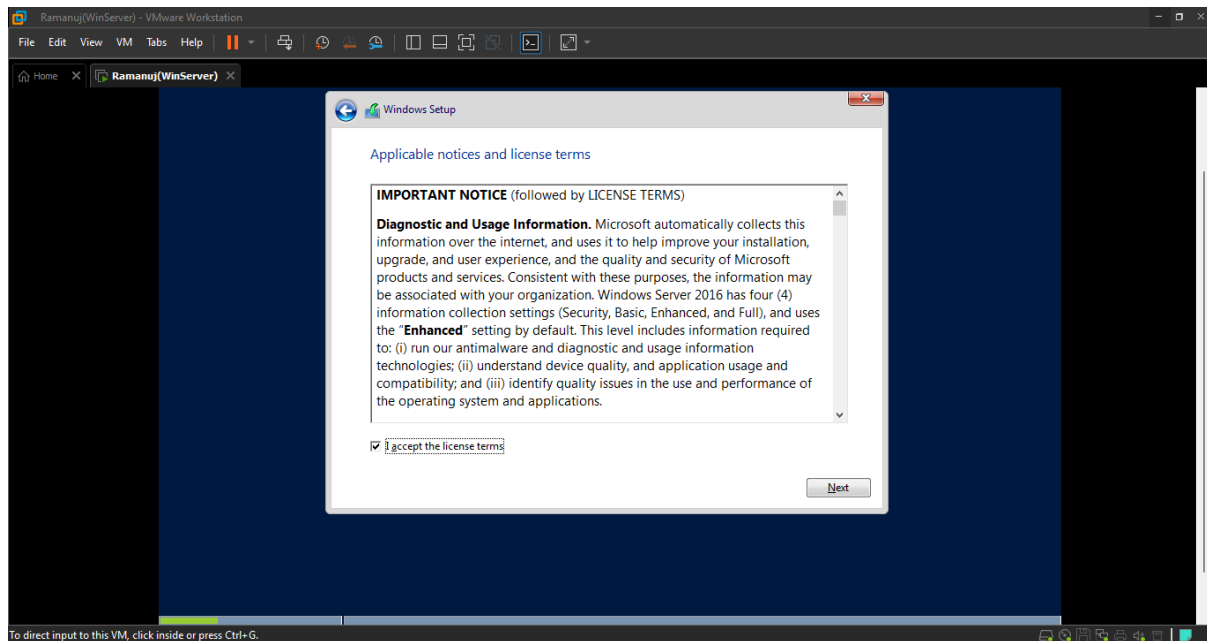
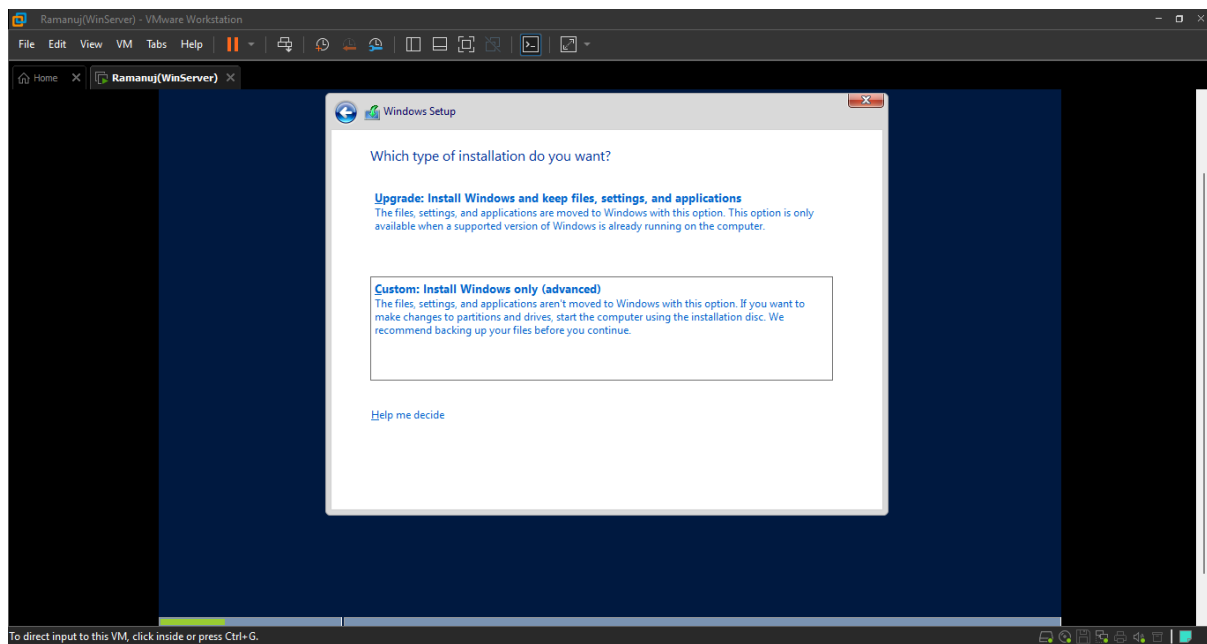


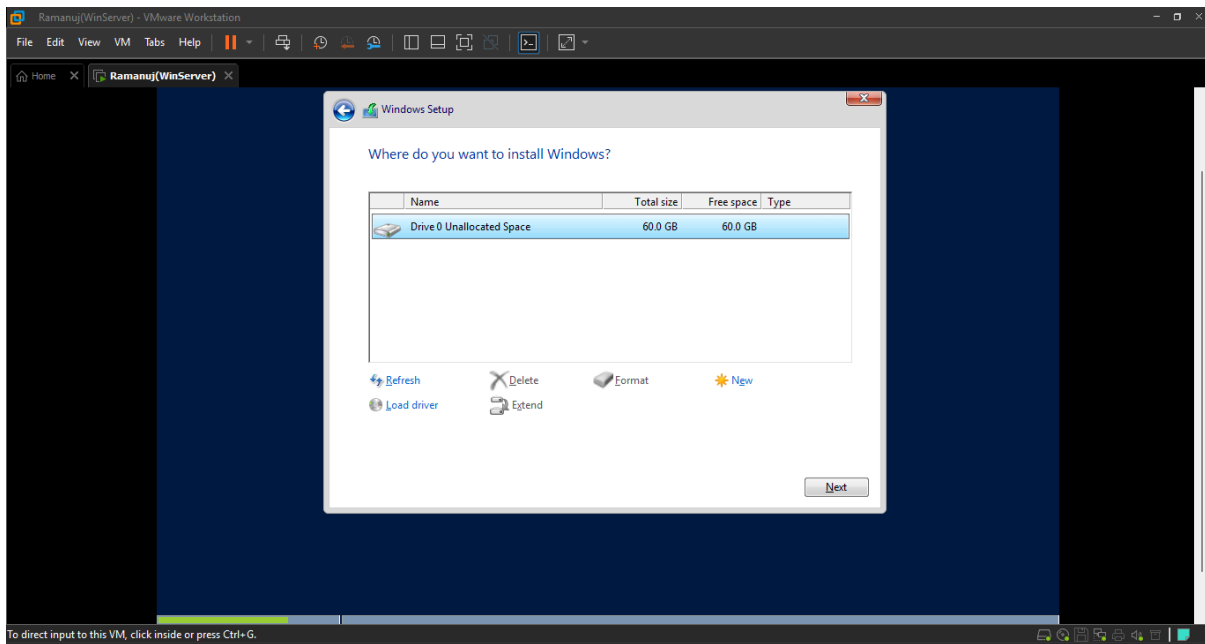
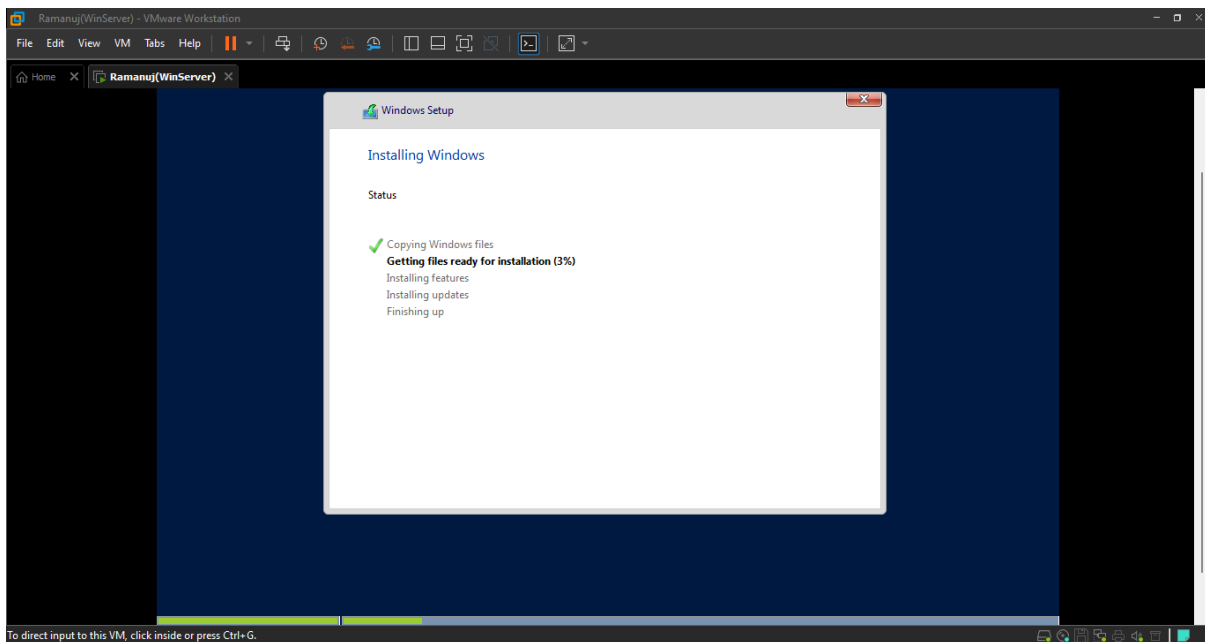
Step 20: Click on **Install Now**



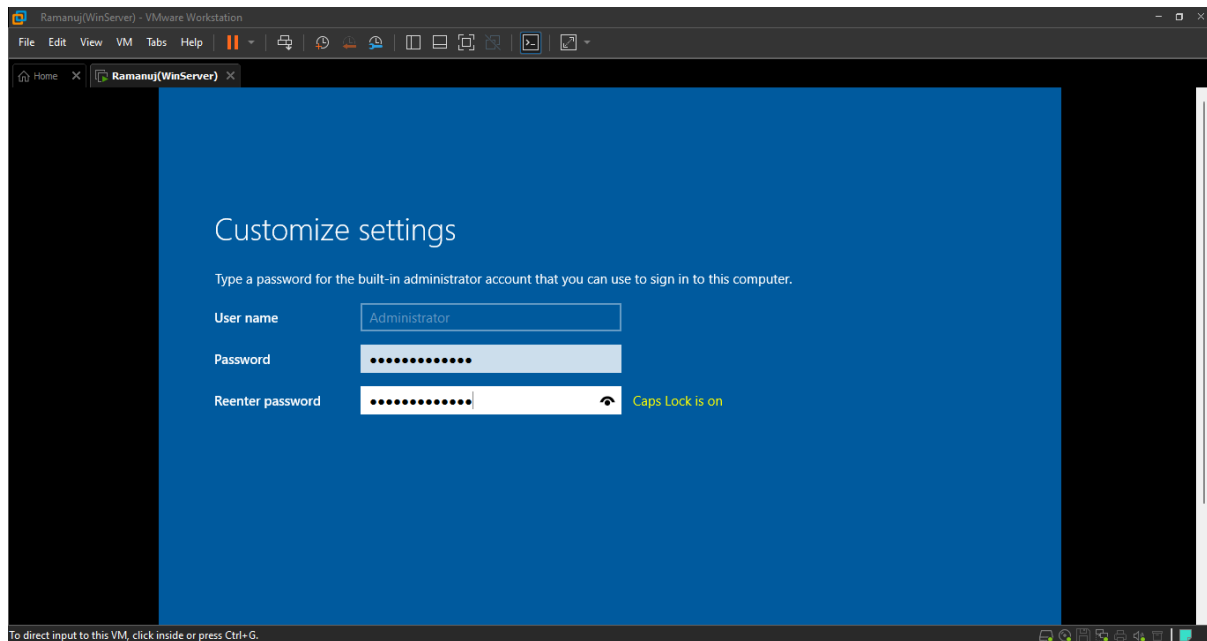
Step 21: Select **Windows Server 2016 Datacenter Evaluation (Desktop Experience x64)**



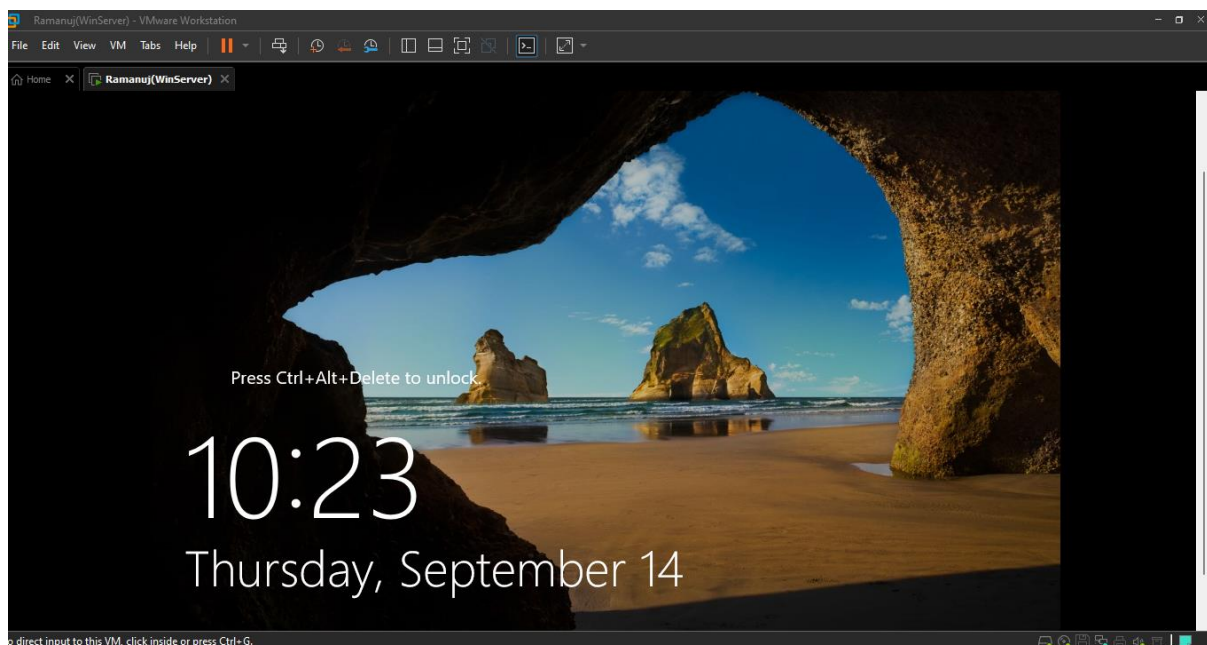
Step 22: Click on I Accept the license terms**Step 23: On type on installation Click Custom**

Step 24: Leave storage as default and Click Next**Step 25: It will start installing, when finished it will auto-restart.**

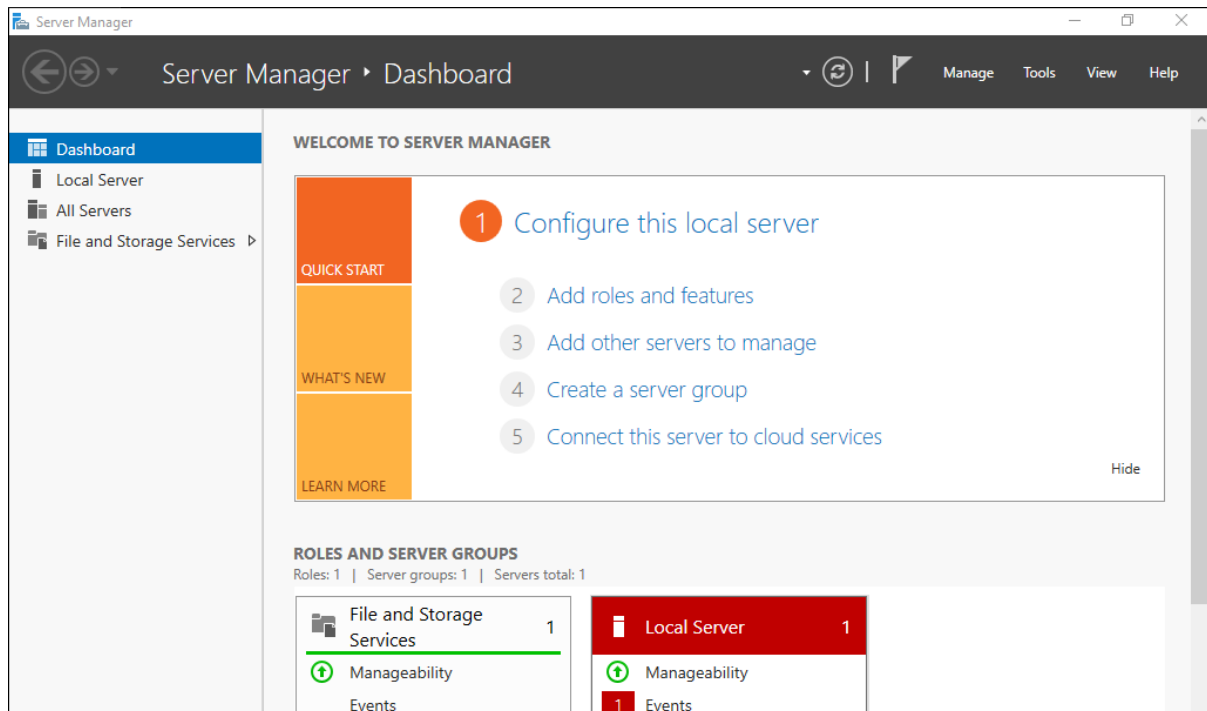
Step 26: After the restart set your password (alphanumeric) and Click on **Finish**



Step 27: Windows Server has been successfully created

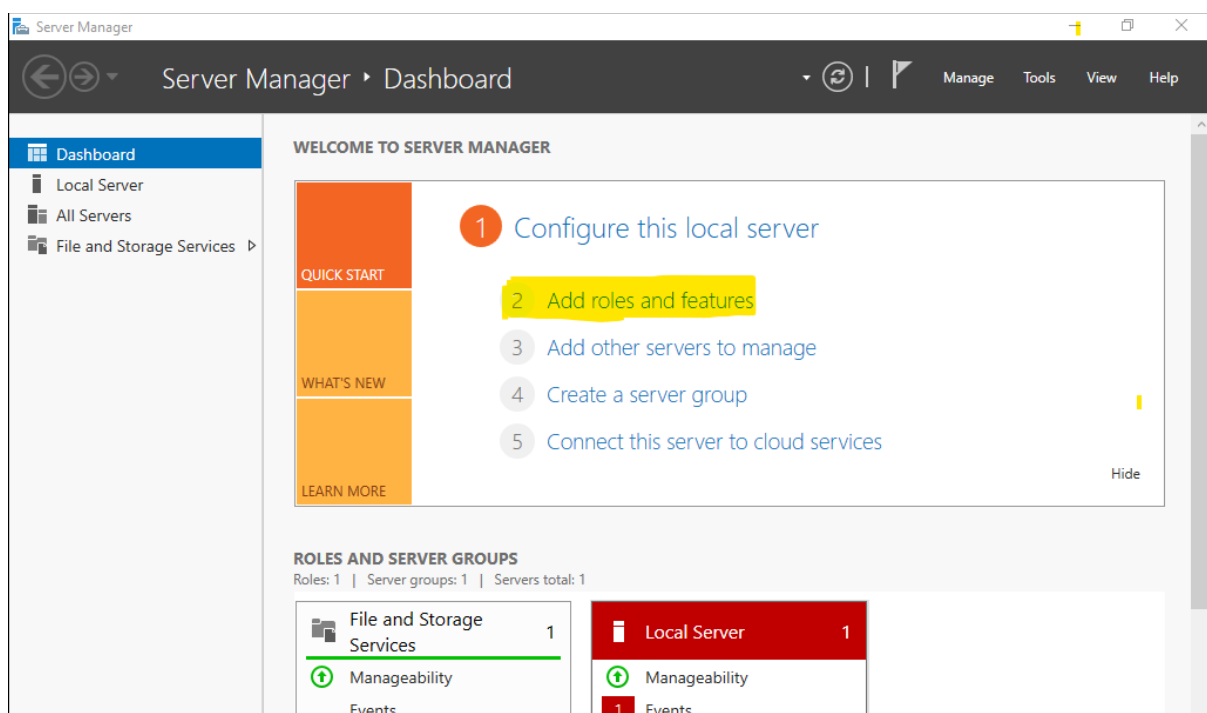


Step 28: Within Windows Open **Server Manager**

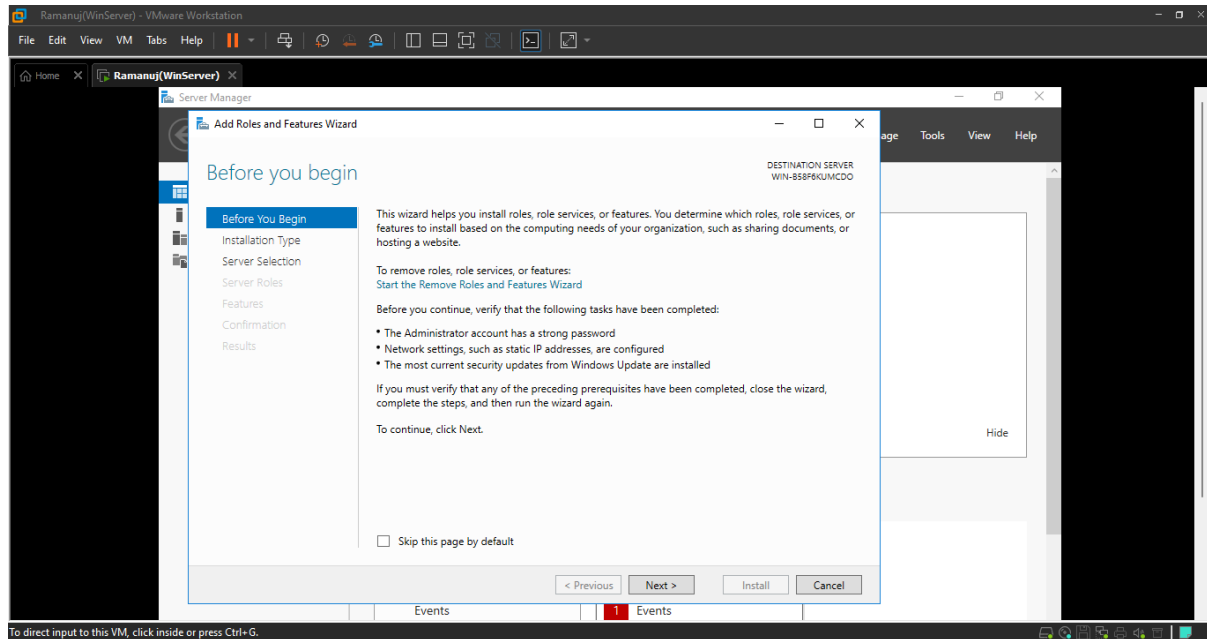


Step 29: Enabling Hyper-V

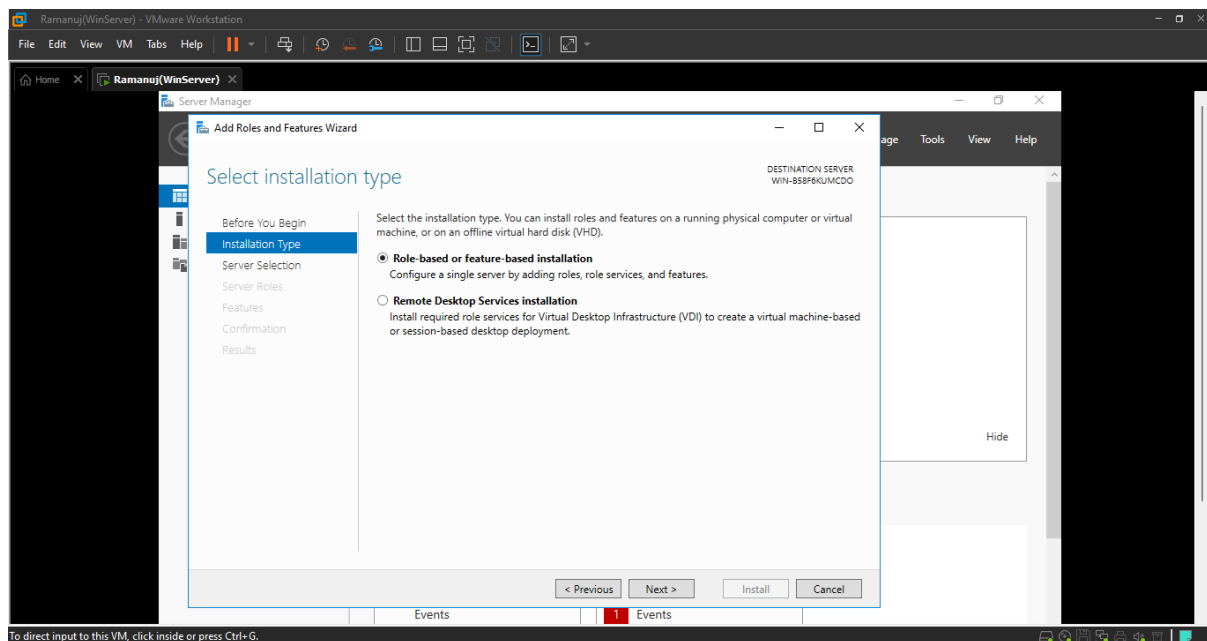
- Within Server Manager Click on **Add Roles and Features**



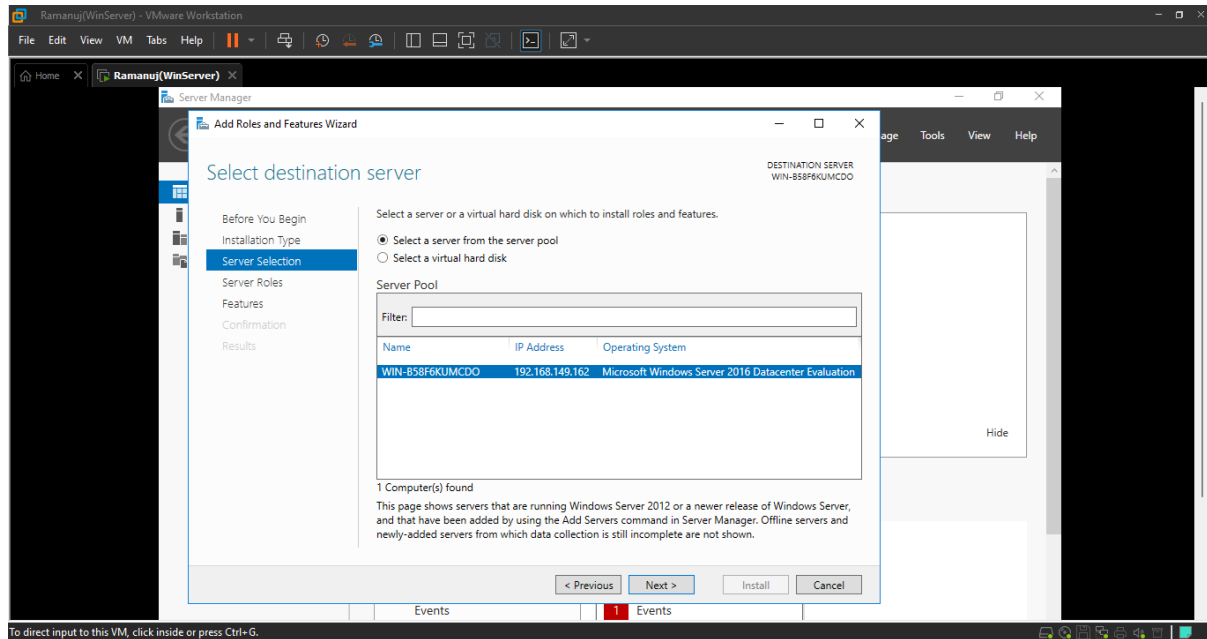
- Click **Next**



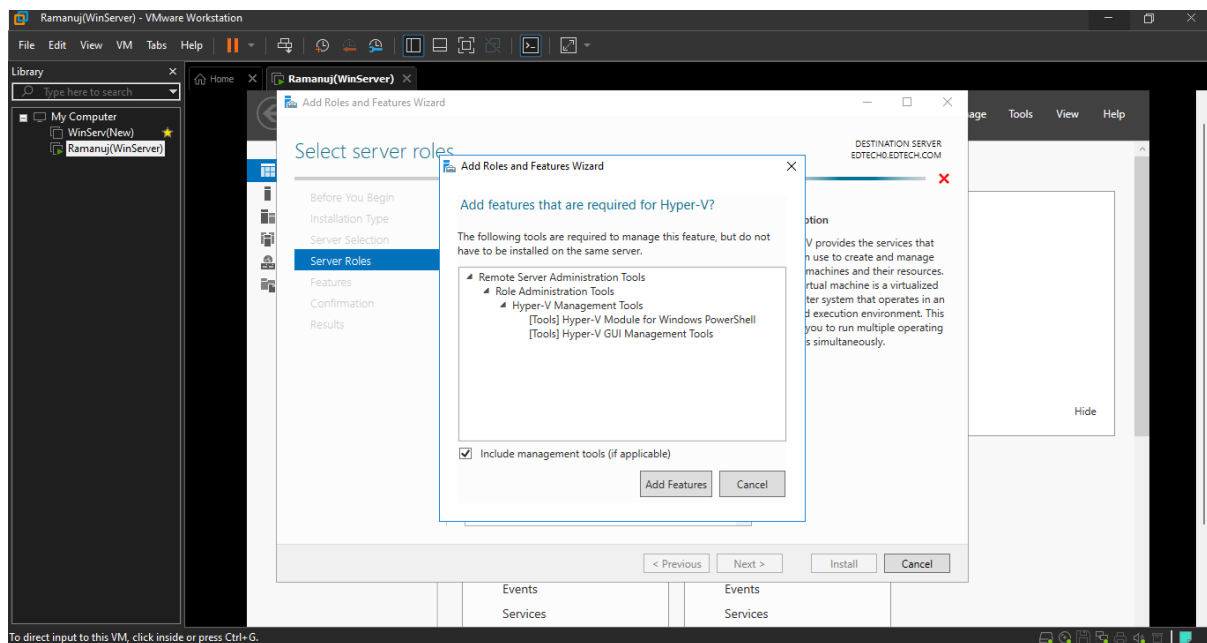
- Make sure **Role-based or Feature-based installation** is selected and Click **Next**



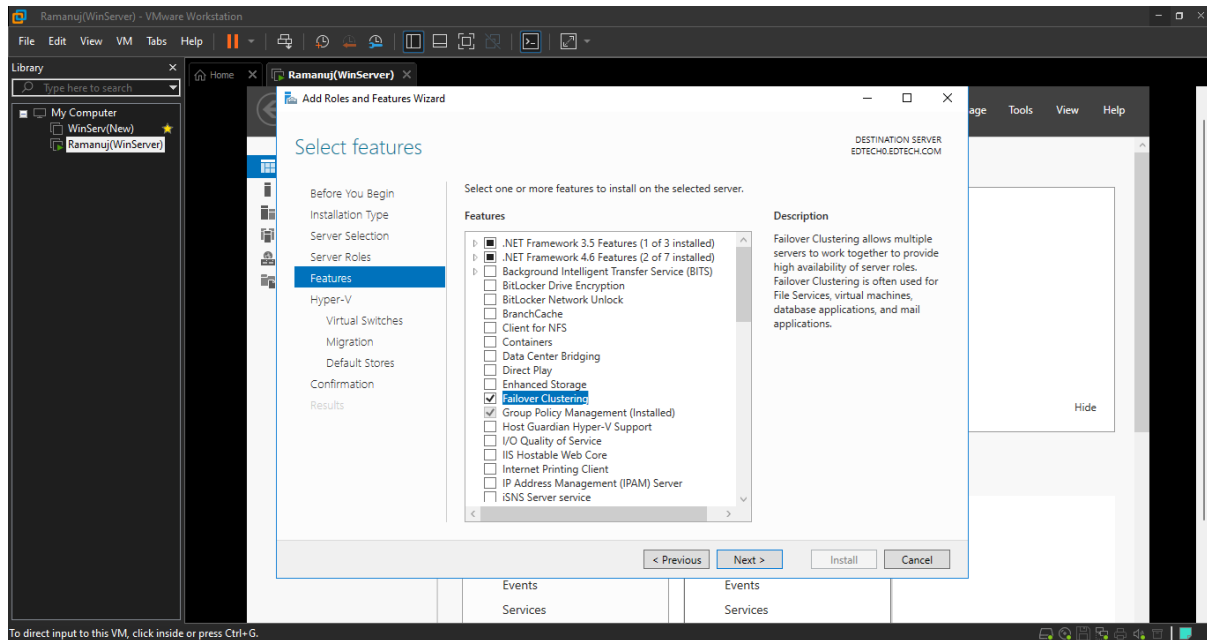
- Keep default settings and Click **Next**



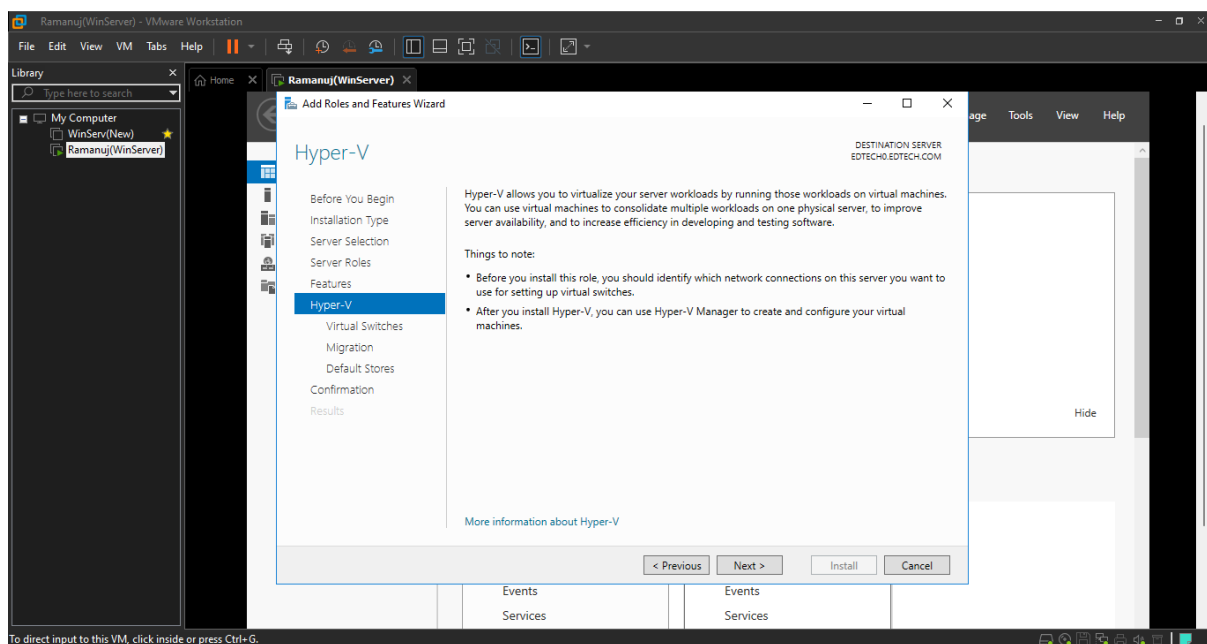
- Select **Hyper-V** and Click **Add Features** and Click **Next**



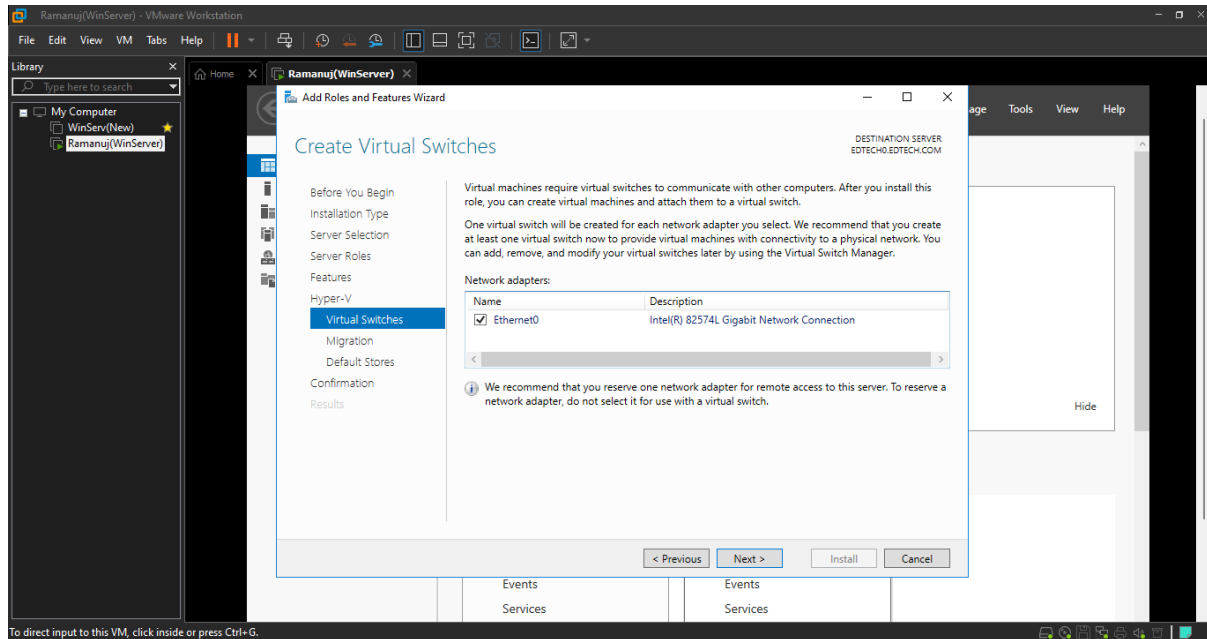
- Select **Failover Clustering** and Click **Add Features** and Click **Next**
- Note: Make sure **Group policy management** is also selected



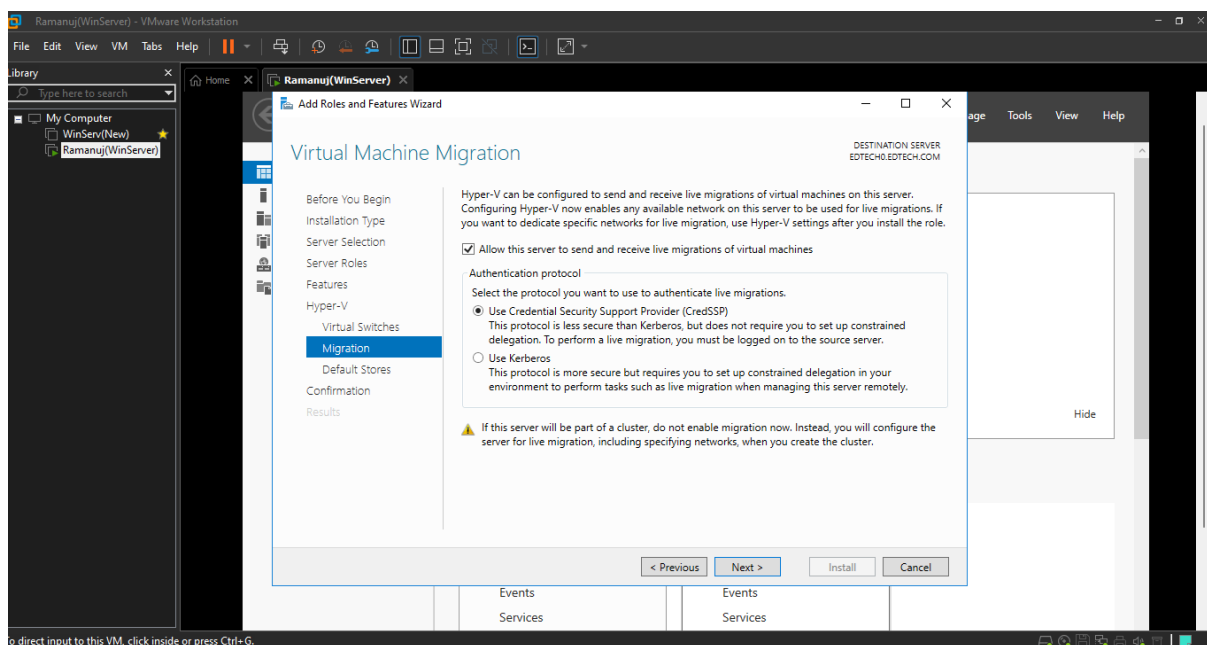
- Keep default settings and Click **Next**



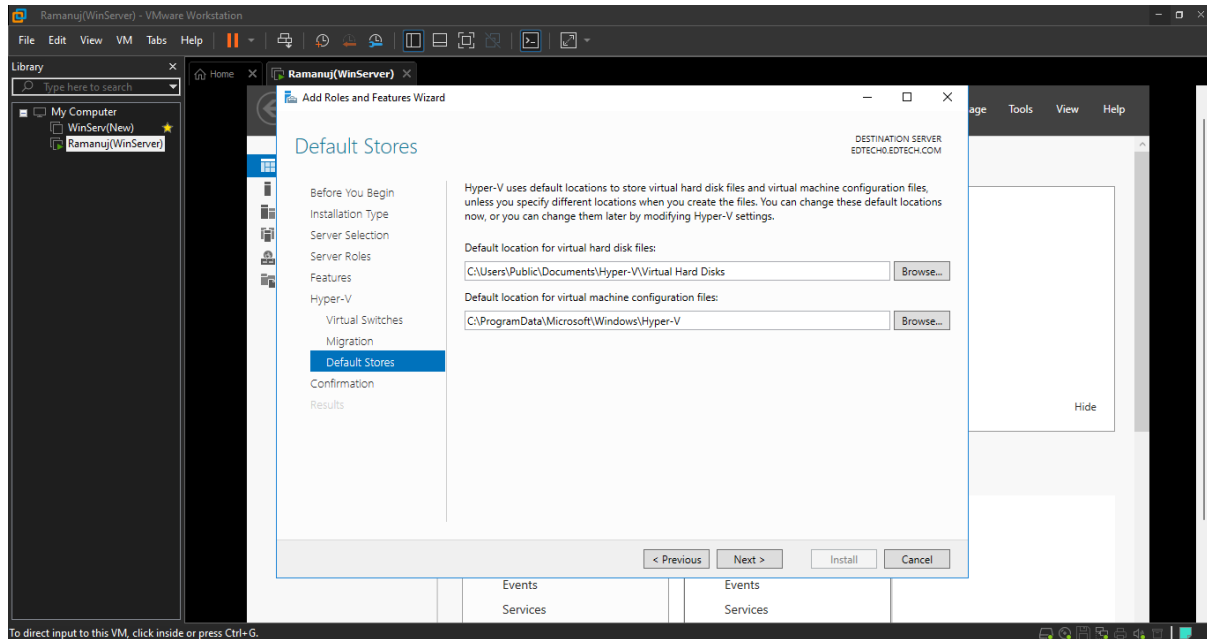
- Select **Ethernet0** and Click **Next**



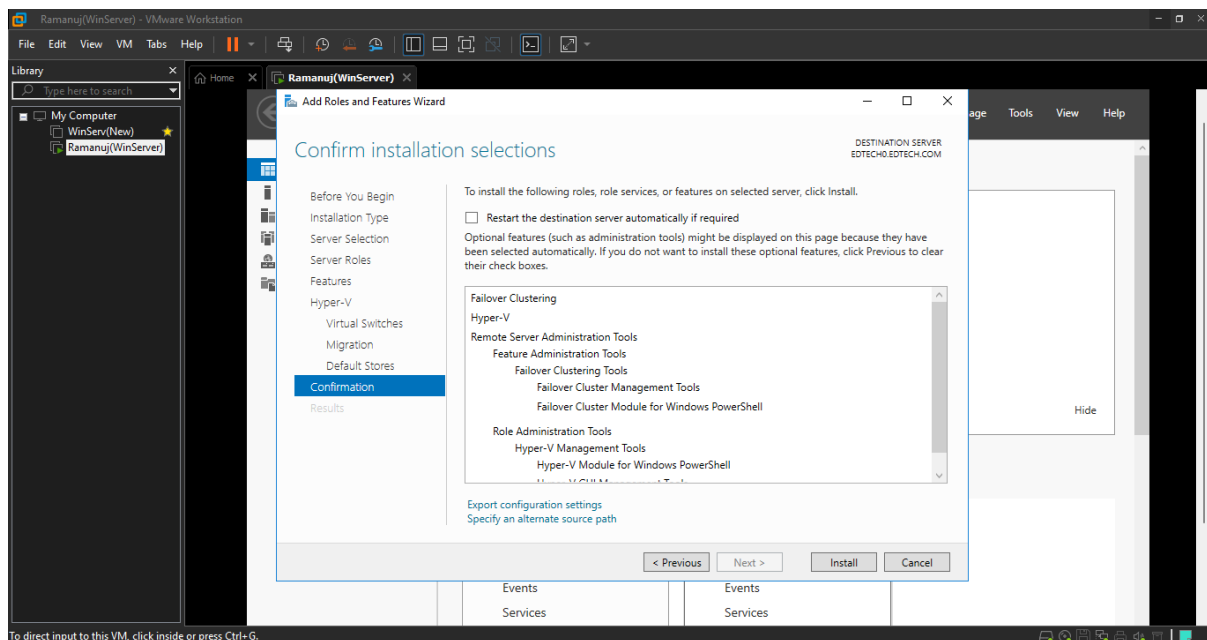
- Select **Allow to server to send and receive Live Migration of virtual machines**



- Keep default setting and Click **Next**

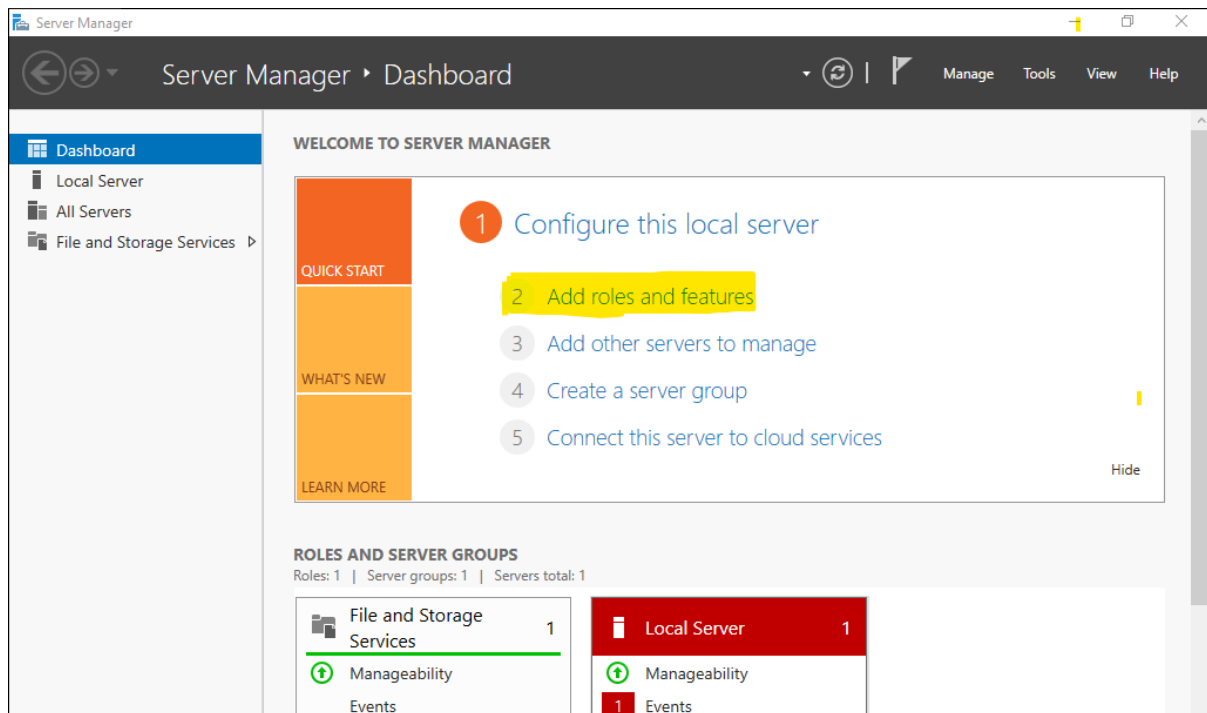


- Click on **Install**, After installation is complete restart the machine for the changes to be taken place.

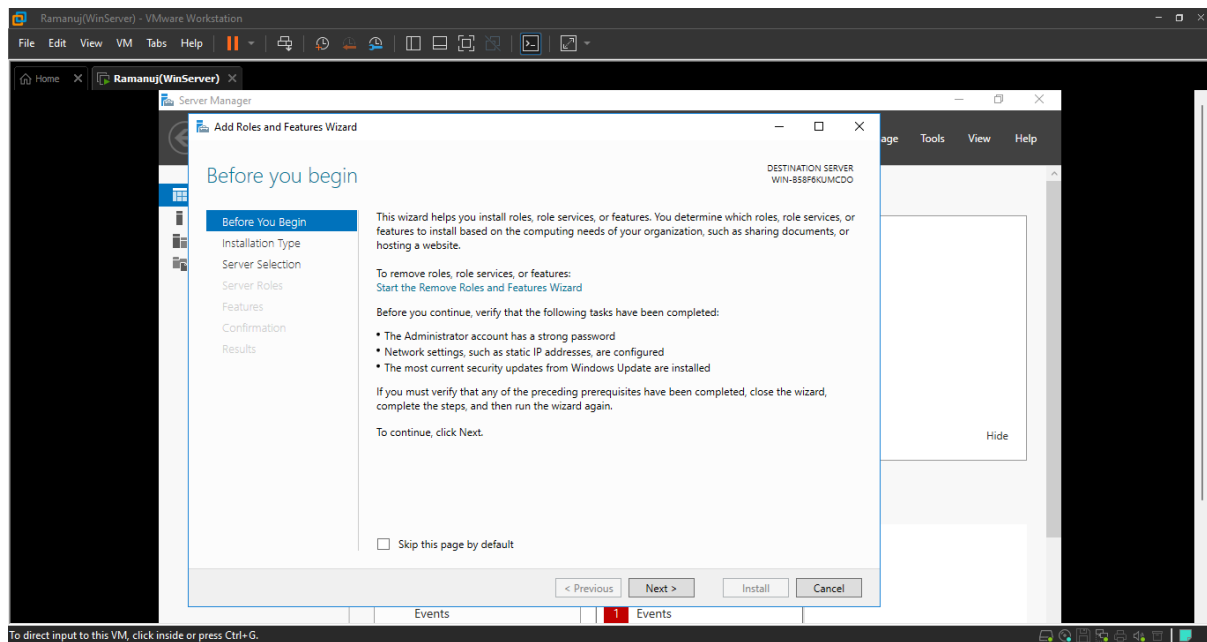


Step 30: Adding Active Domain Directory Services

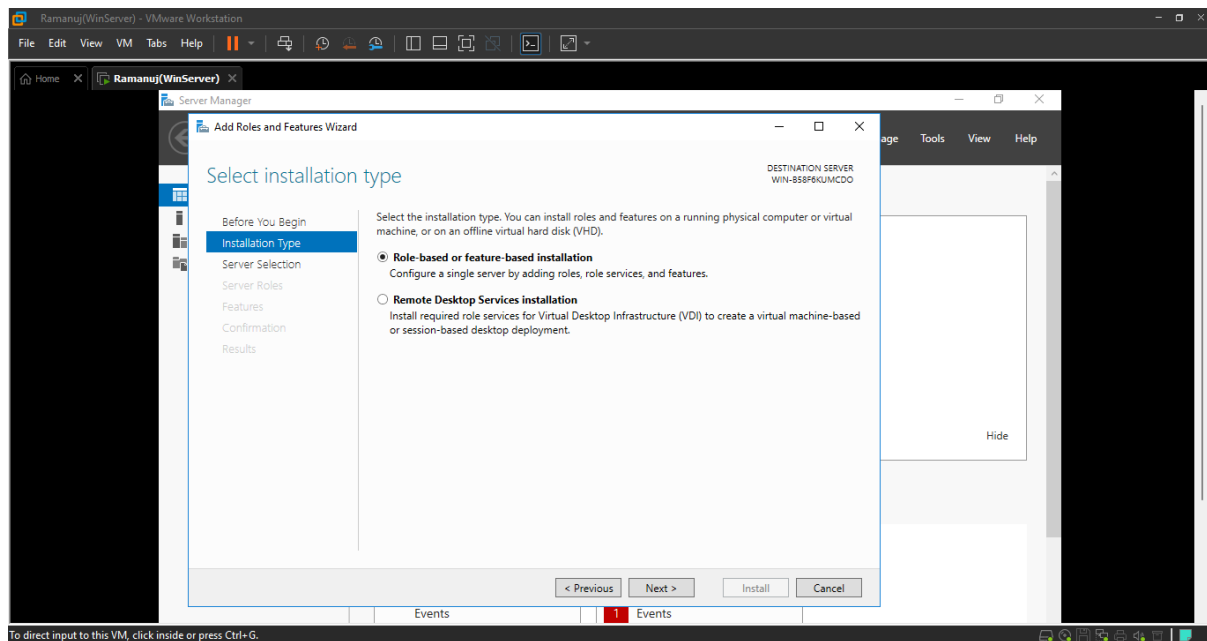
- Within Server Manager Click on **Add Roles and Features**



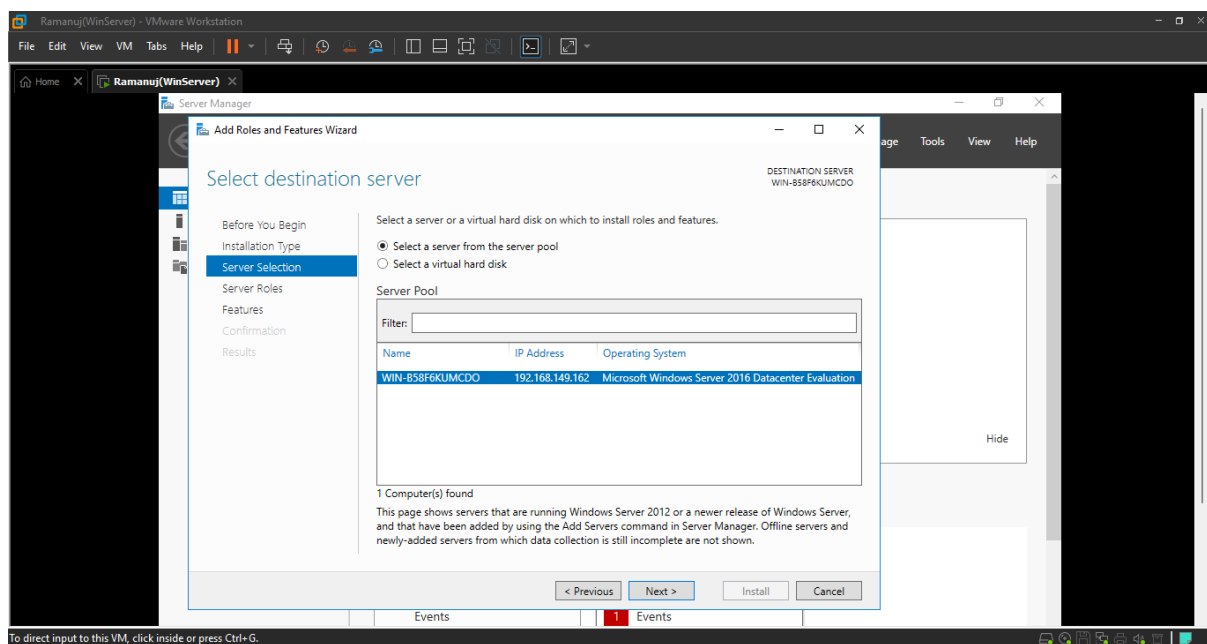
- Click Next



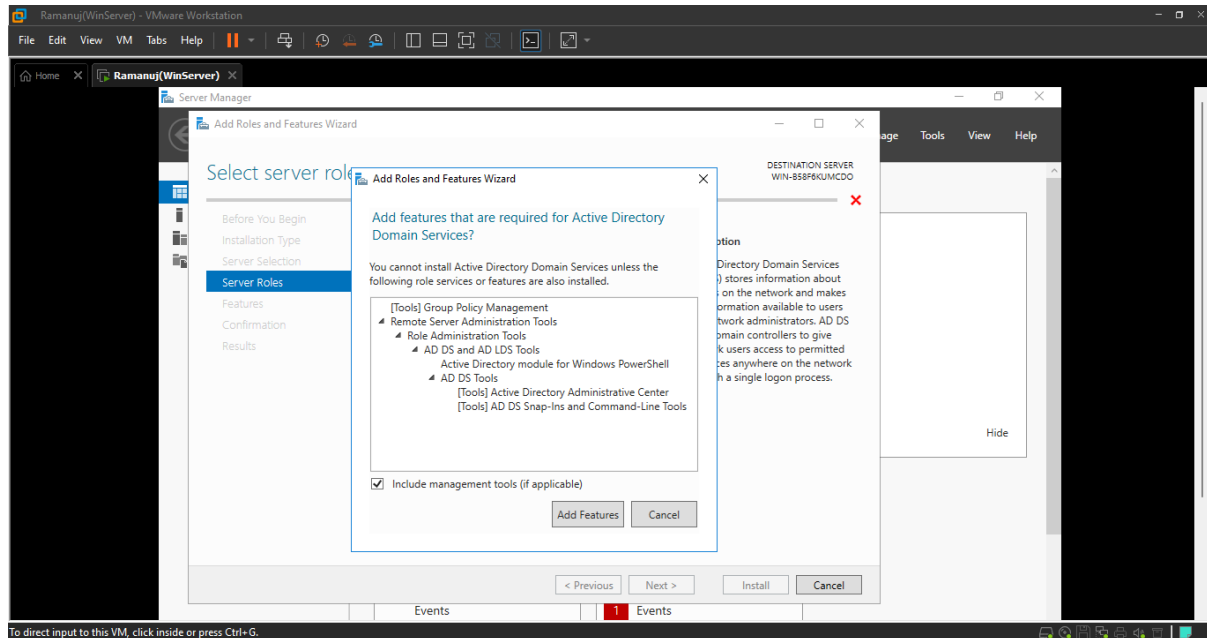
- Make sure **Role-based or Feature-based installation** is selected and Click **Next**



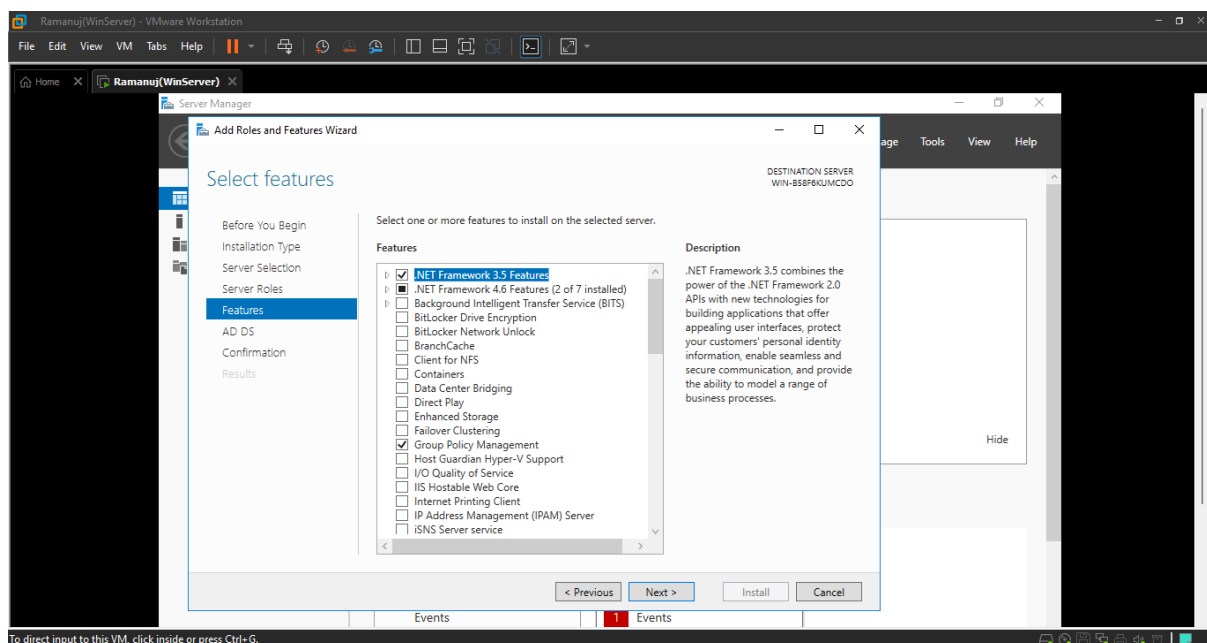
- Keep default settings and Click **Next**



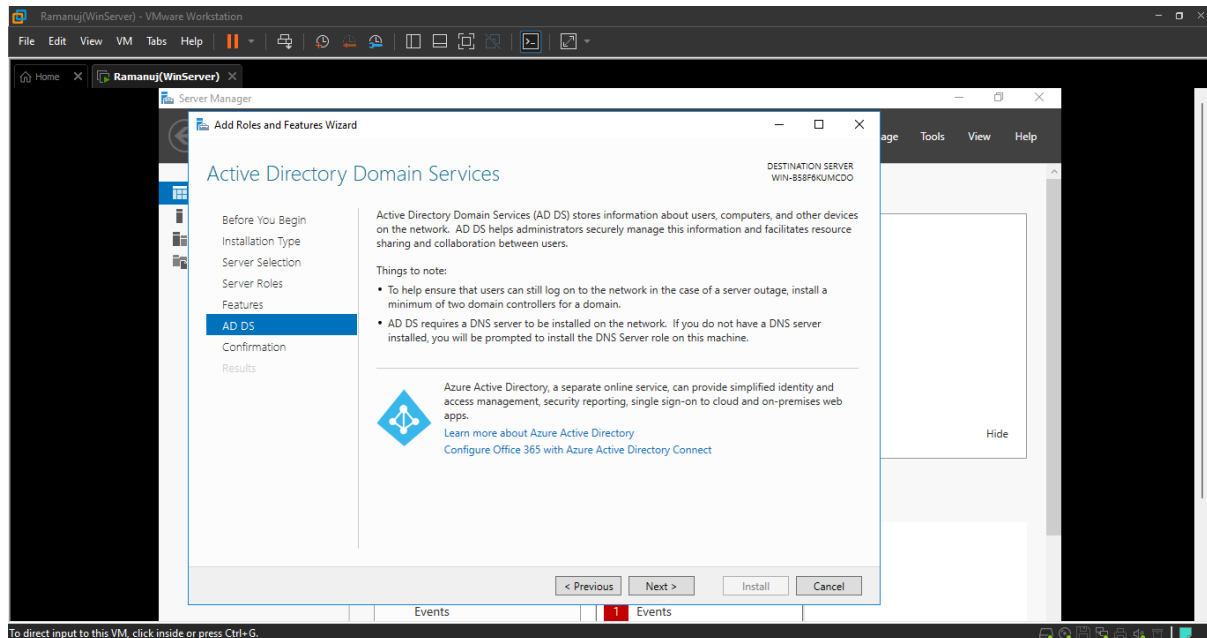
- In Server Roles Click on **Active Directory Domain Services**, a pop-up will appear Click **Add Features**



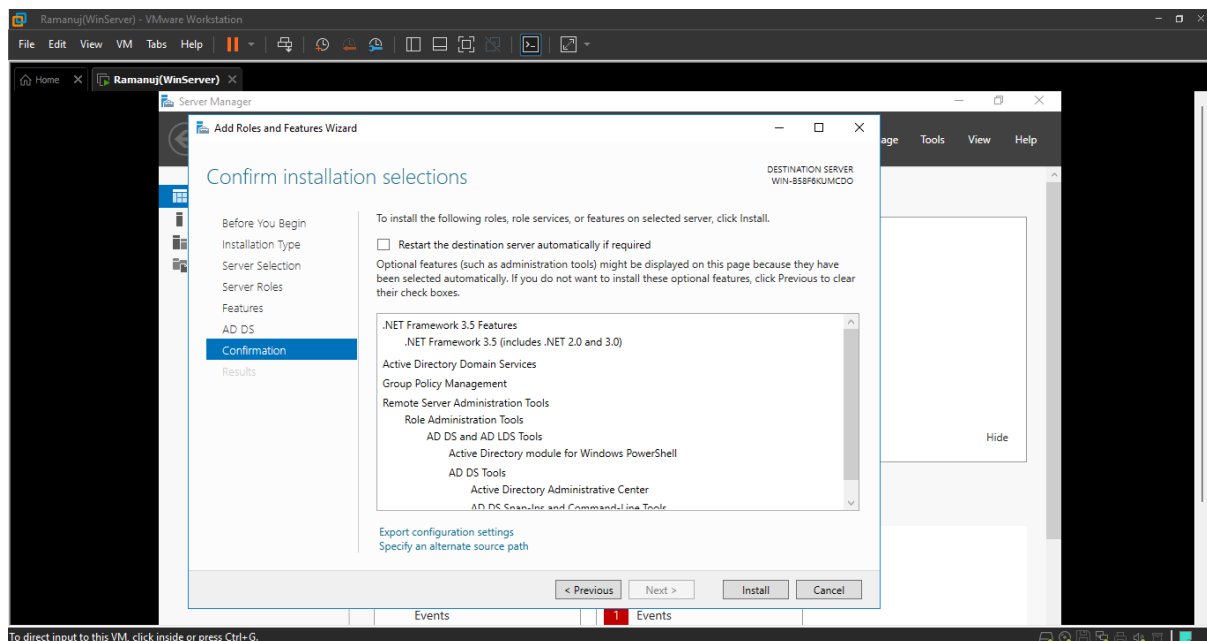
- In Features Select **.NET Framework 3.5 Features** and Click **Next**



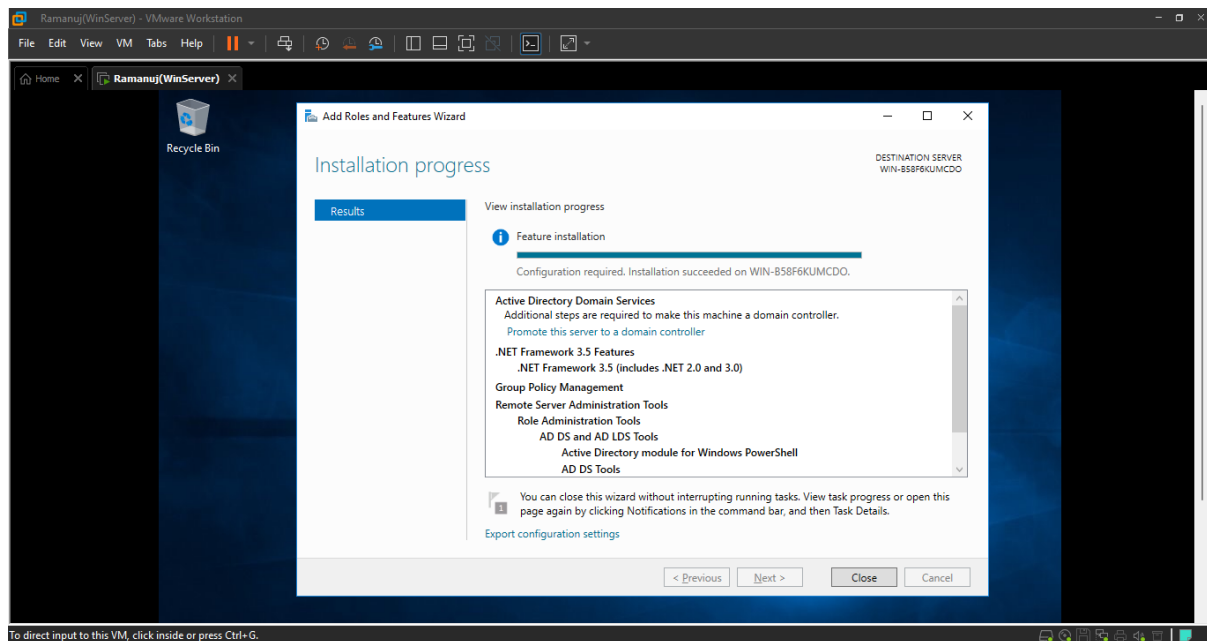
- Click **Next**



- Click on **Install**

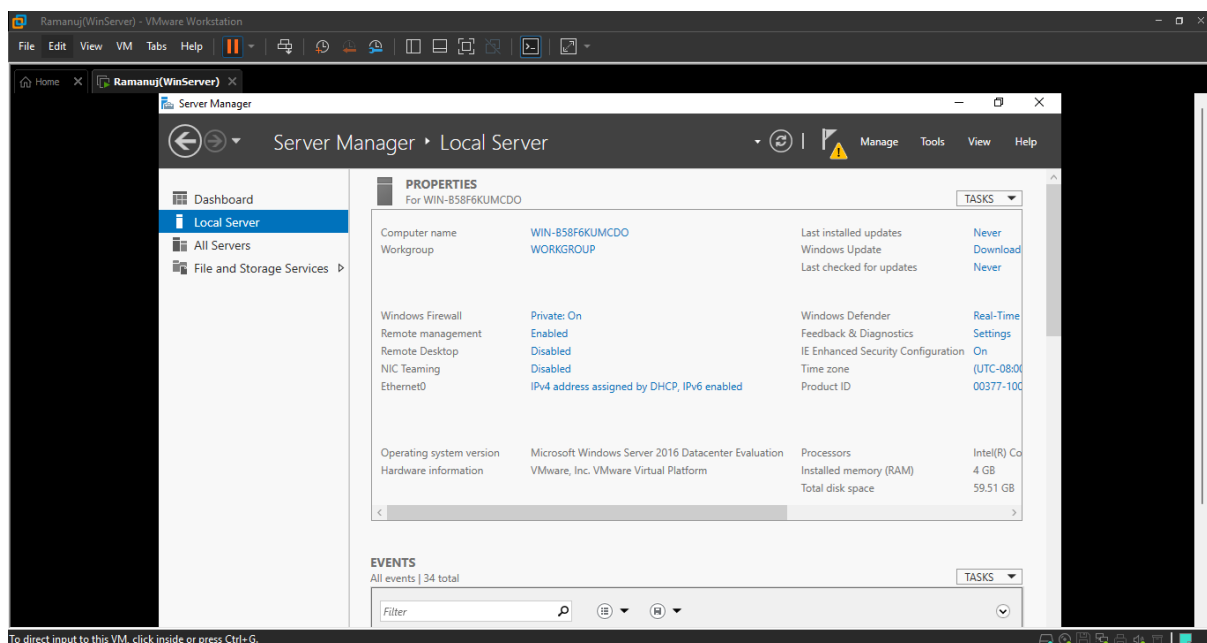


- Features will start installing, will have to restart in the middle to complete the installation

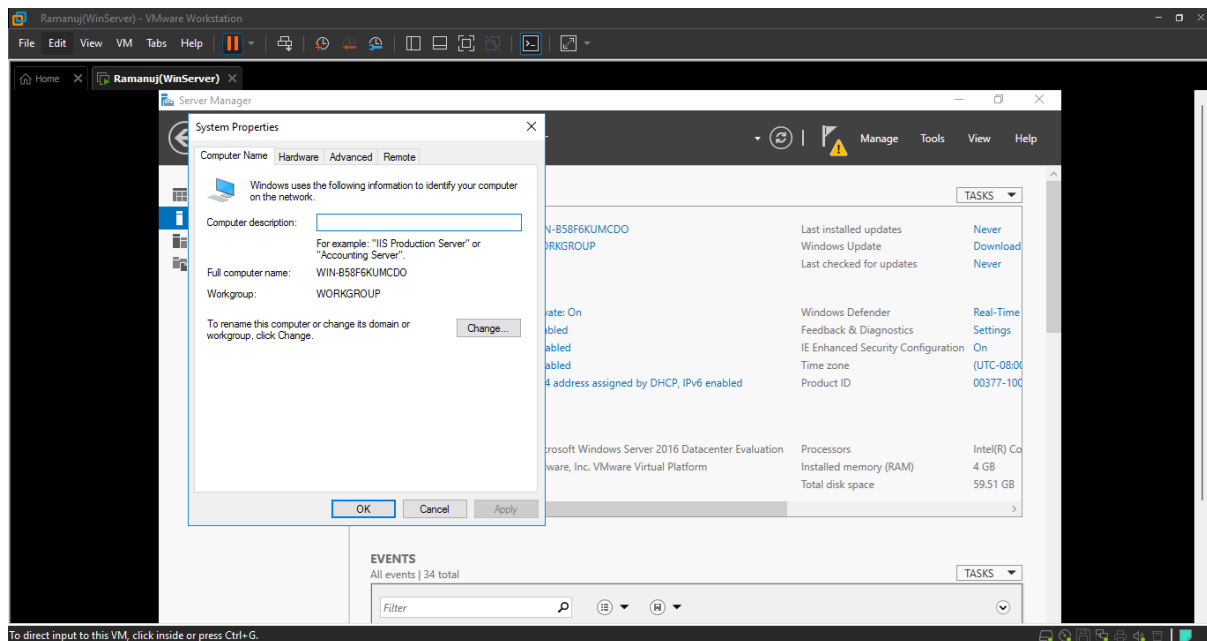


Step 31: Change Server Name

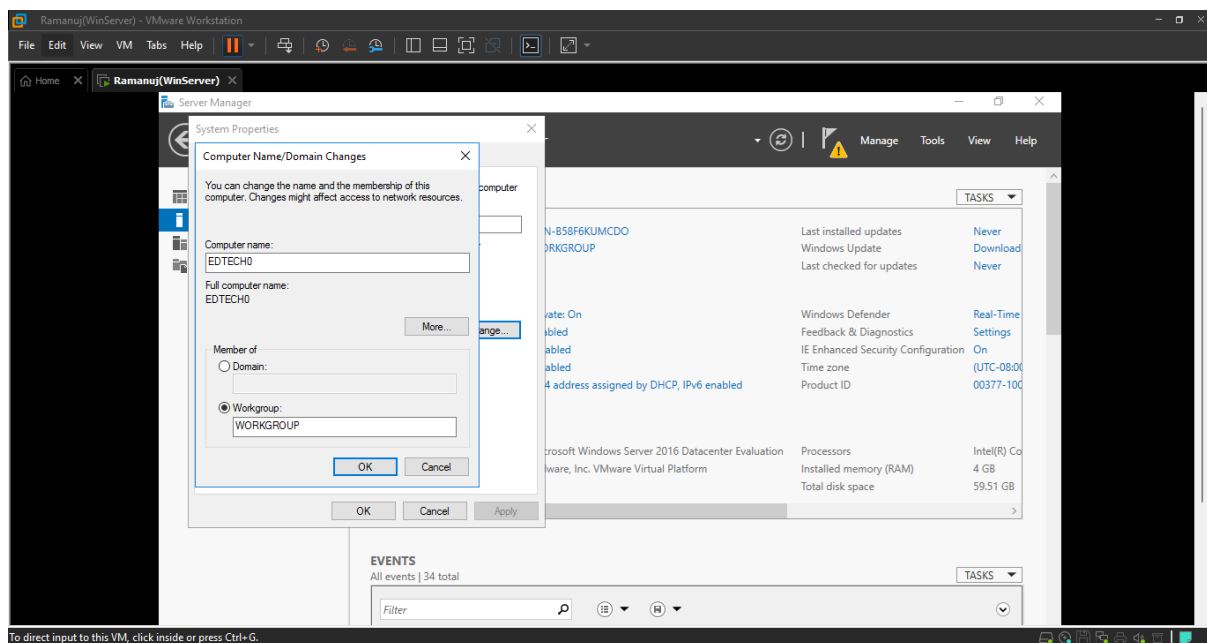
- Open Server Manager and click on **Local Server**



- Click on Computer name and Click **Change**

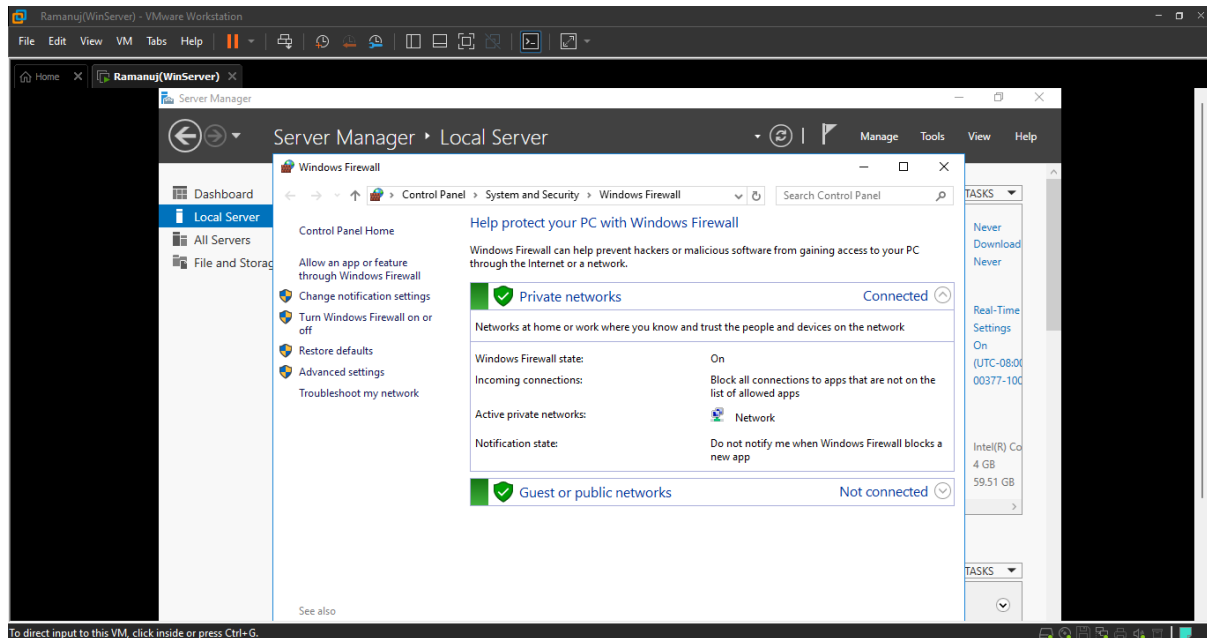


- Change the computer to name to any value (Here it is EDTECH0) and click **OK**, This change will require a restart.

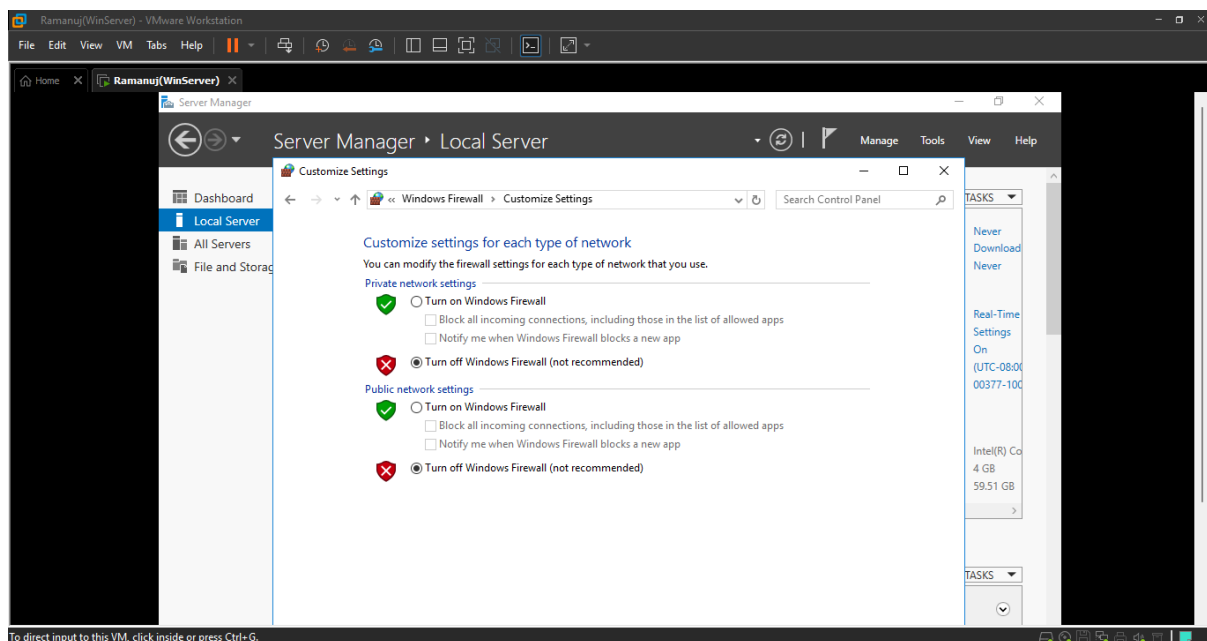


Step 32: Disable Firewall

- Within Server Manager Click on **Local Server** and Click on **Firewall**

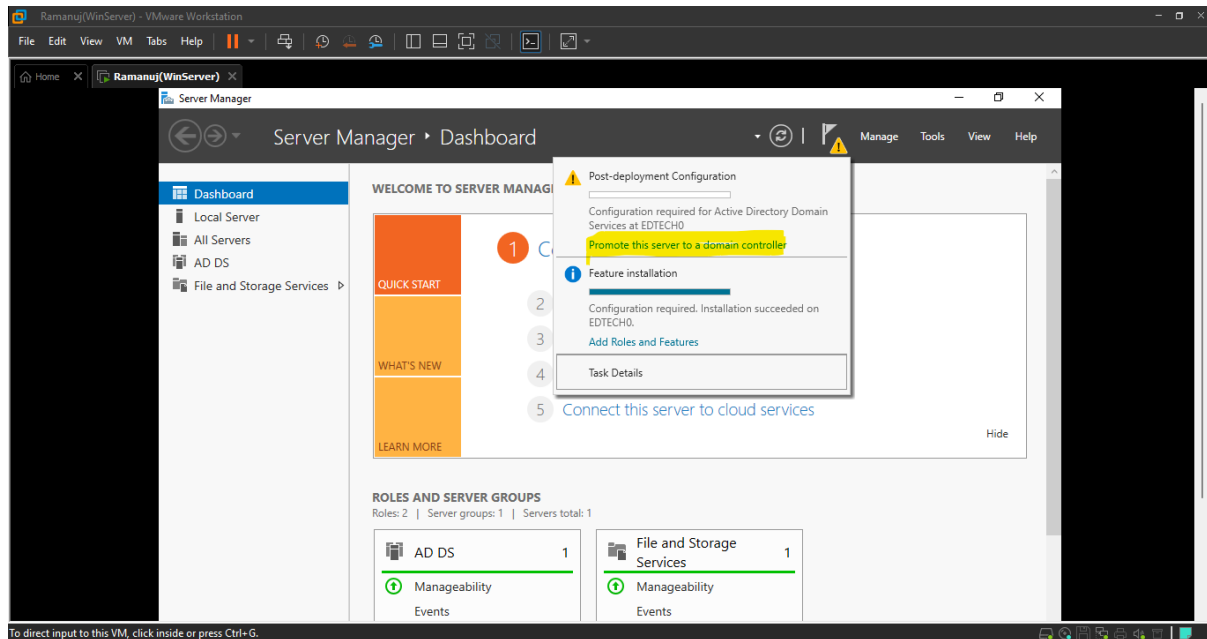


- Click **Turn Windows Firewall On or Off** and Select **Turn off windows Firewall** for both Public and Private. Then Click OK

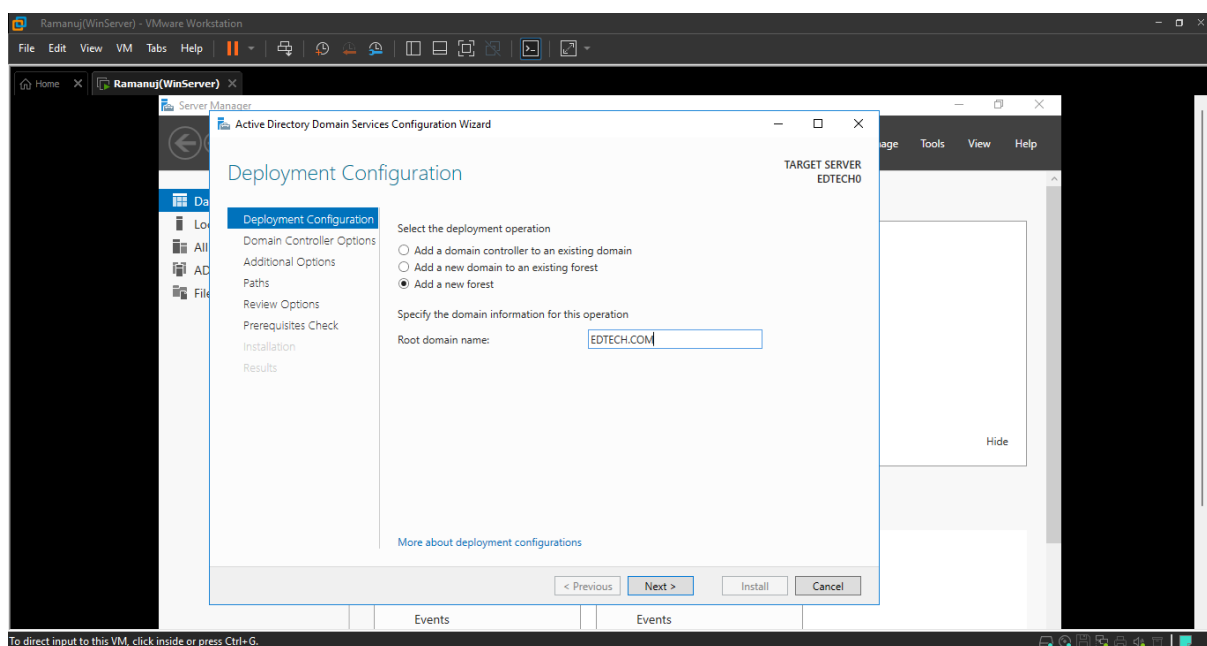


Step 33: Domain Controller Creation

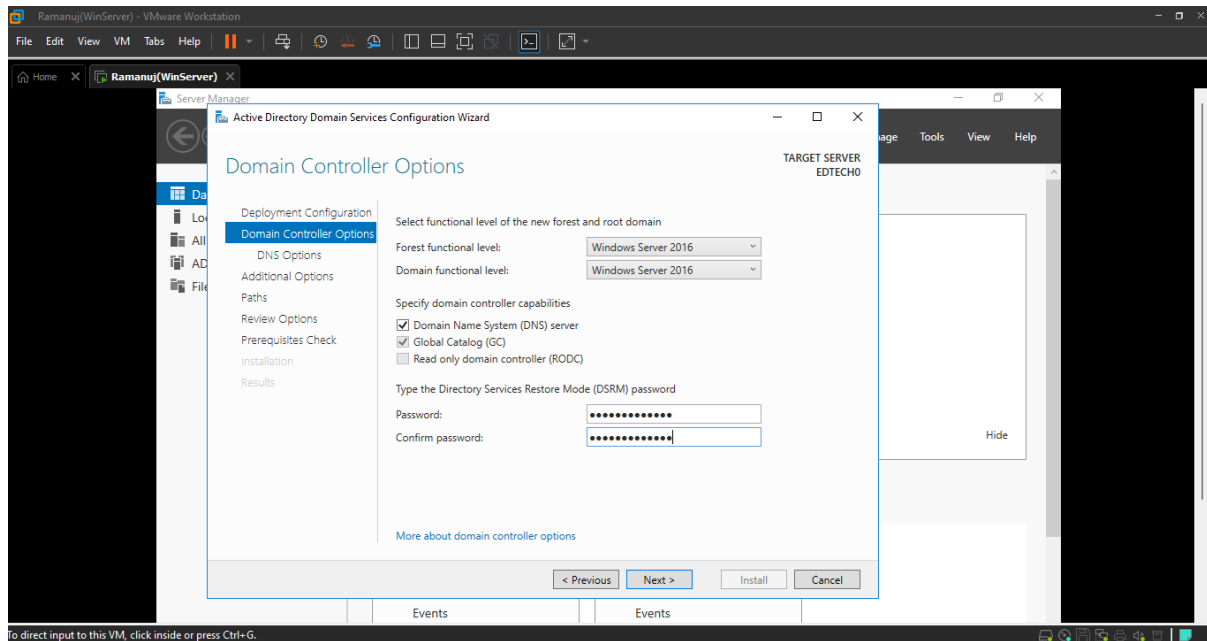
- On the Dashboard of the Server Manager Click on the **Warning Flag** on the top right and Click on **Promote this server to Domain controller**



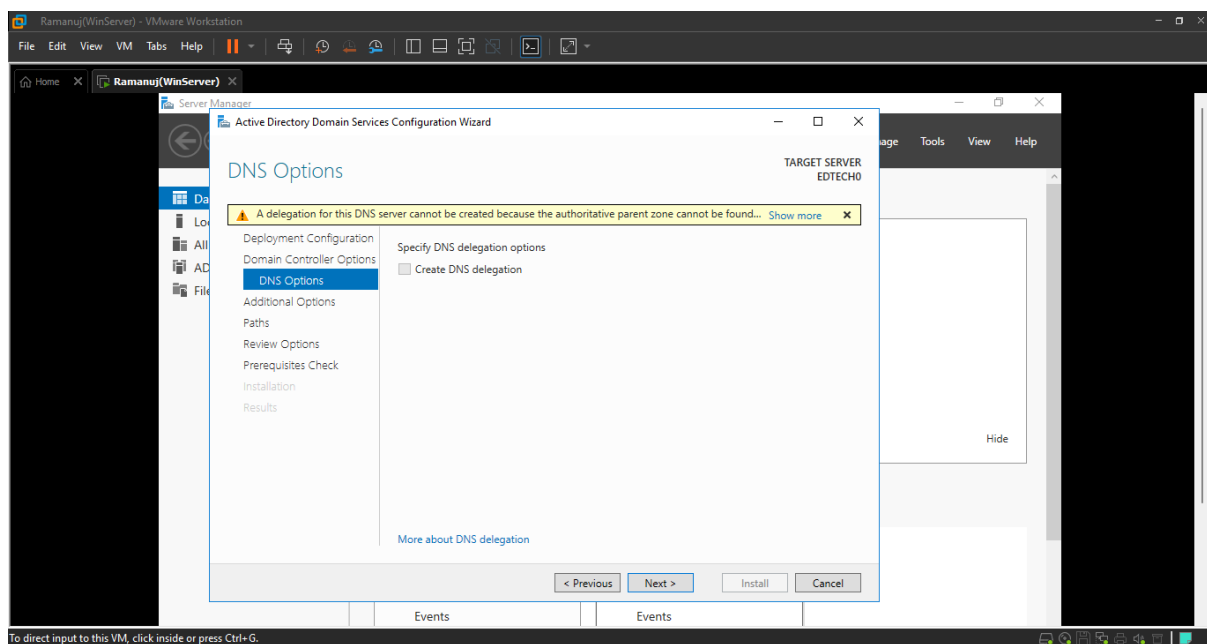
- This opens the Active Directory Domain Services Configuration Wizard, Select **Add a forest** option, enter a name for the new domain (here it is EDTECH.COM) and Click **Next**



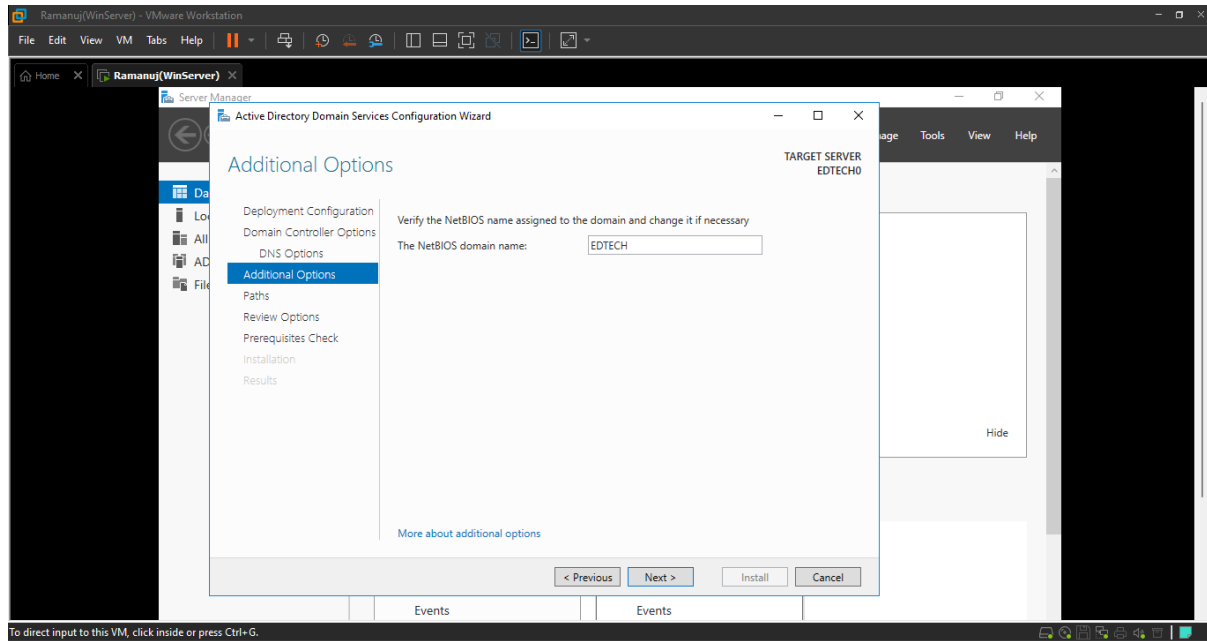
- Enter a password for the domain (keep it same as the administrator password). Click on **Next**



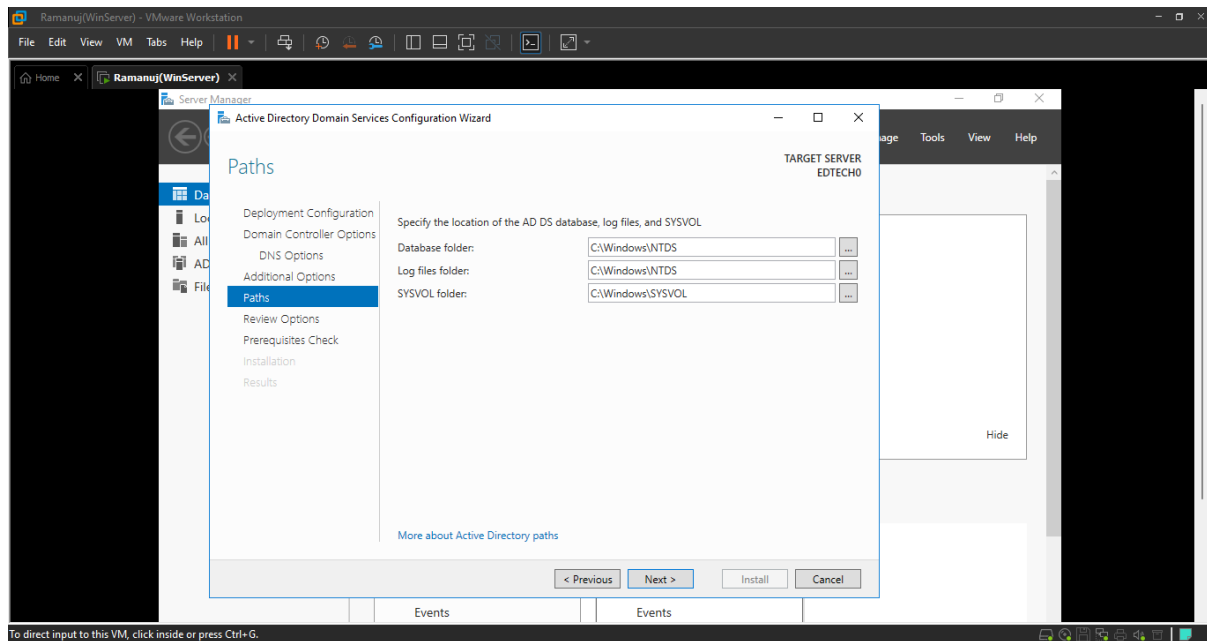
- Keep the default options and Click on **Next**



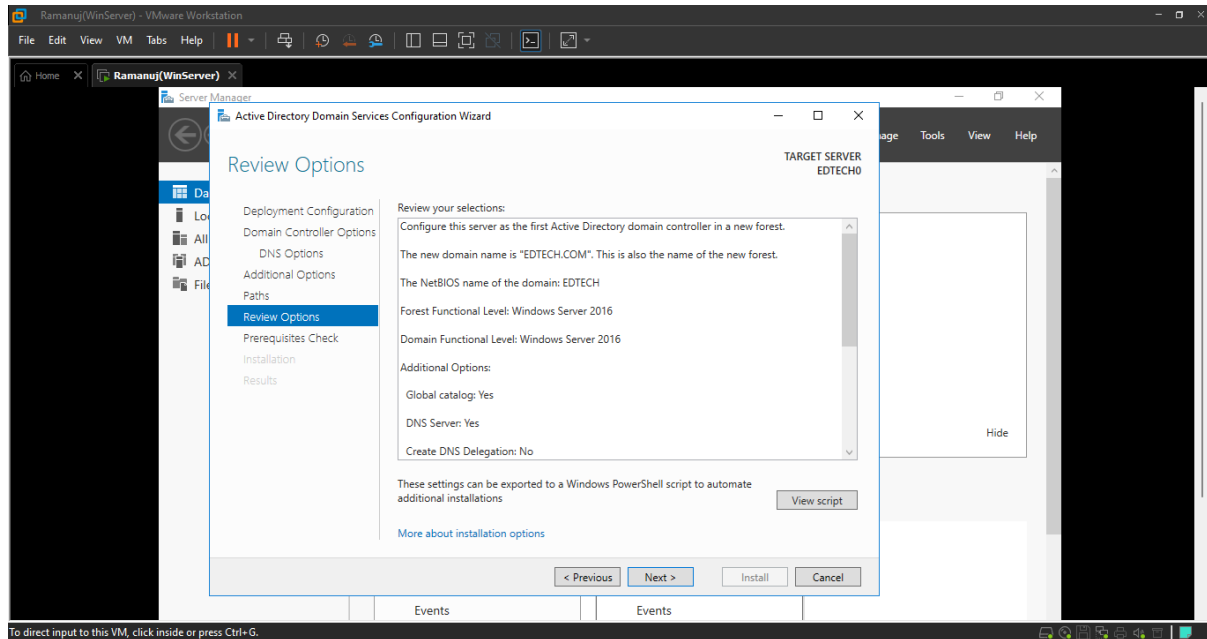
- Keep the default options and Click on **Next**



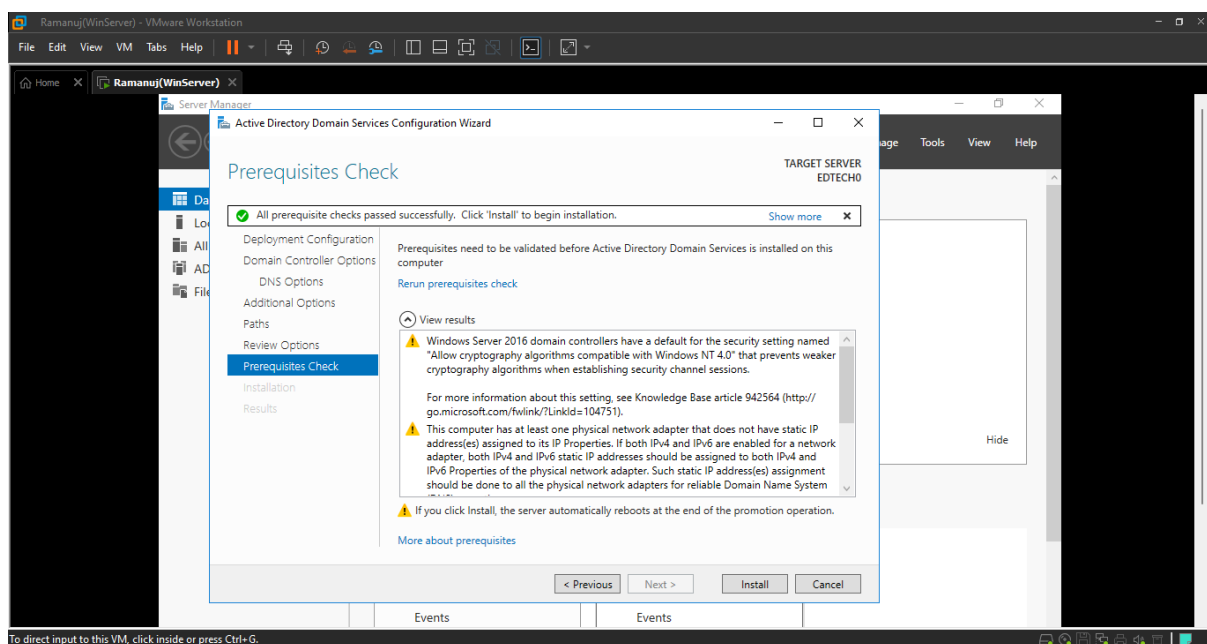
- Keep the default options and Click on **Next**



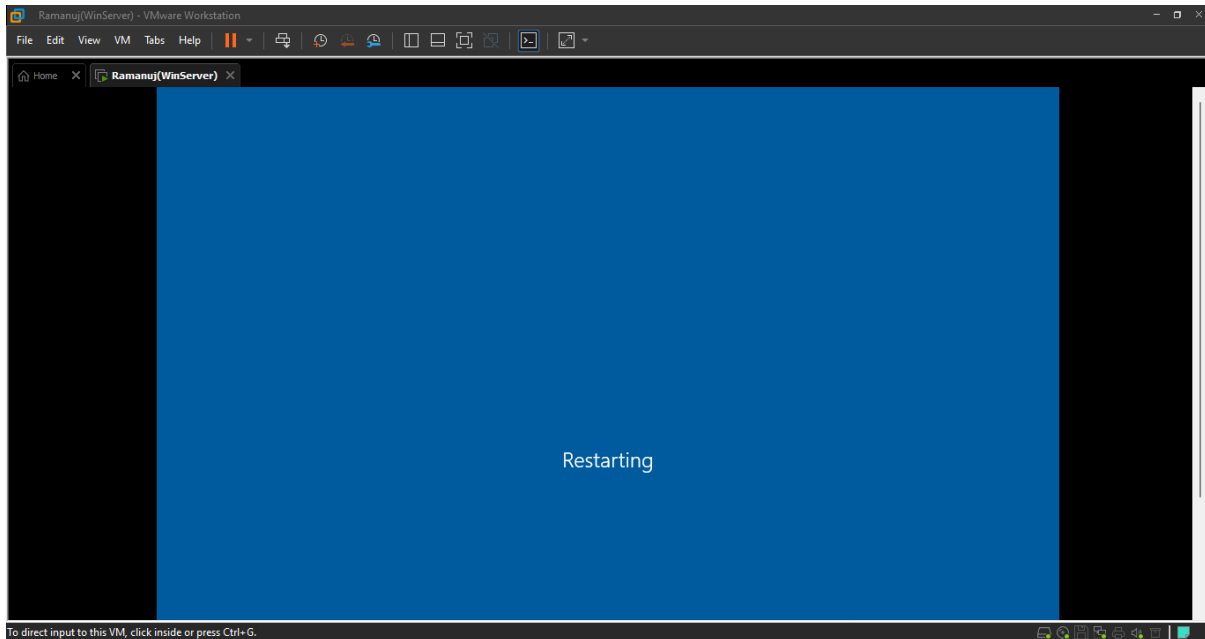
- Keep the default options and Click on **Next**



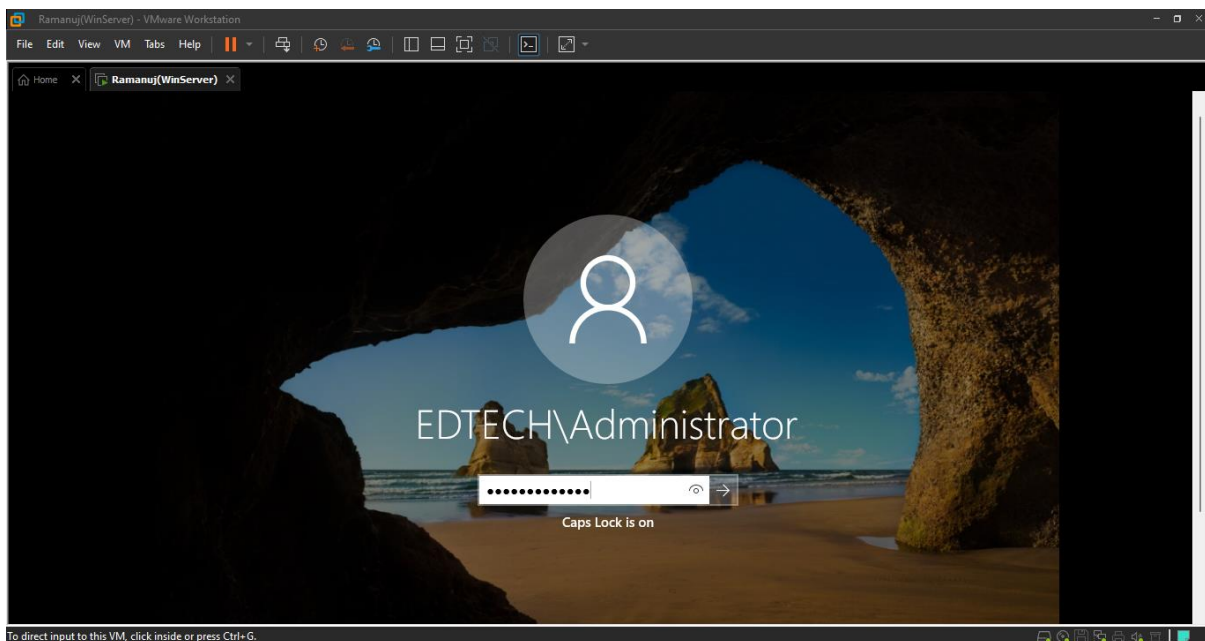
- It checks for pre-requisites. It should pass all the pre-requisites check (Warnings are acceptable) and Click on **Install**



- After Completion of the installation the VM will restart for the changes to take effect

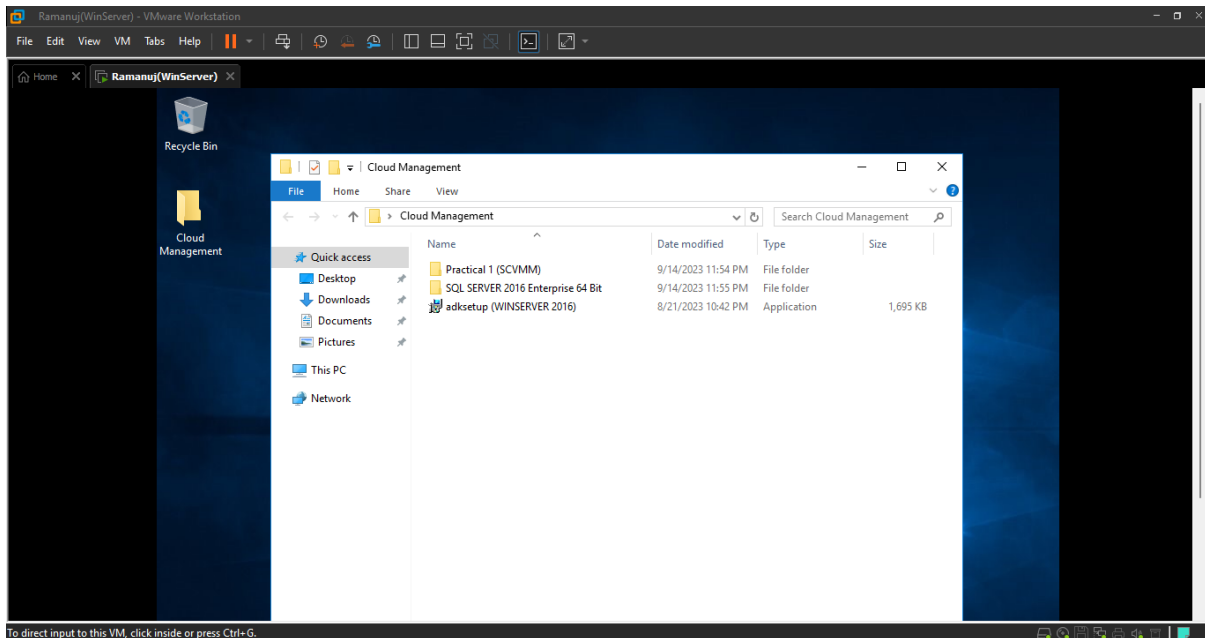


- Now the VM will be signed in as the Domain Controller

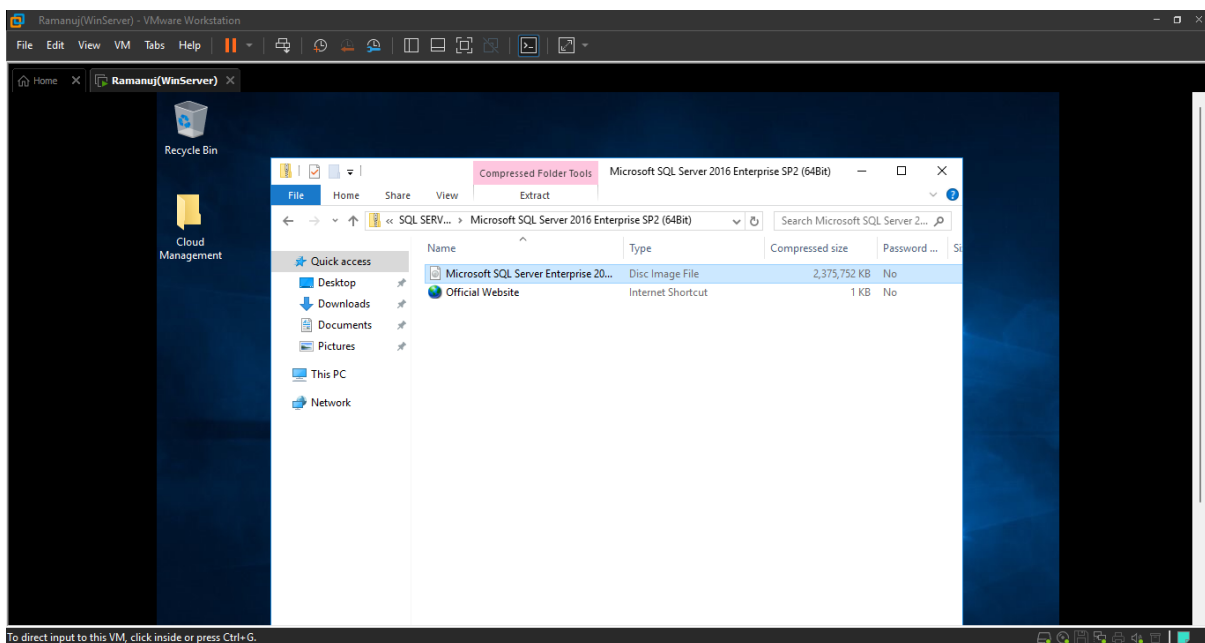


Step 34: Installing and configuring SQL Server

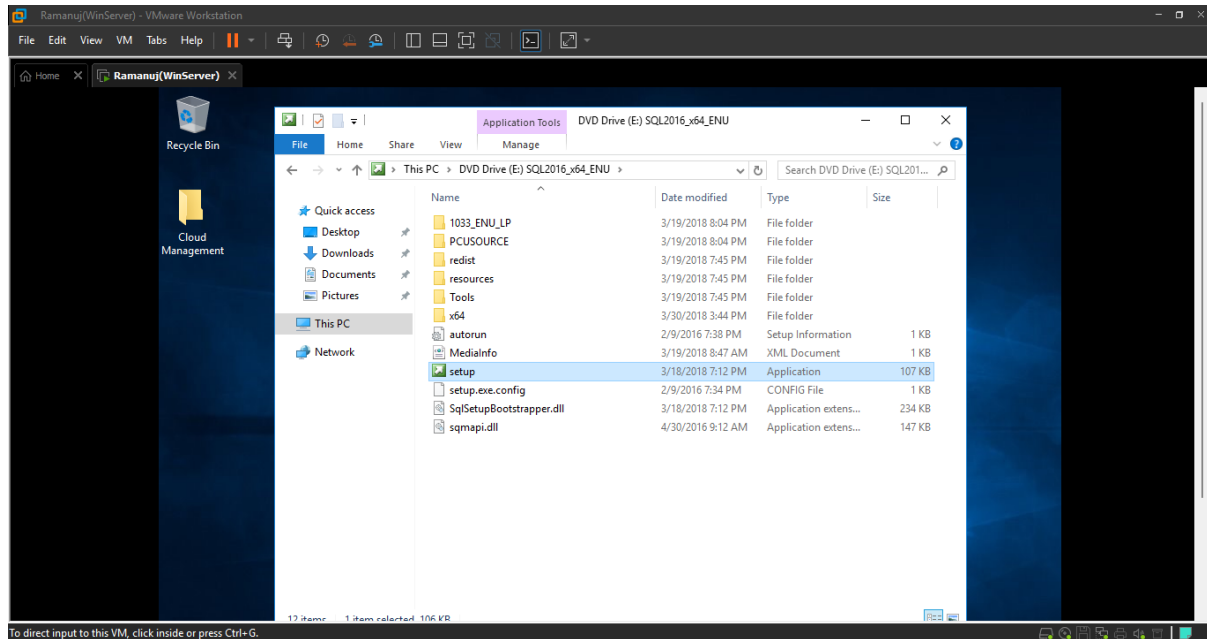
- Drag and Drop the required files for the practical into your VM



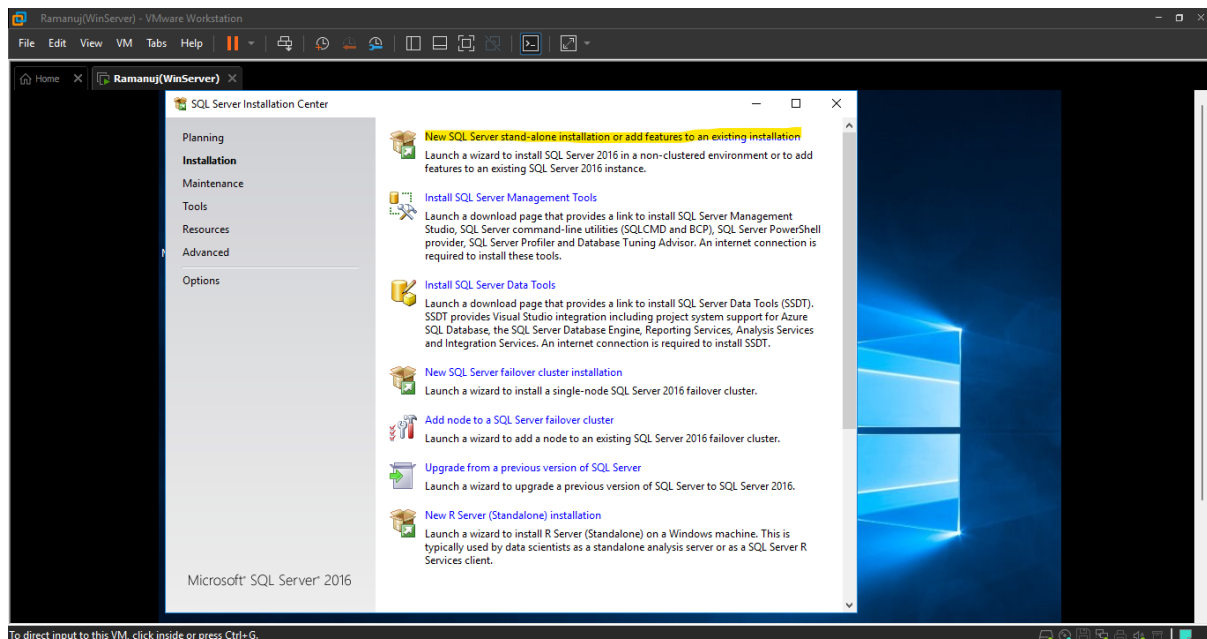
- Mount the SQL Server iso file (Either Double-click the iso file or Right-Click on the iso and Select **Mount**)



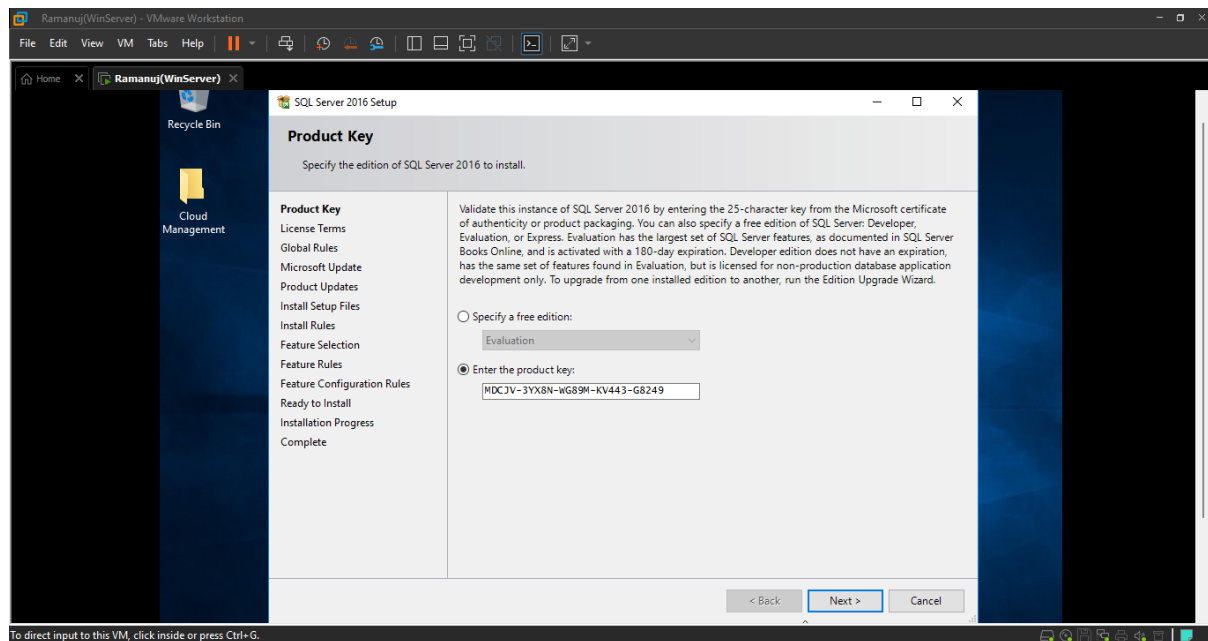
- Click on **Setup.exe**



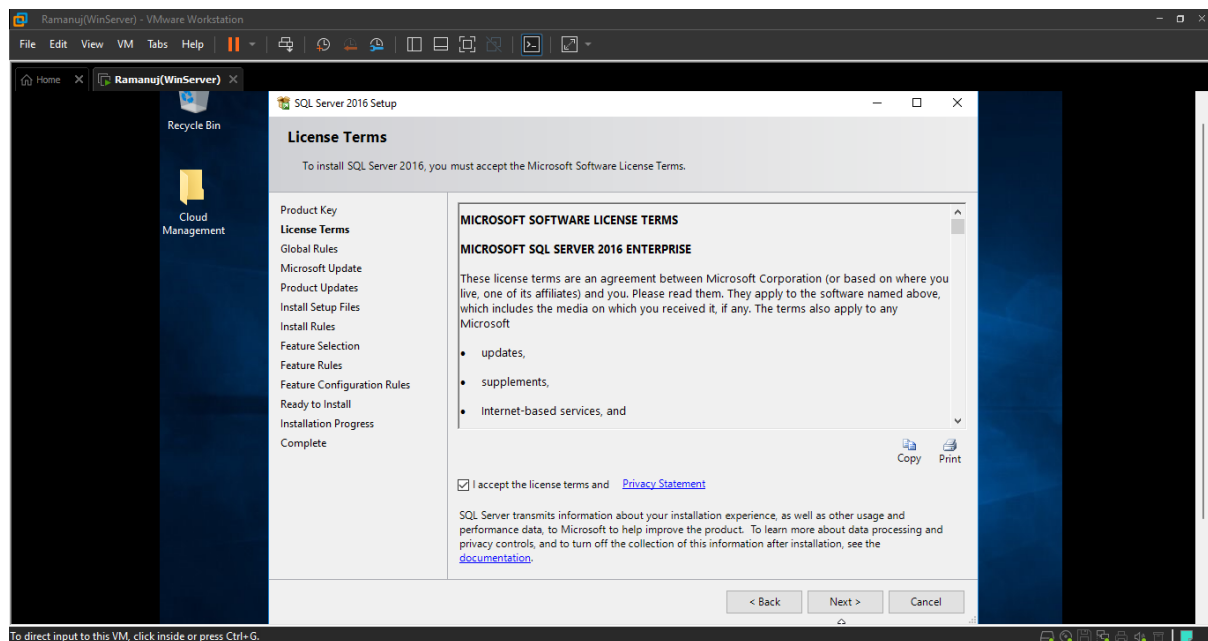
- The SQL Server Installation Center will open, Click on **Installation**
- Within Installation Click on **New SQL Server stand-alone installation or add features to an existing installation**



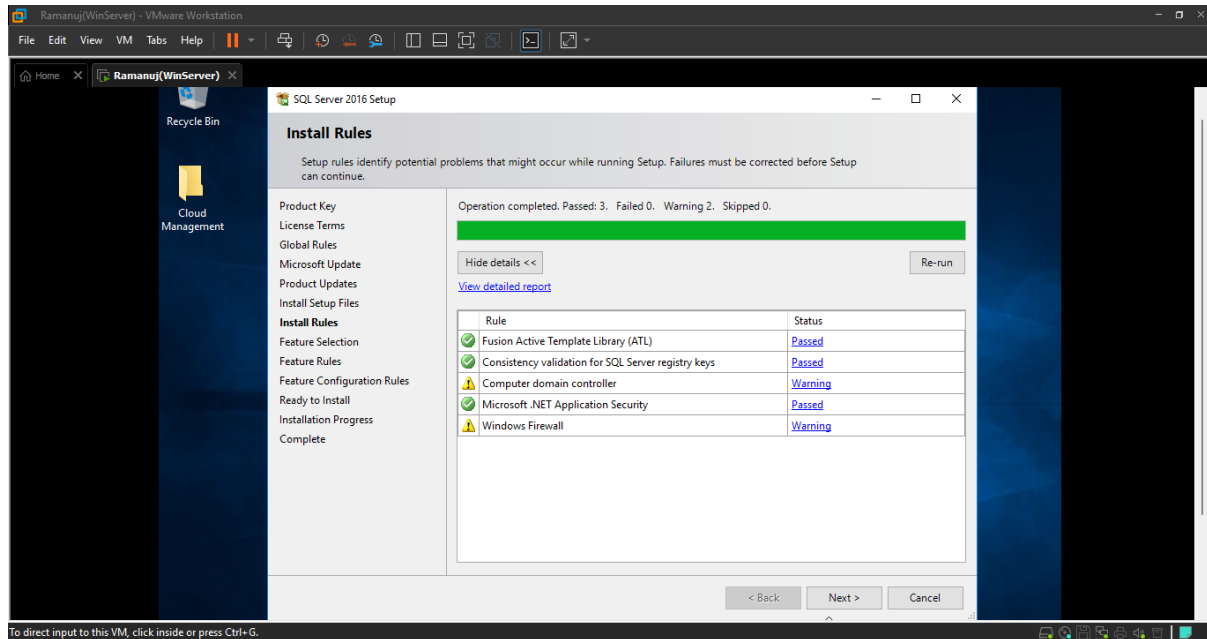
- Make sure Enter Product Key is selected and the default value is entered, Click **Next**



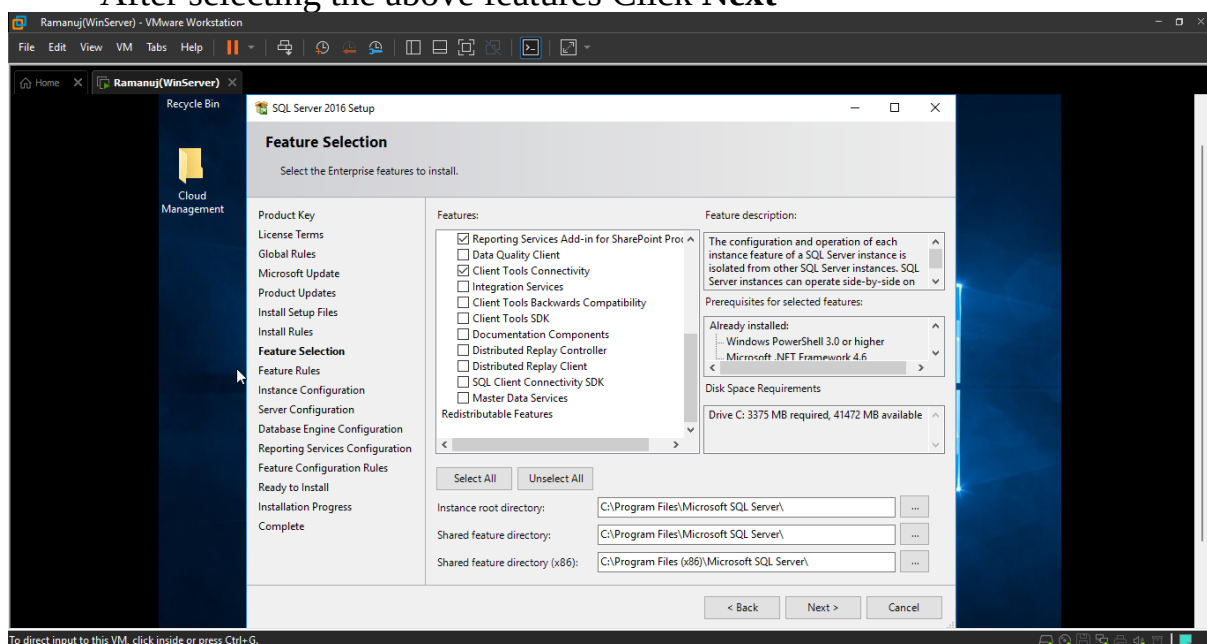
- Select **Accept license and terms** and Click **Next**



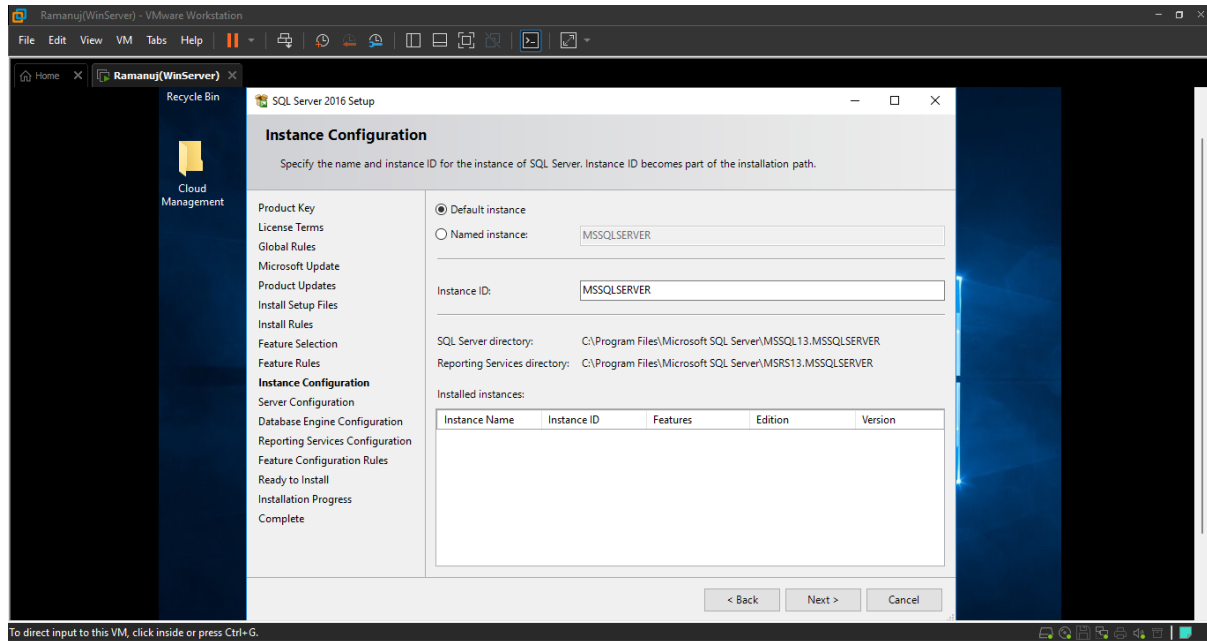
- Click Next on Microsoft Updates and Install Rules which should pass with only a few warnings



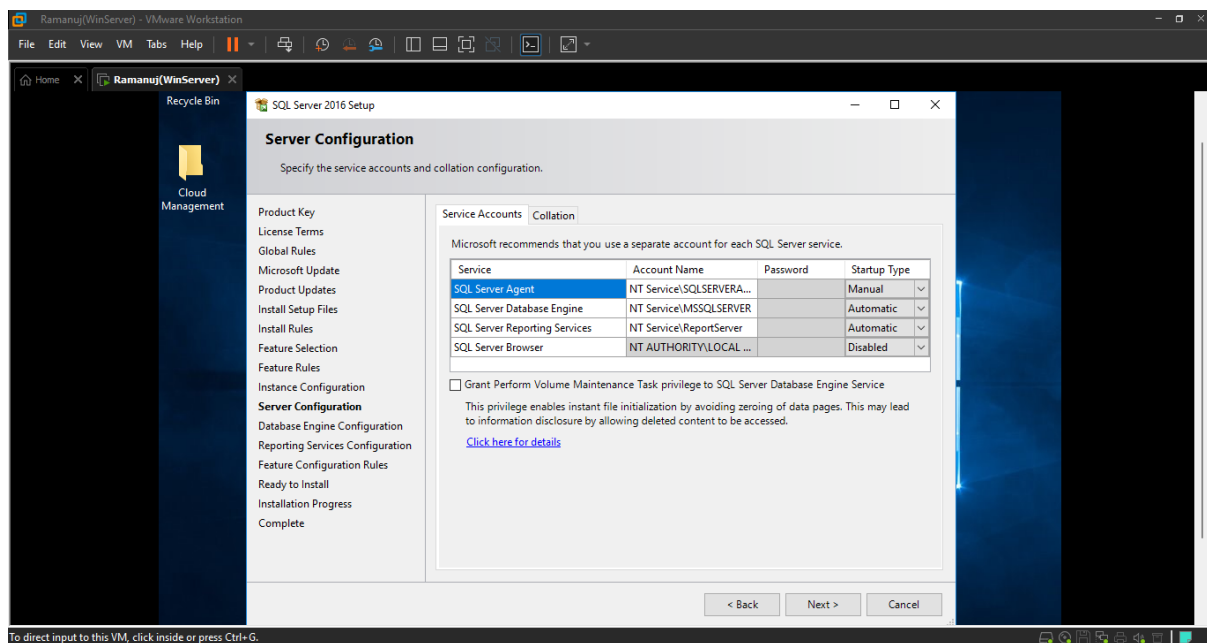
- In Feature Selection select the features given below:
 - Database Engine Services
 - Reporting Services – Native
 - Reporting Services SharePoint
 - Reporting Services Add-in for SharePoint products
 - Client Tools Connectivity
 - Management Tools – Basic (If not there its ok)
 - Management Tools – Complete (If not there its ok)
- After selecting the above features Click **Next**



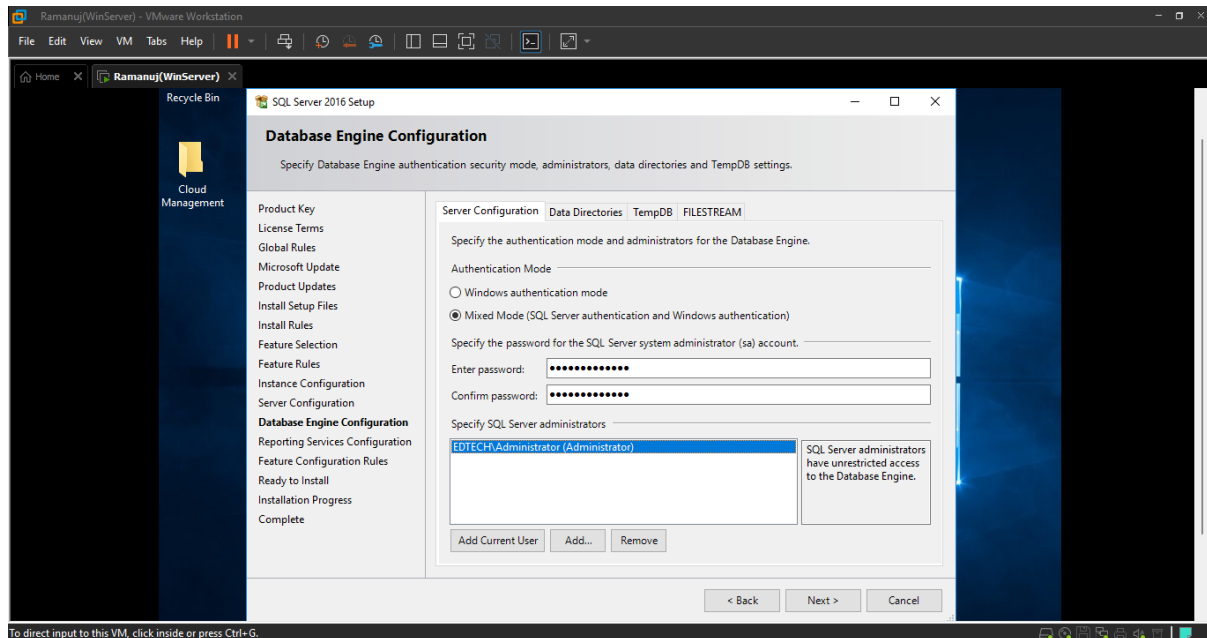
- Select Default instance and Click **Next**



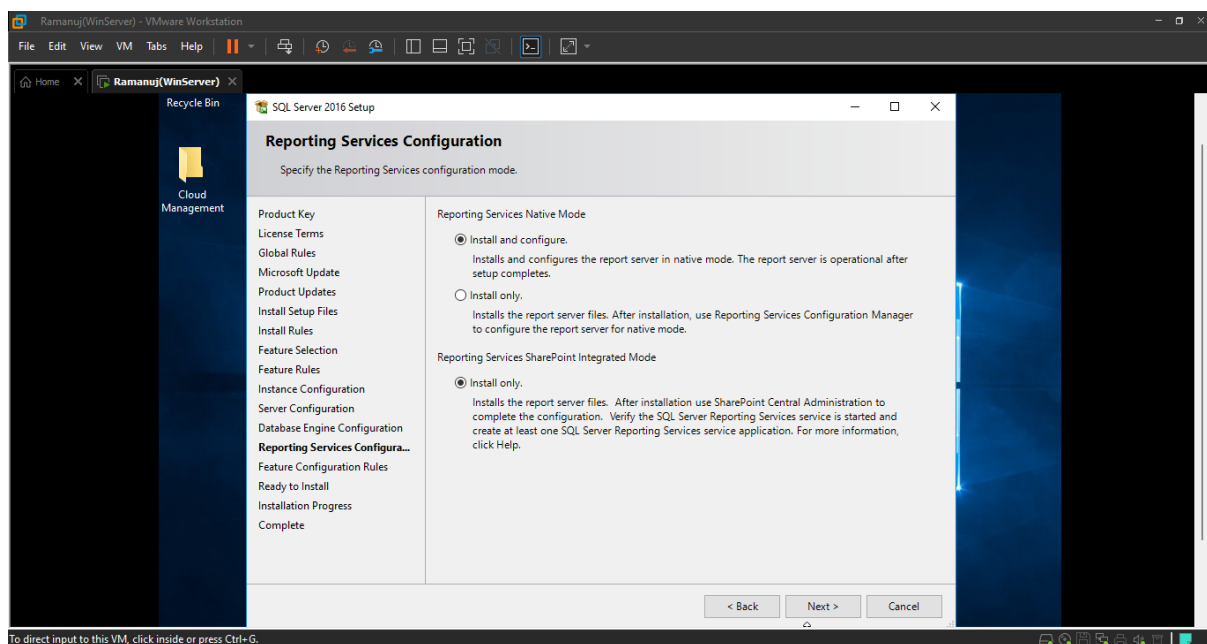
- Keep default values and Click **Next**



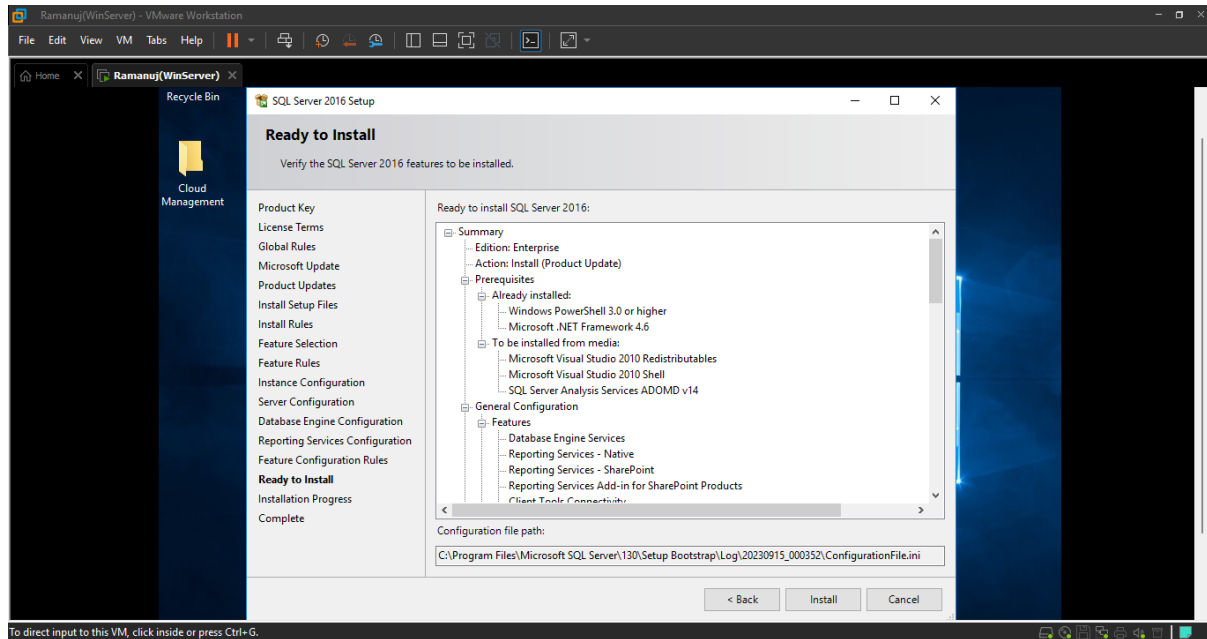
- In Database Engine Configuration, Select **Mixed Mode** and enter the same password as set while creating the VM
- Click on **Add Current User**, it should display the current Domain User and Click **Ok**



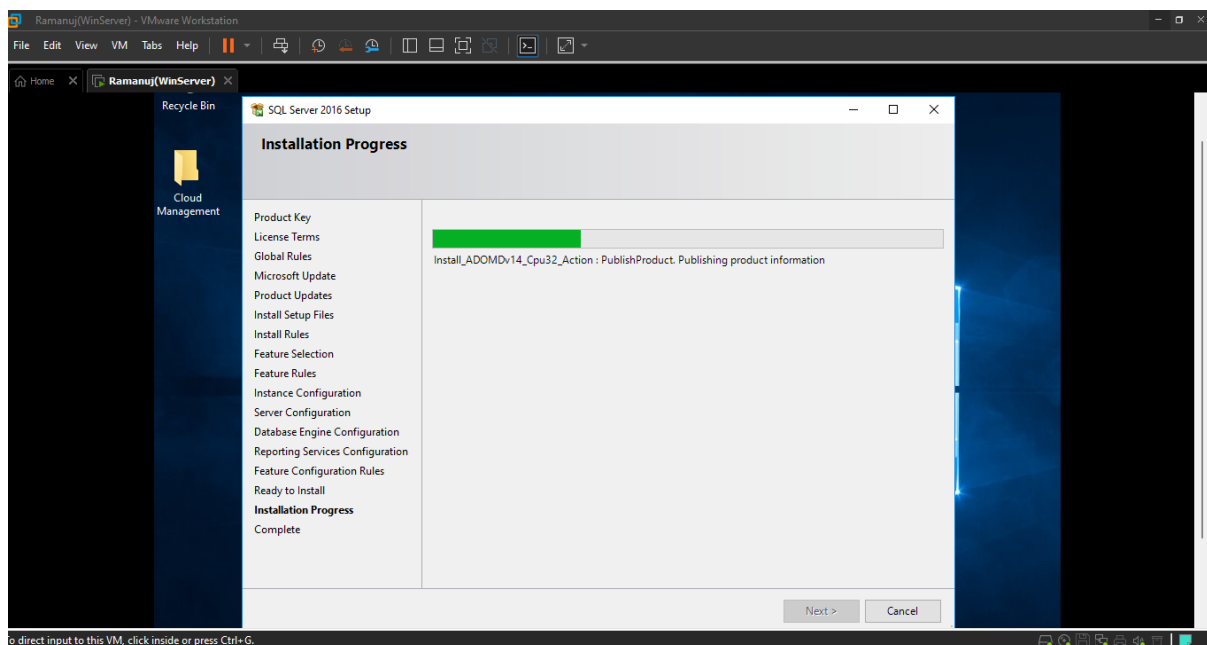
- Keep default values and Click **Next**



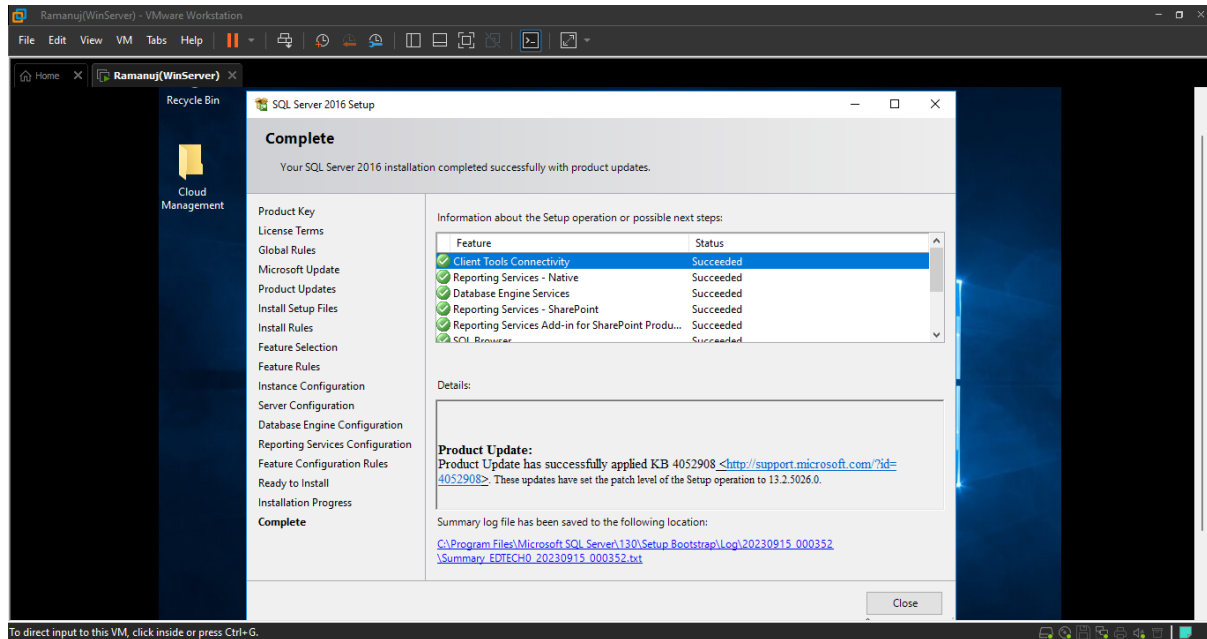
- Click on **Install**



- It will start installing

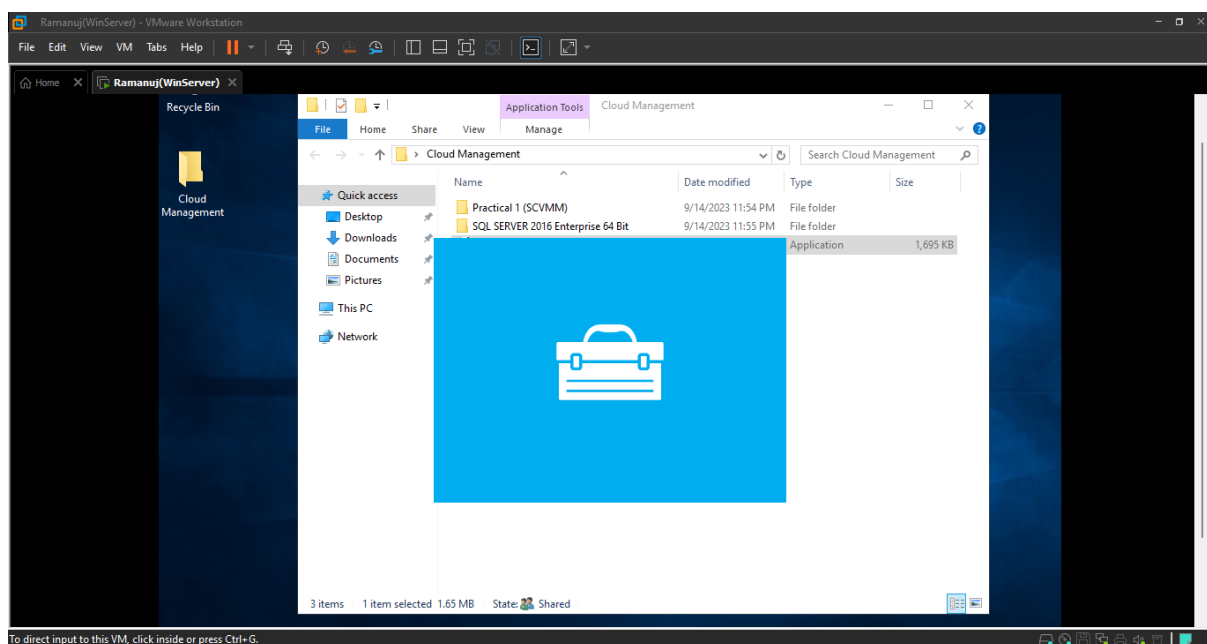


- It should show Installation succeeded

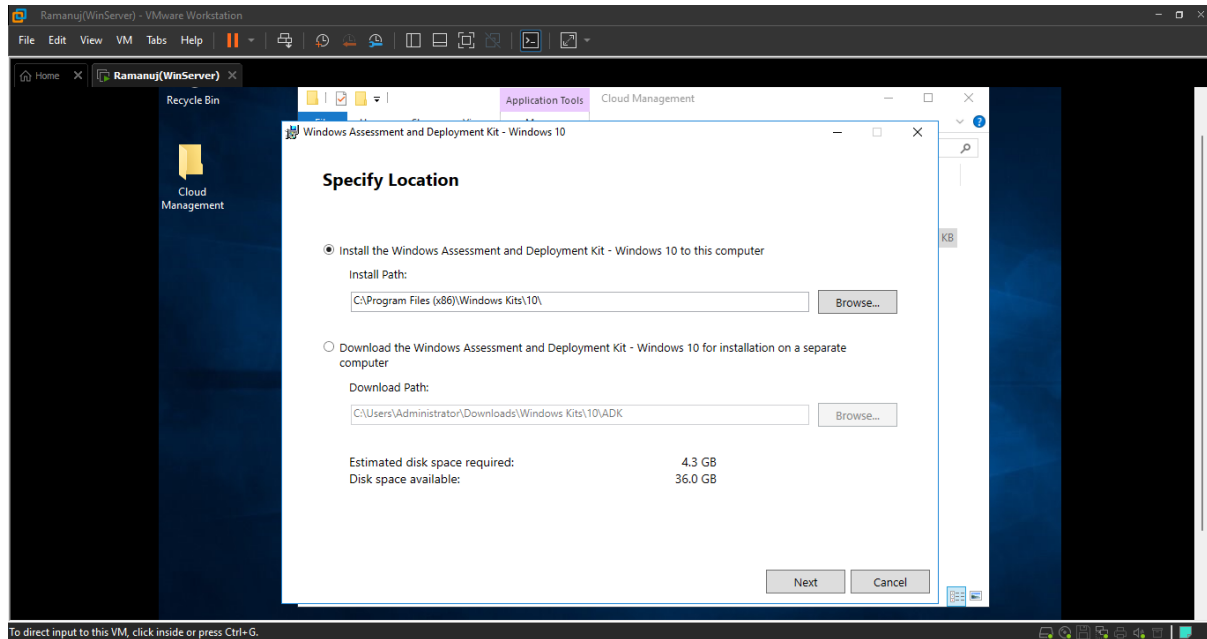


Step 35: Installing ADK-setup 2016 (Windows Toolkit)

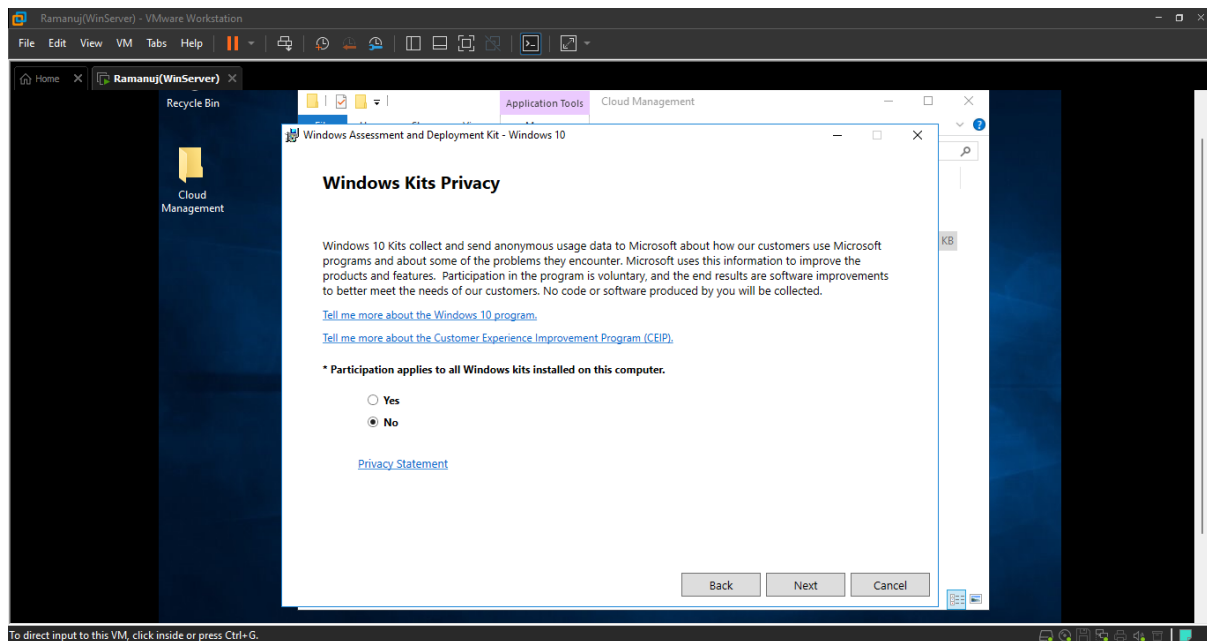
- run the ADK setup from the imported files in your VM



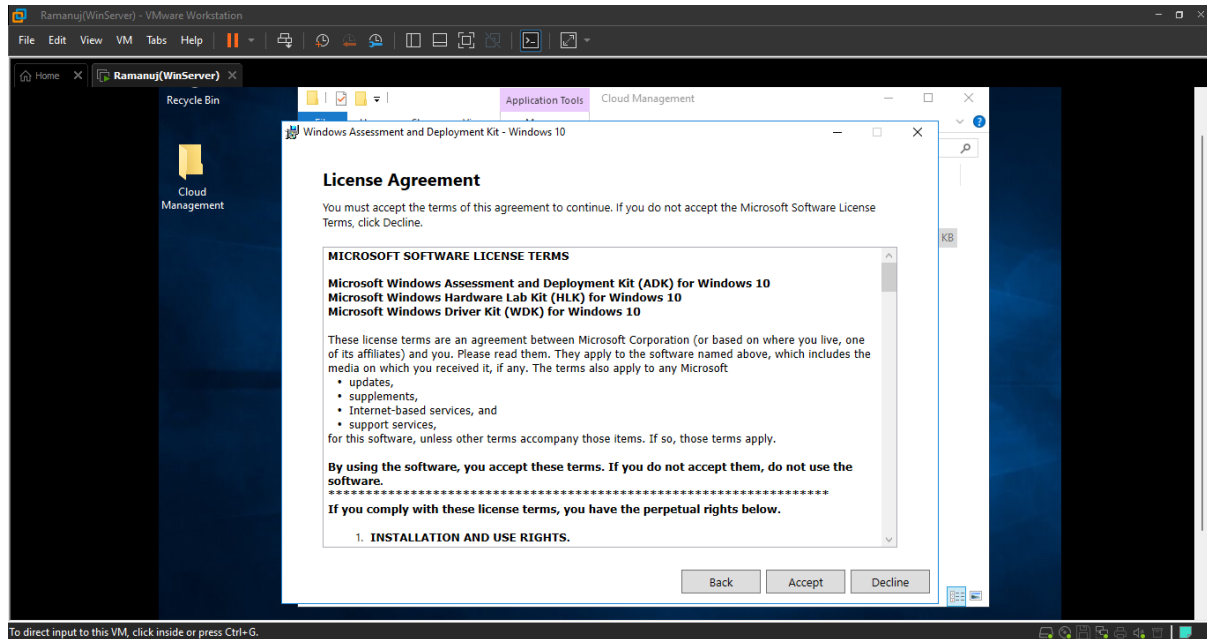
- Keep Default and Click **Next**



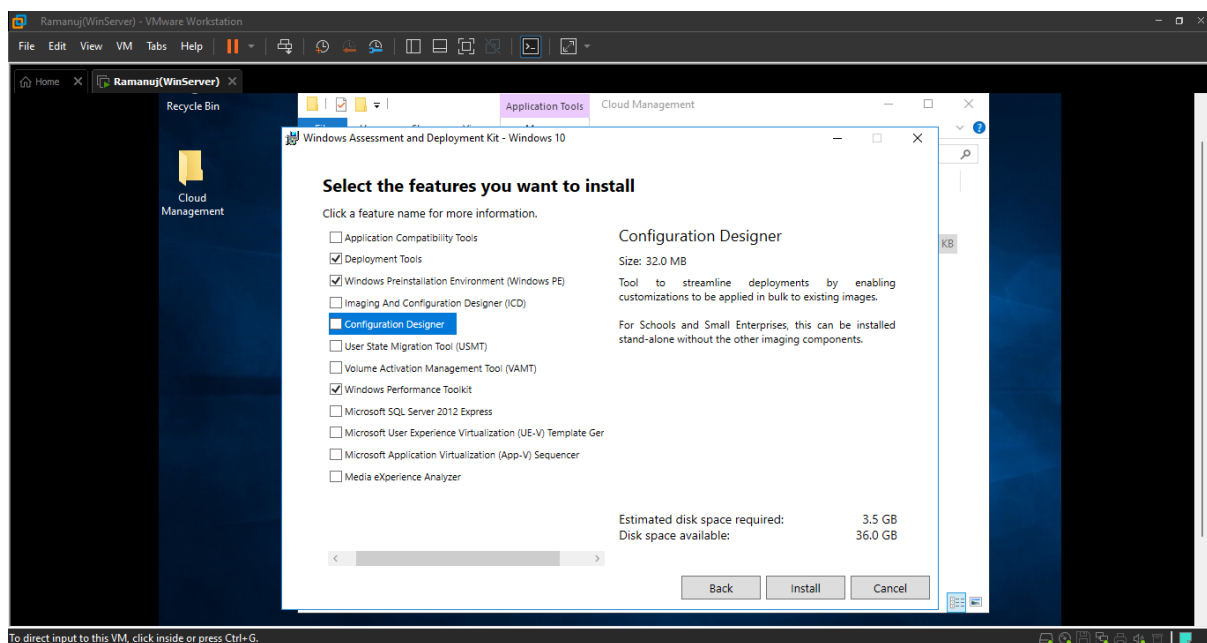
- Select **No** for participation in Windows Kit Privacy and Click **Next**



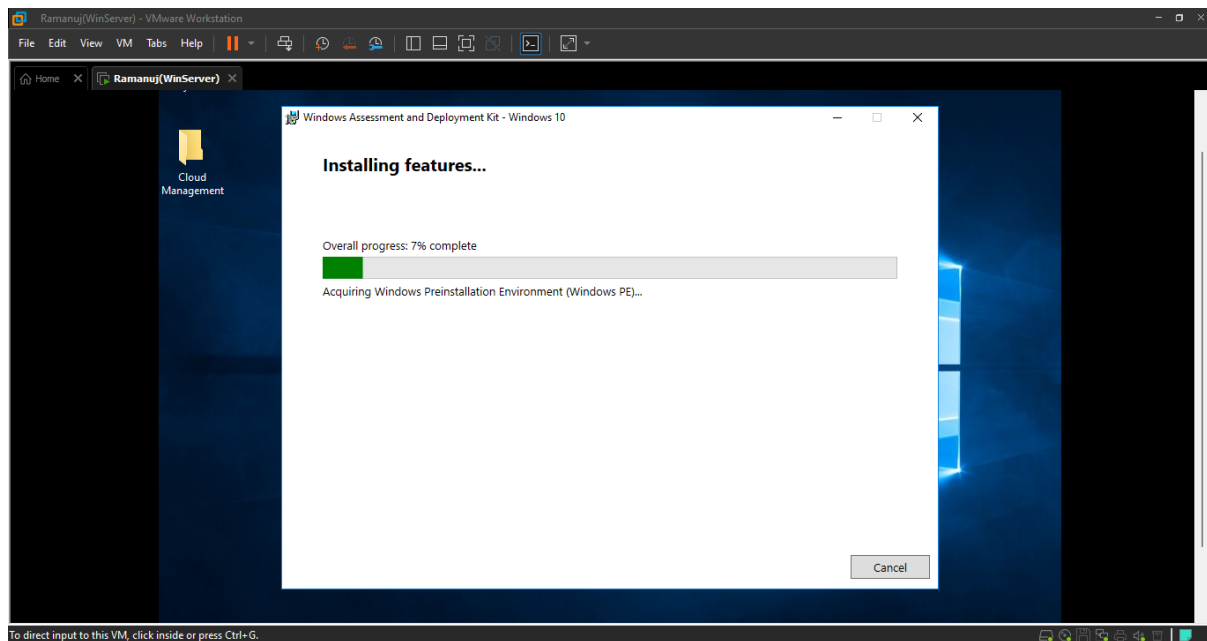
- Click **Accept**



- In the Selecting Features part click on the options given below:
 1. Deployment Tools
 2. Windows Pre-installation Environment (Windows PE)
 3. Windows Performance Toolkit

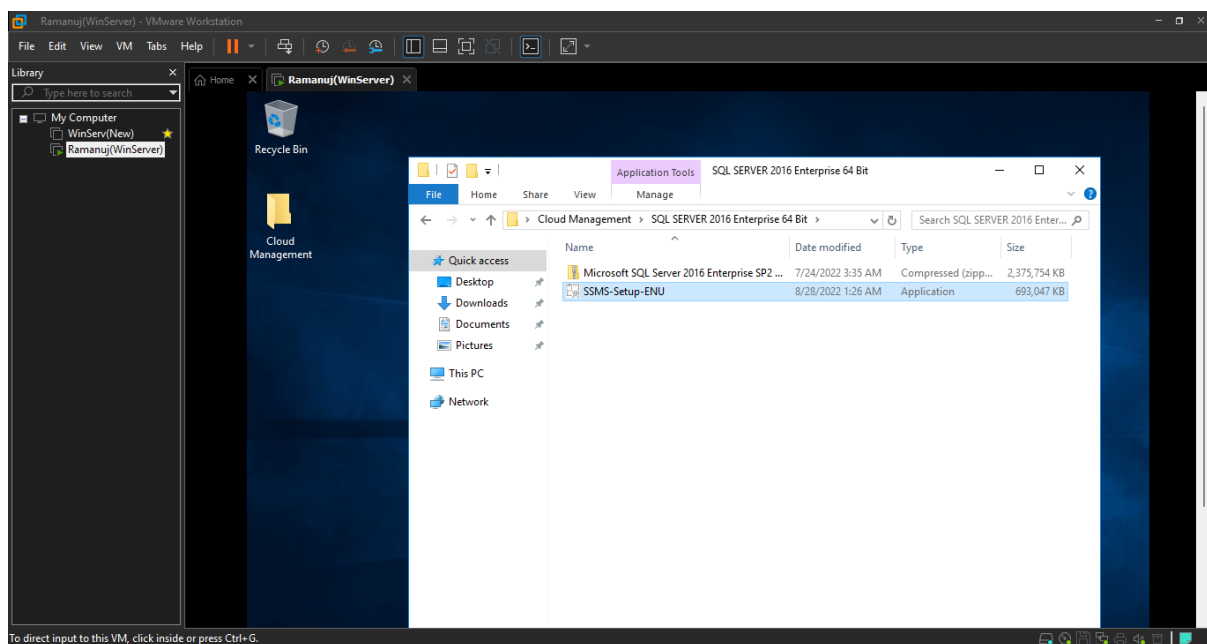


- Installation of Features will now start, after installation close the setup wizard.

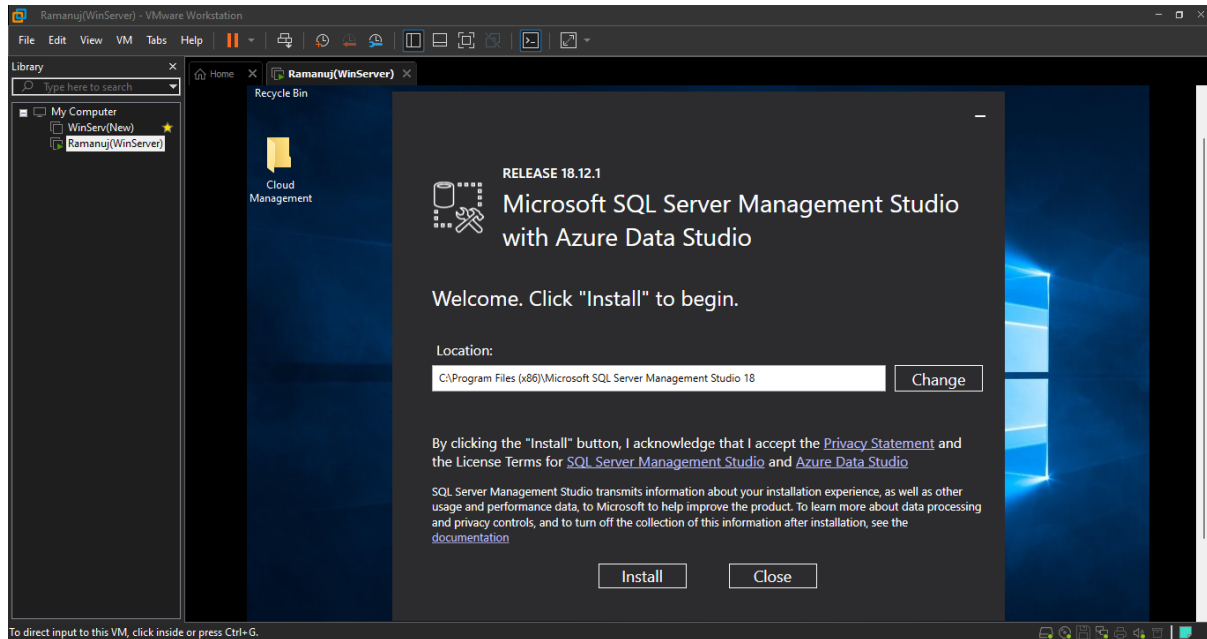


Step 36: Installing SQL Server Management Studio

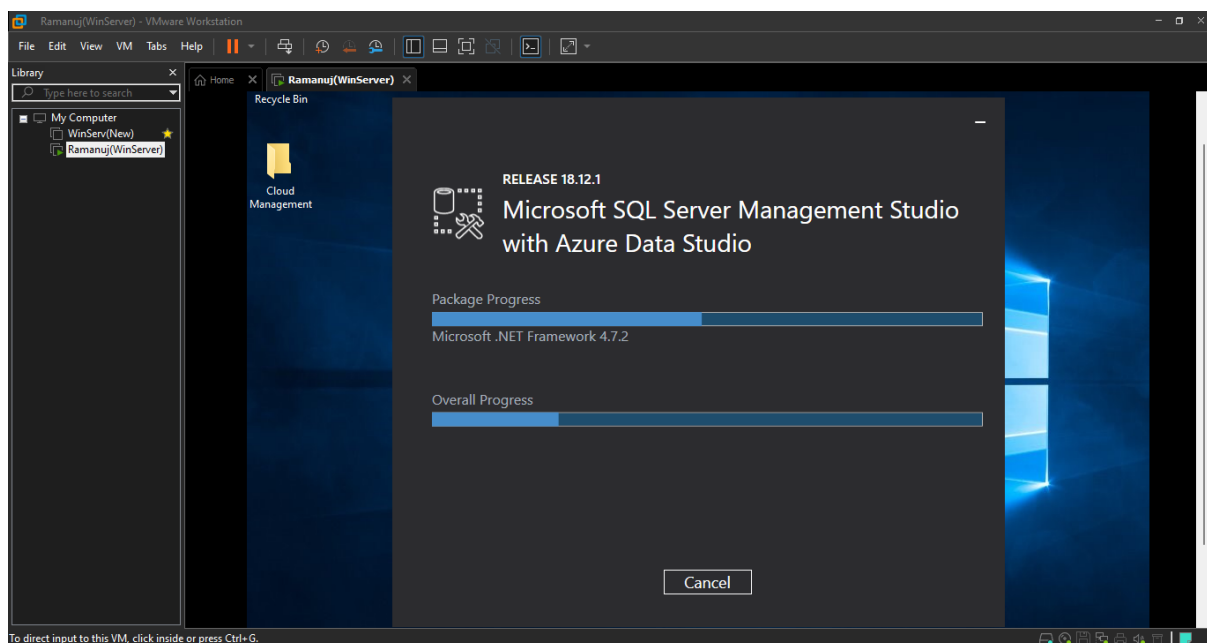
- Run the SSMS setup



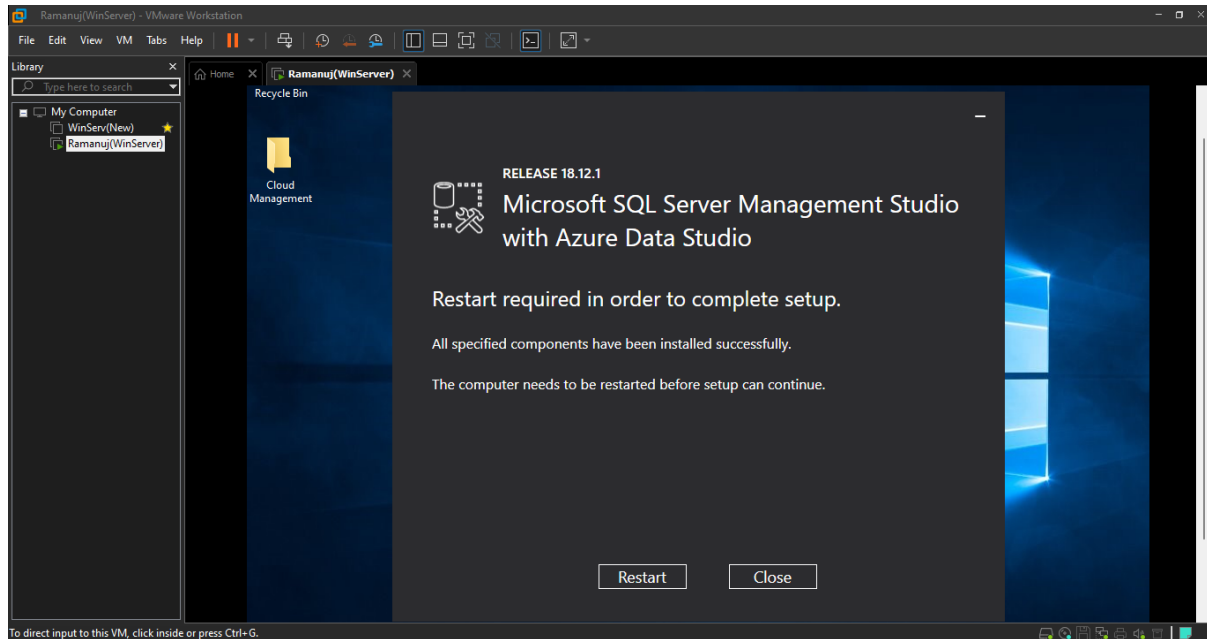
- Keep Default settings and Click on **Install**



- It will start installing the necessary files

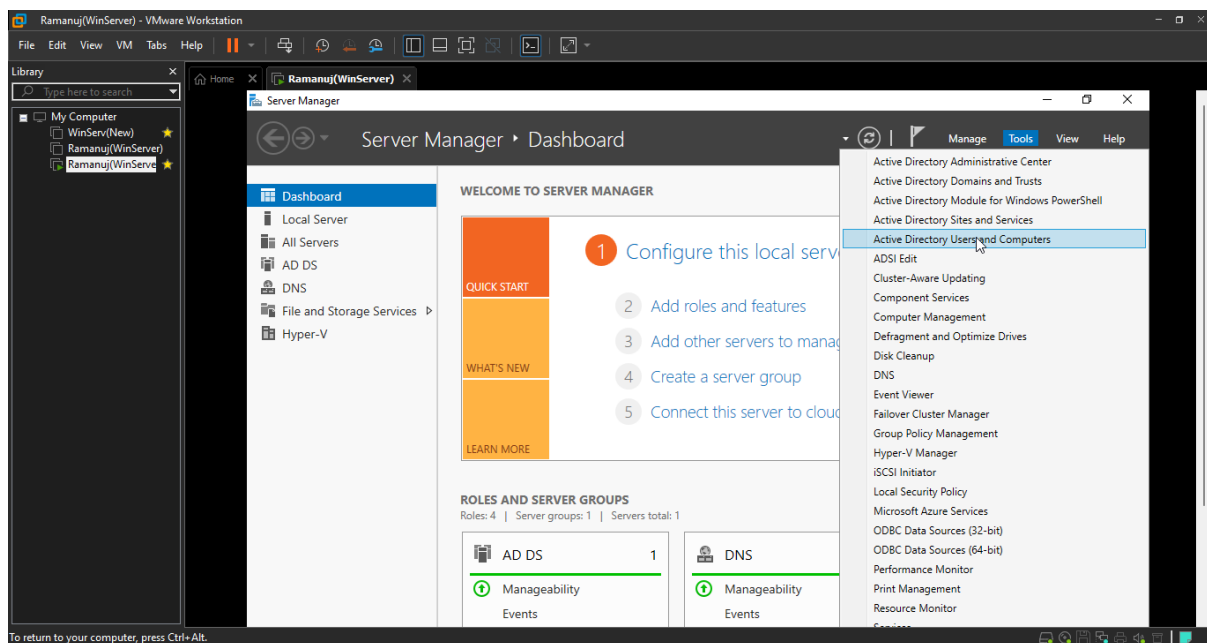


- Upon completion of the installation you will have to restart the VM

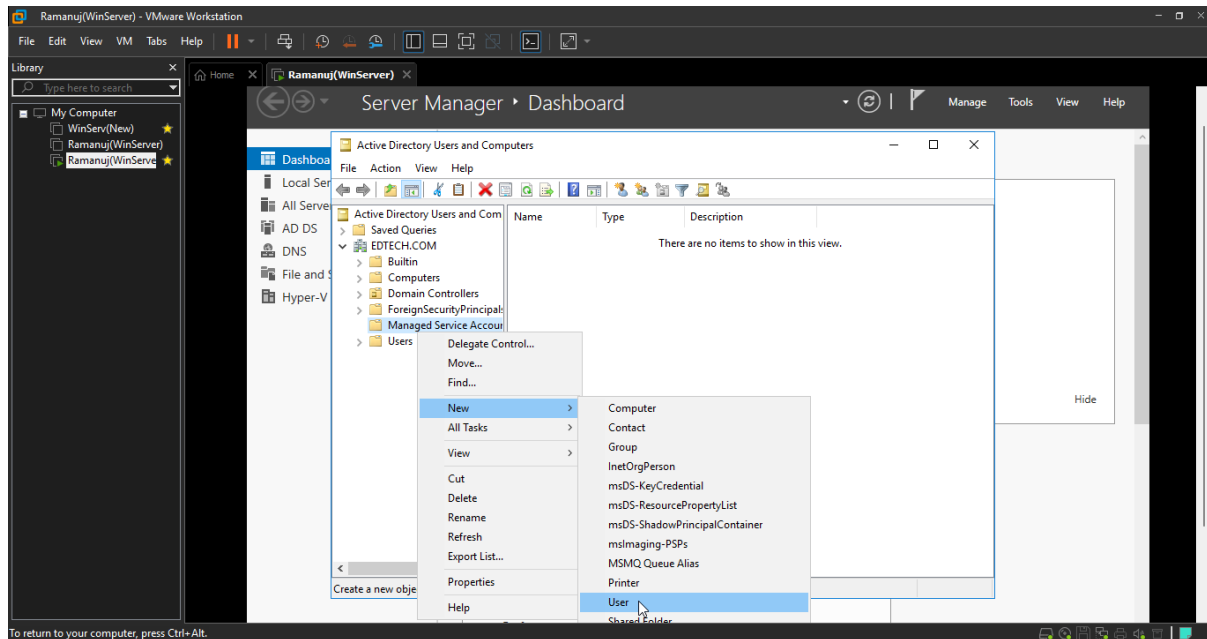


Step 37: Creation of User

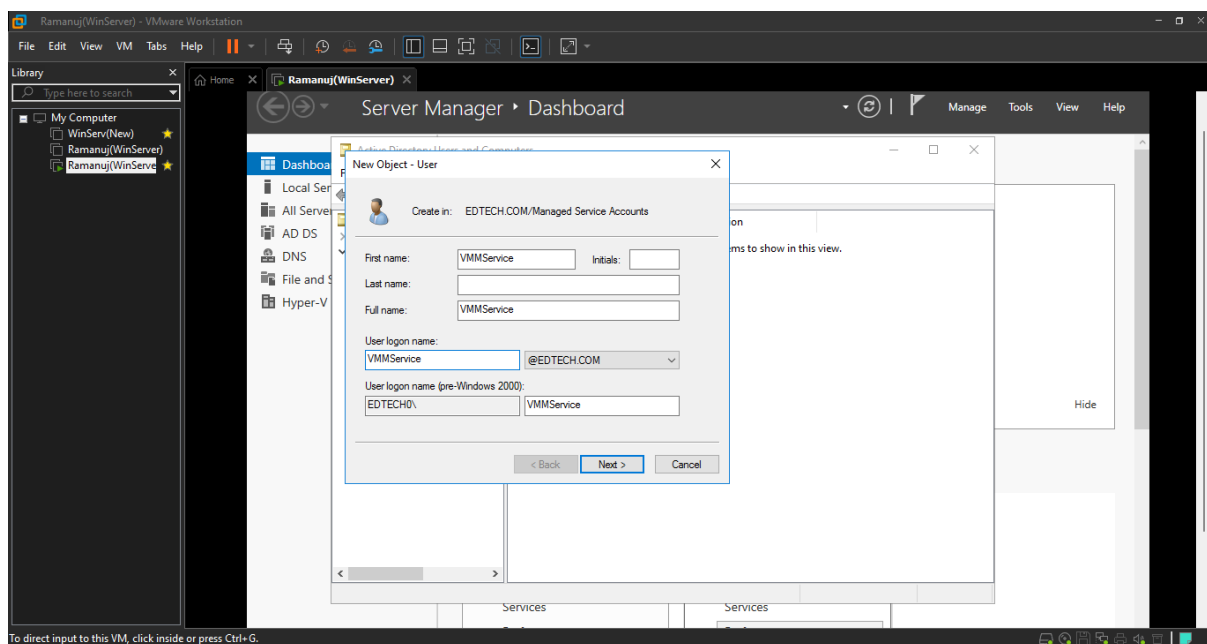
- Open **Server Manager** and Click on **Tools**
- Within Tools Click on **Active Directory Users and Computers**



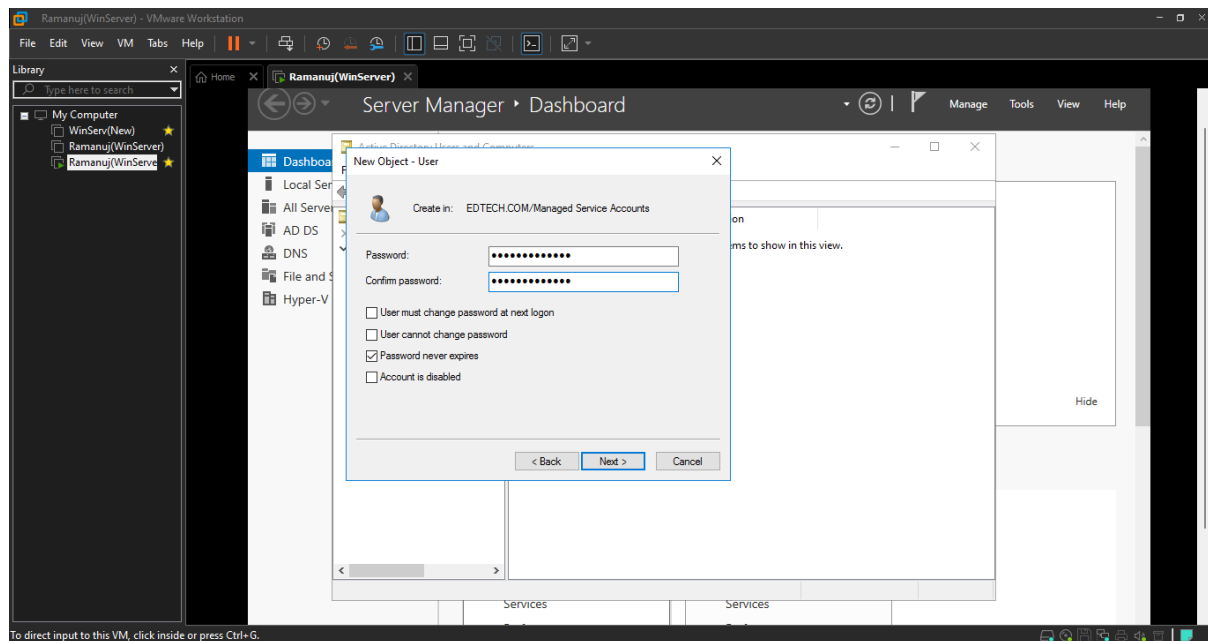
- Now Right Click on **Managed Service Accounts** and Select **New**
- Within New Select **User**



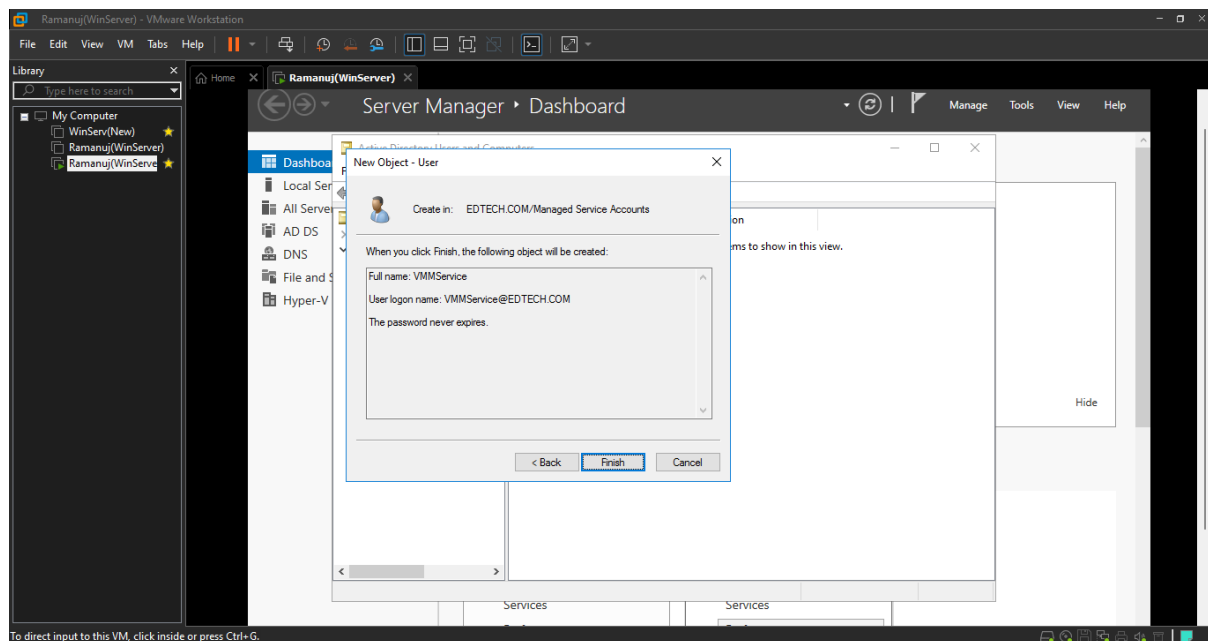
- In New Object -> User
- Enter a Username (Here it is VMMSERVICE) on both **First Name** and **Login Name** and Click **Next**



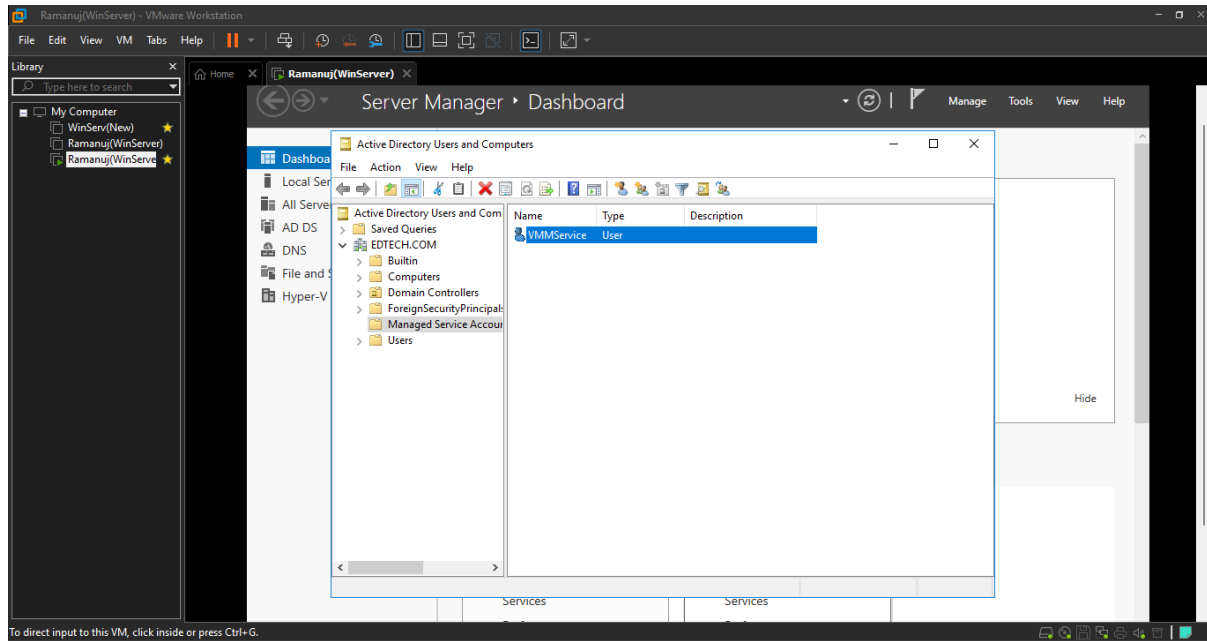
- In the next step enter a password (preferably the same as the admin) and Select **Password never expires** in the options and Click **Next**



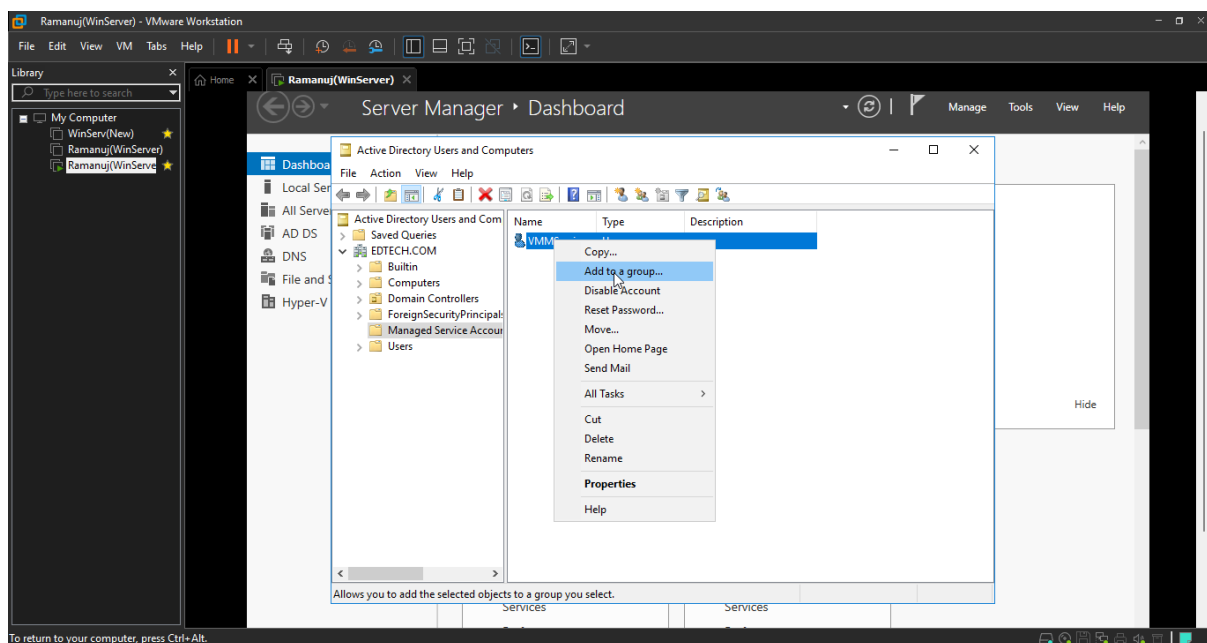
- Click on **Finish**



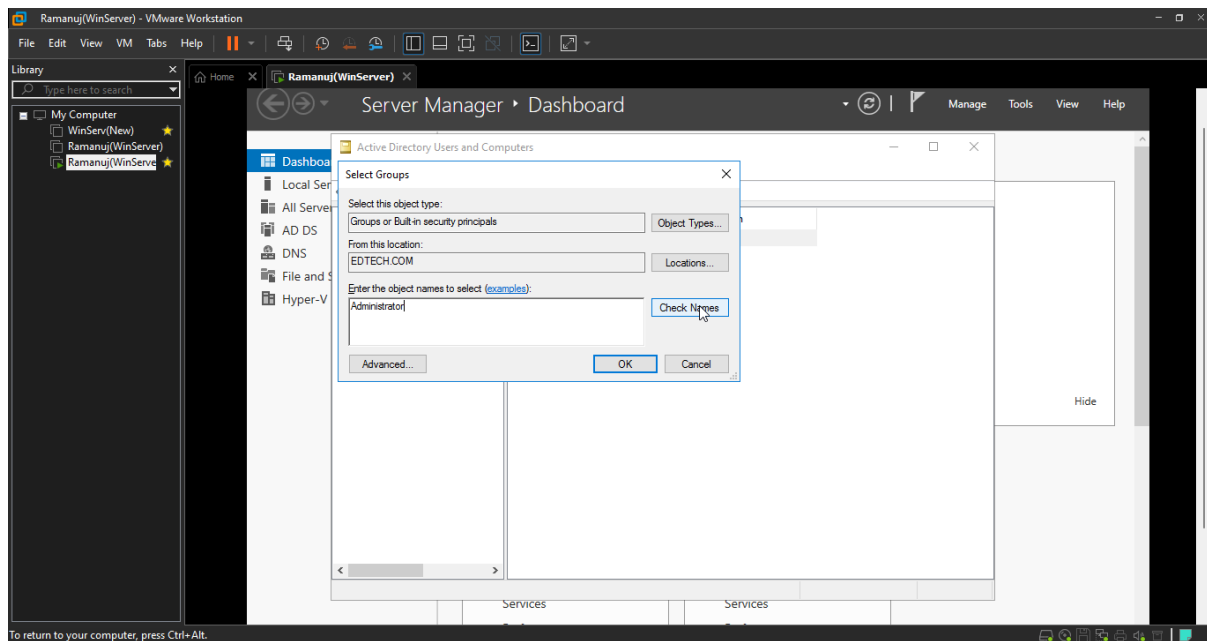
- Now **VMMService** will show up under Active Users and Computers



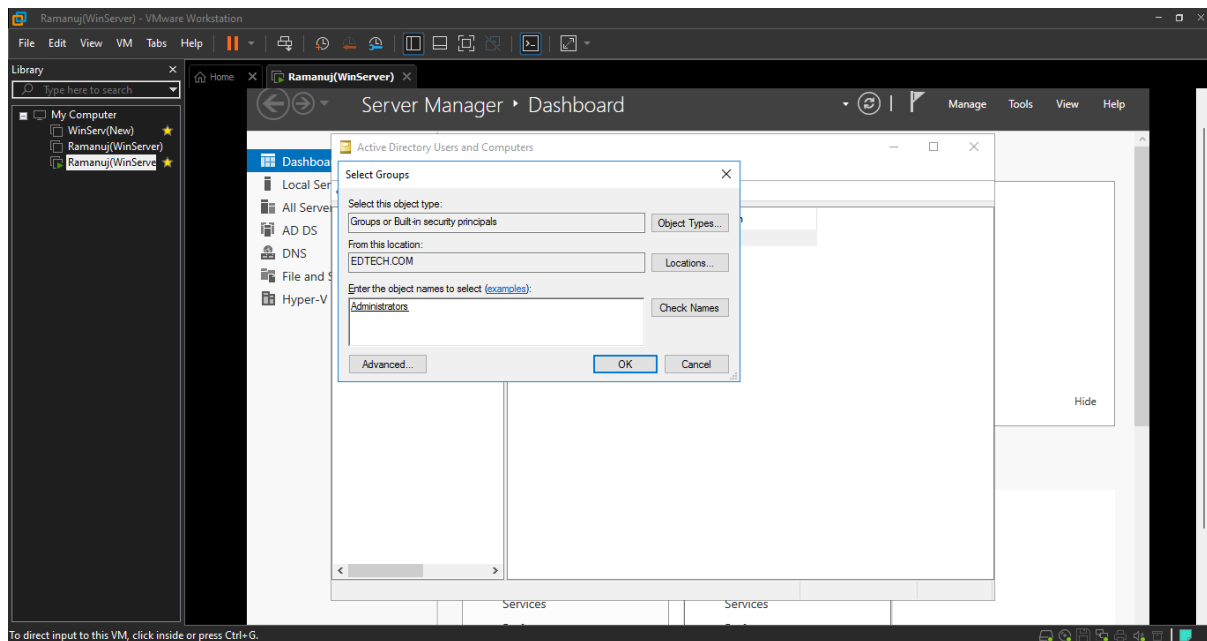
- Now Right-Click on **VMMService** and Select **Add to a Group**



- Under Object name to select type the beginning words of **Administrator** and click on **Check Names**

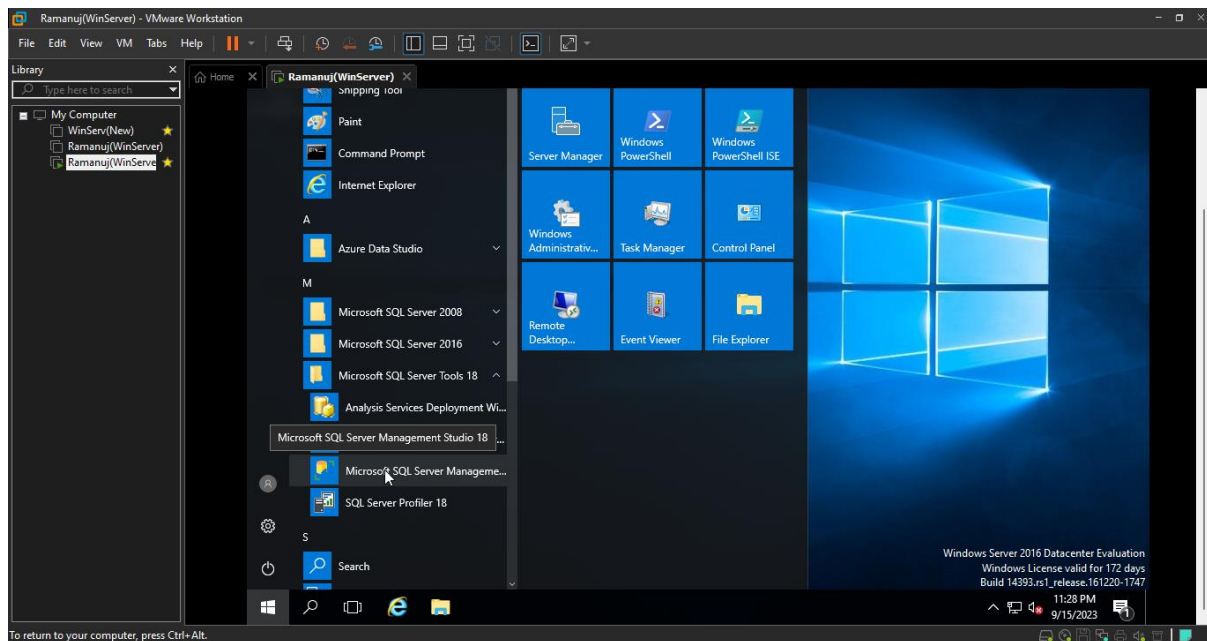


- Now **Administrators** will show up, Click OK
- Now the User will be successfully added to the group

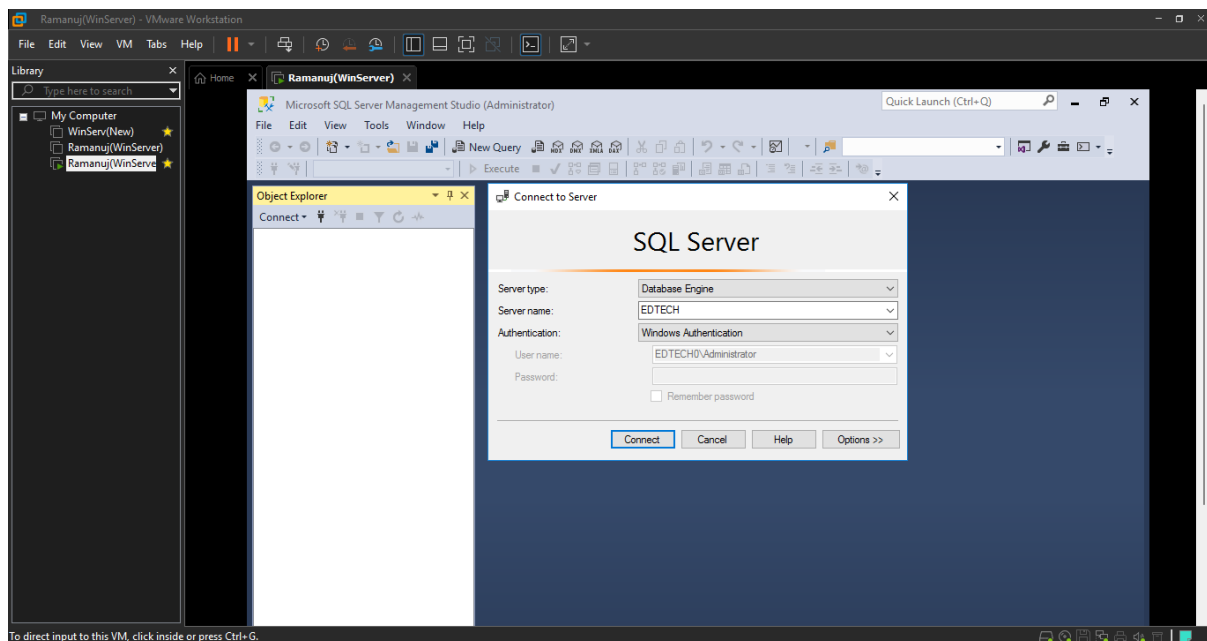


Step 38: Connecting SQL Server to Windows Server

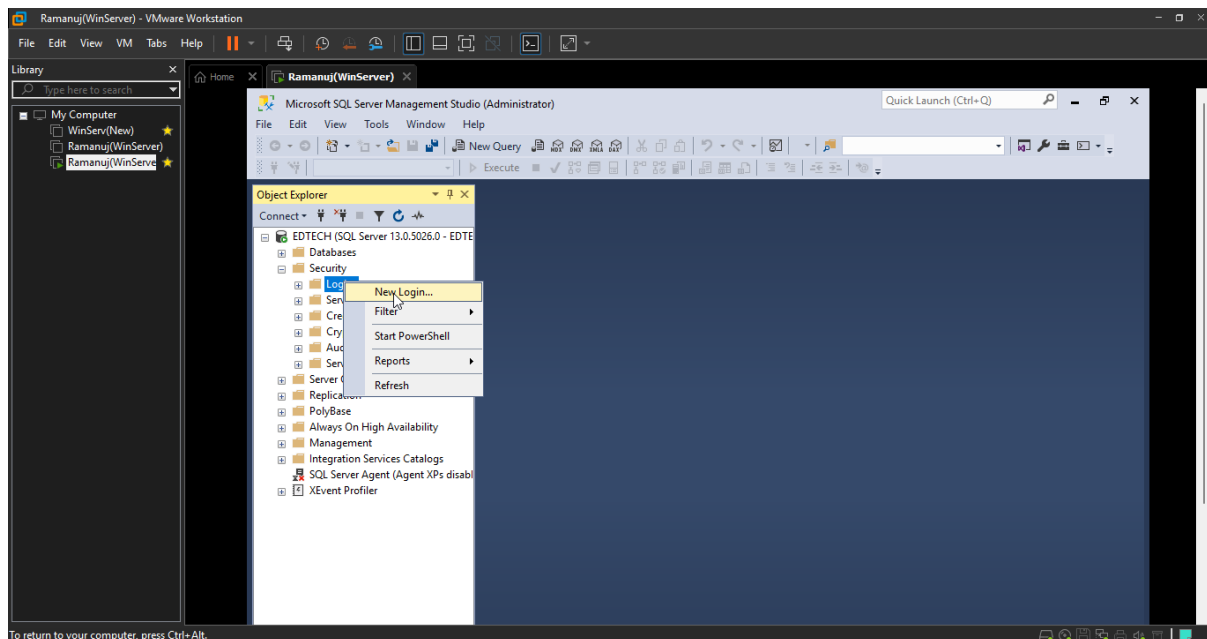
- Open the SSMS application



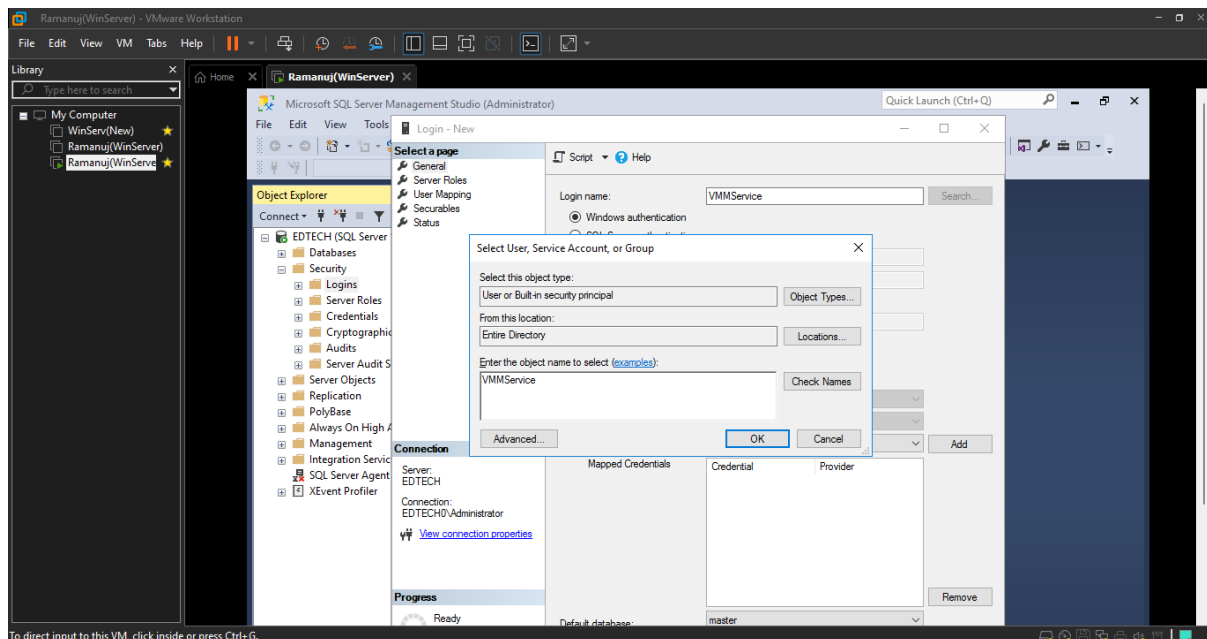
- Keep default values and Click on **Connect**



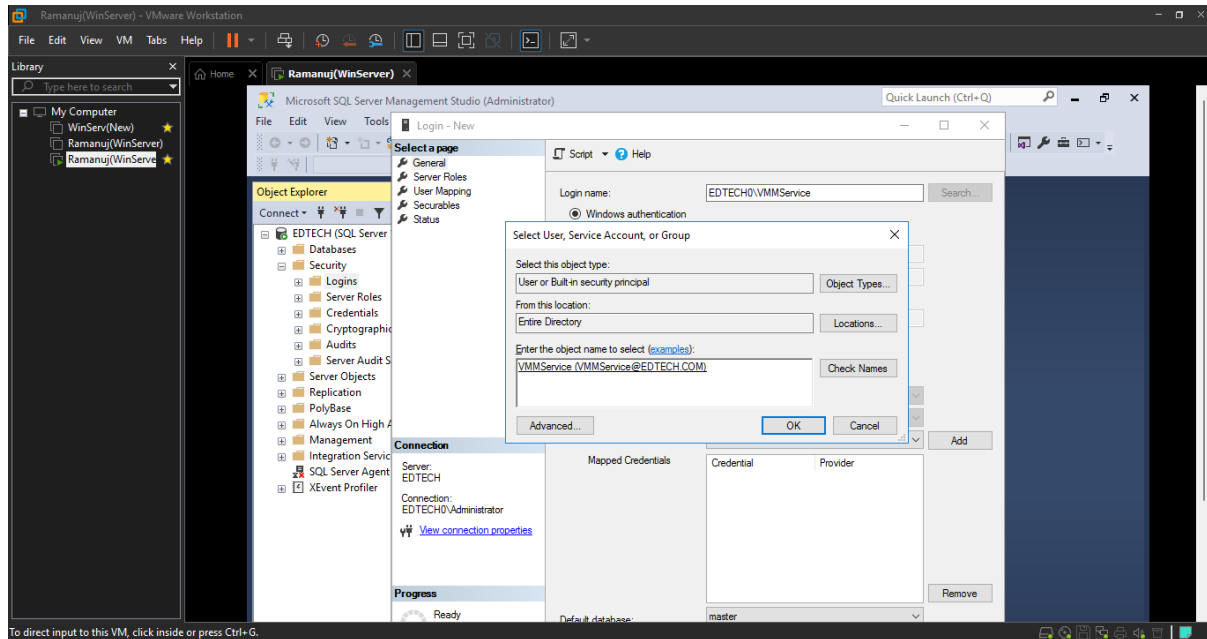
- Now within the instance Click on **Security** and Right Click on **Login** and Select **New Login**



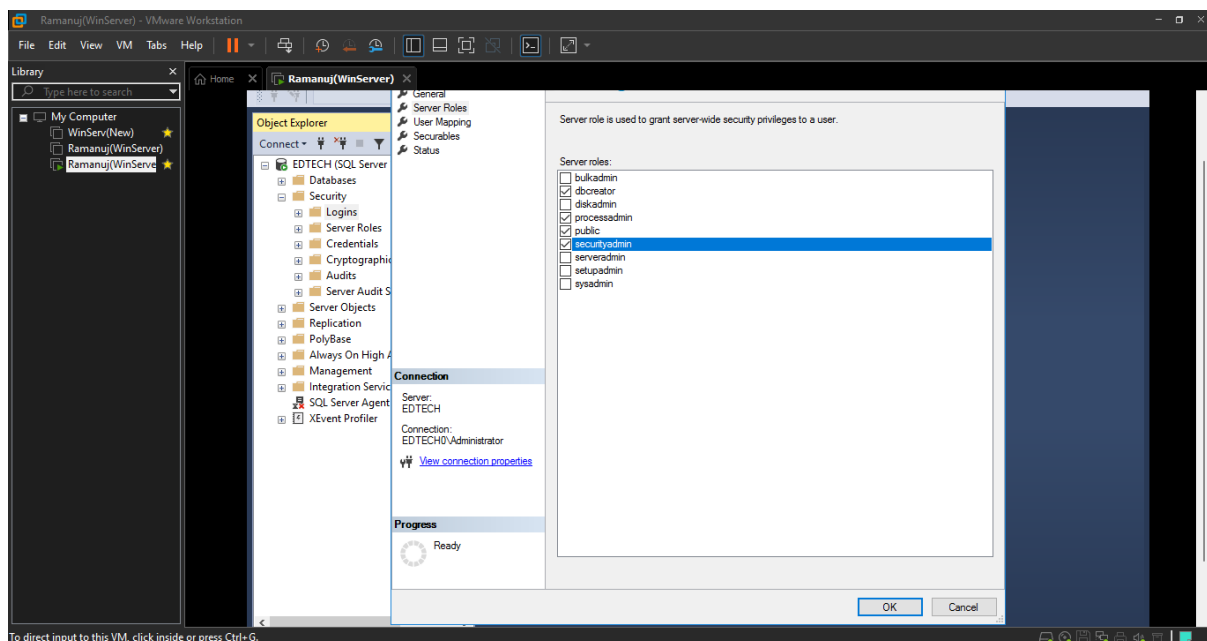
- In the Login Name type the User Created (**VMMService**)
- Type **VMMService** in the Small Window and Click on **Check Names**



- It will show the full value of the User and Click **Ok**

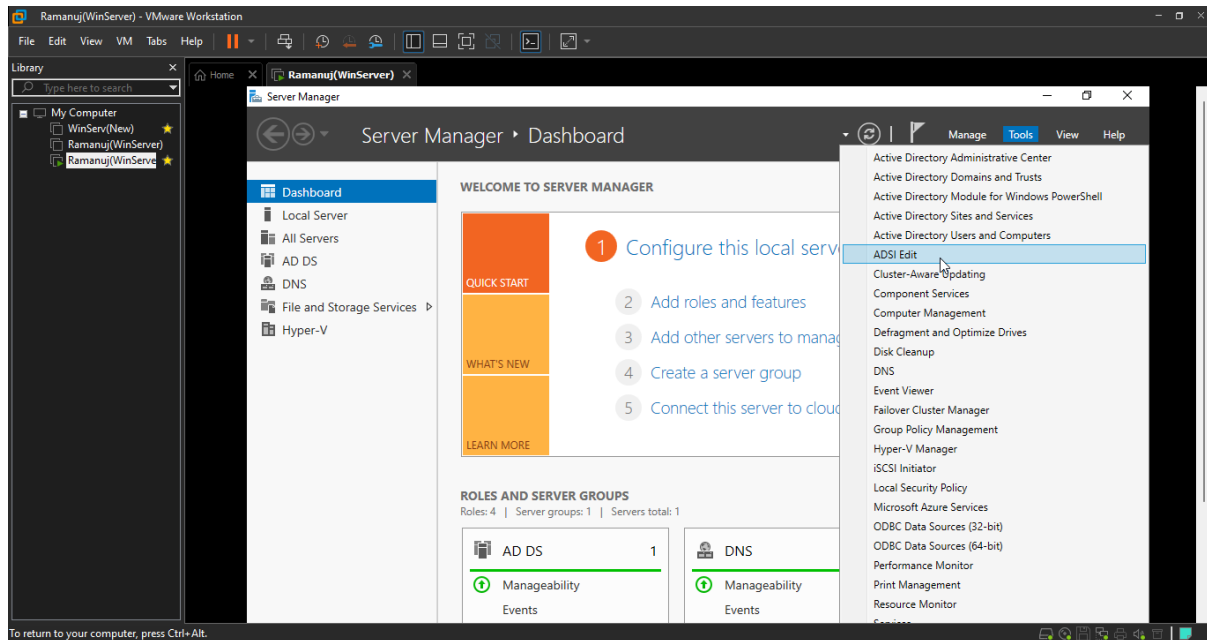


- Click on **Server Roles**, including **Public** select these options:
dbcreator
processadmin
securityadmin
- Click on **Ok** and Close SSMS

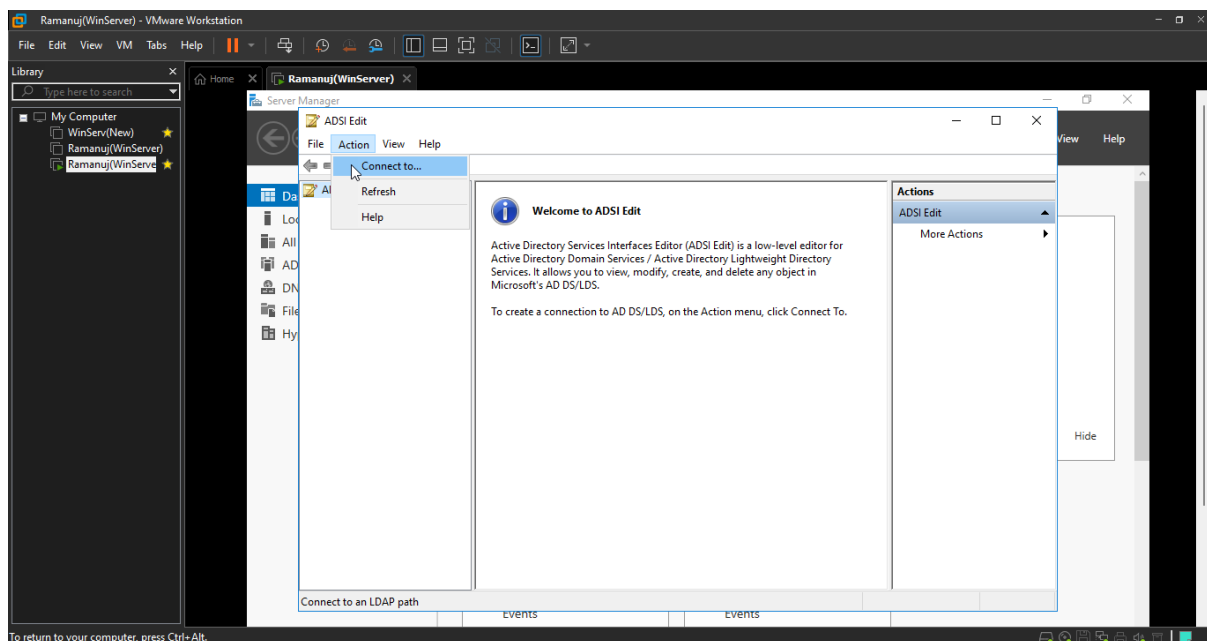


Step 39: Creating an Object in ADSI Edit

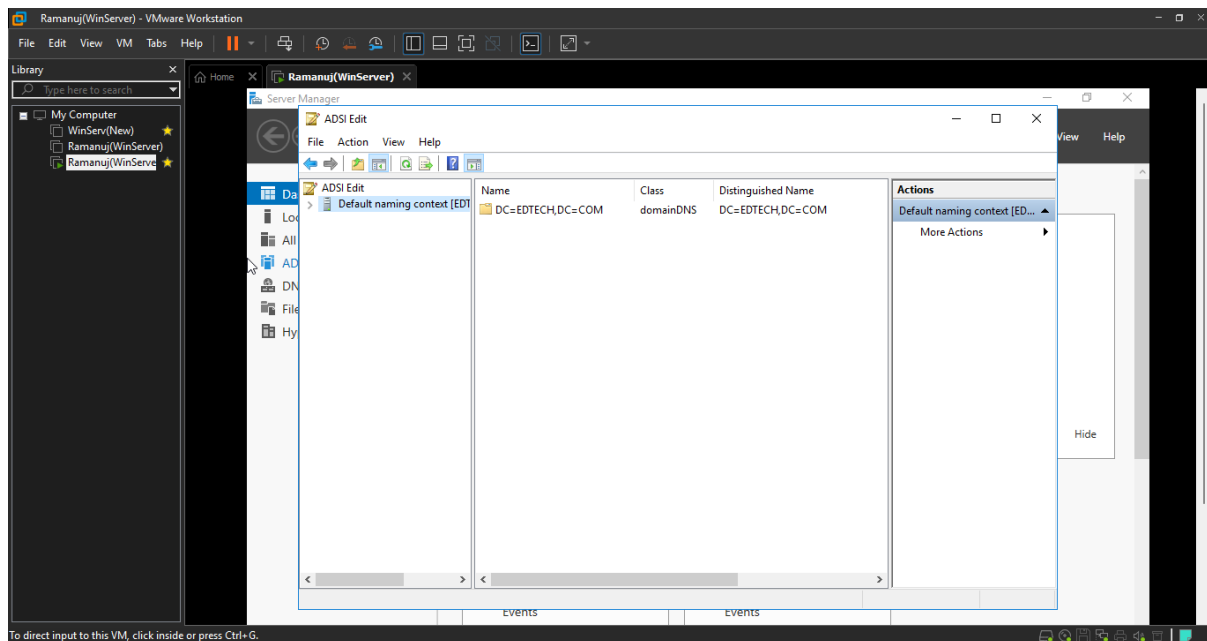
- Open **Server Manager** and Click on **Tools**
- Within Tools Click on **ADSI Edit**



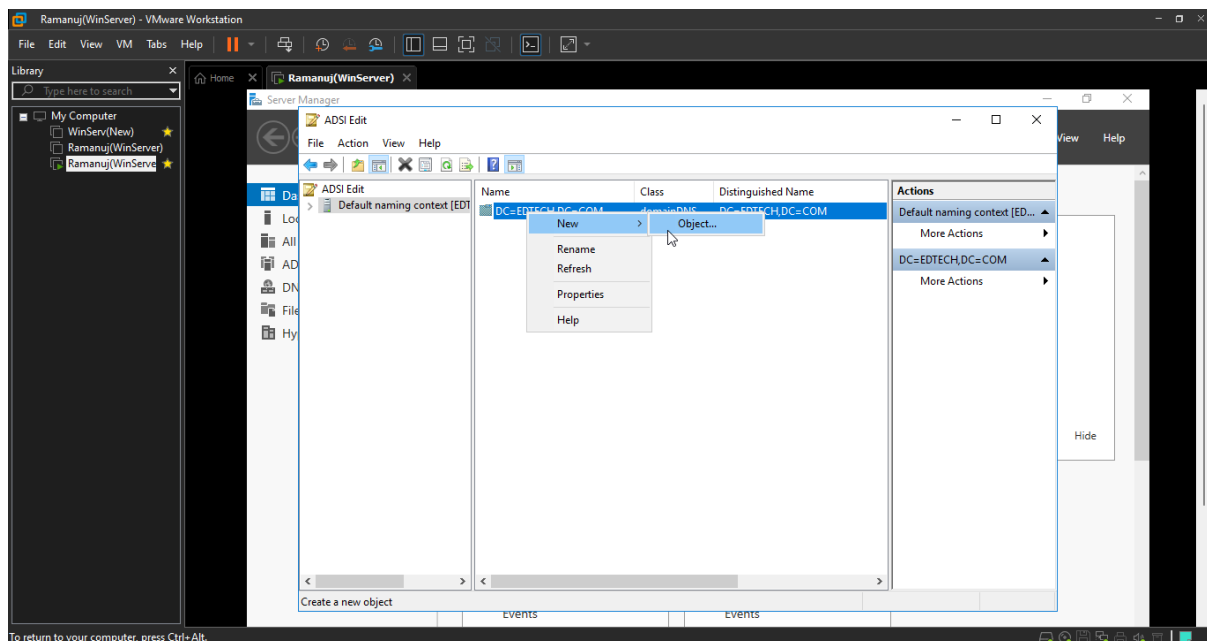
- Within ADSI Edit Click on **Actions** and Select **Connect To**



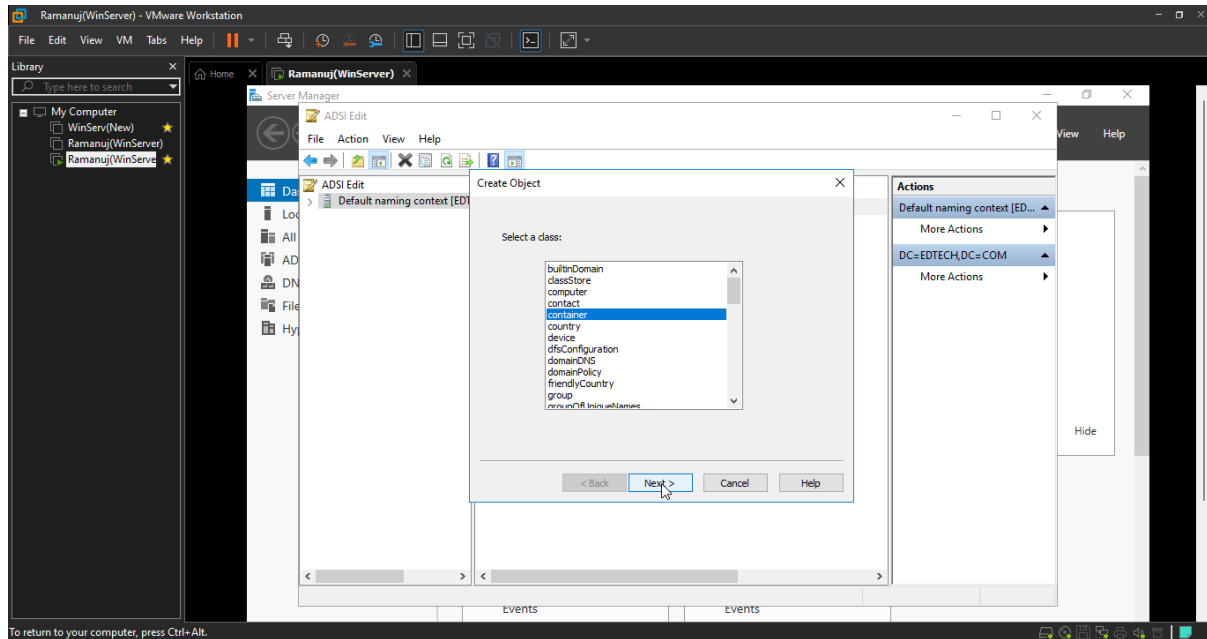
- In Connection Settings Keep the Default Values and Click **Ok** and a default value is created



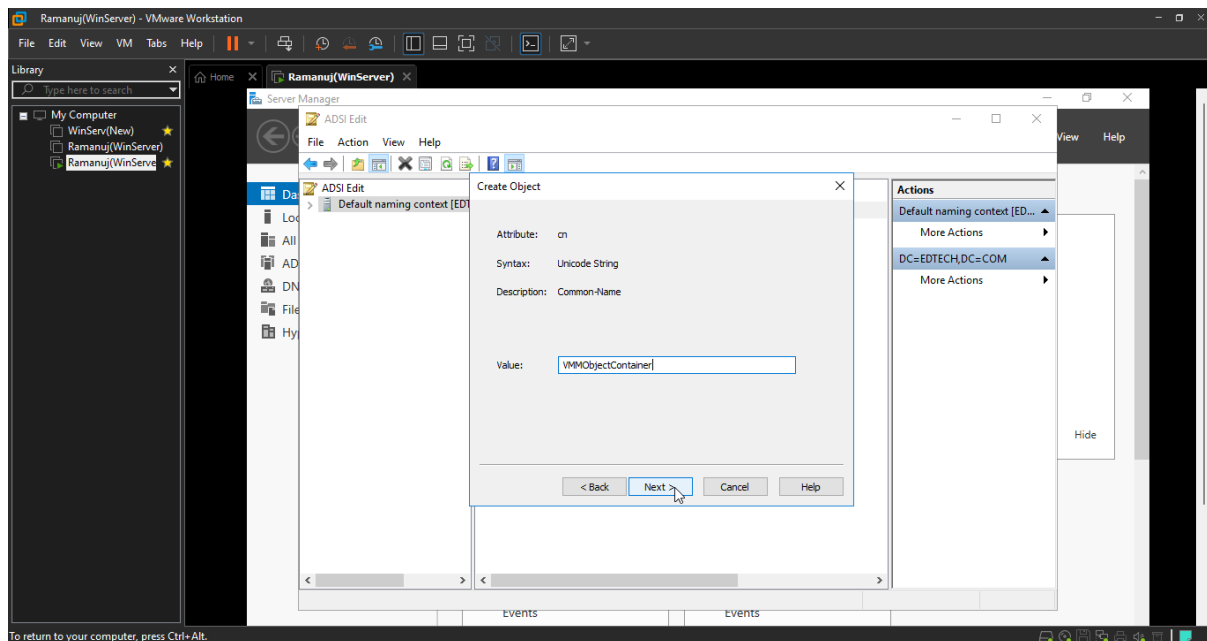
- Right Click on the **DC=EDTECH** Folder and Click on **New** and Select **Object...**



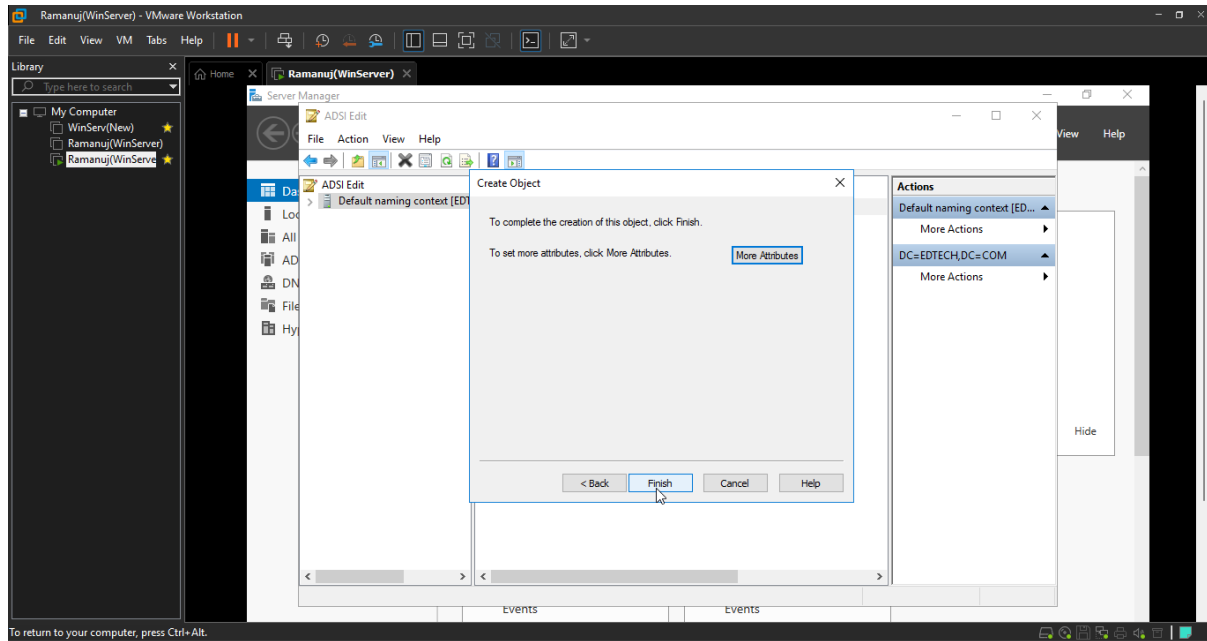
- Select **container** and Click on **Next**



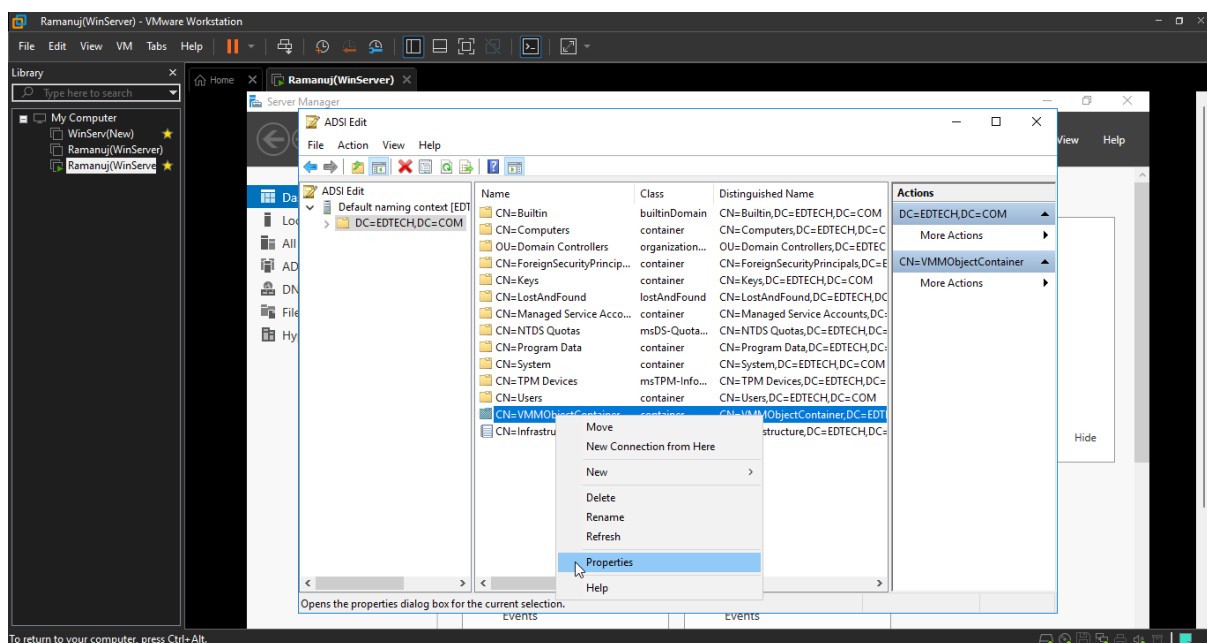
- Enter the value for the object. (Here it is VMXObjectContainer)
- Click on **Next**



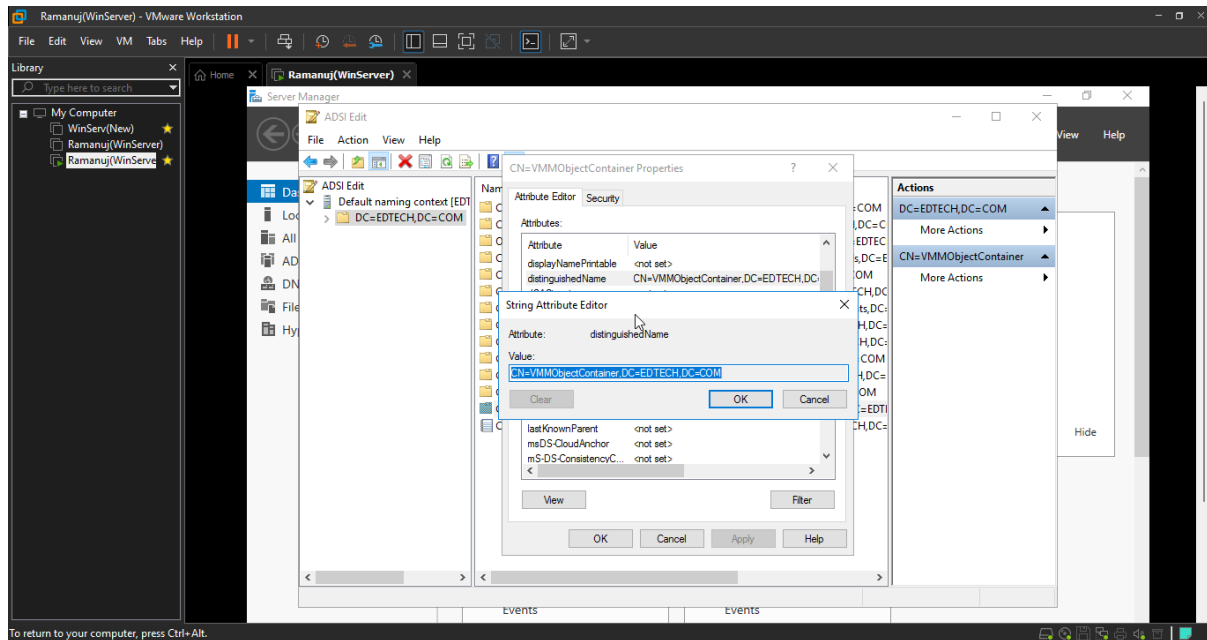
- Click on **Finish**



- Double Click on **DC=EDTECH** and Right Click on the **CN=VMMObjectiContainer** Folder and Select **Properties**

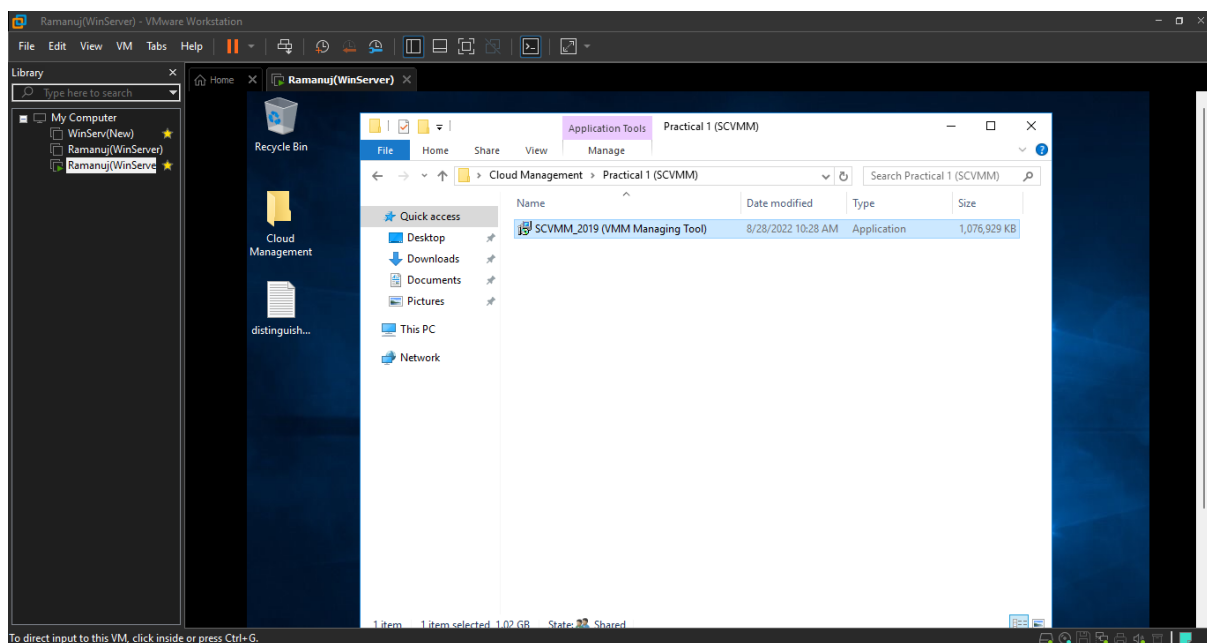


- Within Properties Click on **distinguishedName** and Click on **View**
- Copy the **distinguishedName** onto a Notepad as it is needed for the SCVMM setup
- Close the ADSI setup

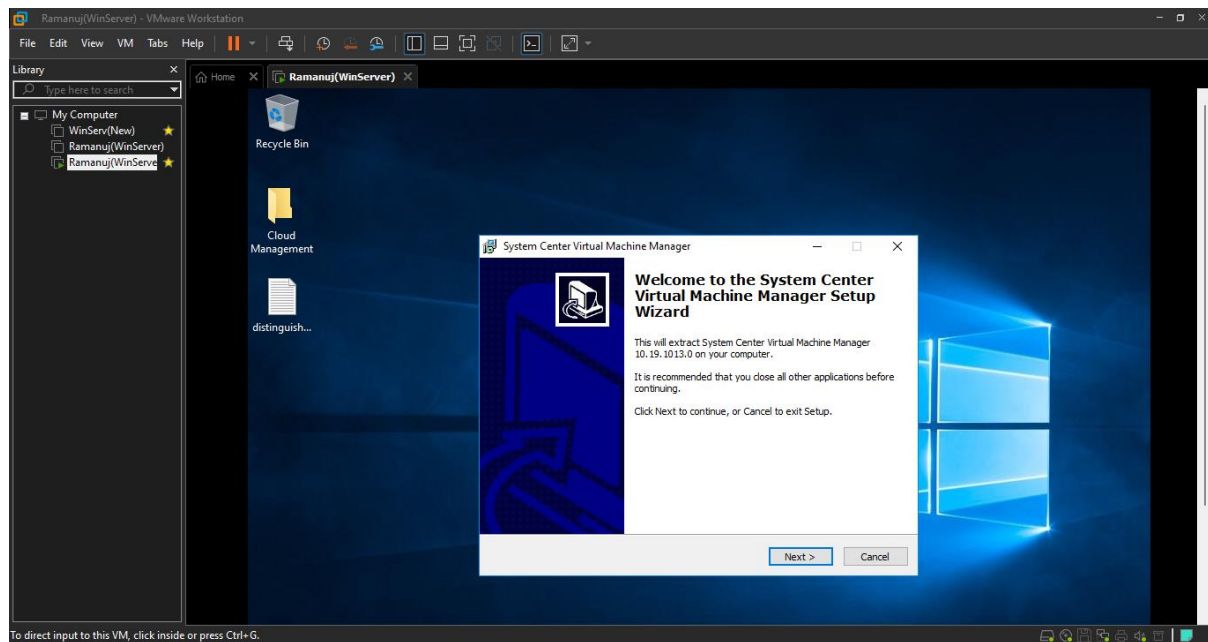


Step 40: SCVMM Setup

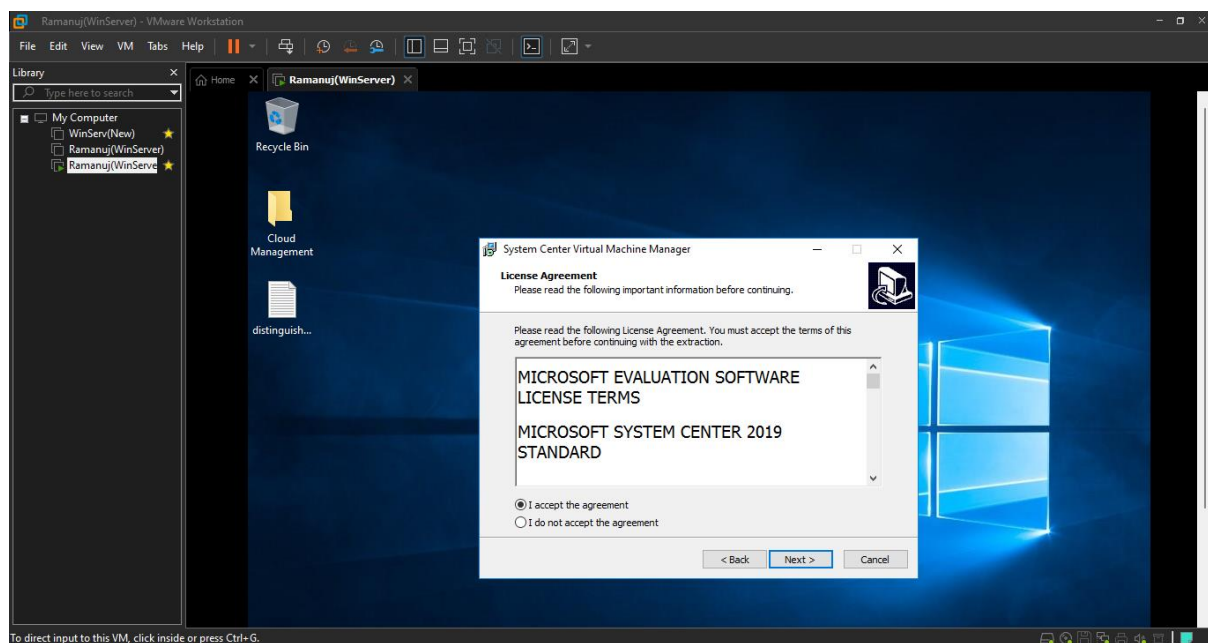
- From your Cloud Management Folder and Mount the SCVMM Setup file.



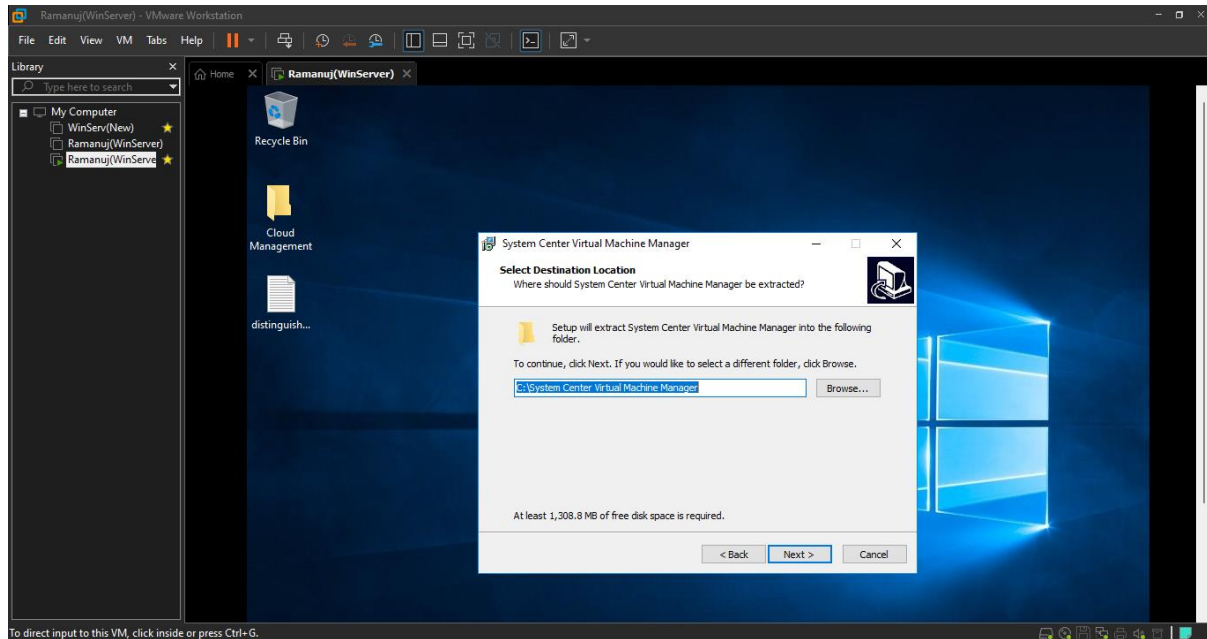
- Click **Next**



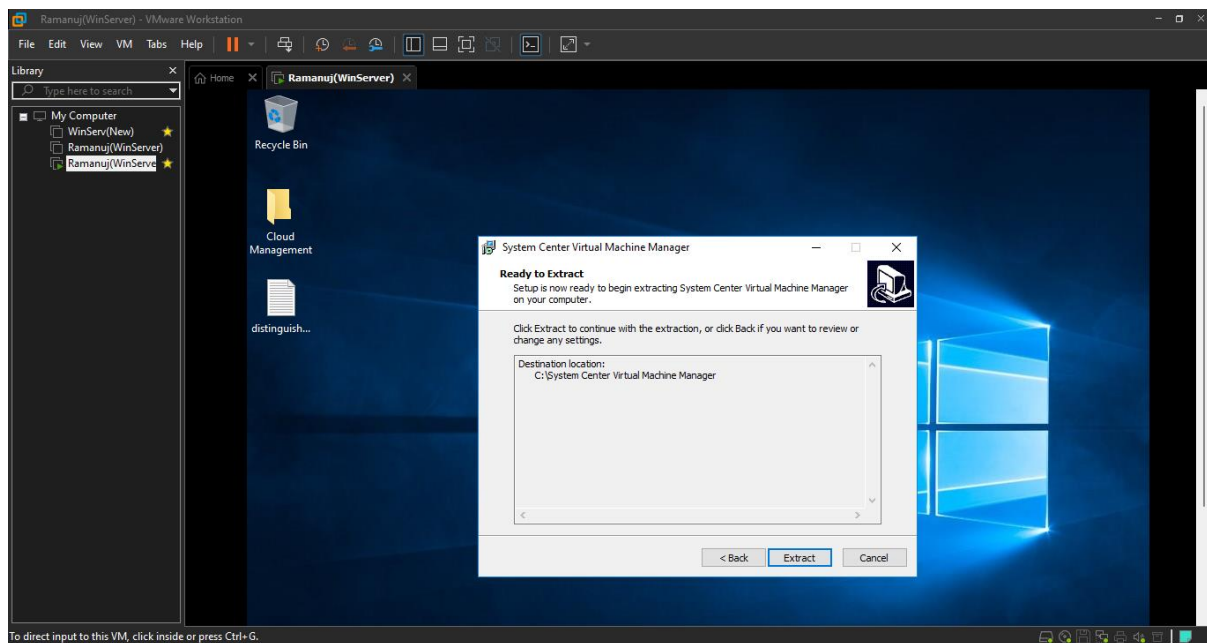
- Select **I accept the agreement** and Click on **Next**



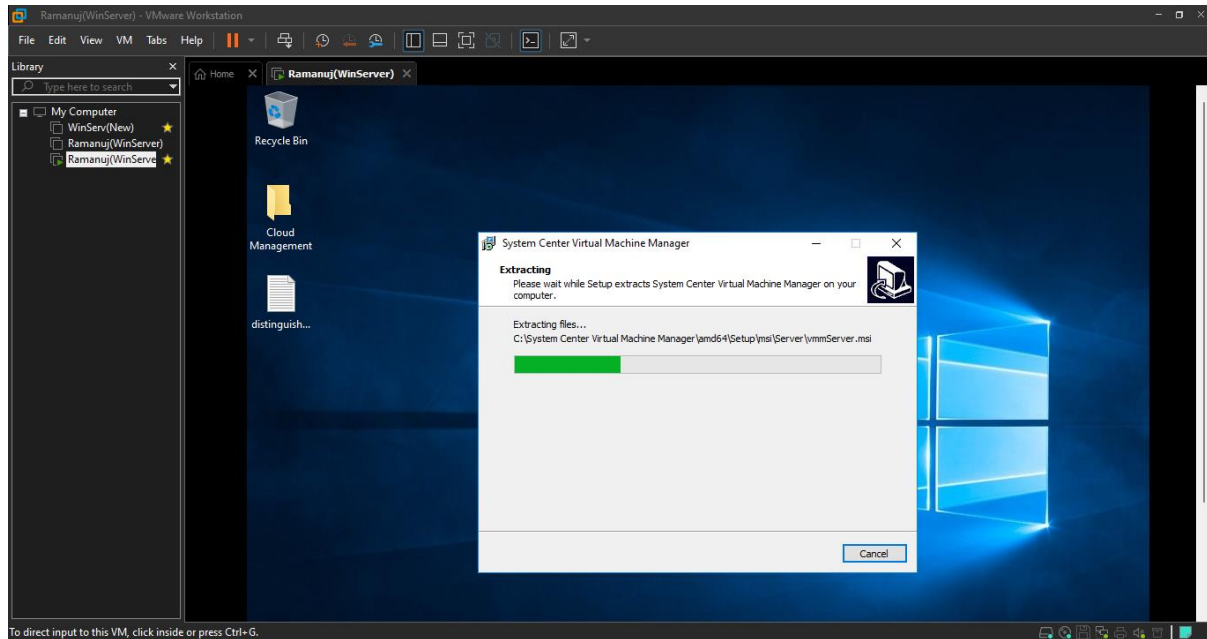
- Keep default values and Click **Next**



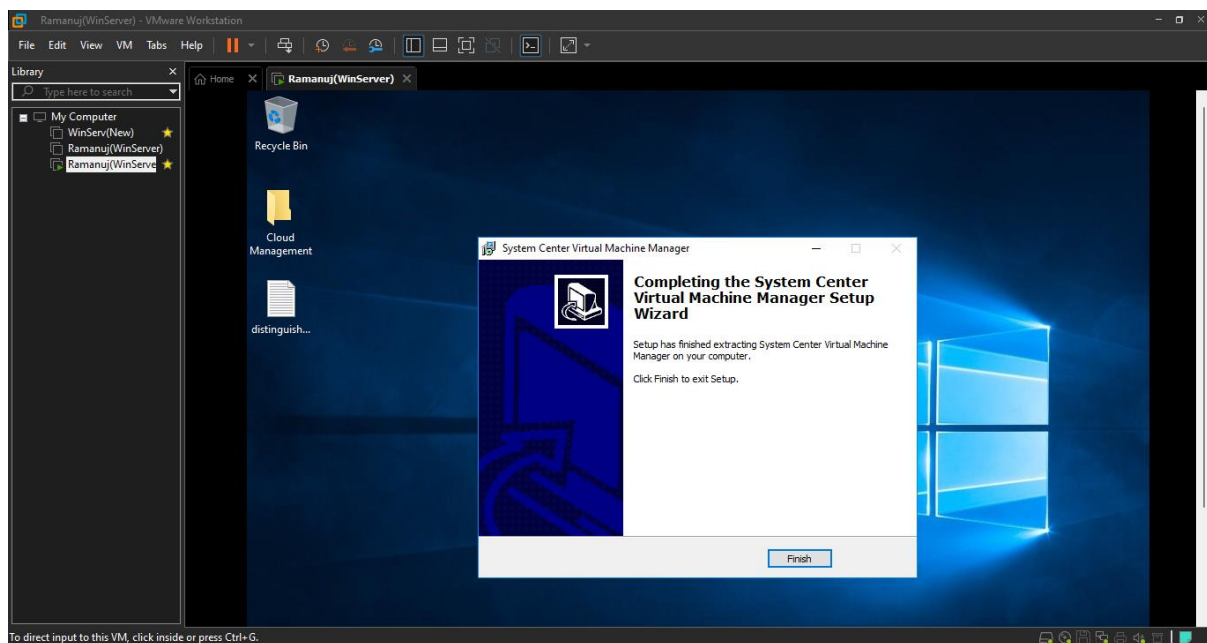
- Click on **Extract**



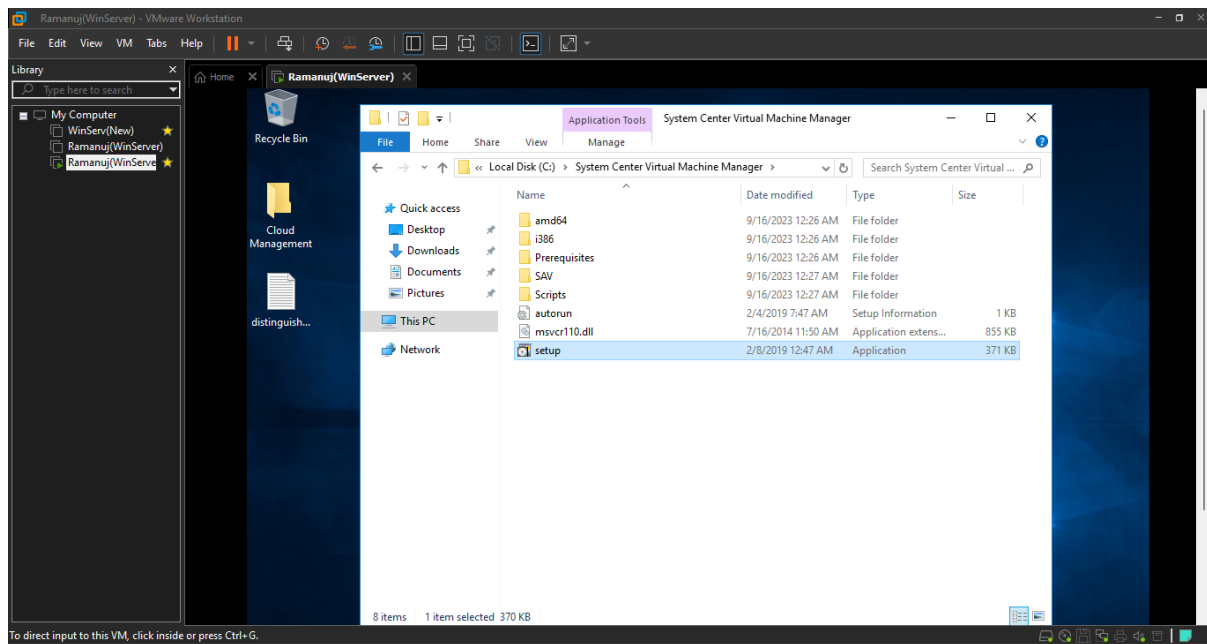
- It will extract the necessary files



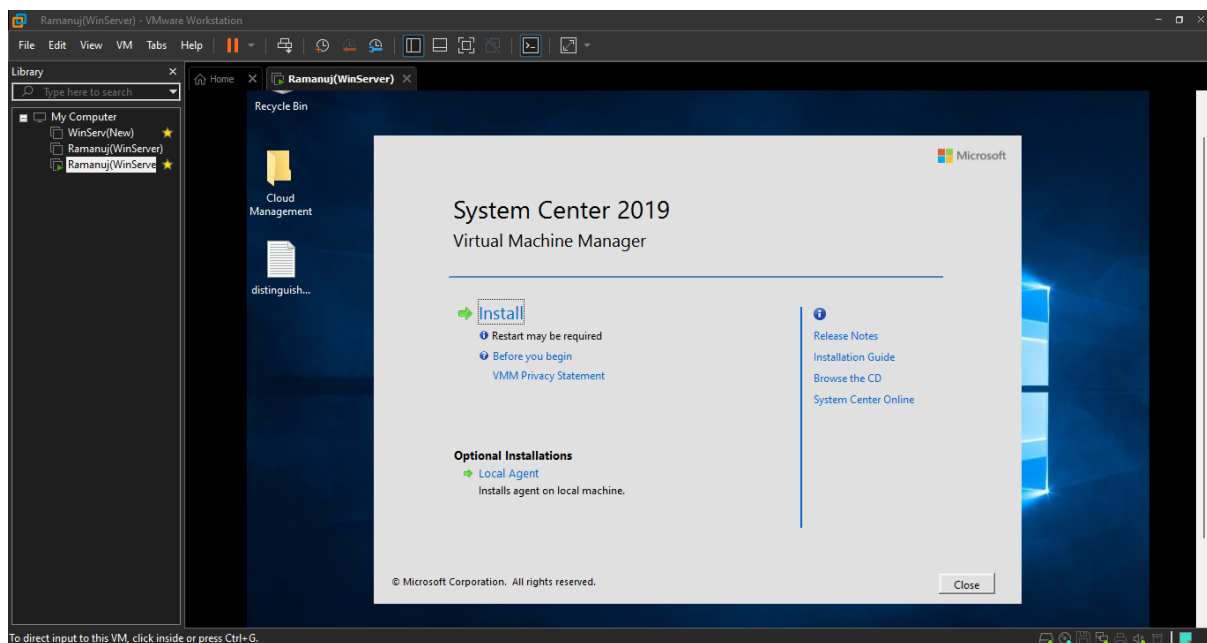
- Click on **Finish**



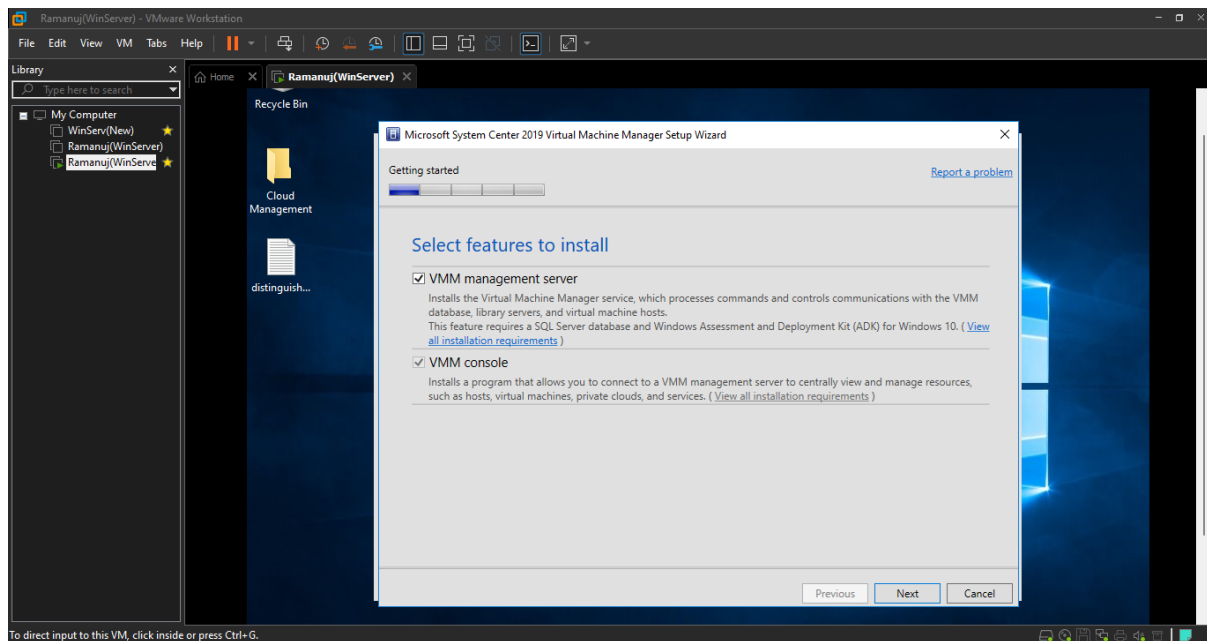
- Now within your **C: Drive** Click on **System Center Virtual Machine Manager** and Click on **Setup**



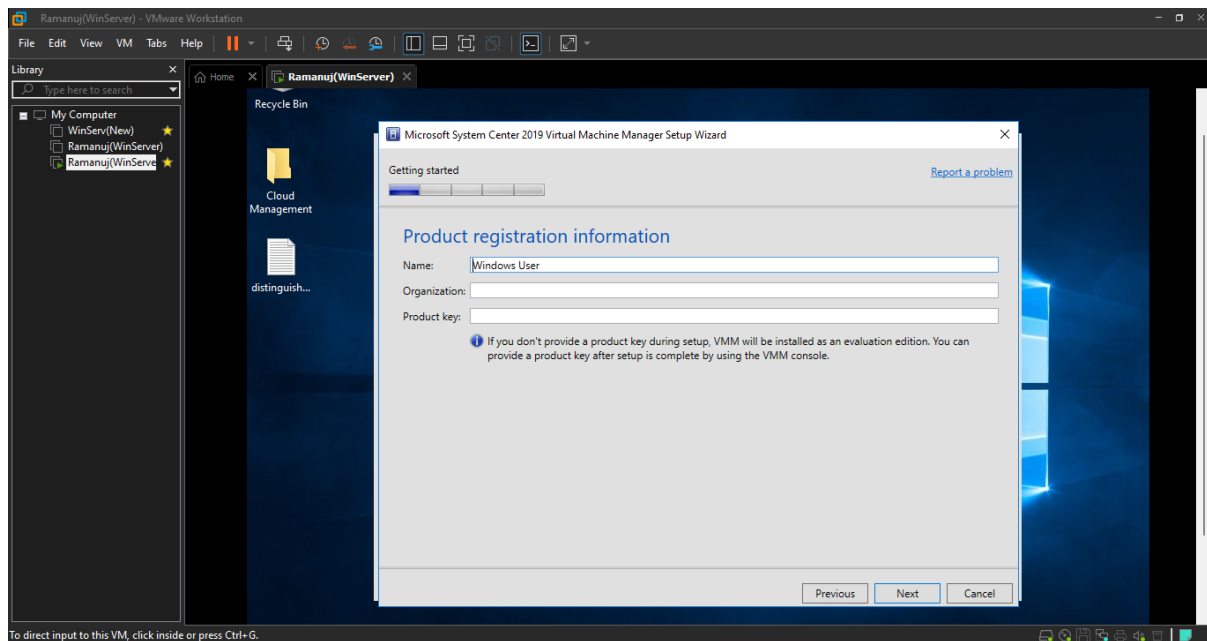
- Click on **Install**



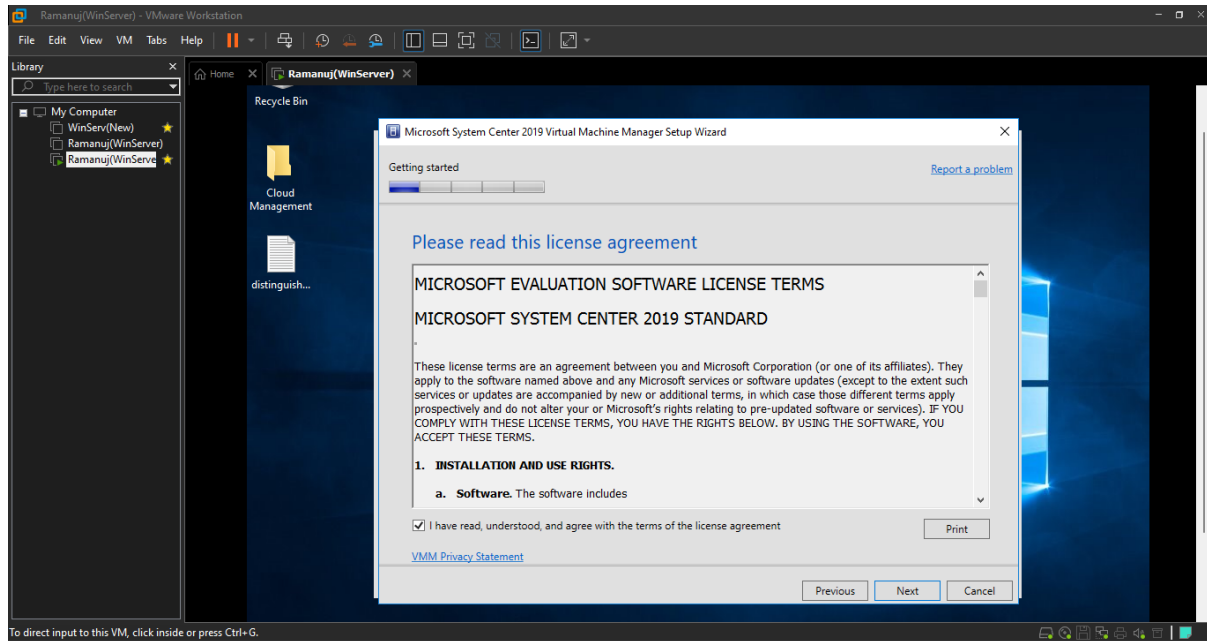
- Click on **VMM Management Server** and **VMM Console** gets auto-selected and Click **Next**



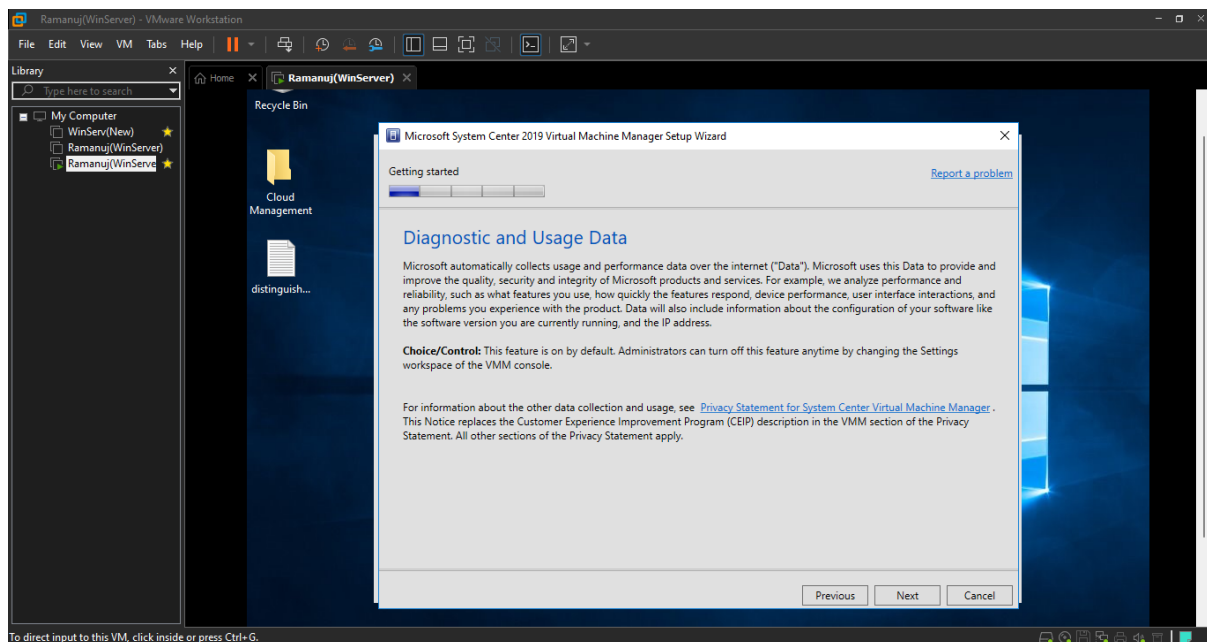
- Keep default values and Click **Next**



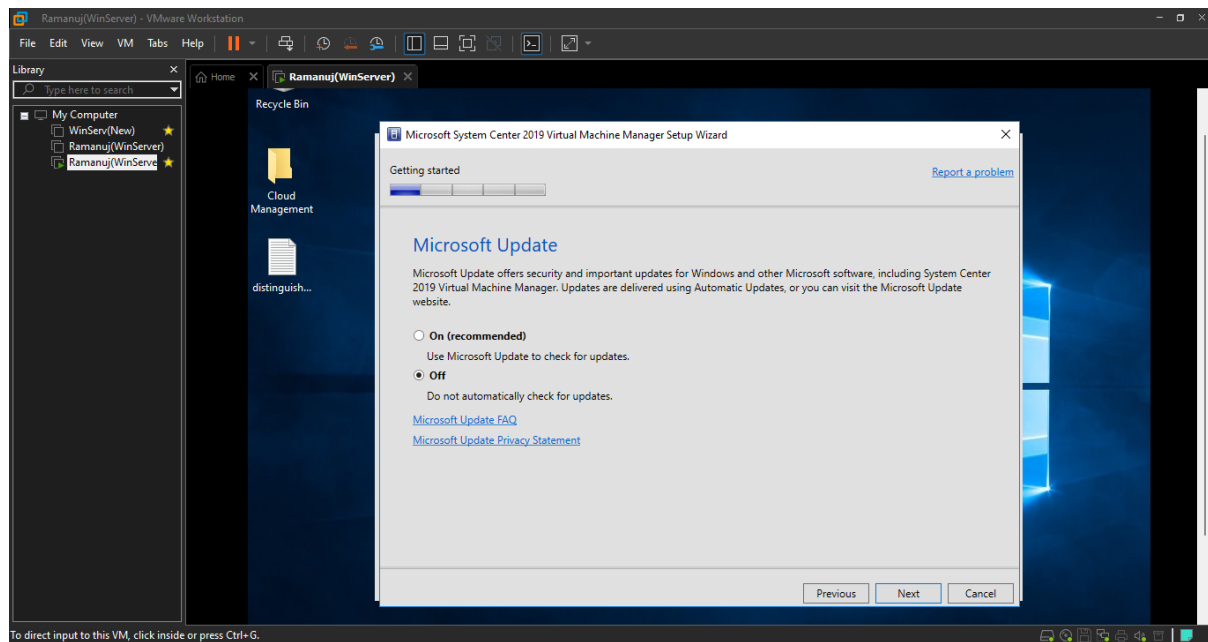
- Select **I have read and accept the license agreement** and Click **Next**



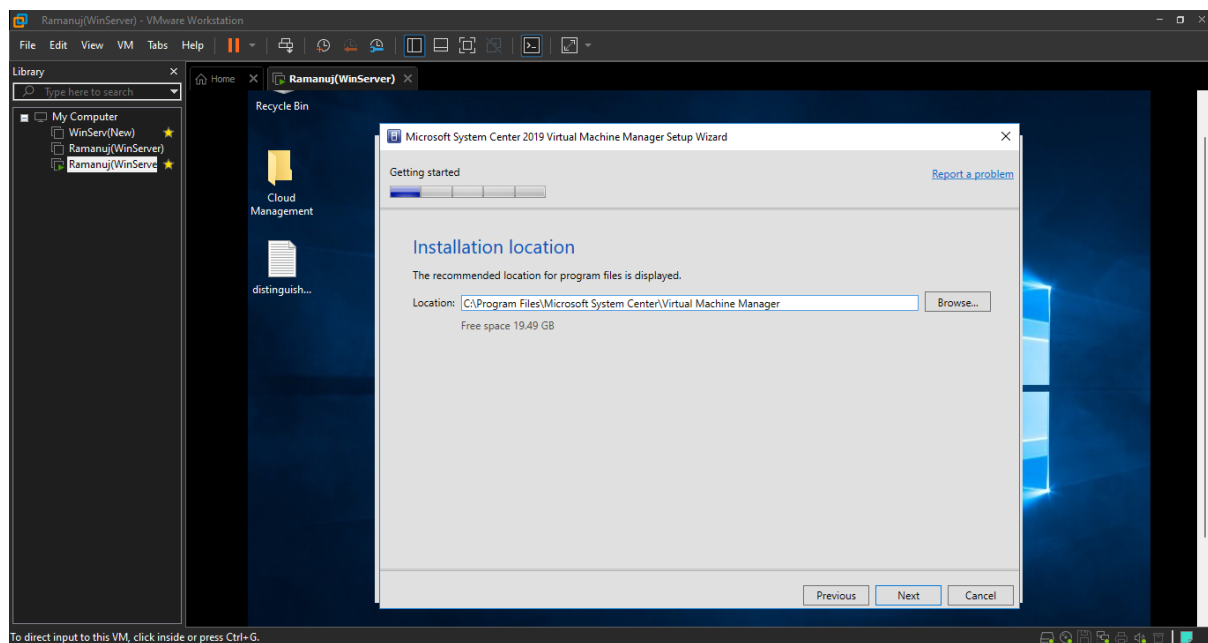
- Click **Next**



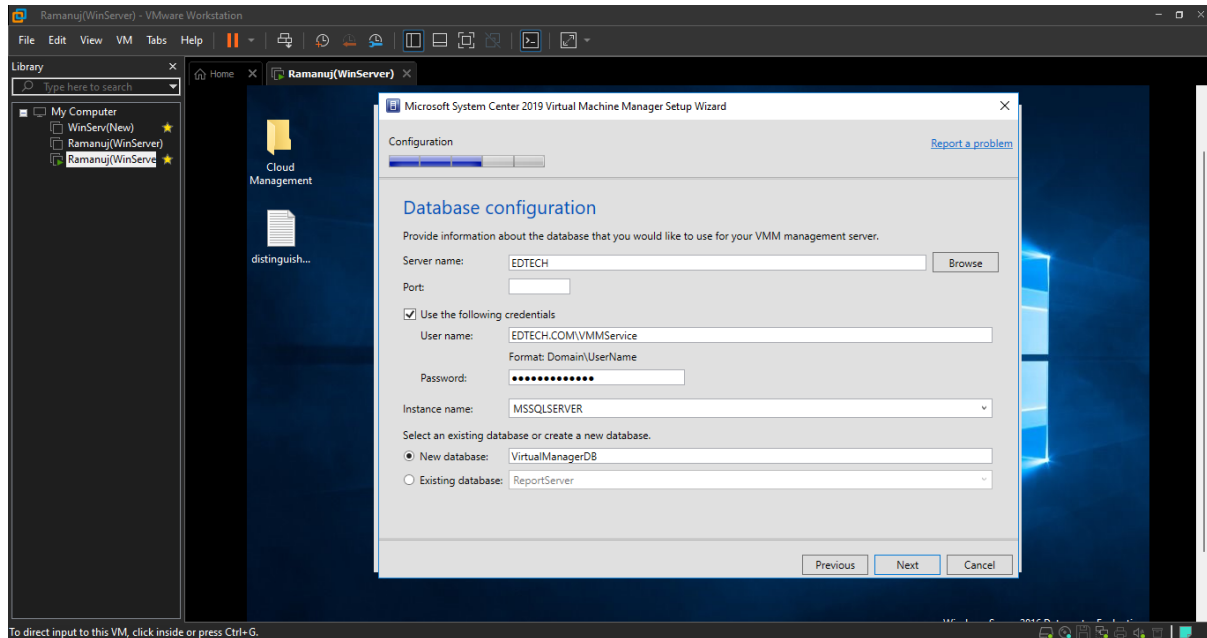
- Select **Off** for Microsoft Updates and Click **Next**



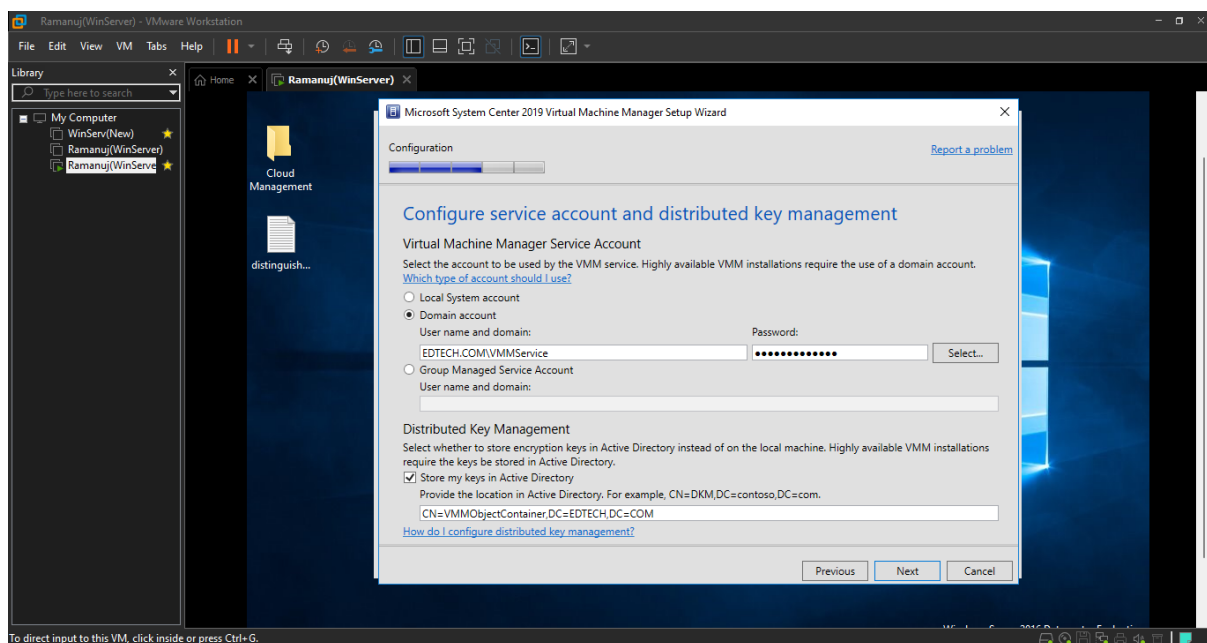
- Keep default values and Click **Next**



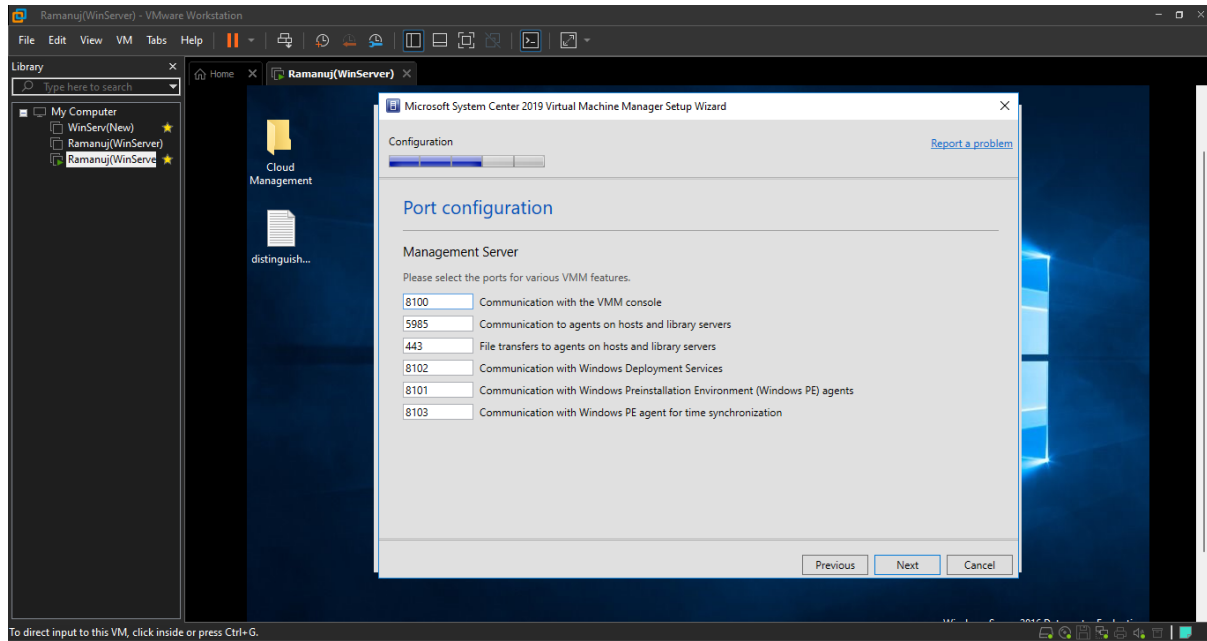
- Now in database configuration, type the same user and domain which was created (In this case its **VMMService**) and keep the instance name as default **MSSQLSERVER**



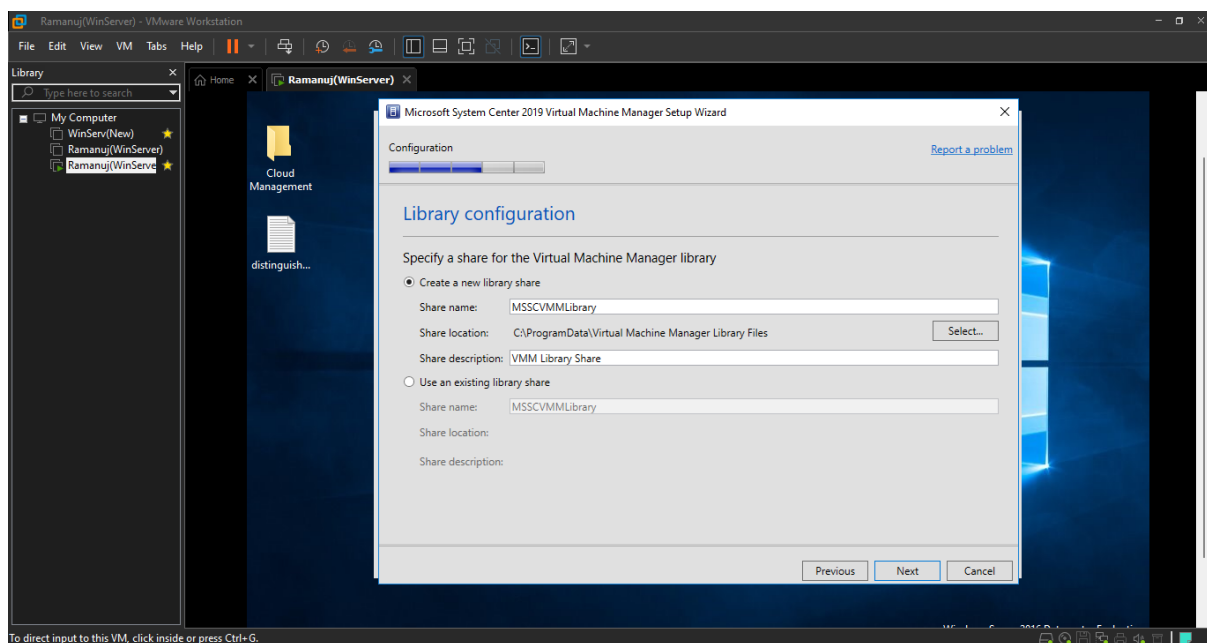
- Here in service account configuration, Select Domain Account and type in the User and Password (For VMMService)
- Select Store my keys in Active Directory and paste the **distinguishedName** that we have saved from before.



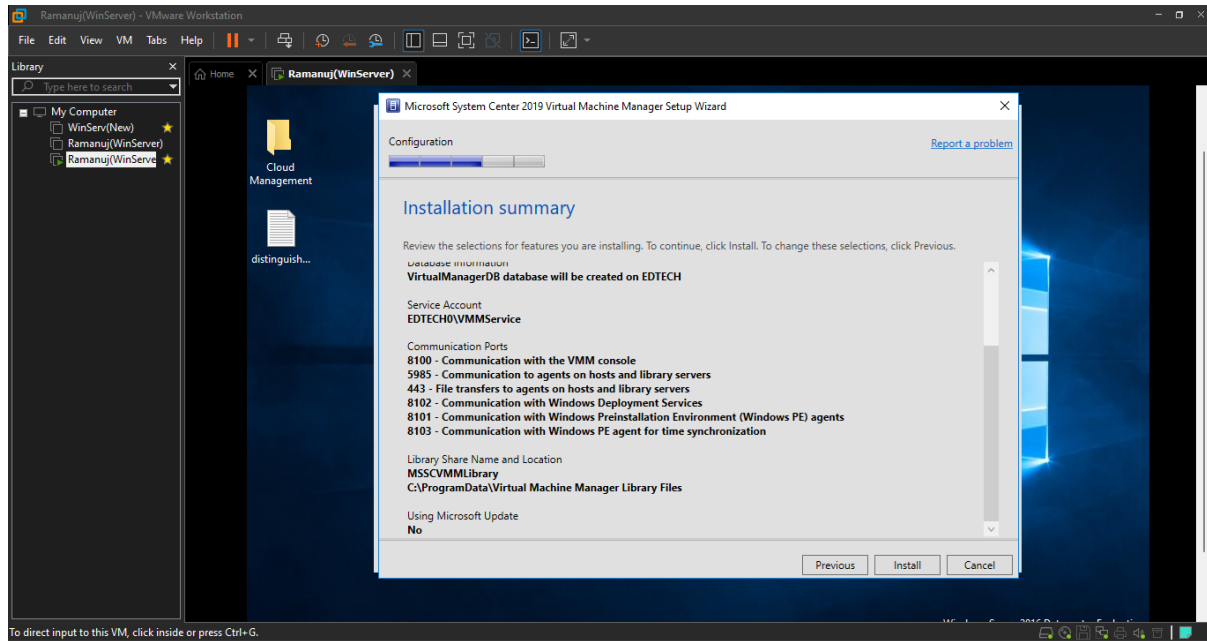
- Keep default values and Click **Next**



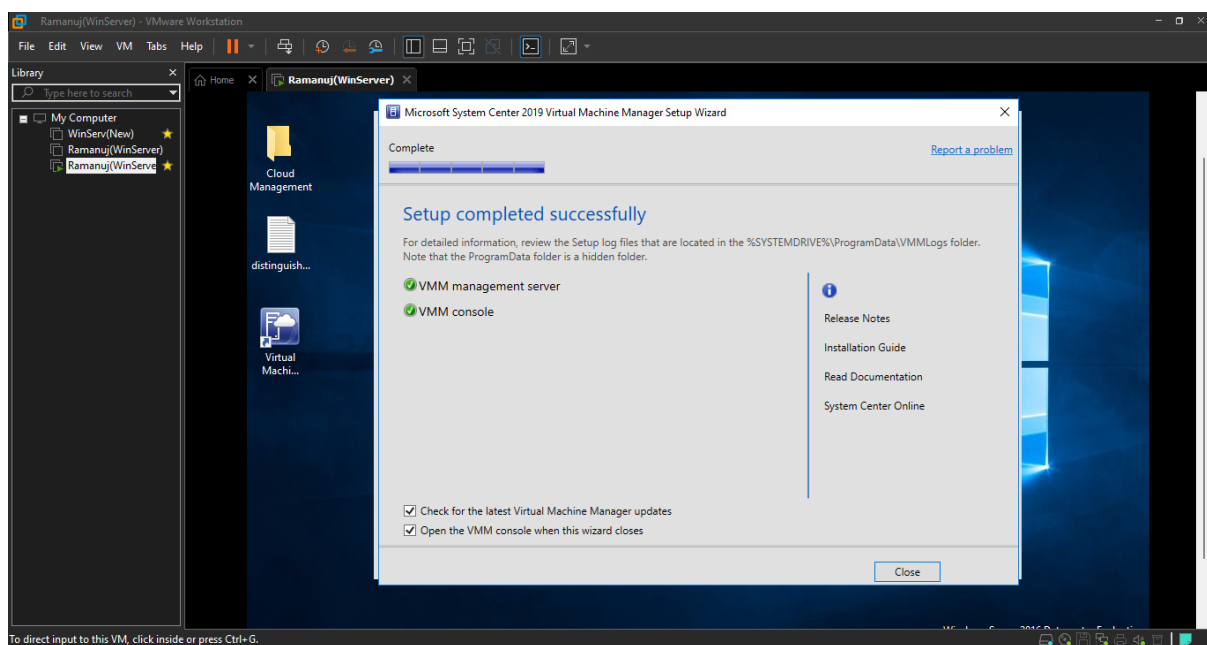
- Click on Next



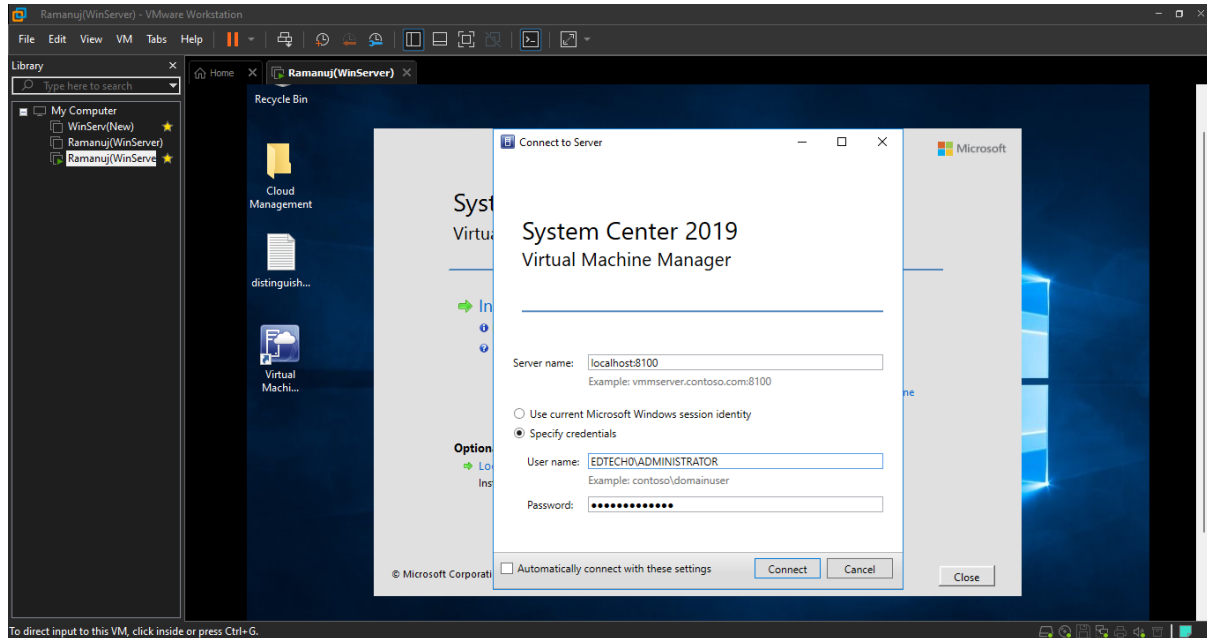
- Review the Installation Summary and Click on **Install**



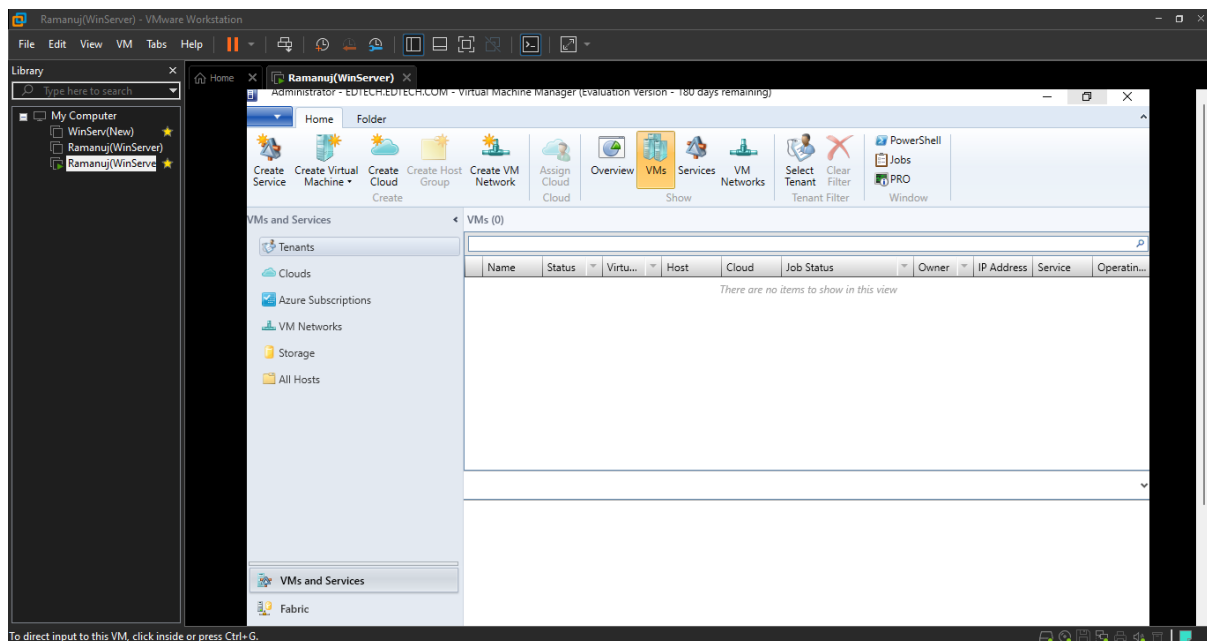
- After installation Click on **Close**



- Now when SCVMM starts enter your Domain controller credentials (In this case its EDTECH0\ADMINISTRATOR) with its password.
- After this SCVMM will start

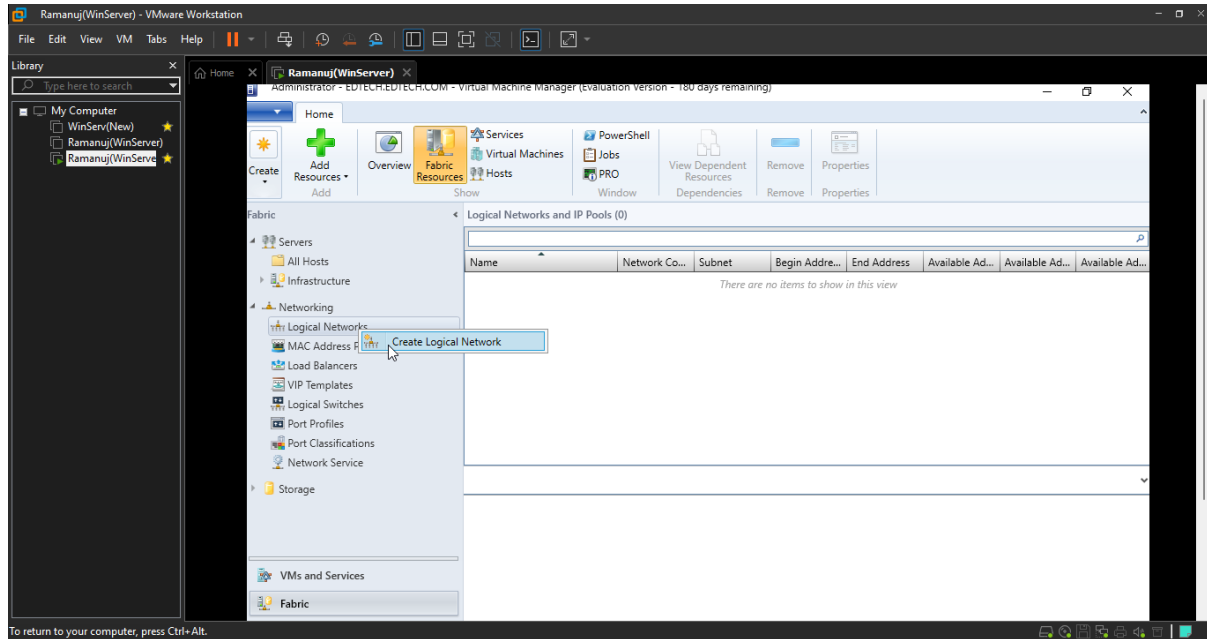


- This is the starting screen of SCVMM

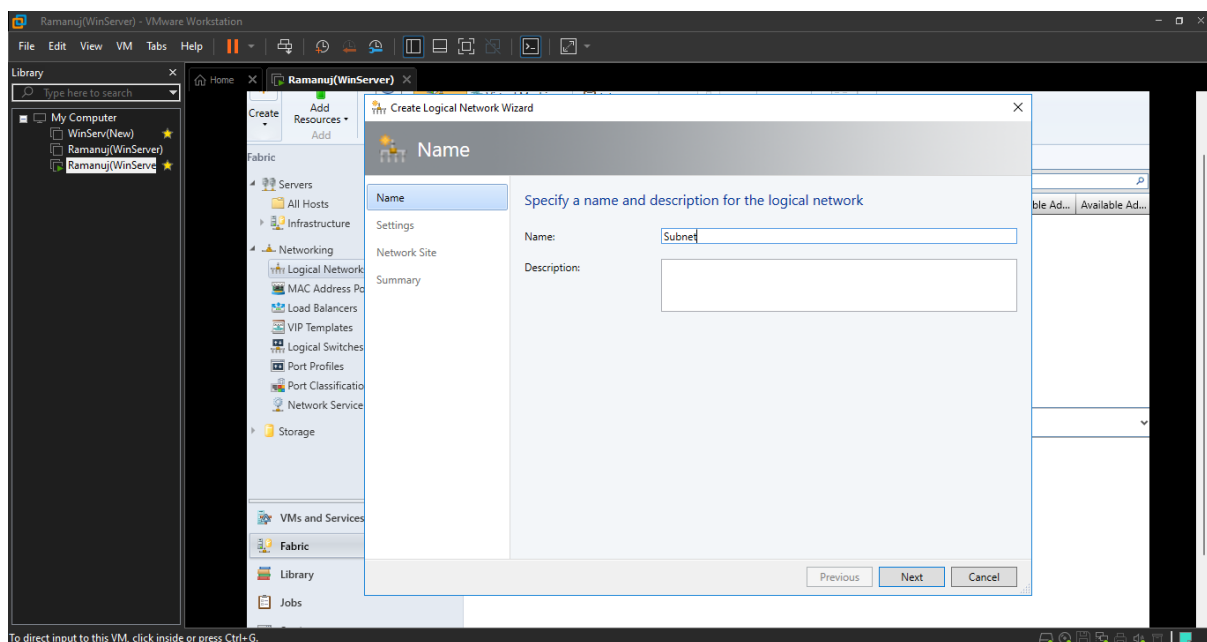


Step 41: Creation on Logical Networks

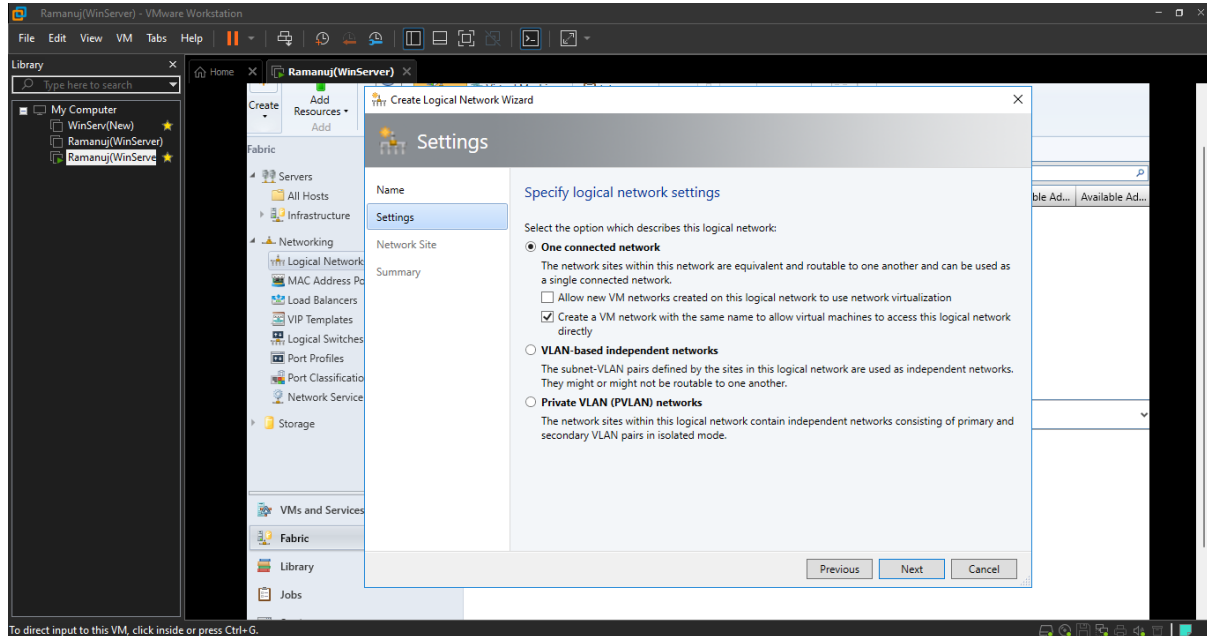
- Click on **Fabric**
- Click on **Networking**
- Within Networking Right-Click on **Logical Network** and Select **Create Logical Network**



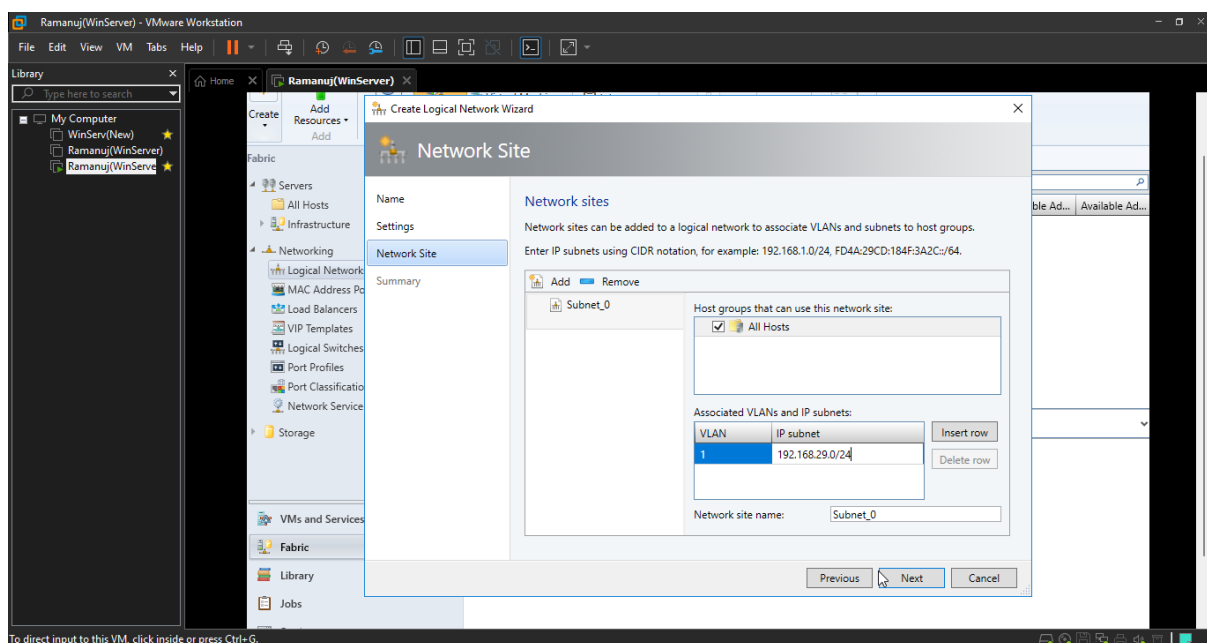
- Call this Subnet and Click **Next**



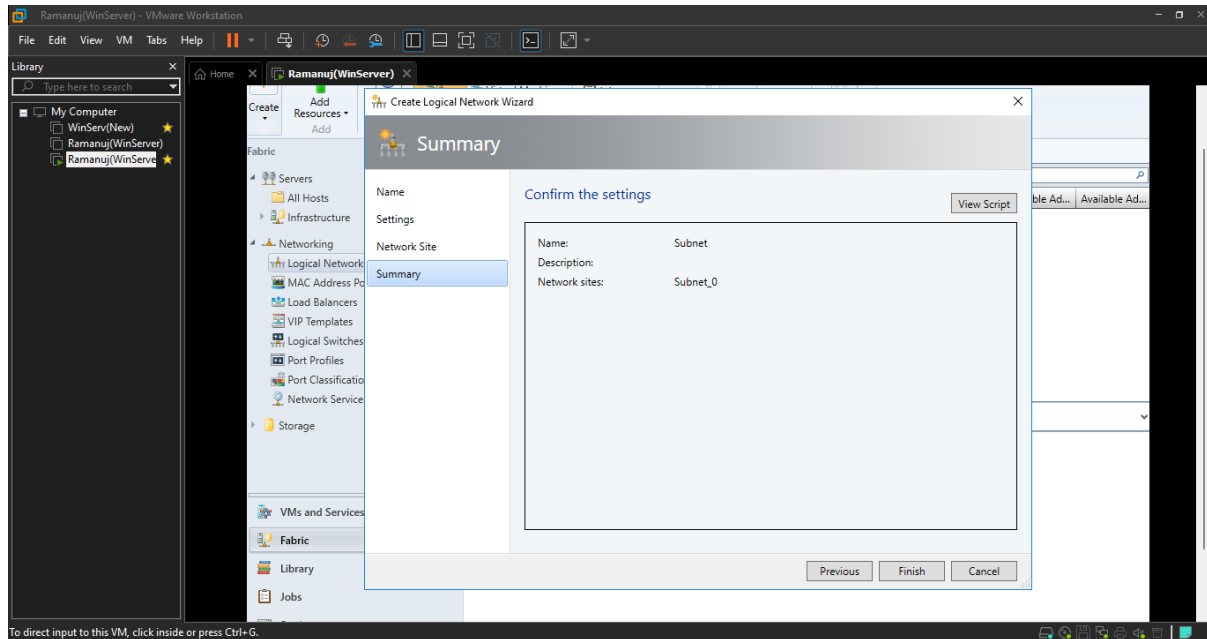
- Select **One Connected Network** and within that Tick **Create a VM network with the same name to allow virtual machines to access this logical network directly** and Click **Next**



- In Network Sites and Click **Add**
- Click on **All hosts**
- Click on **Insert Now** and Type **1** in VLAN and a sample IP (Here it is 192.168.29.0/24) and Click **Next**

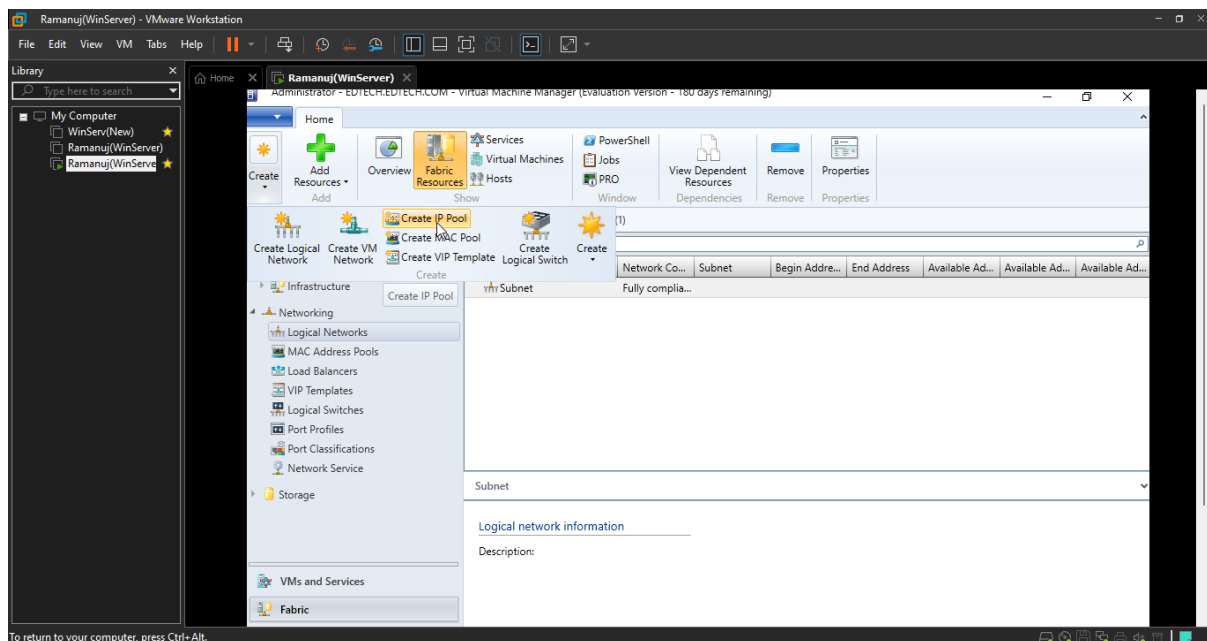


- Check the Summary and Click on **Finish**

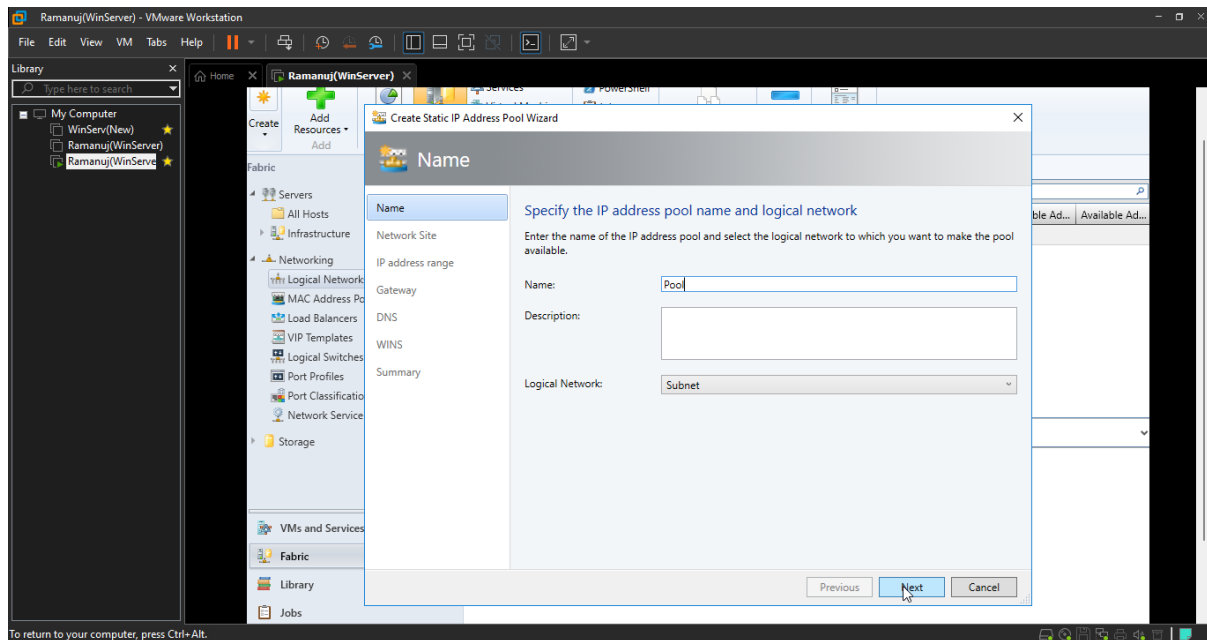


Step 42: Creating IP Pools

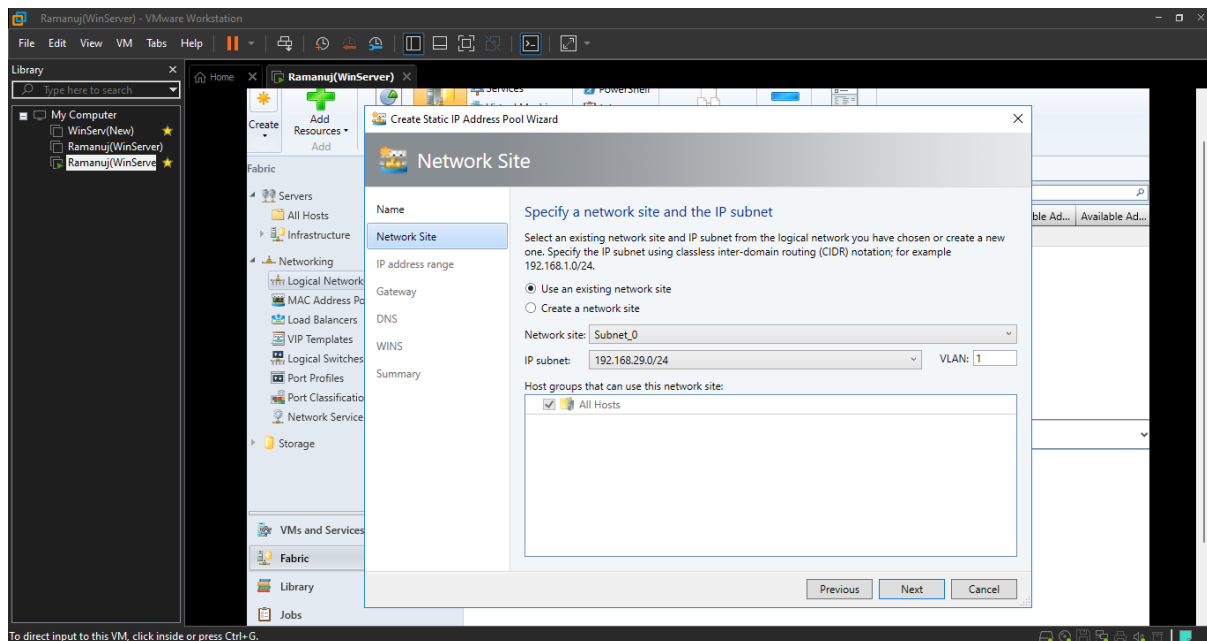
- Now in SCVMM on the Top Click on **Create**
- Within Create Click on **Create IP Pool**



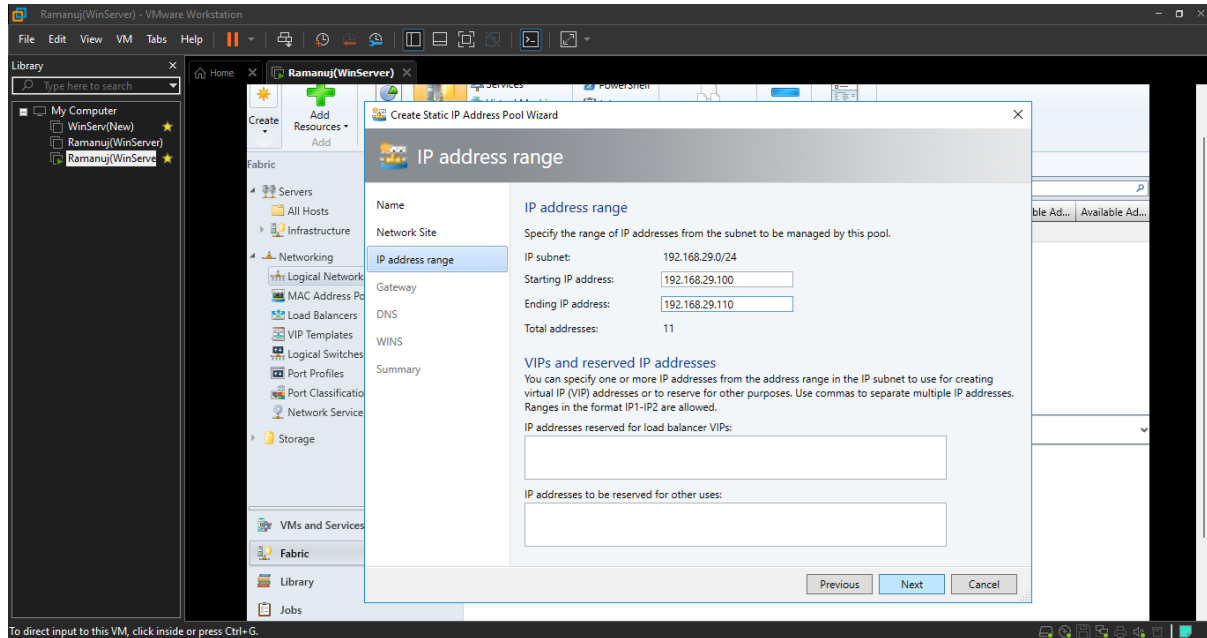
- Call it **Pool** and Make sure Logical Network is **Subnet**
- Then Click **Next**



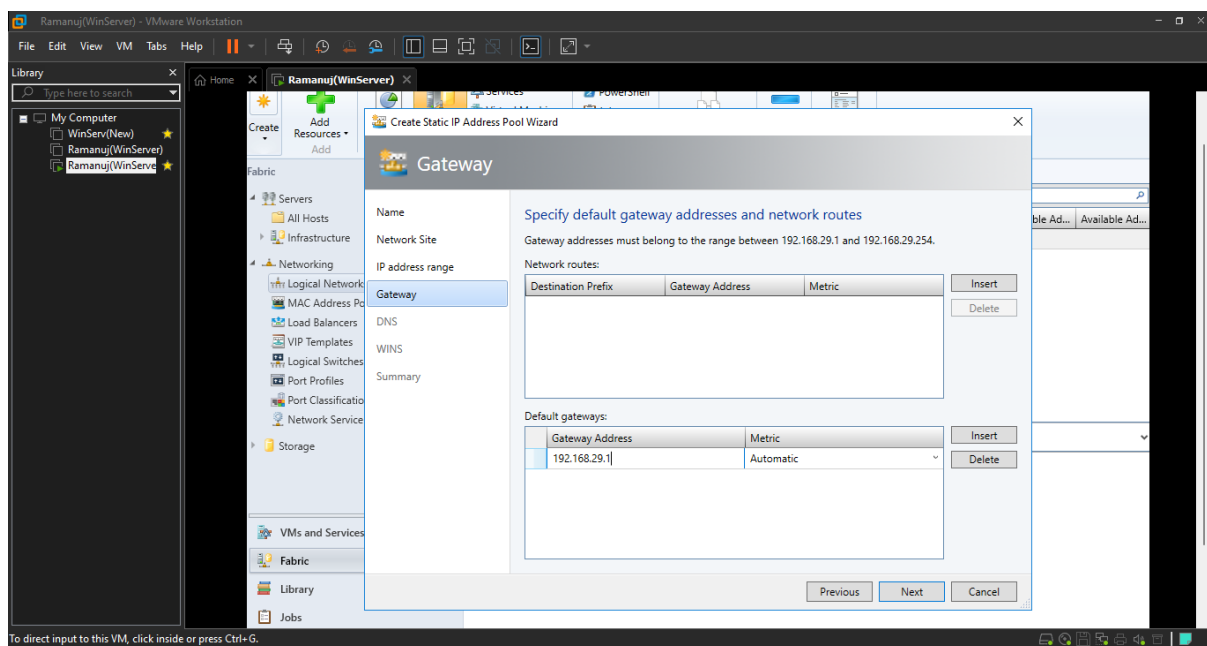
- Keep default values and Click **Next**



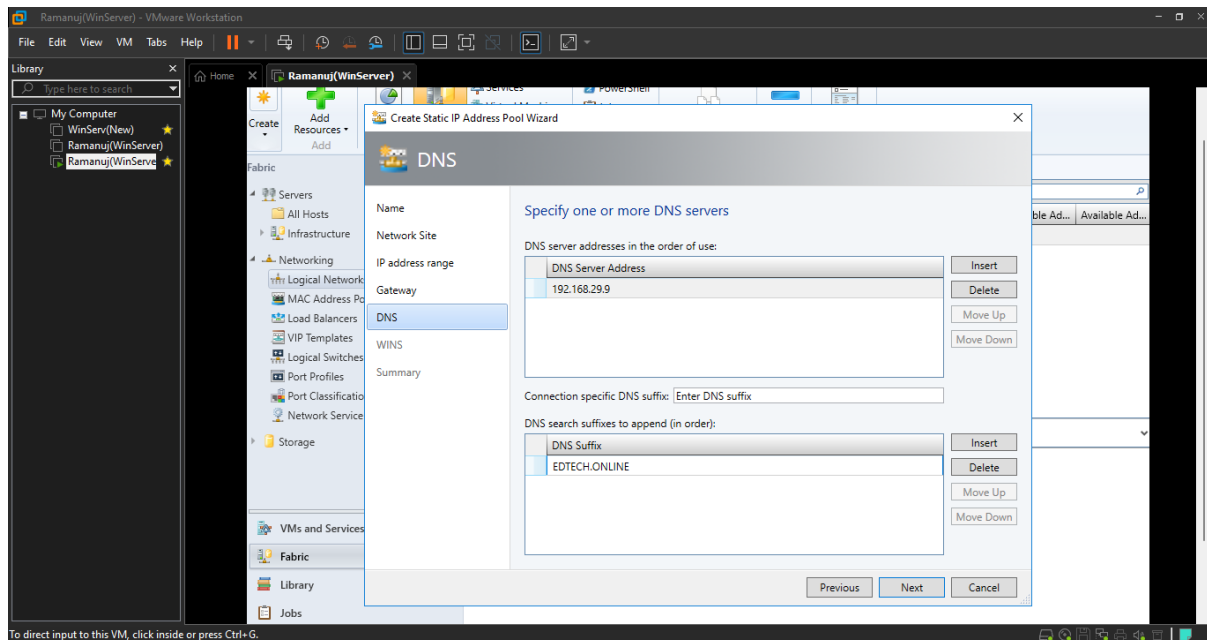
- In IP Address Range change the **Starting IP Address** to **192.168.29.100** and **Ending IP Address Range** to **192.168.29.110**
- Click **Next**



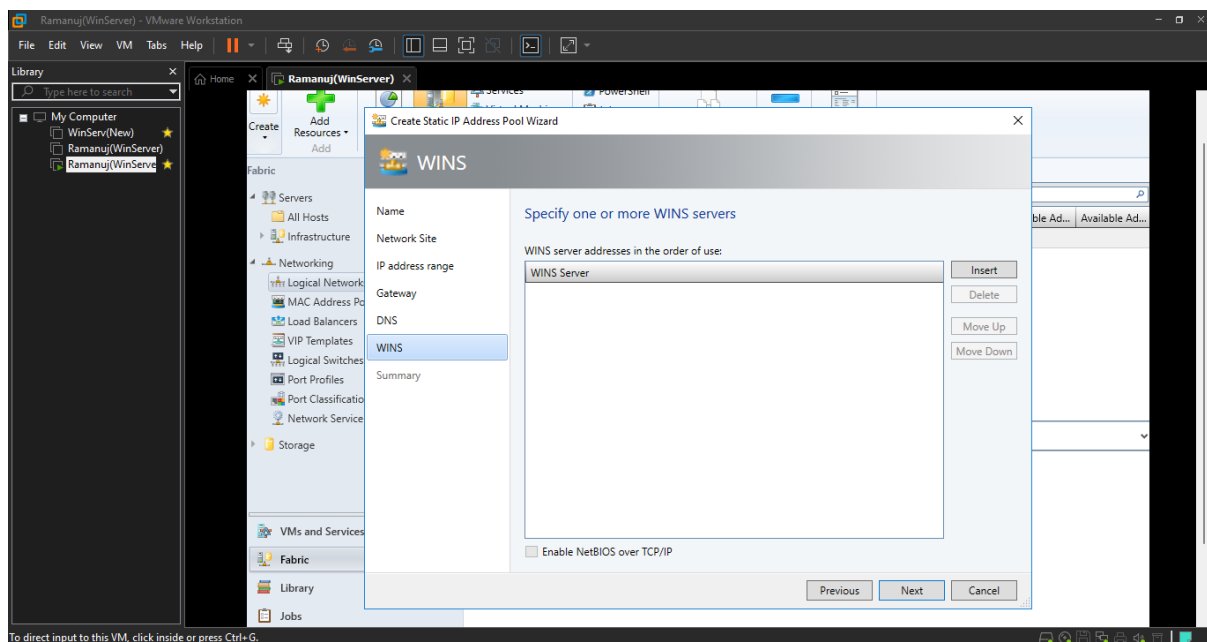
- In Gateway,
- Within Default Gateways Click on **Insert**
- Type the Gateway Address as **192.168.29.1** and Keep the metric as **Automatic**
- Click **Next**



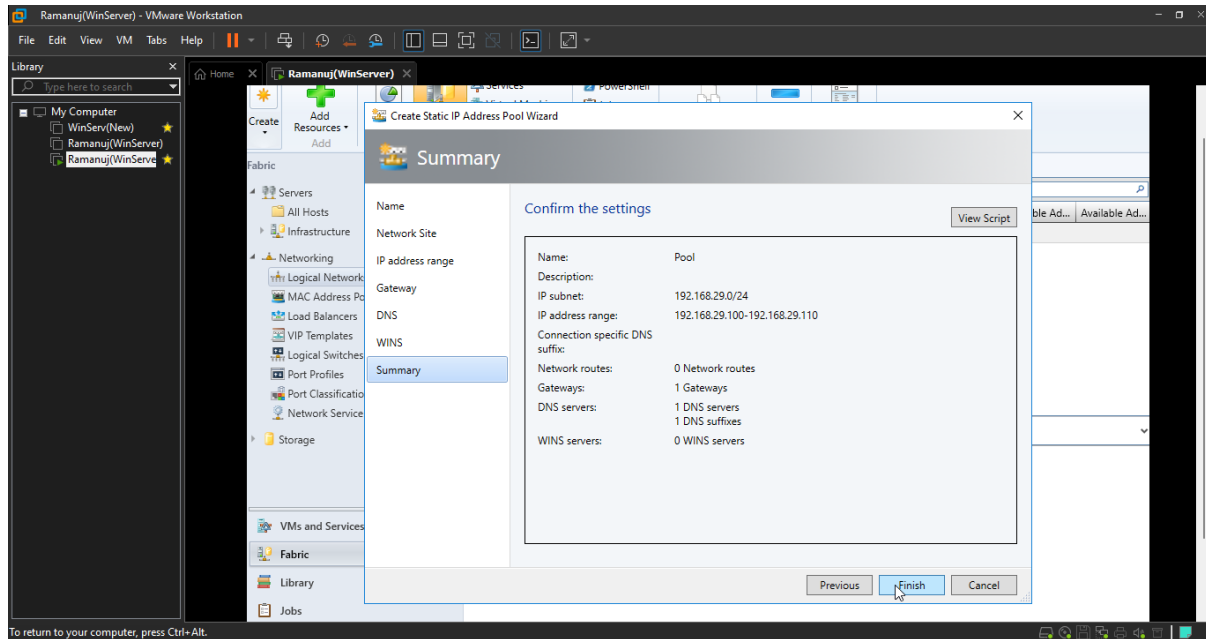
- In DNS, within **DNS server address in the order of use** Click on **Insert**
- Type in the IP Address **192.168.29.9**
- In **DNS Suffix** type your **Domainname.Online** i.e in this case it is **EDTECH.ONLINE**
- Click **Next**



- In WINS, keep it as is and Click **Next**

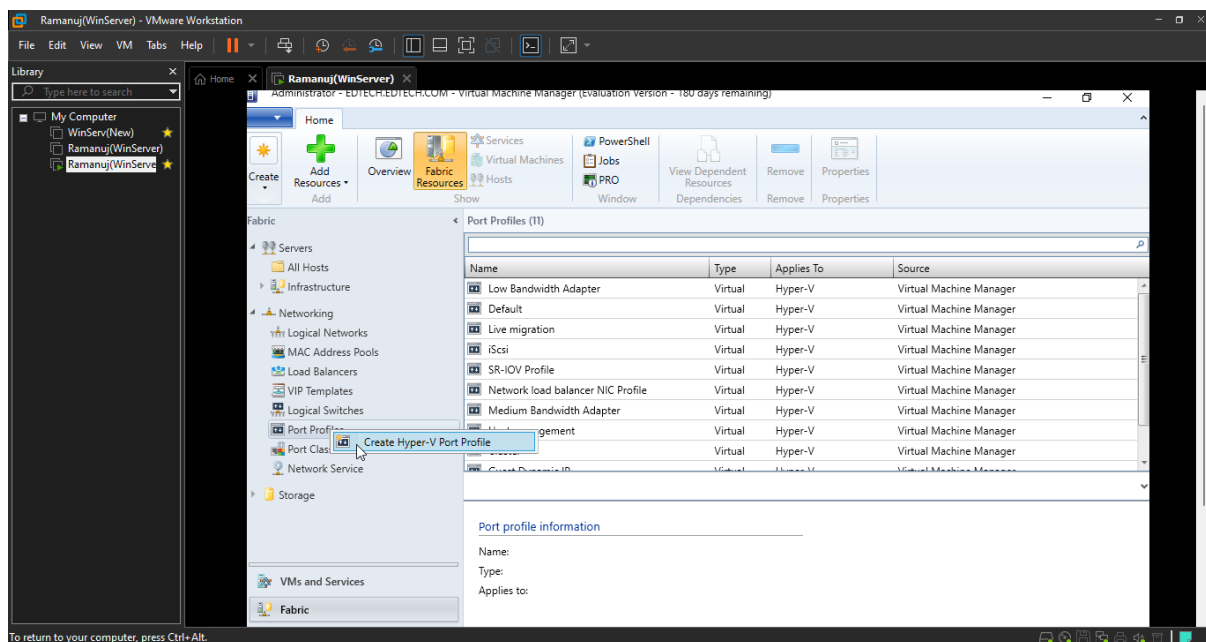


- View the Summary and Click **Finish**

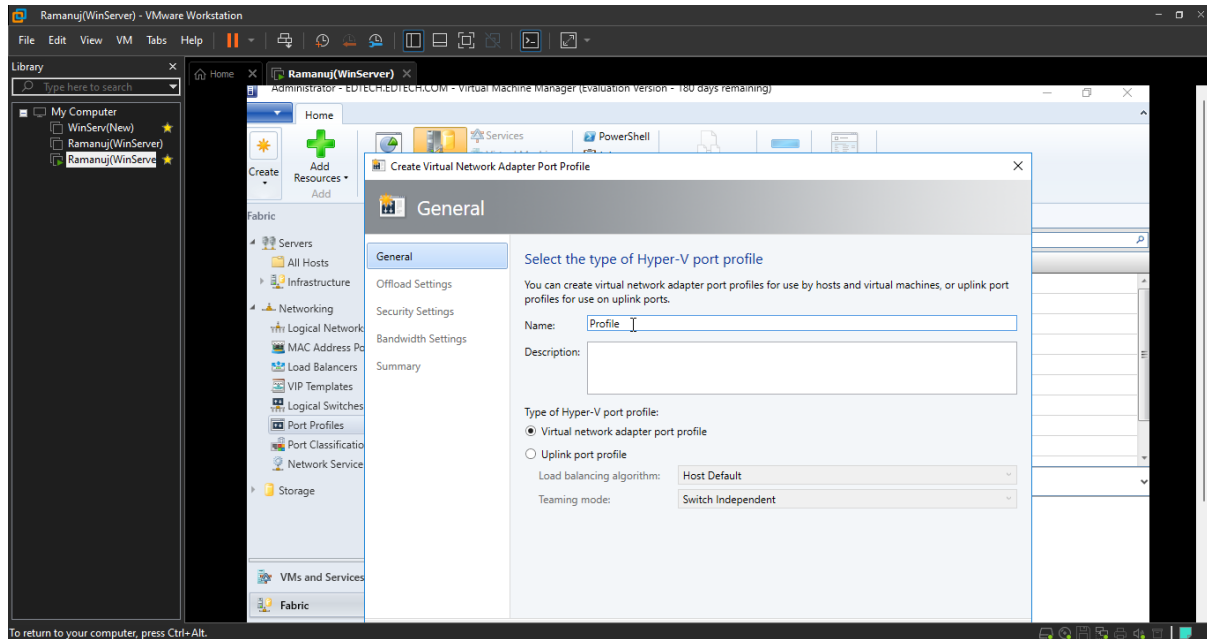


Step 43: Create IP Port Profiles

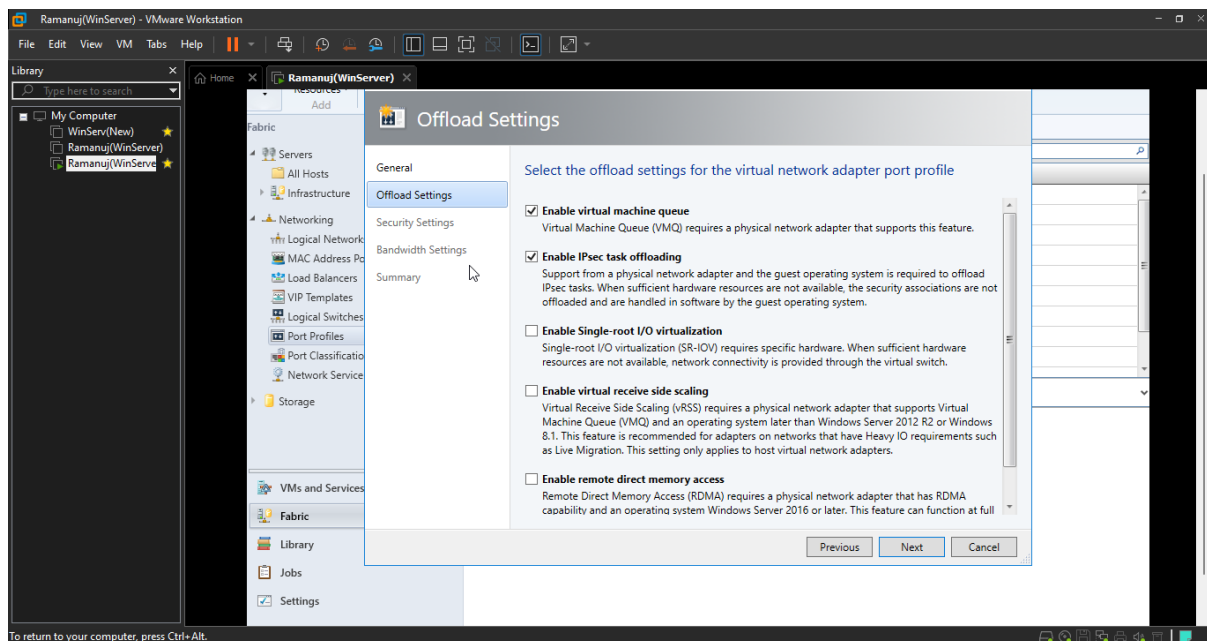
- In SCVMM Click on **Fabric**
- Within Fabric Click on **Networking**
- Within Networking Right-Click on **Port Profiles** and Select **Create Hyper-V Port Profile**



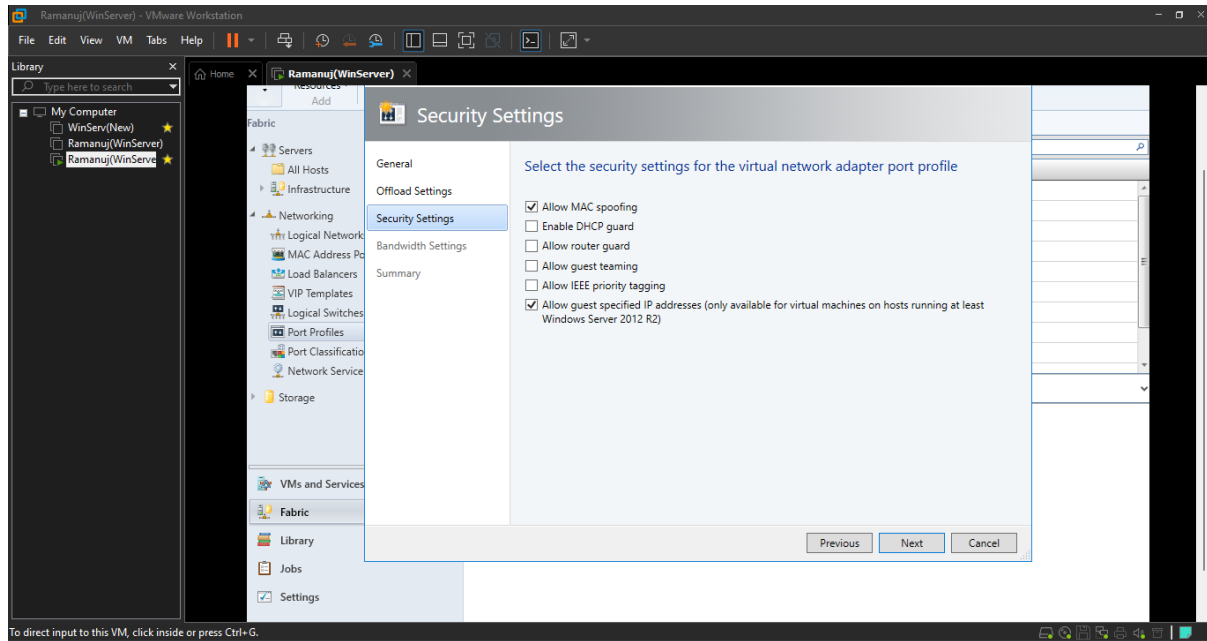
- Call this **Profile** and Click **Next**



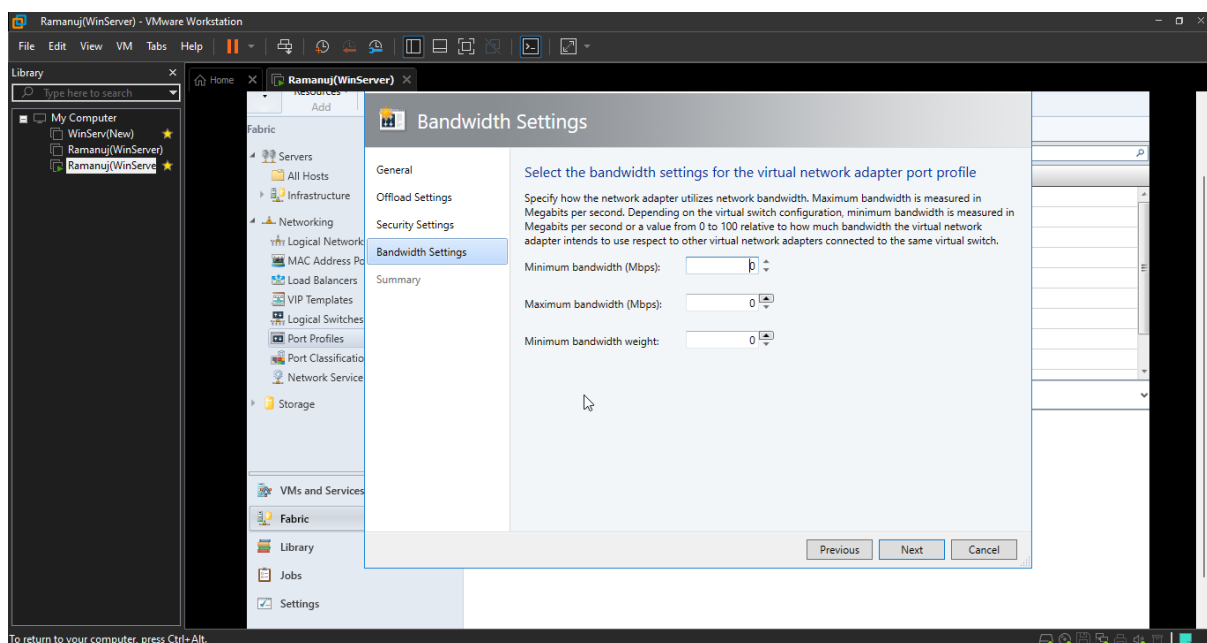
- In Offload Setting Select the following options:
Enable virtual machine queue
Enable IPsec task offloading
- Click on **Next**



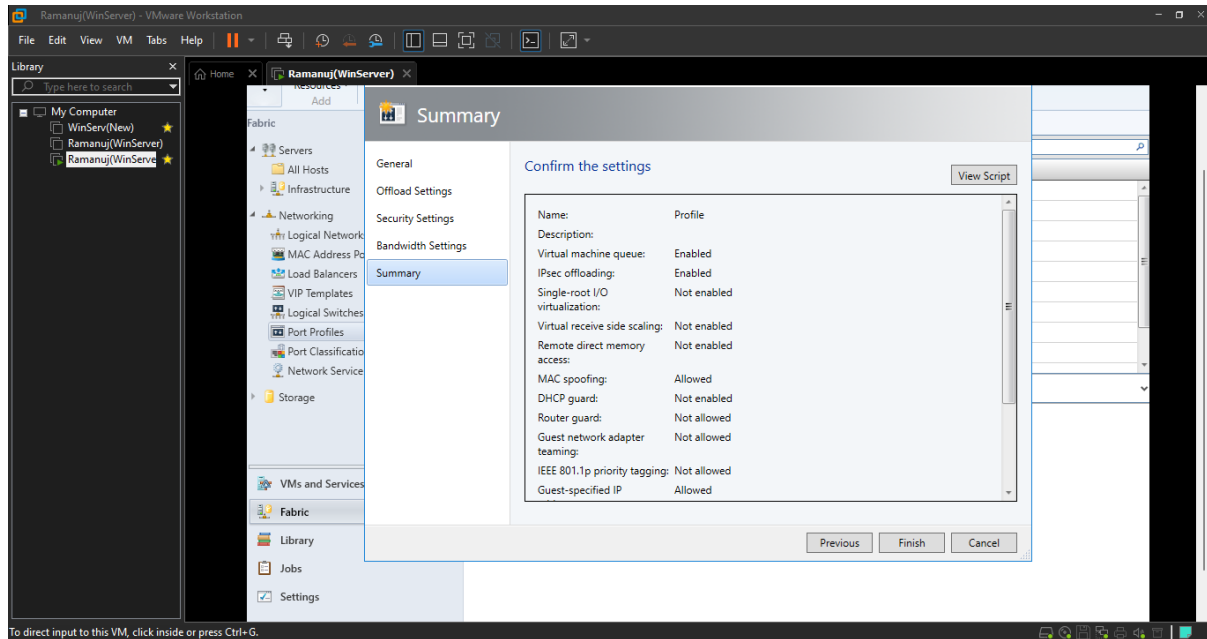
- In Security Settings Select **Allow MAC spoofing** and Click **Next**



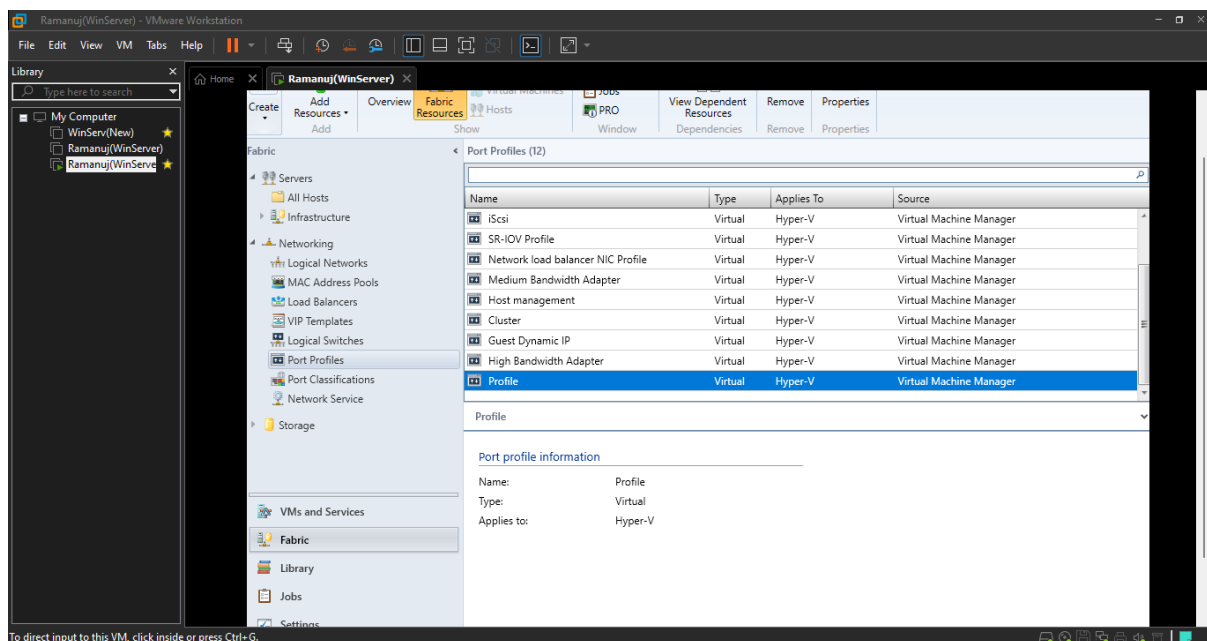
- Keep default values and Click **Next**



- View Summary and Click **Finish**

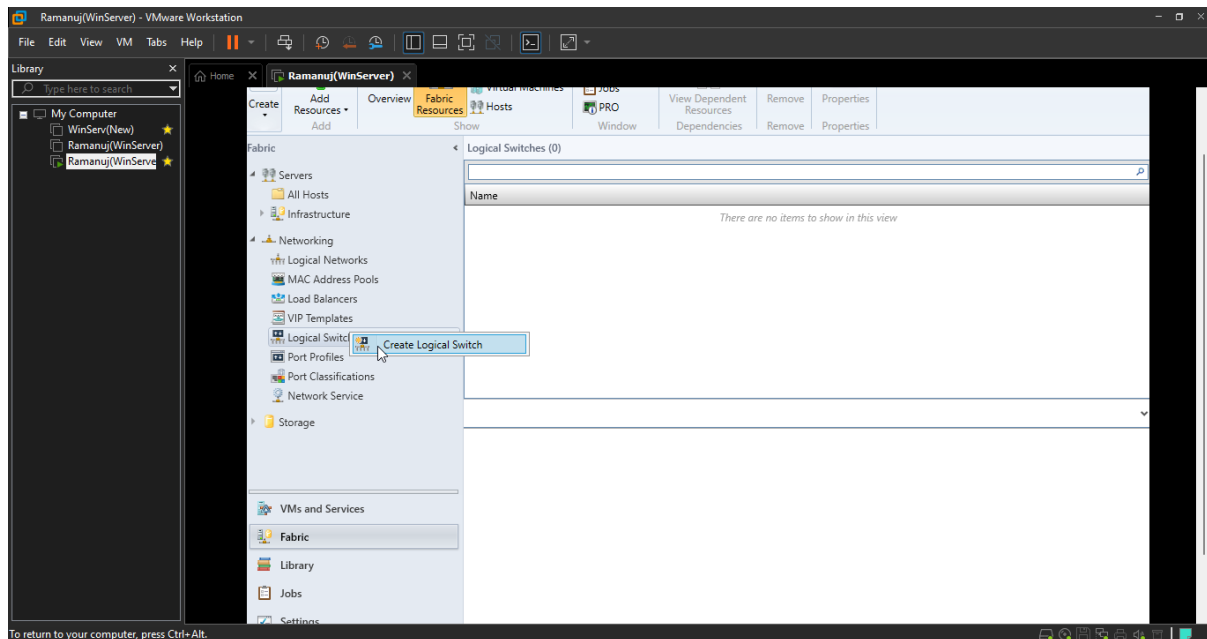


- New Port Profile called **Profile** is created

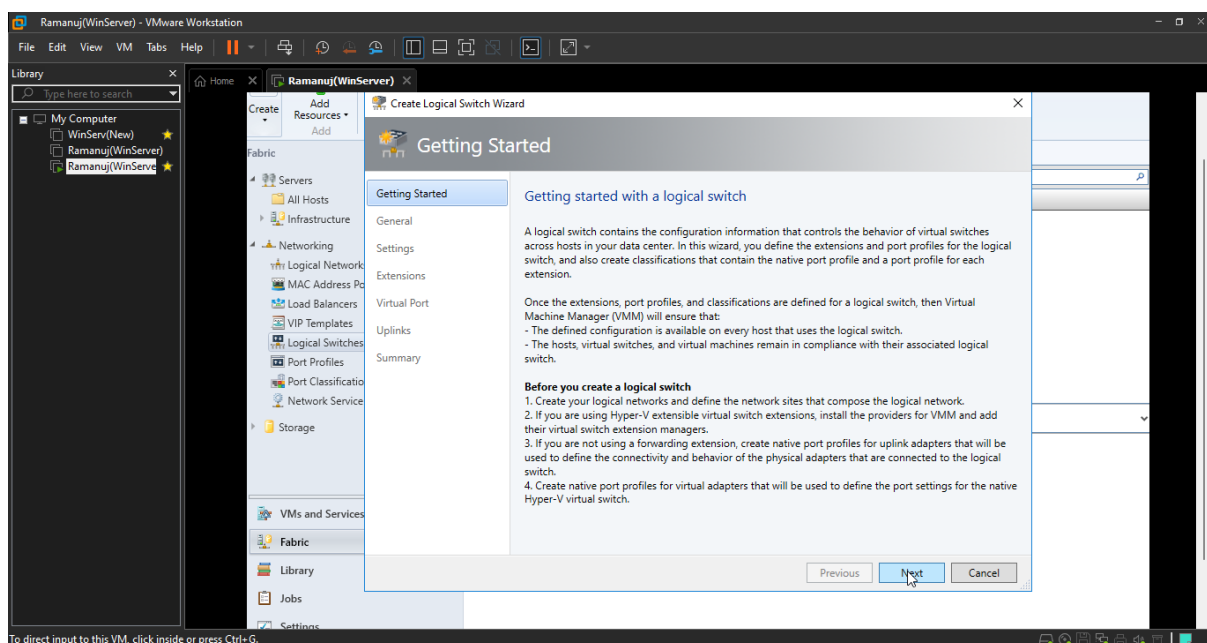


Step 44: Create Logical Switches

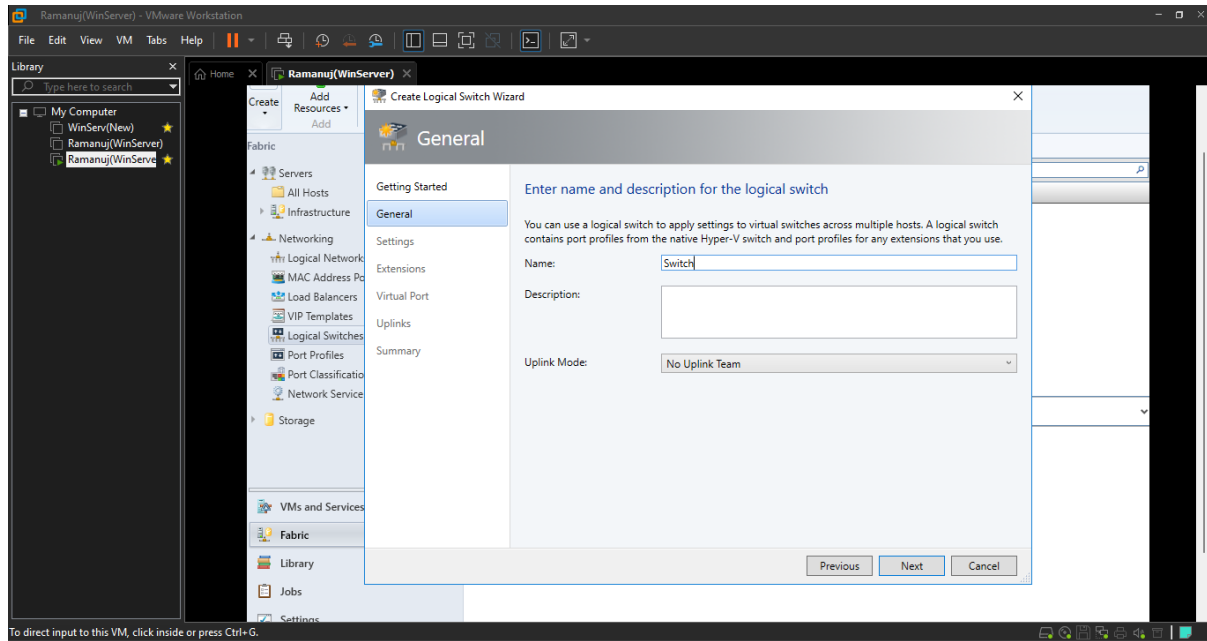
- In SCVMM Click on **Fabric**
- Within Fabric Click on **Networking**
- Within Networking Right-Click on **Logical Switches** and Select **Create Logical Switch**



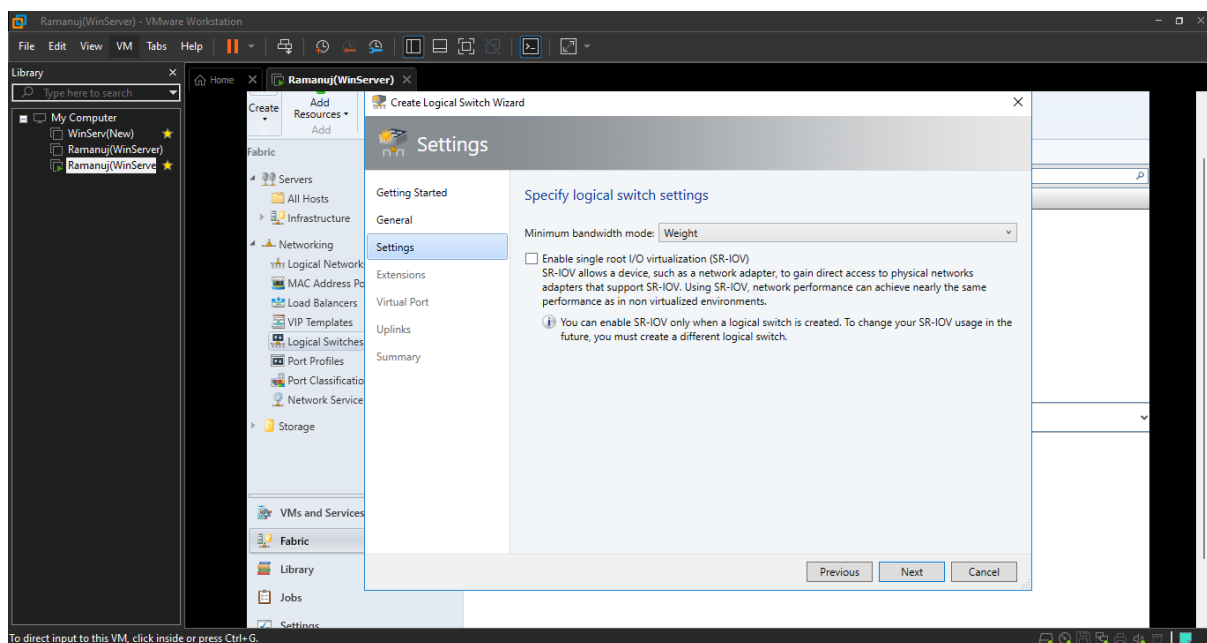
- Click **Next**



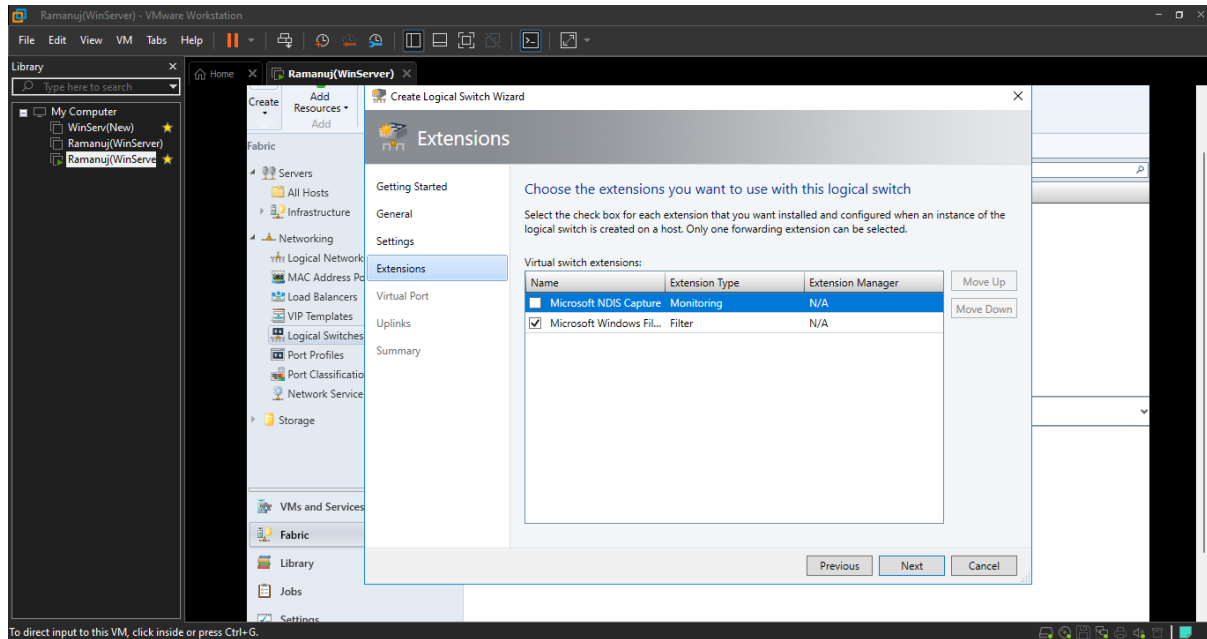
- Call it **Switch** and Click **Next**



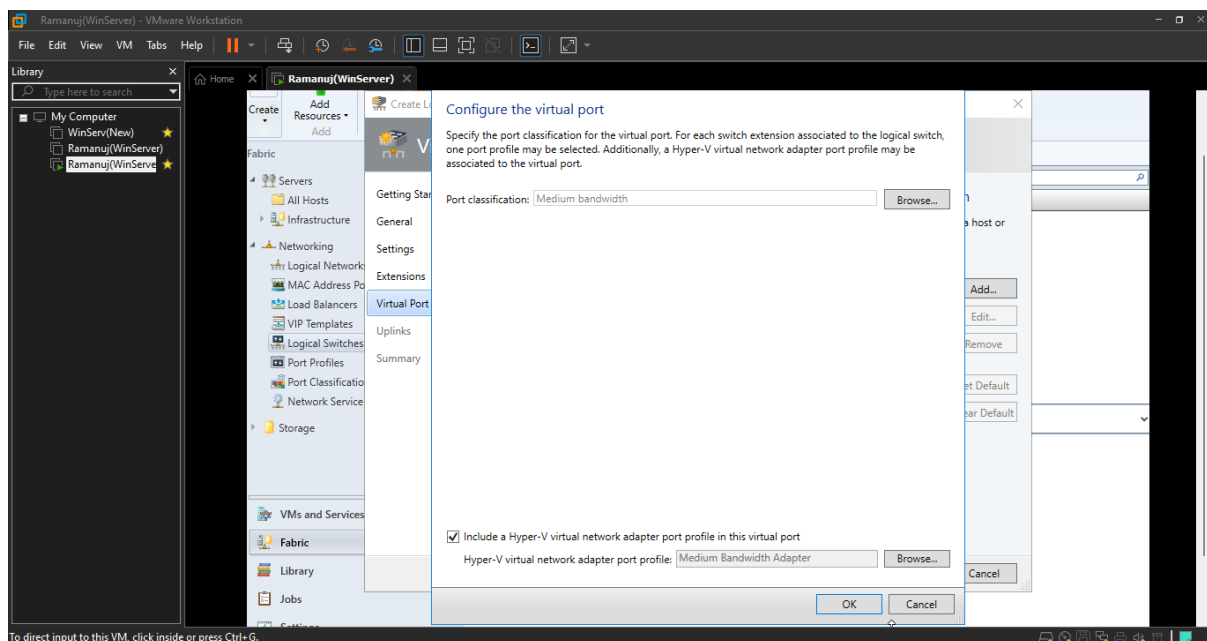
- Keep default values and Click **Next**



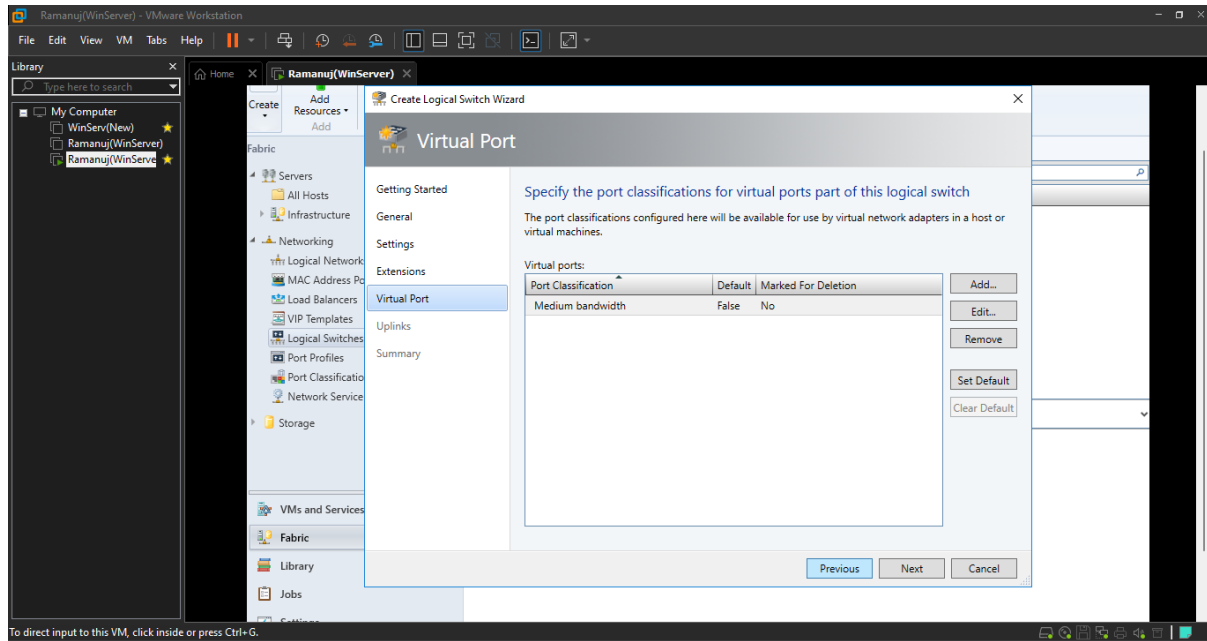
- Keep default values and Click **Next**



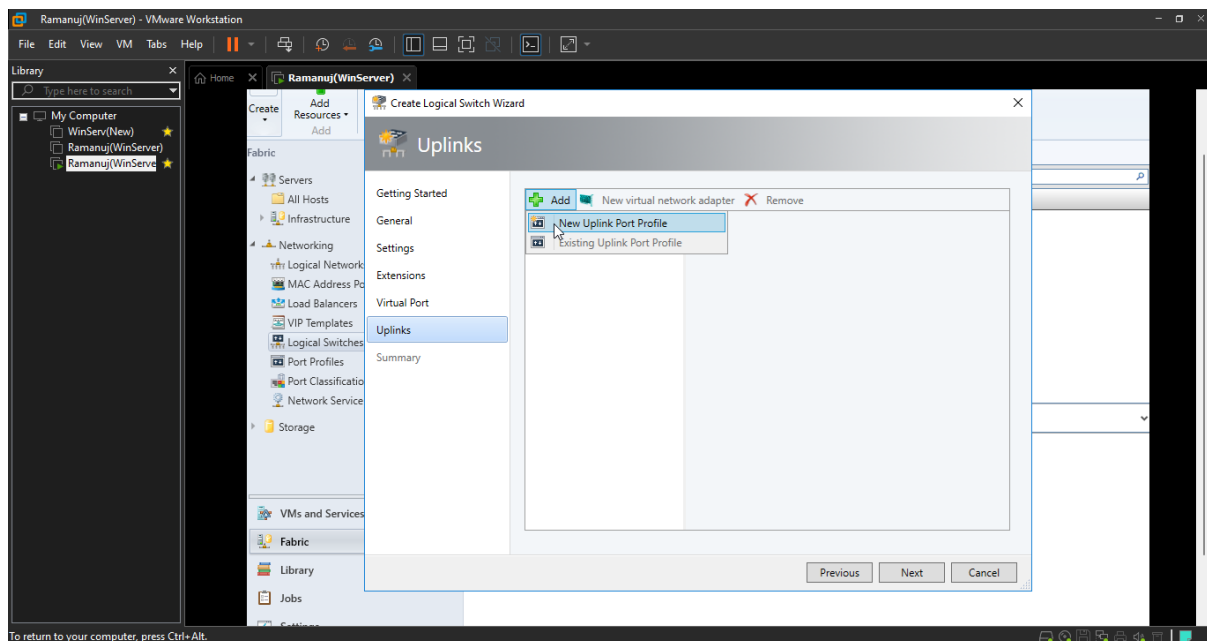
- In Virtual Port Click **Add**
- In Port Classification Select **Medium Bandwidth**
- Click on **Include a Hyper-V Virtual Network adapter port profile in this virtual port** and Select **Medium Bandwidth**
- Click on Ok



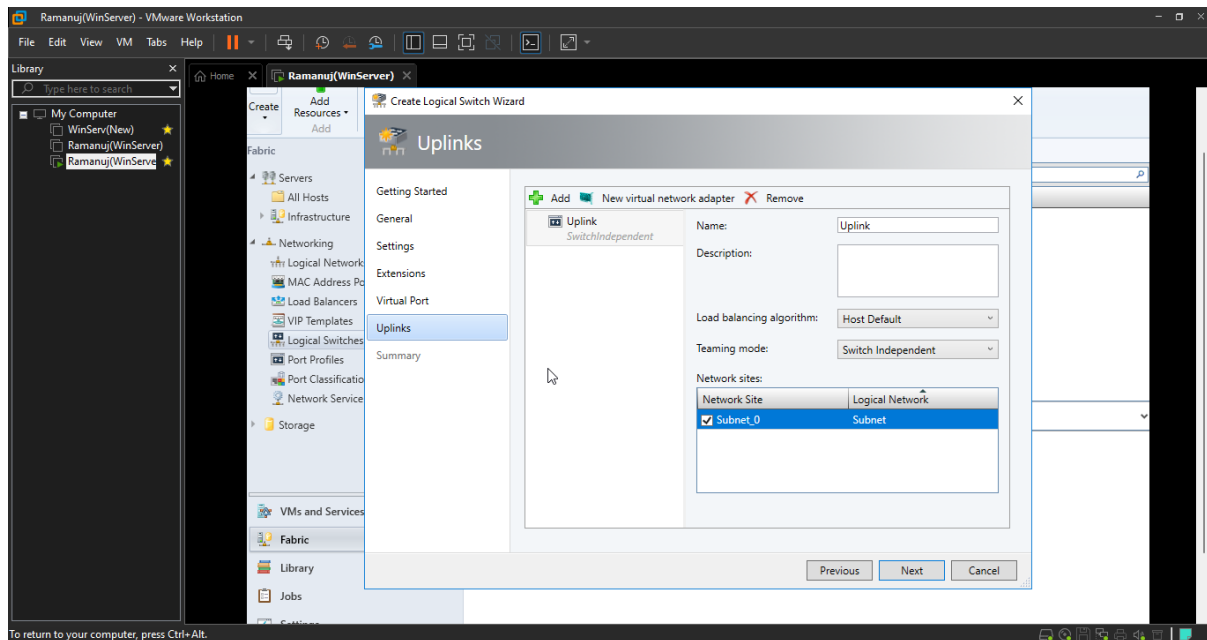
- On Virtual Port Click **Next**



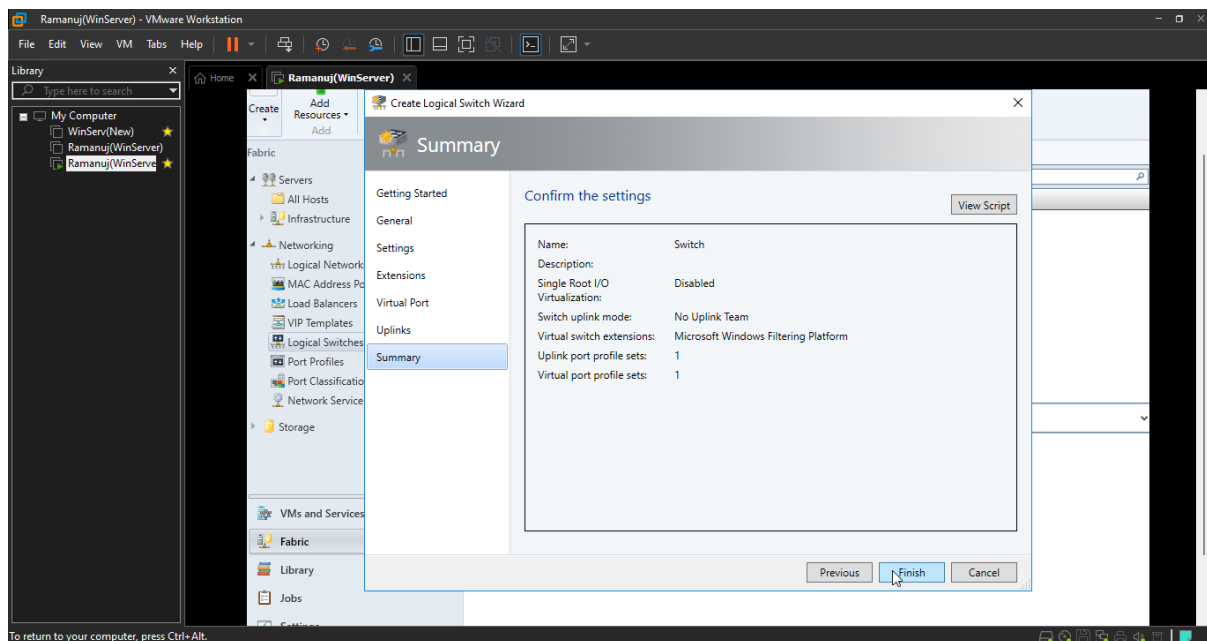
- On Uplink Click on **Add**
- Select **New Uplink Port Profile**



- In Name type **Uplink**
- Select the **Subnet_0**

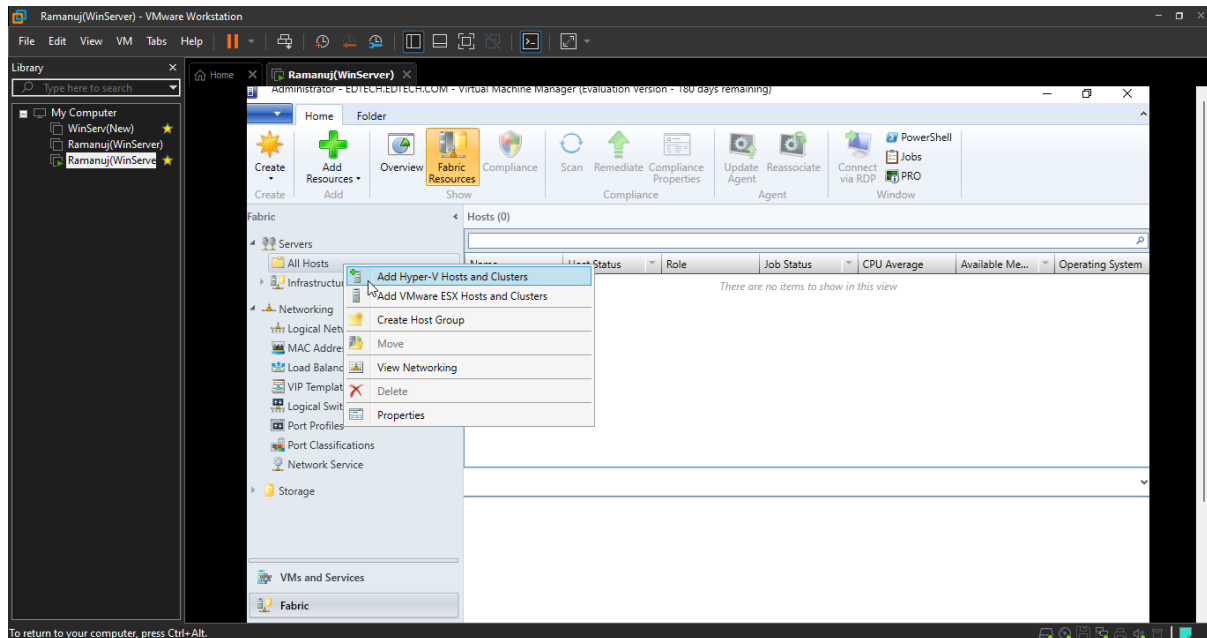


- View Summary and Click **Finish**

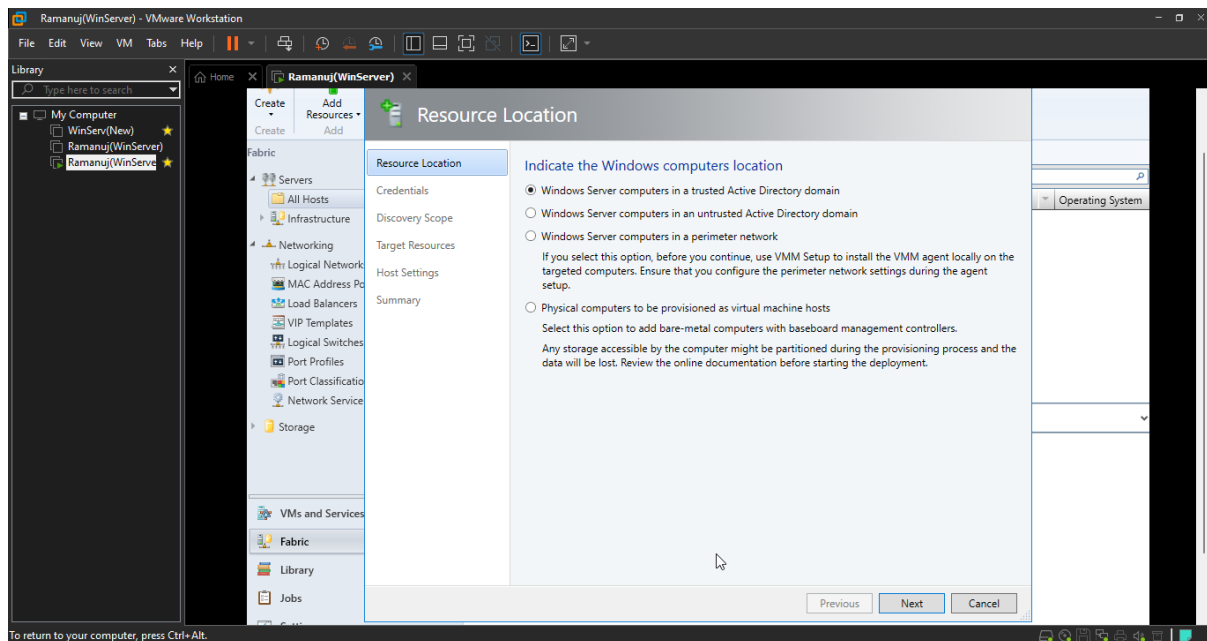


Step 45: Create Hyper-V Hosts & Clusters

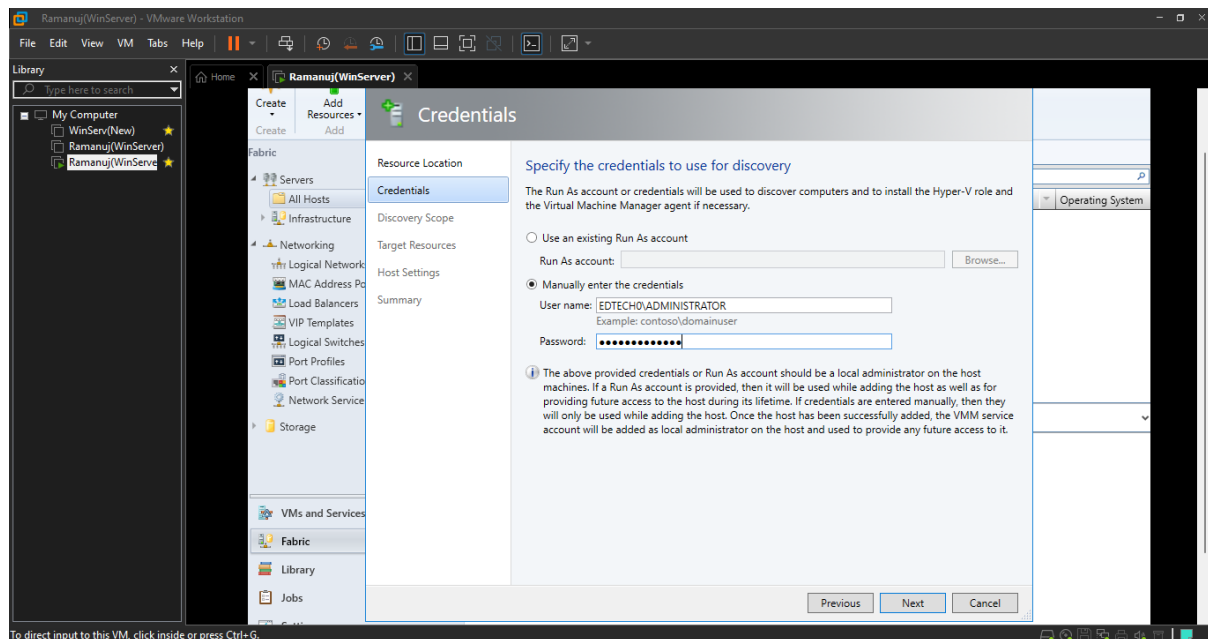
- In SCVMM Click on **Fabric**
- Right-Click on **All Host** and Select **Add Hyper-V Hosts and Clusters**



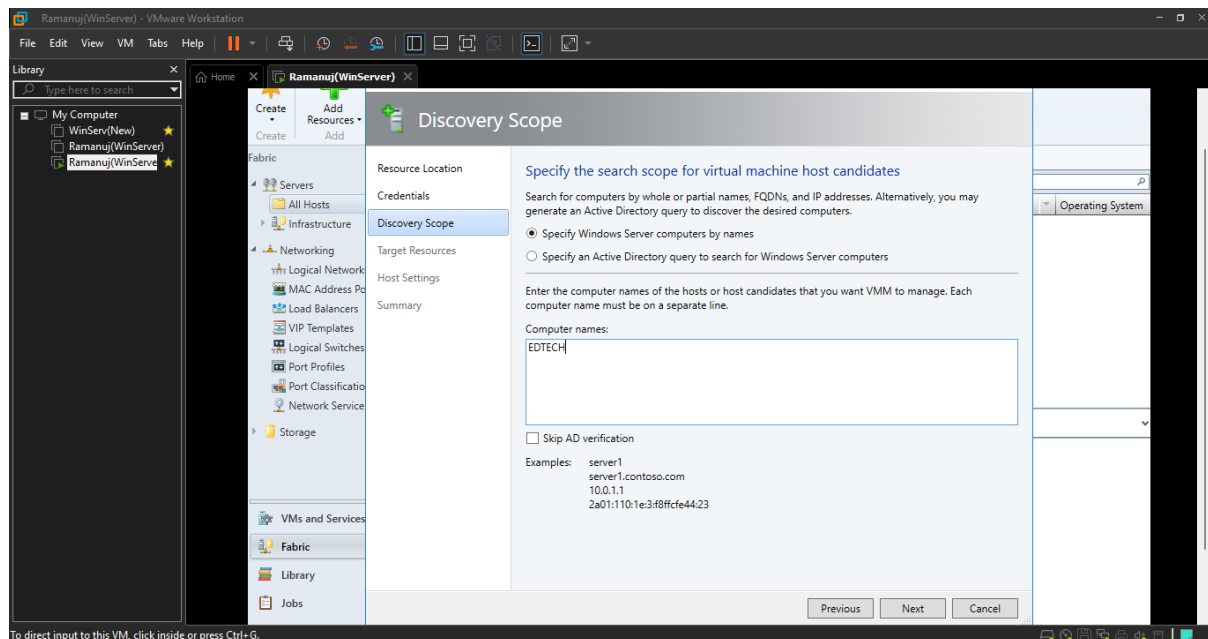
- Keep the default selected values and Click **Next**



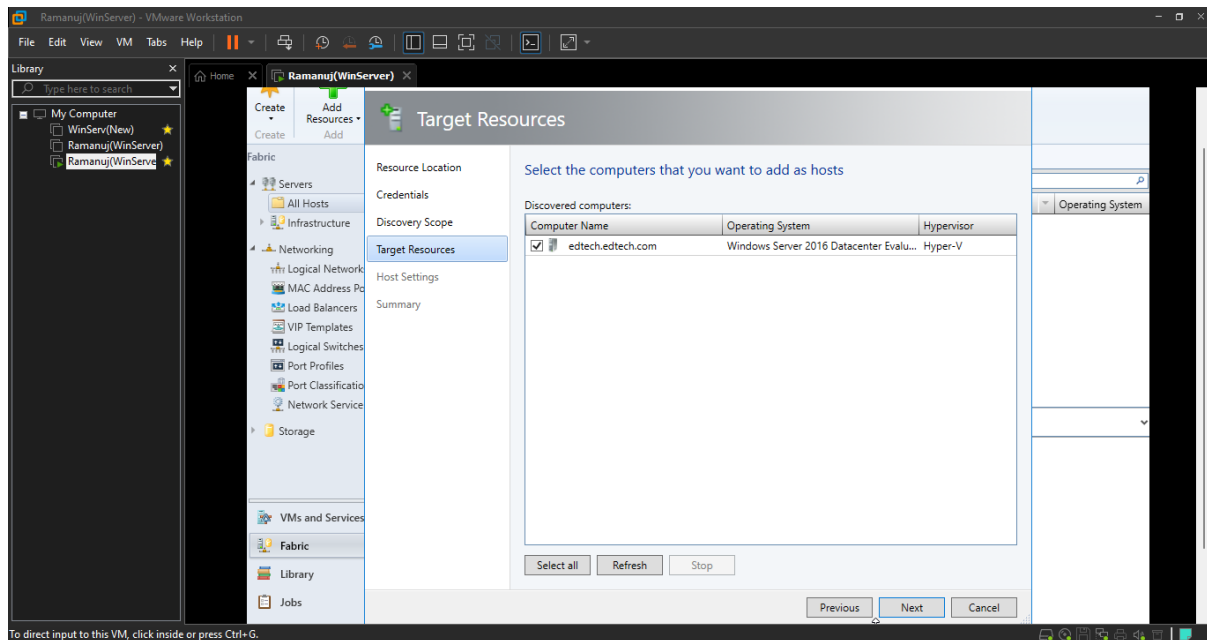
- Select **Manually enter credentials** and enter your Administrator credentials(Here it is EDTECH0\ADMINISTRATOR) and Click on **Next**



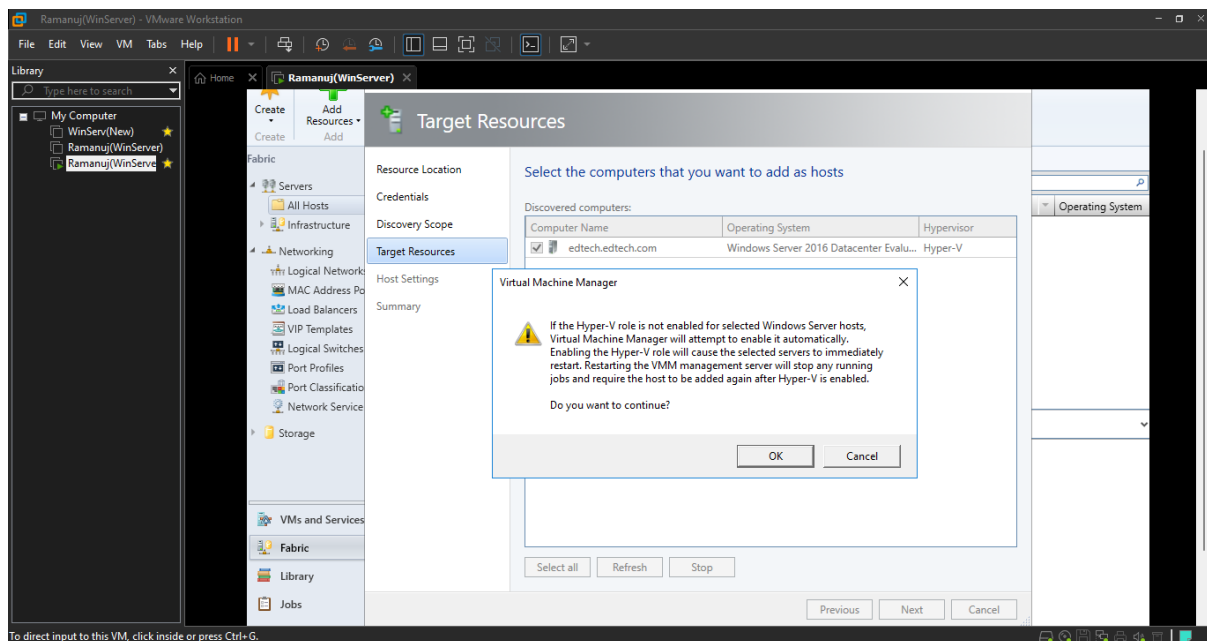
- In **Computer Names** specify your computer name (Here it is EDTECH)
- Click **Next**



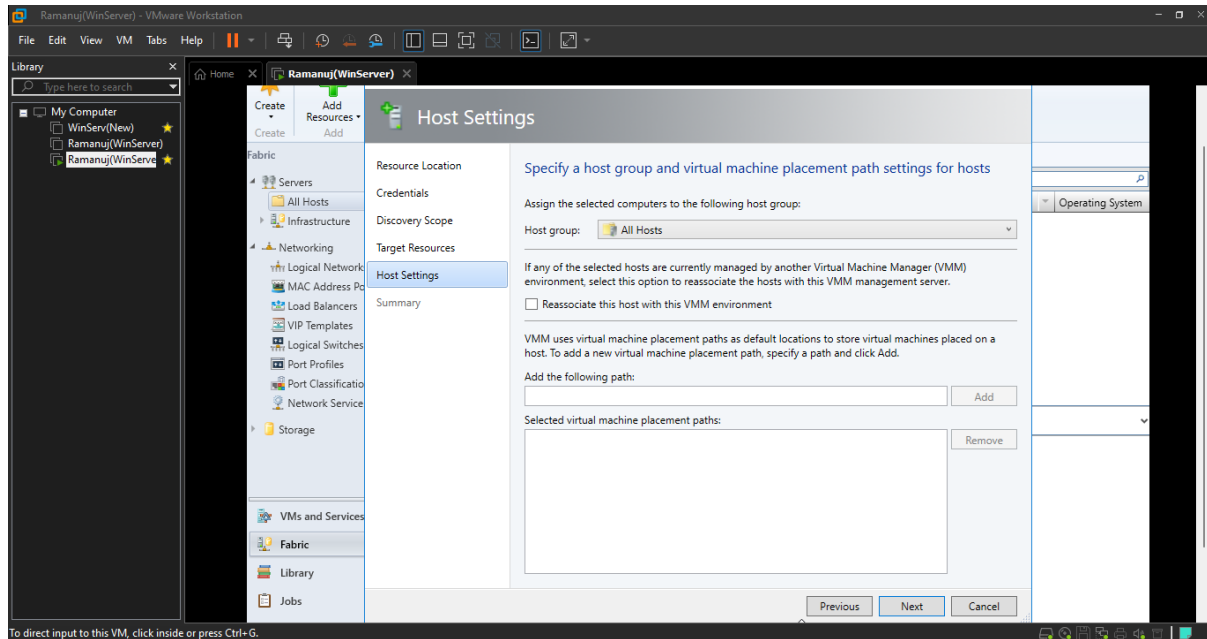
- In Target Resources Tick the computer (Here it is **edtech.edtech.com**)
- Click **Next**



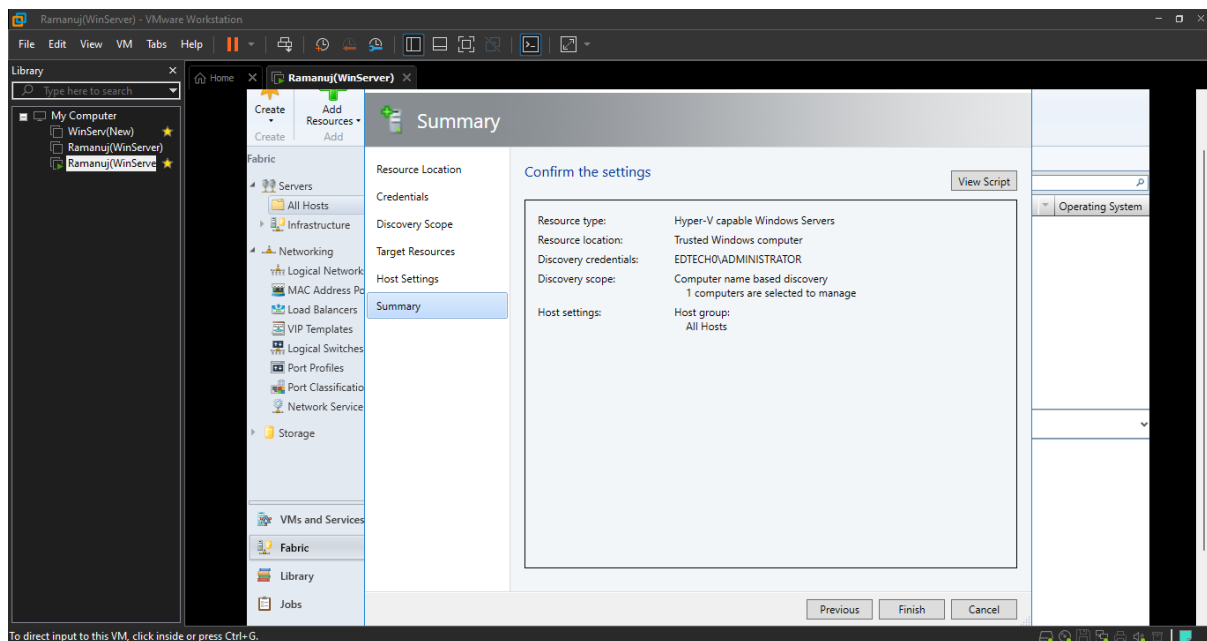
- After you Click Next you will see a pop-up Click **Ok**



- Keep default values and Click **Next**

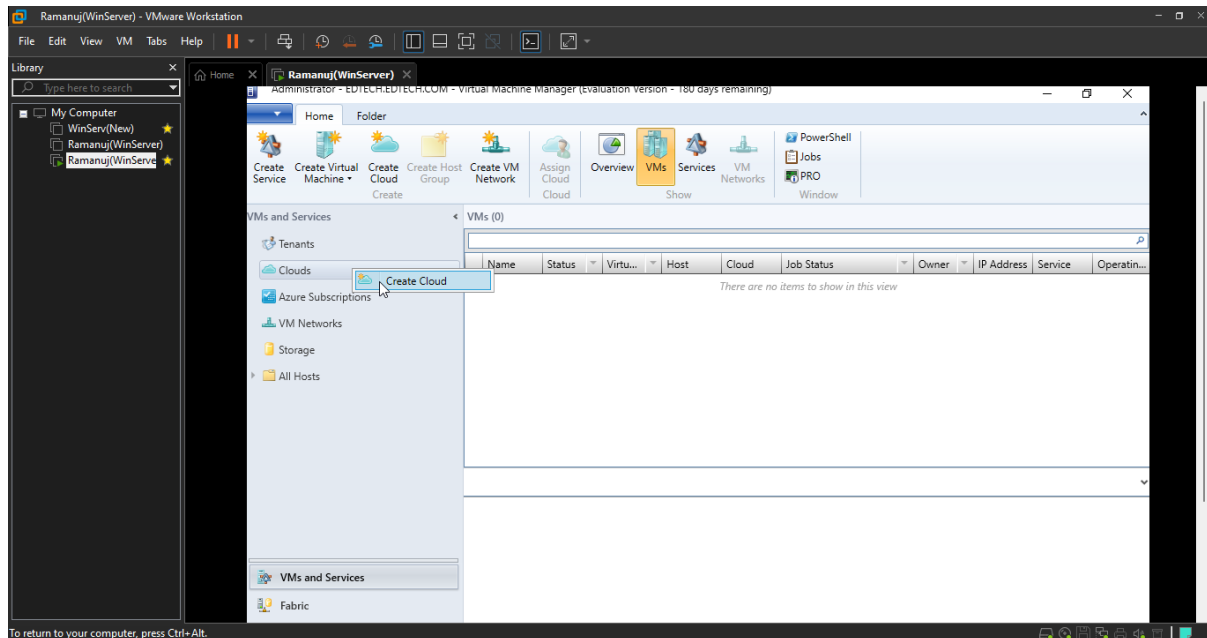


- View the Summary and Click **Finish**

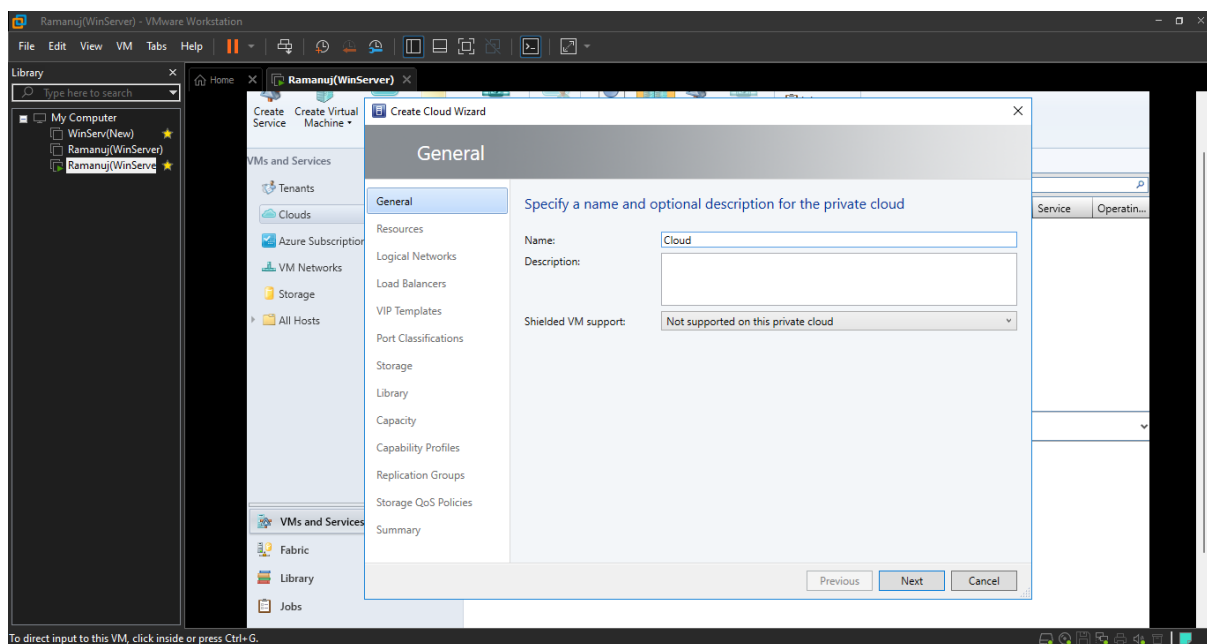


Step 45: Create a Cloud

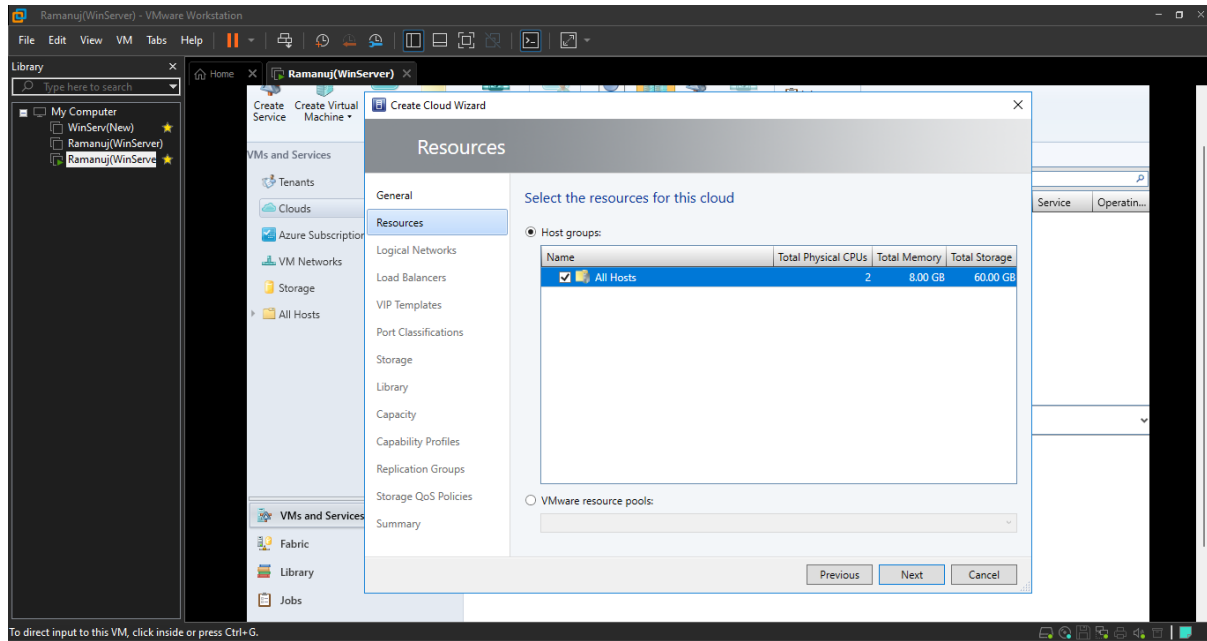
- In SCVMM Click on **VMs and Services**
- In VMs and Services Right-Click on **Cloud** and Select **Create Cloud**



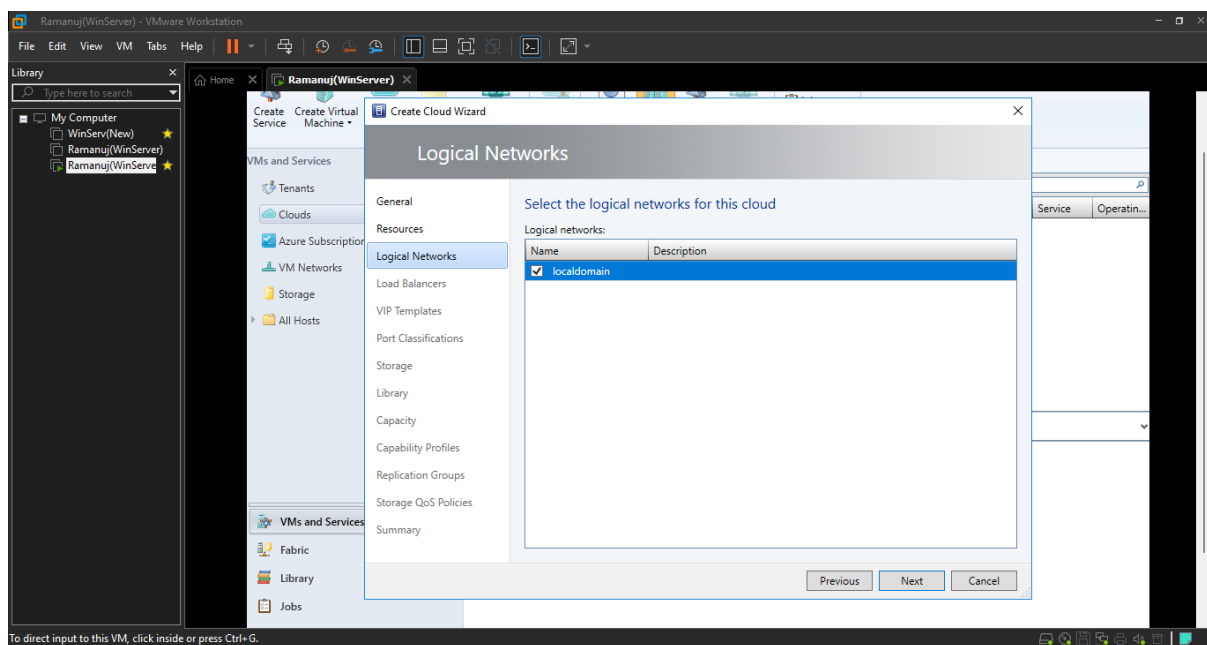
- Type in a name (Here it is **Cloud**) and Click **Next**



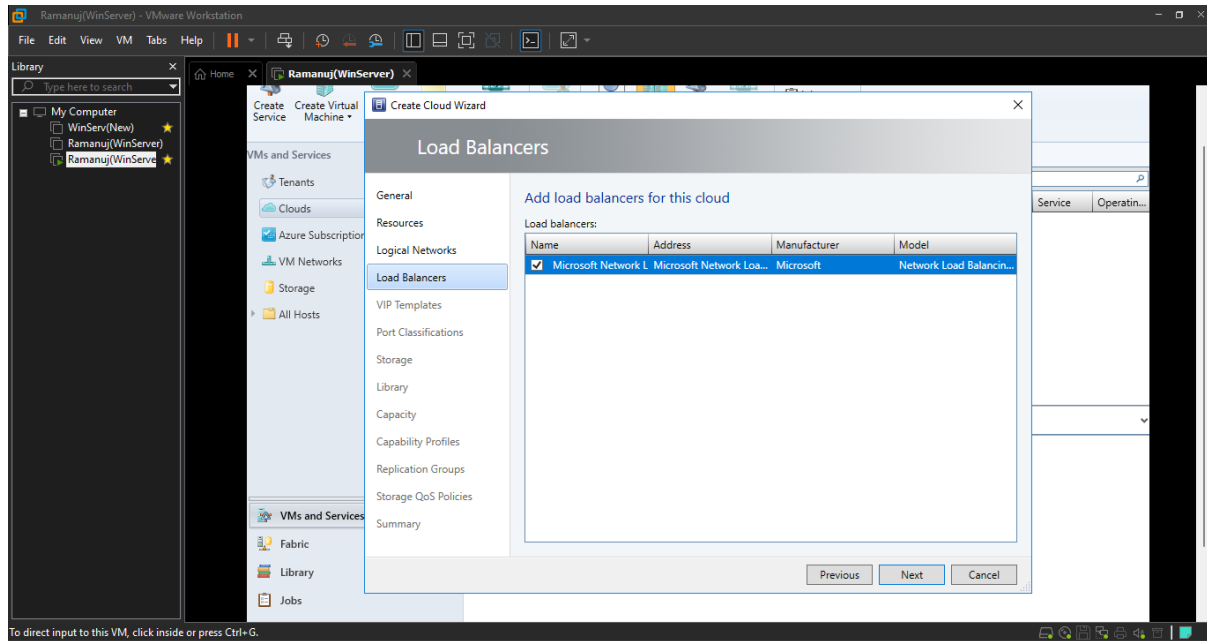
- Select **All hosts** and Click **Next**



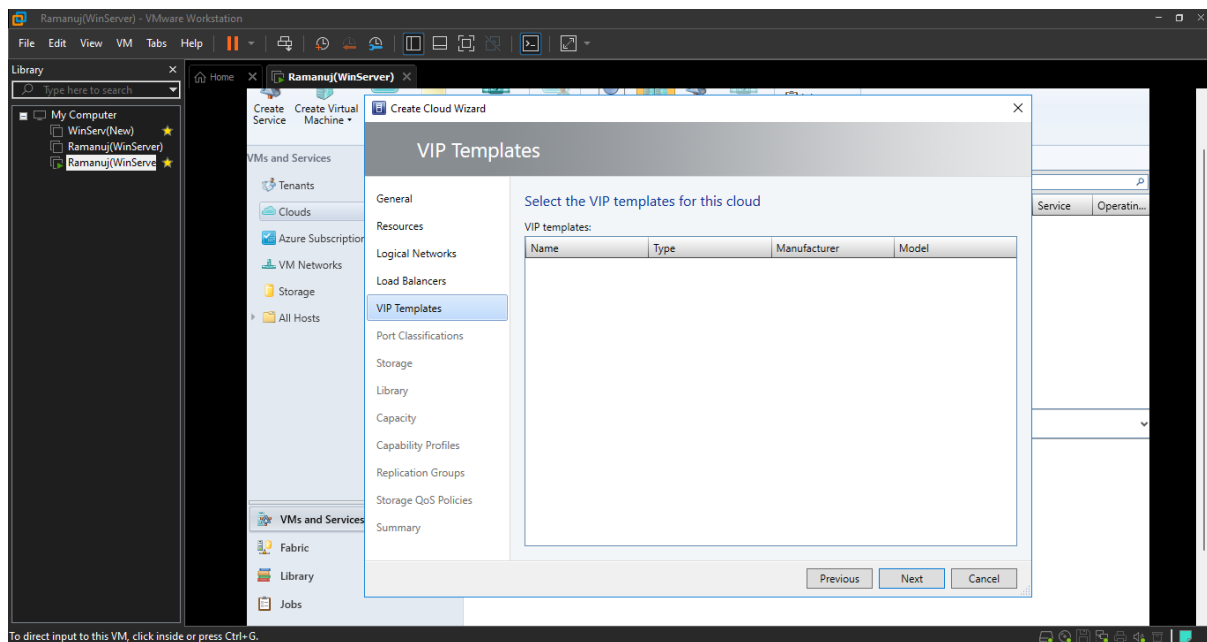
- Select **localdomain** and Click **Next**



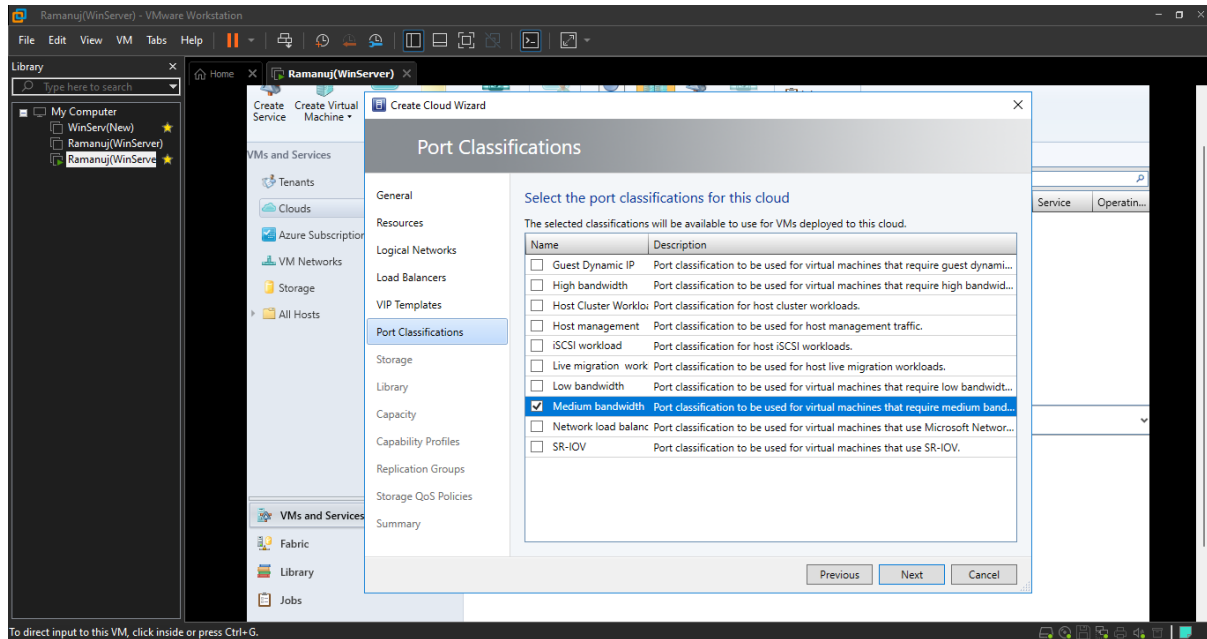
- Select **Microsoft Network Load Balancer** and Click **Next**



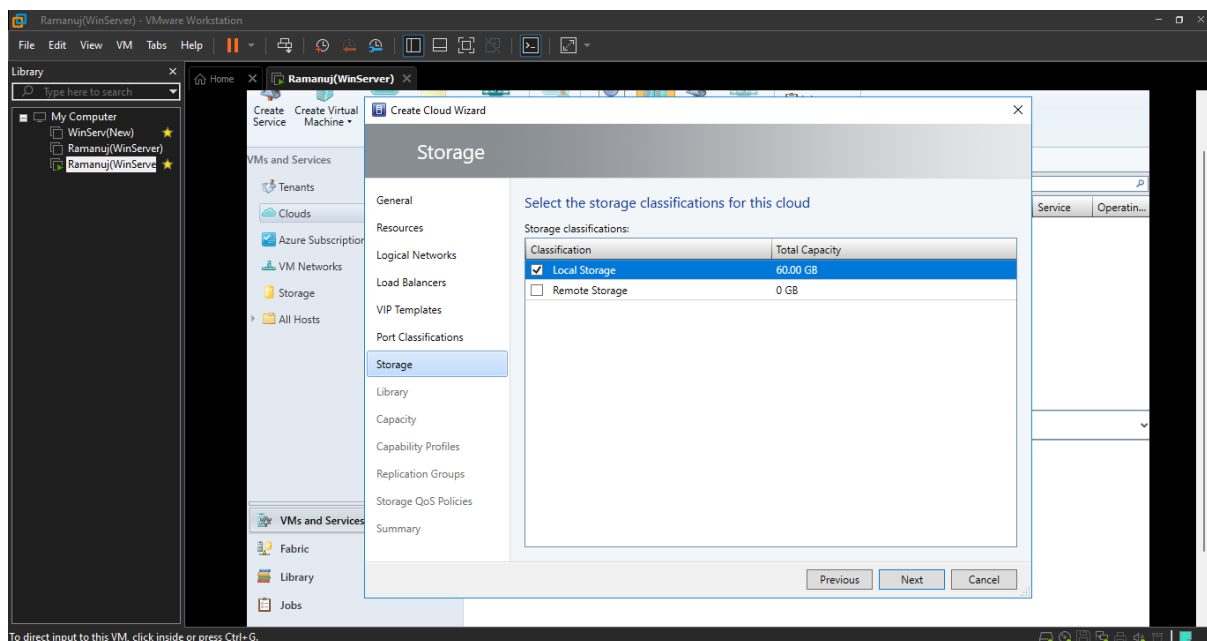
- Keep default values and Click **Next**



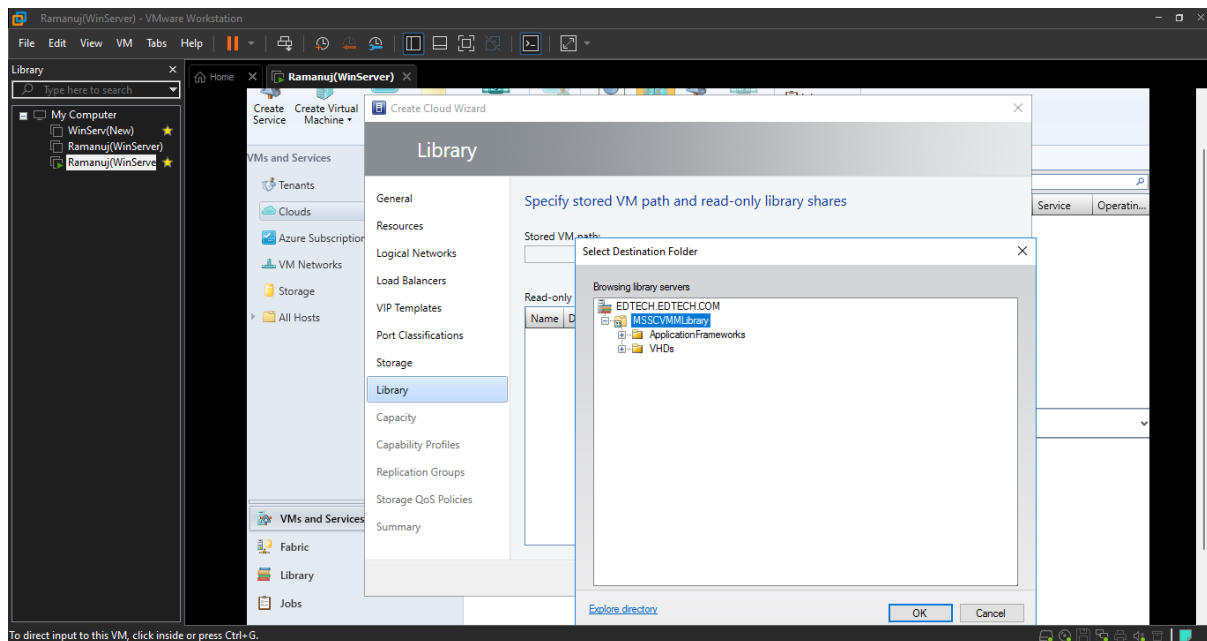
- In Port Classification select **Medium Bandwidth** and Click **Next**



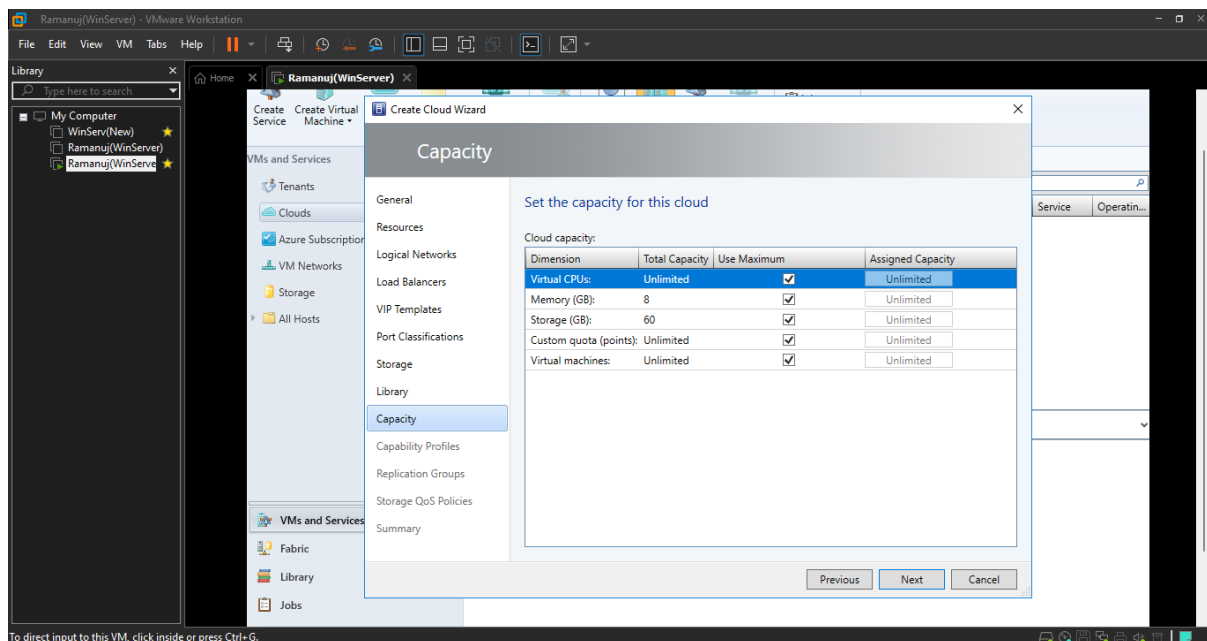
- In Storage select **Local Storage** and Click **Next**



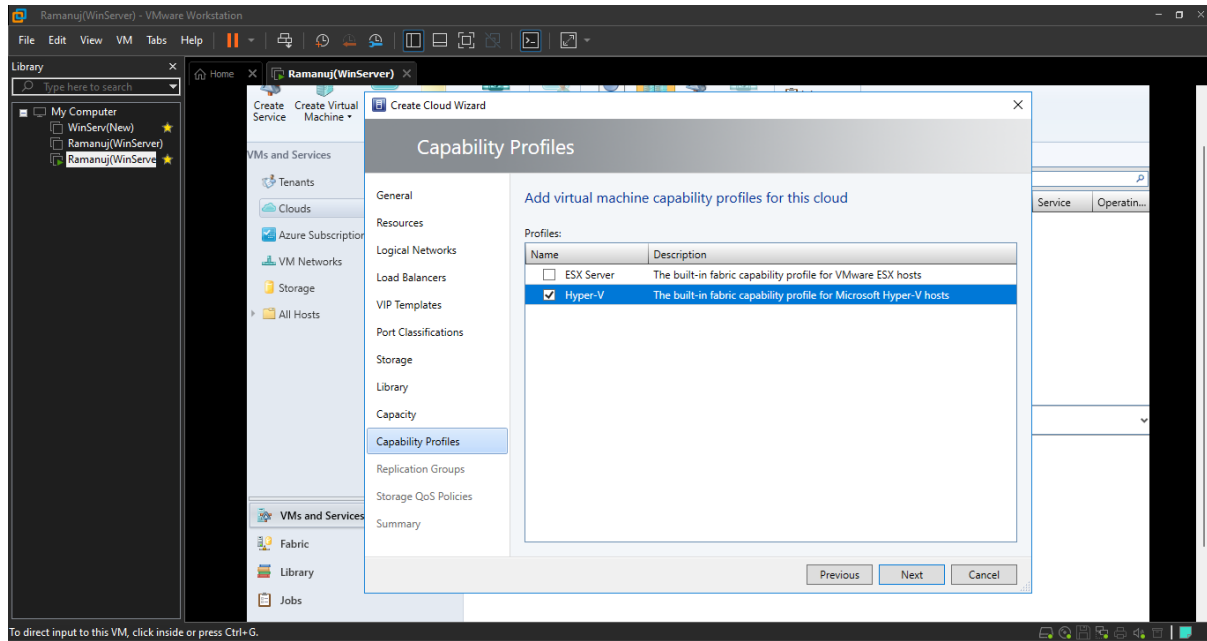
- In Library browse stored VM path and select **MSSCVMMLibrary** and Click **Ok**



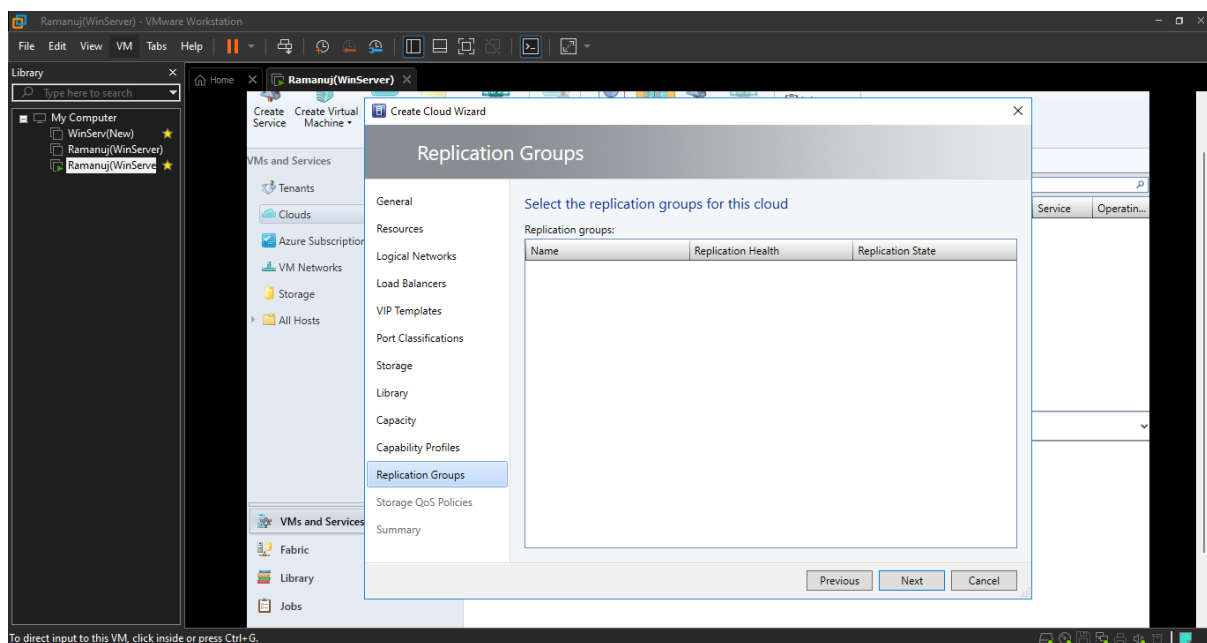
- Keep default values and Click **Next**



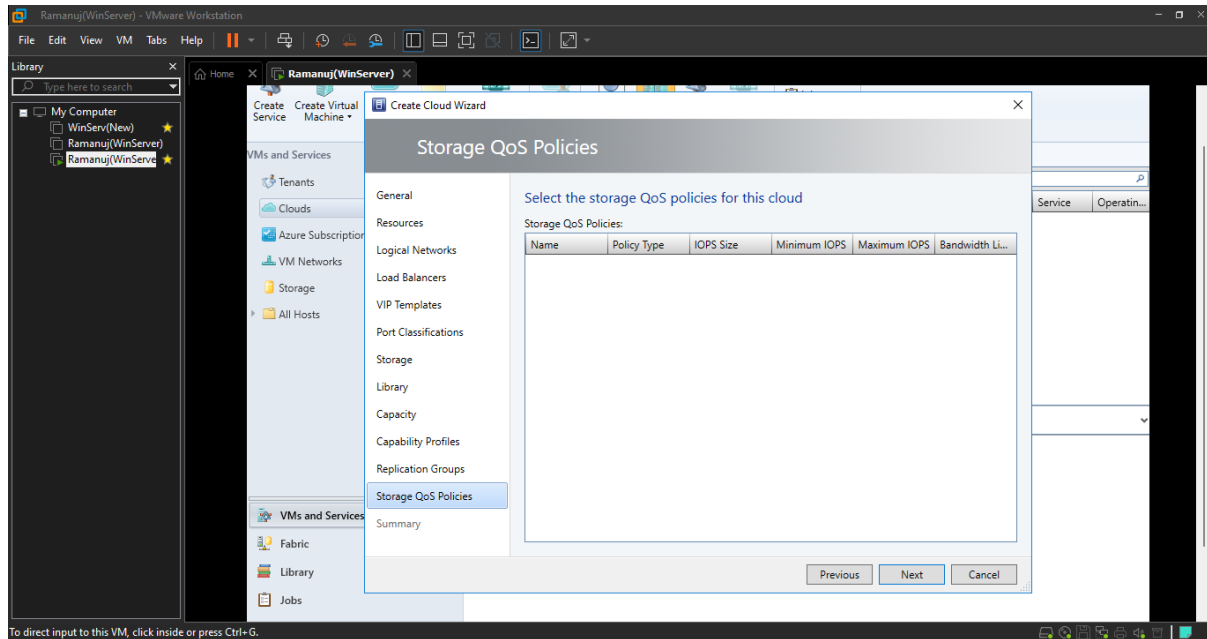
- In Capability Profiles Select **Hyper-V** and Click **Next**



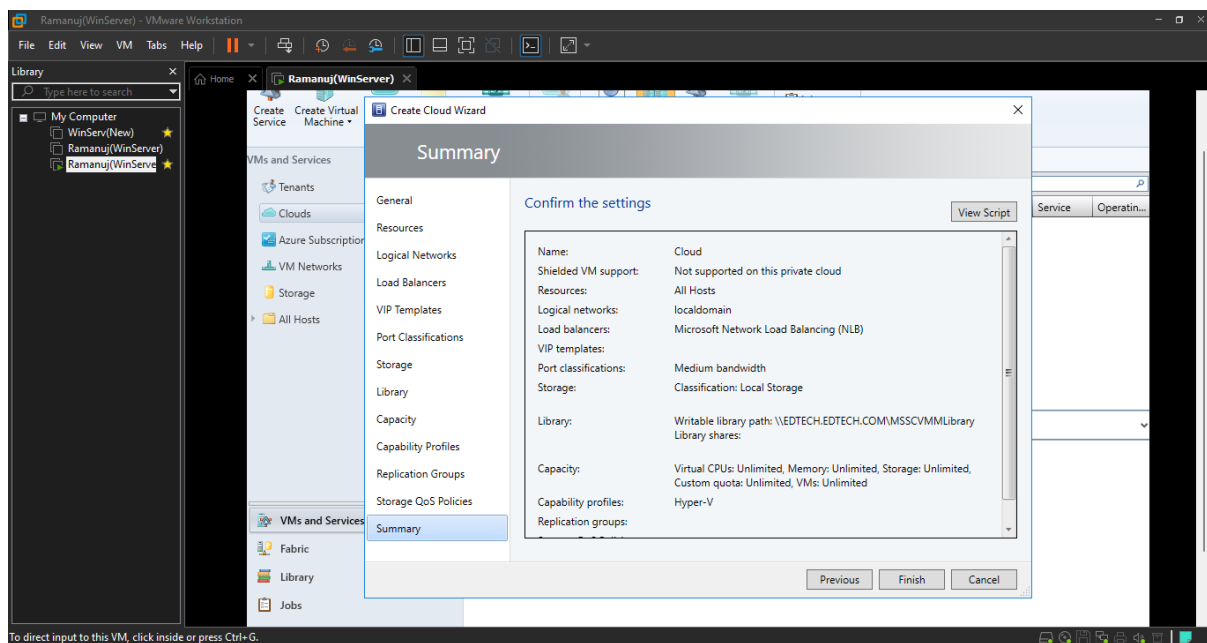
- Keep default values and Click **Next**



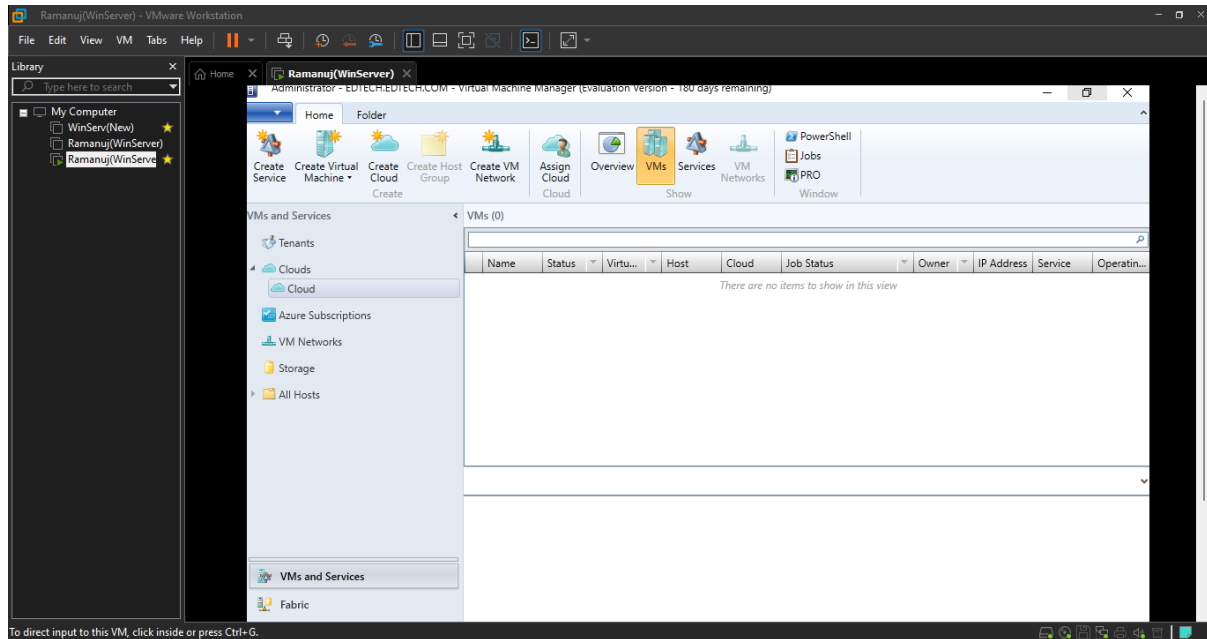
- Keep default values and Click **Next**



- View the Summary and Click **Finish**

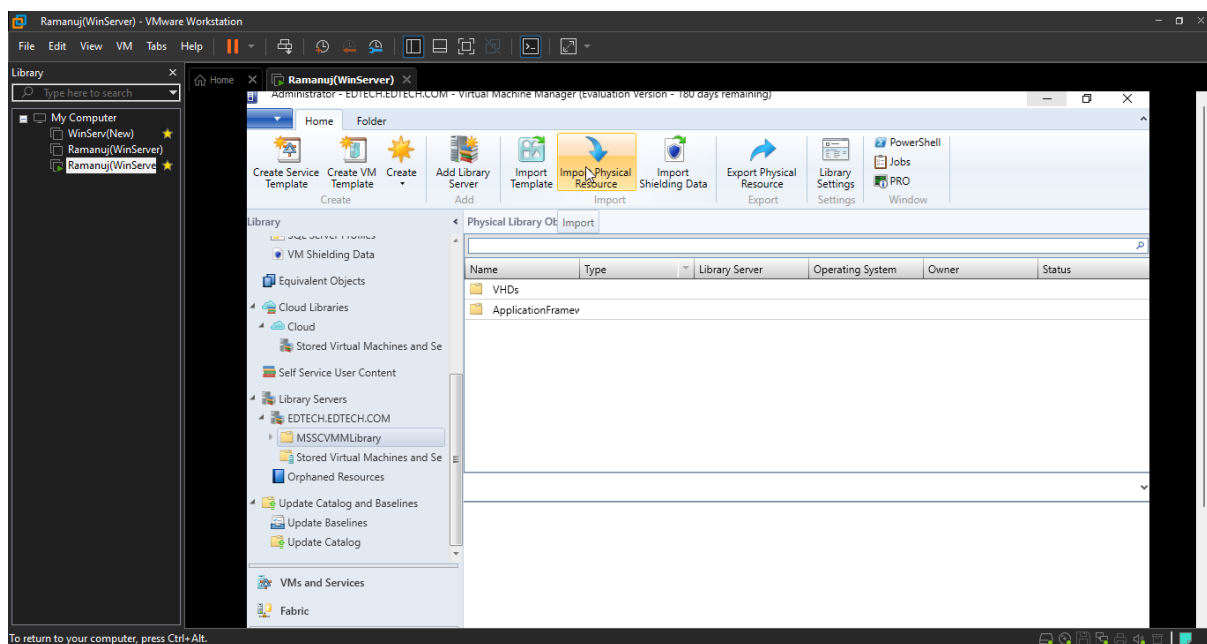


- Cloud has been created

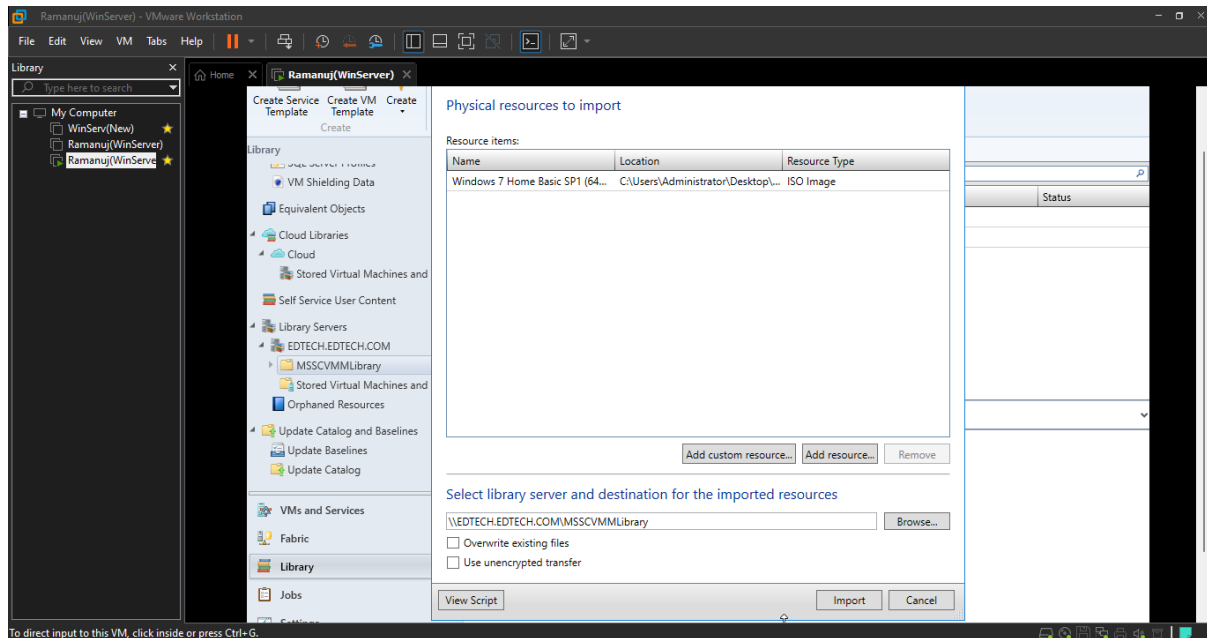


Step 46: Add Windows iso to SCVMM

- In SCVMM Click on **Library**
- Within Library Click on **Library Servers**
- Within Library Servers Click on **Domain Name** (Here it is EDTECH.EDTECH.COM)
- Within that Click on **MSSCVMMLibrary**
- On Top Click the **Import Physical Resource**

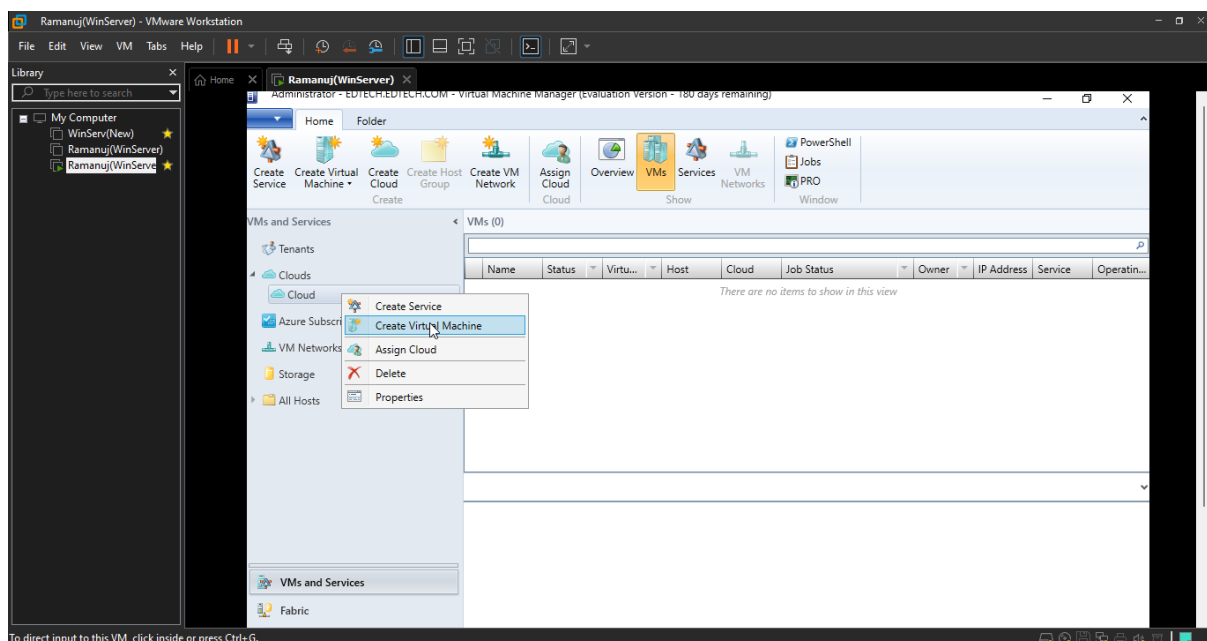


- In Import Physical Resource Click on **Add resource** and Select the Windows 7 iso file on your Desktop
- In Select Library Server Click **Browse** and Select **MSSCVMMLibrary** and Click **Import**

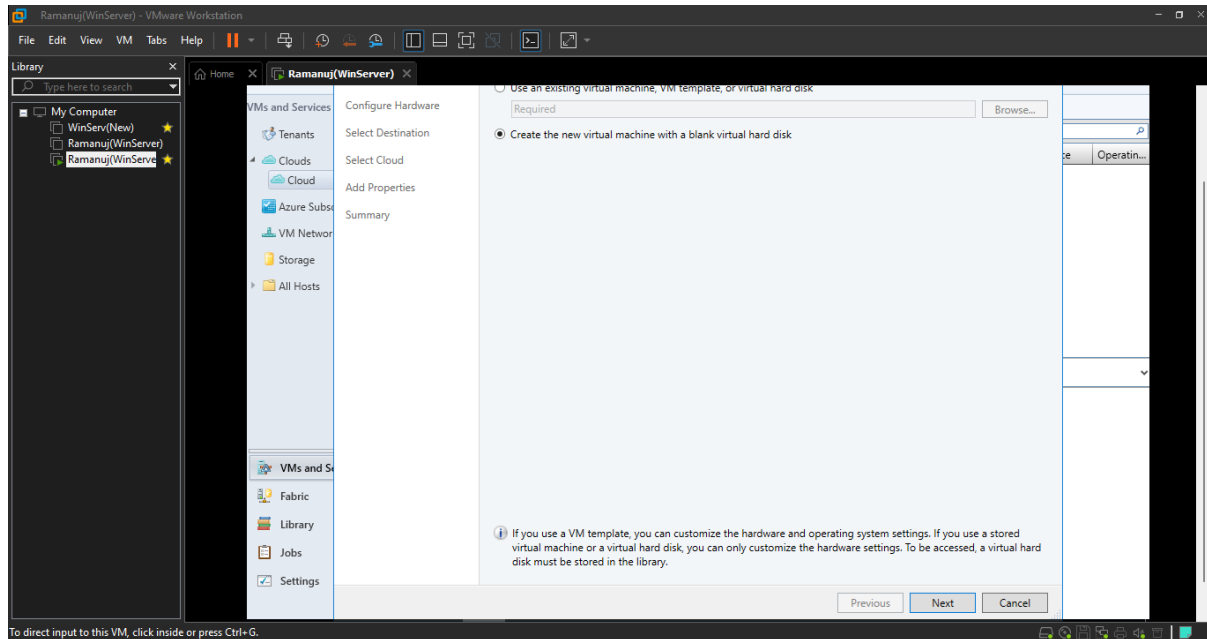


Step 47: Create a Cloud

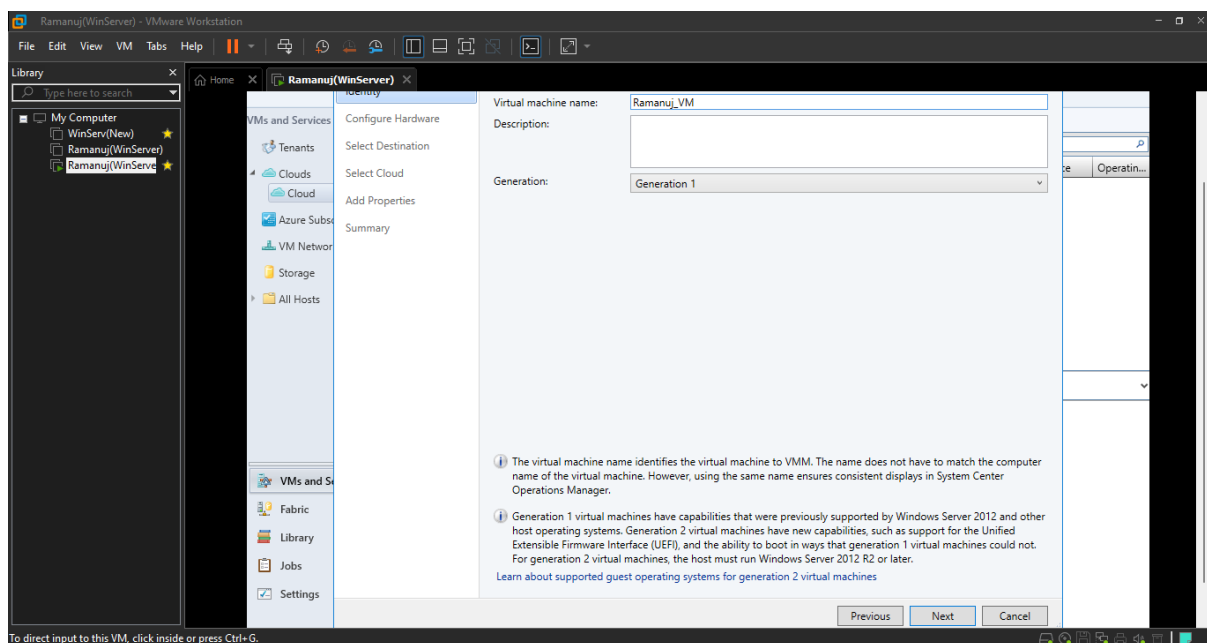
- In SCVMM Click on **VMs and Services**
- In VMs and Services Right-Click on **Cloud** instance and Select **Create a Virtual Machine**



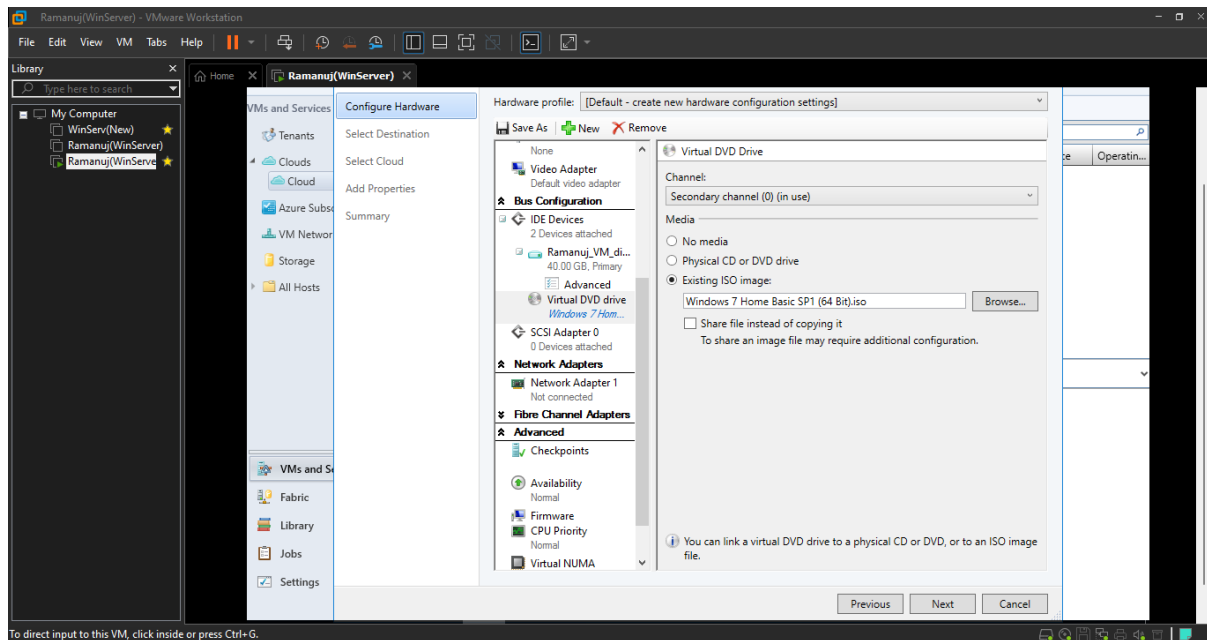
- In Select Source, Select **create the new virtual machine with a blank virtual hard disk** and Click **Next**



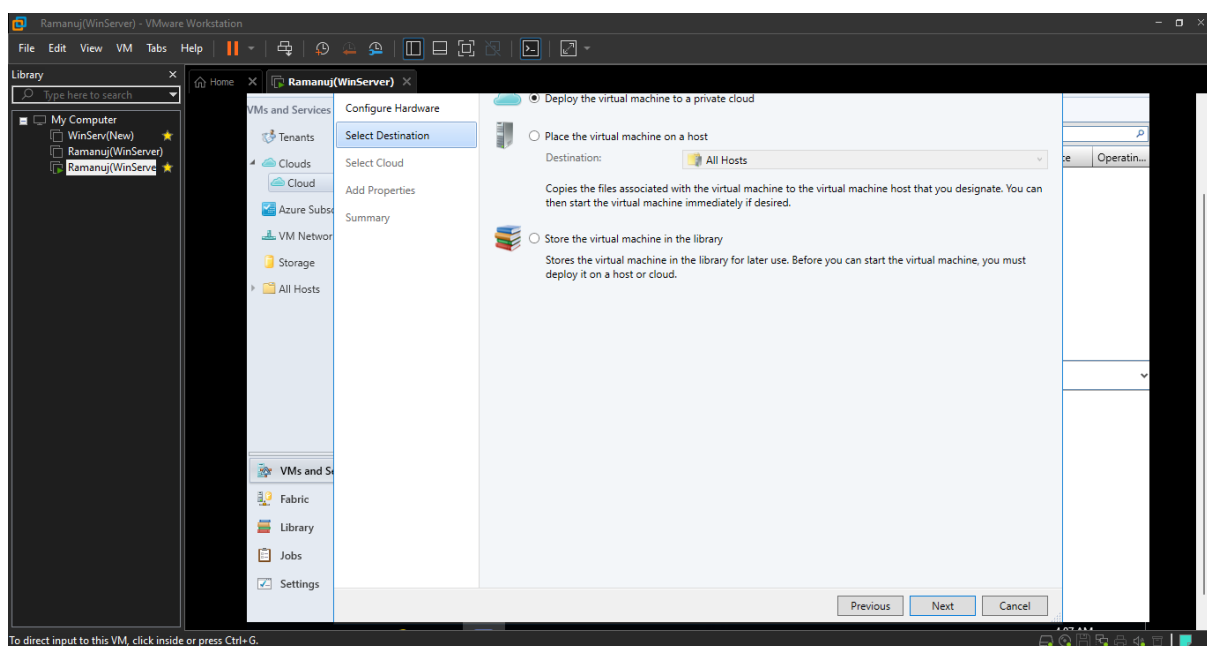
- Type a name for your VM (Here it is Ramanuj_VM) and Select Generation as **Generation 1** and Click **Next**



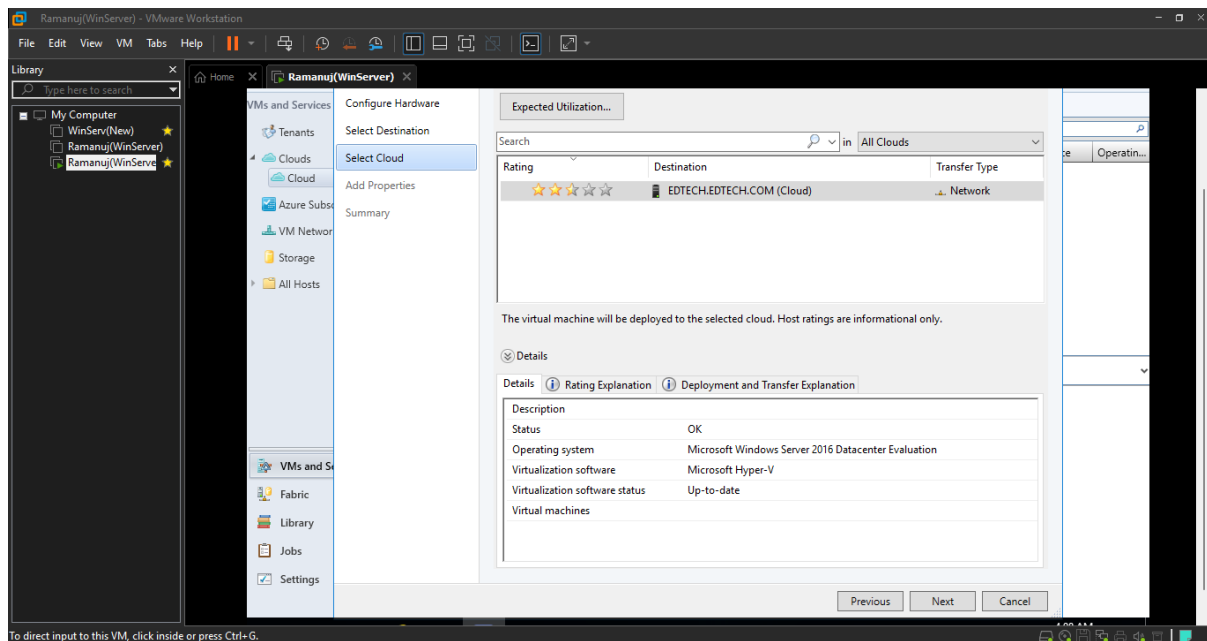
- In Configure Hardware Select the Following Options:
 In **Compatibility** Select **Hyper-V**
 In **Memory** Change the value to **2048MB**
 In **Bus Configuration** Click on **Virtual DVD Drive**
 In Virtual DVD Drive Click on **Existing ISO Image** and **Browse** for the imported Windows 7 iso file
 After the above steps Click **Next**



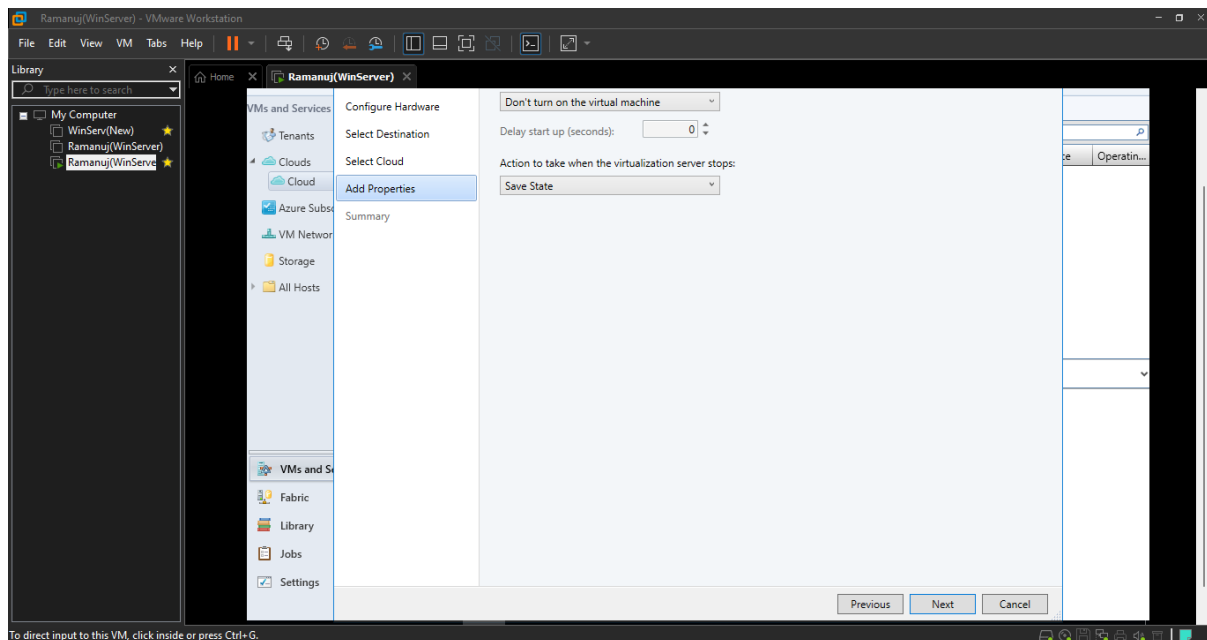
- Leave default settings and Click **Next**



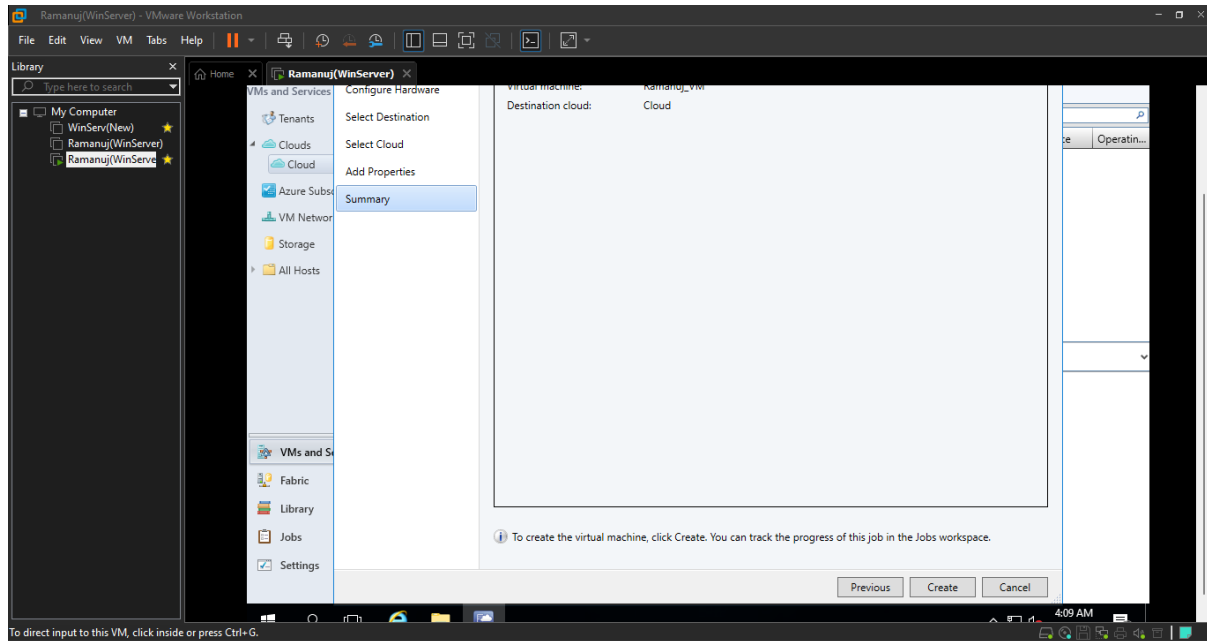
- Check the rating of your Virtual machine (Here it is 2 and ½ stars) and Click **Next**



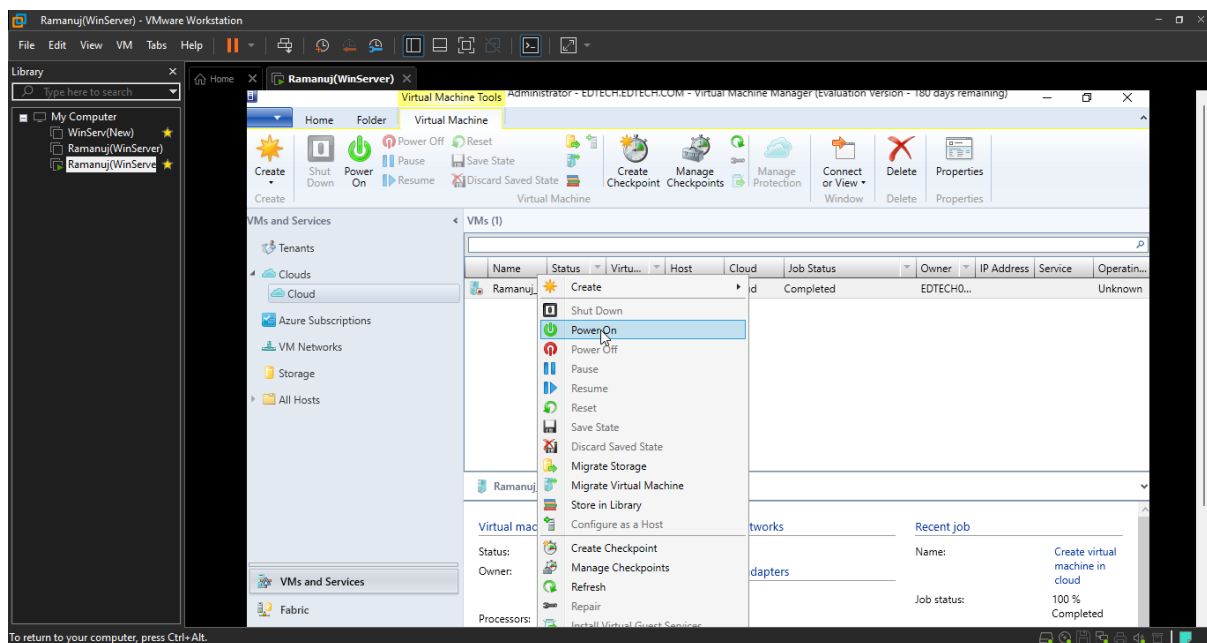
- Keep default settings and Click **Next**



- View the Summary and Click **Create**

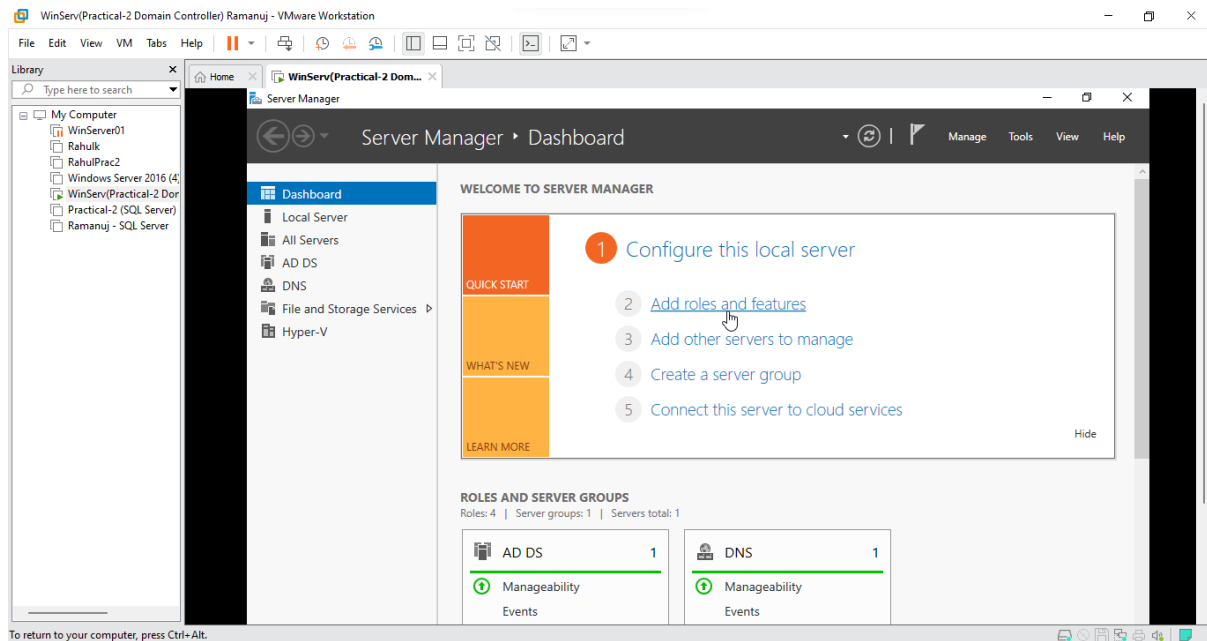


- Now the Virtual Machine is created, Right-Click on the VM and Select **Power On**

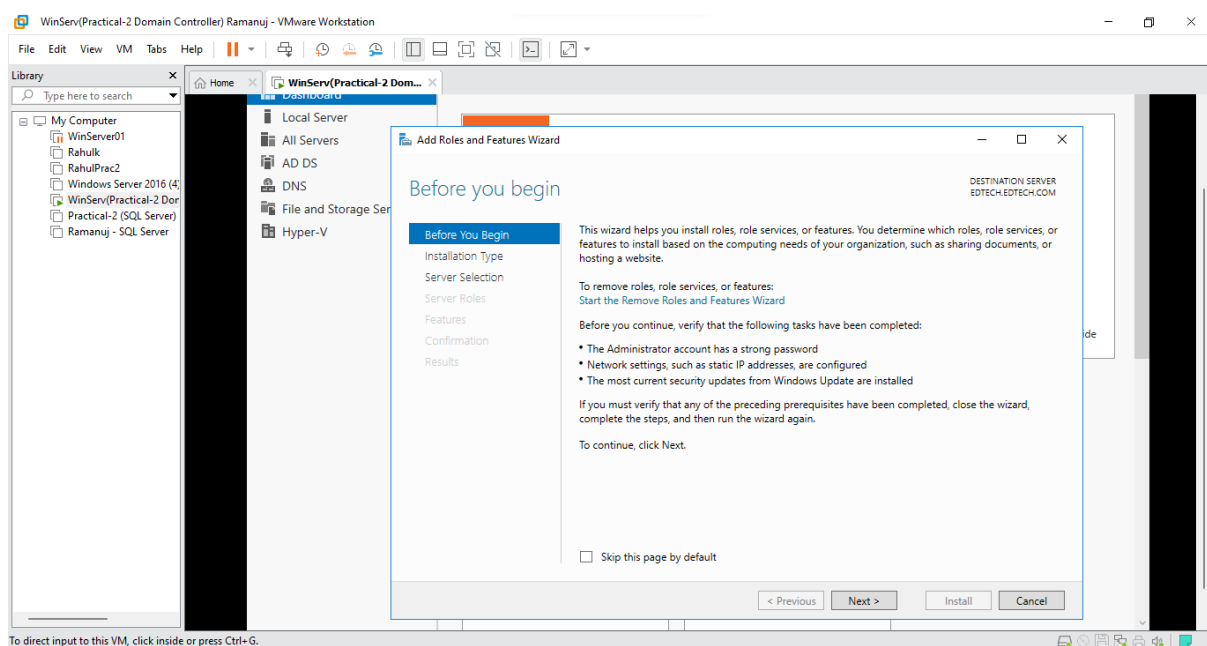


Practical 1-B:**Aim:** Installing and Enabling Host Guardian in a Cloud**Step 1:** Adding the Host Guardian Role in Roles and Features

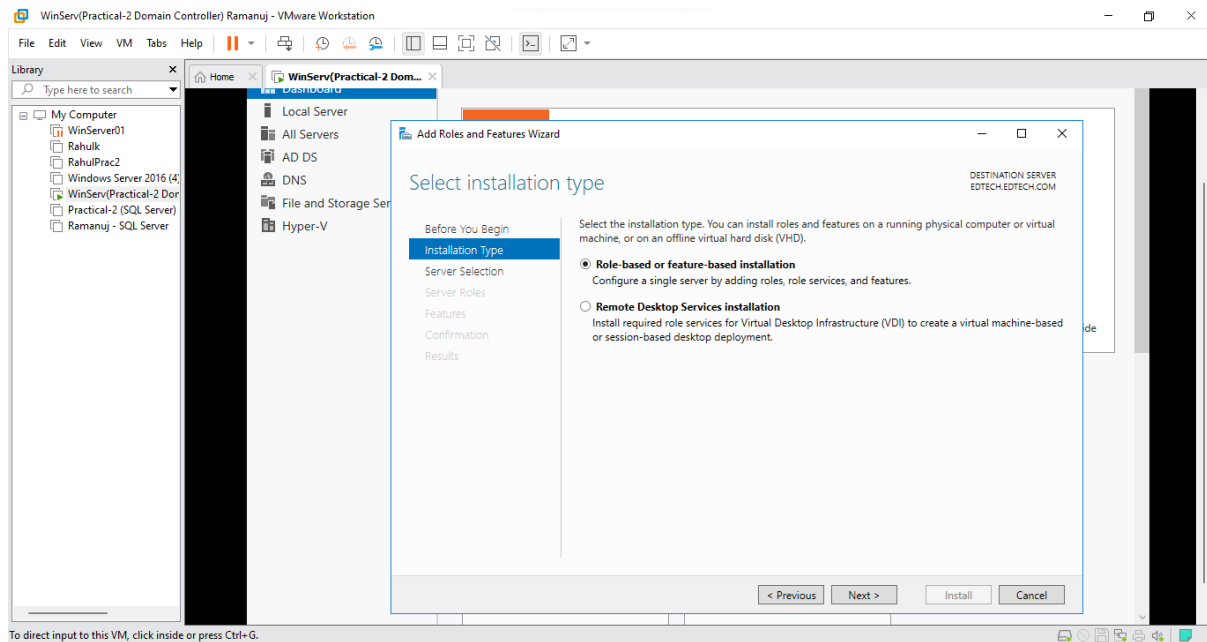
- Open **Server Manager** and Click on **Add Roles and Features**



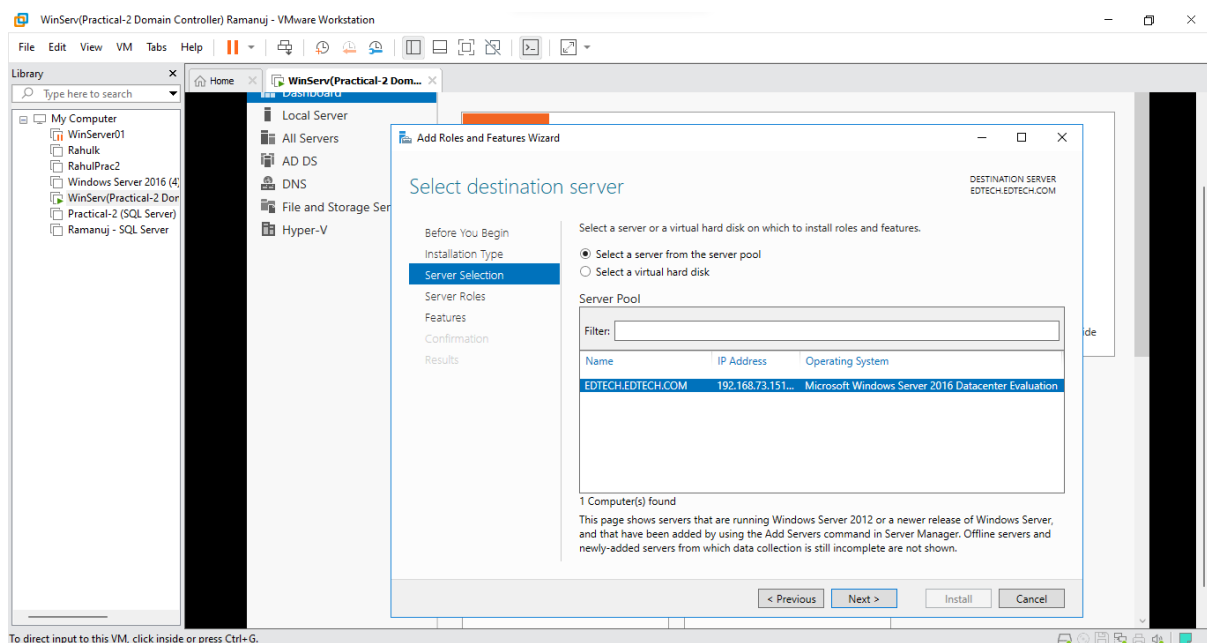
- Click **Next**



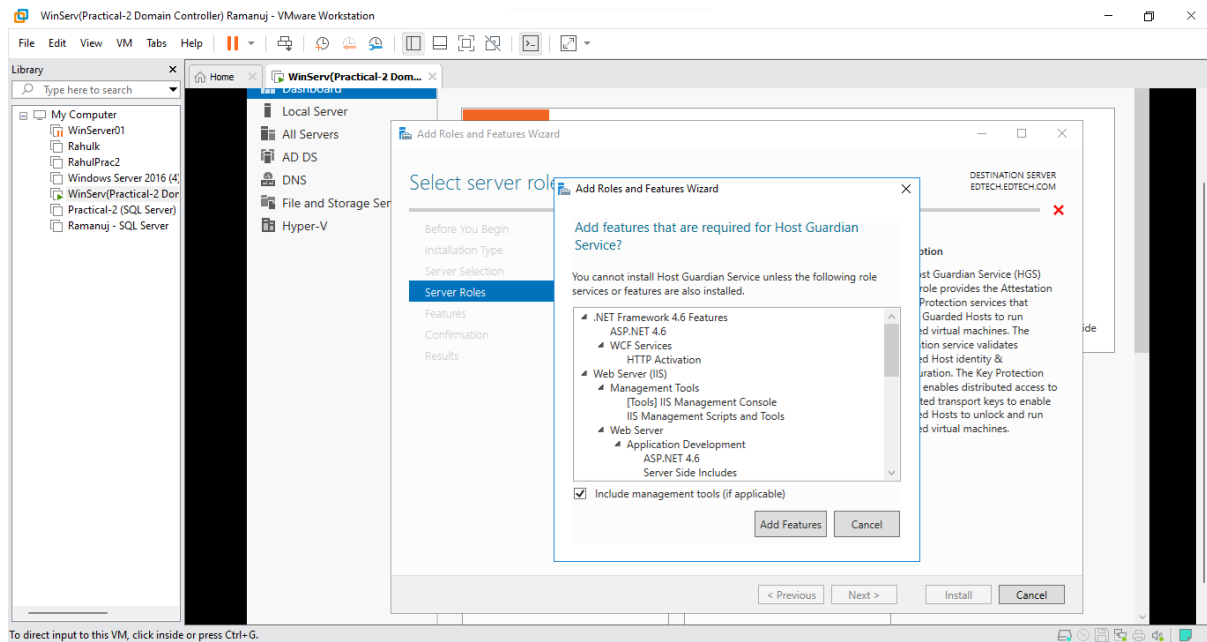
- Keep default values and Click **Next**



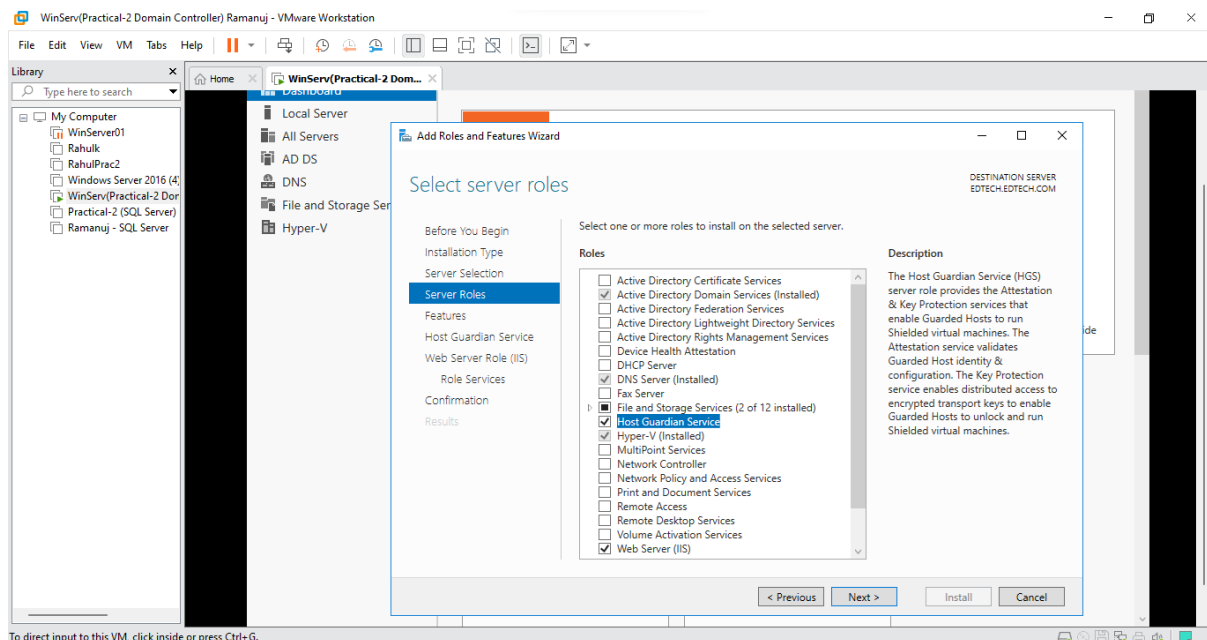
- Keep default values and Click **Next**



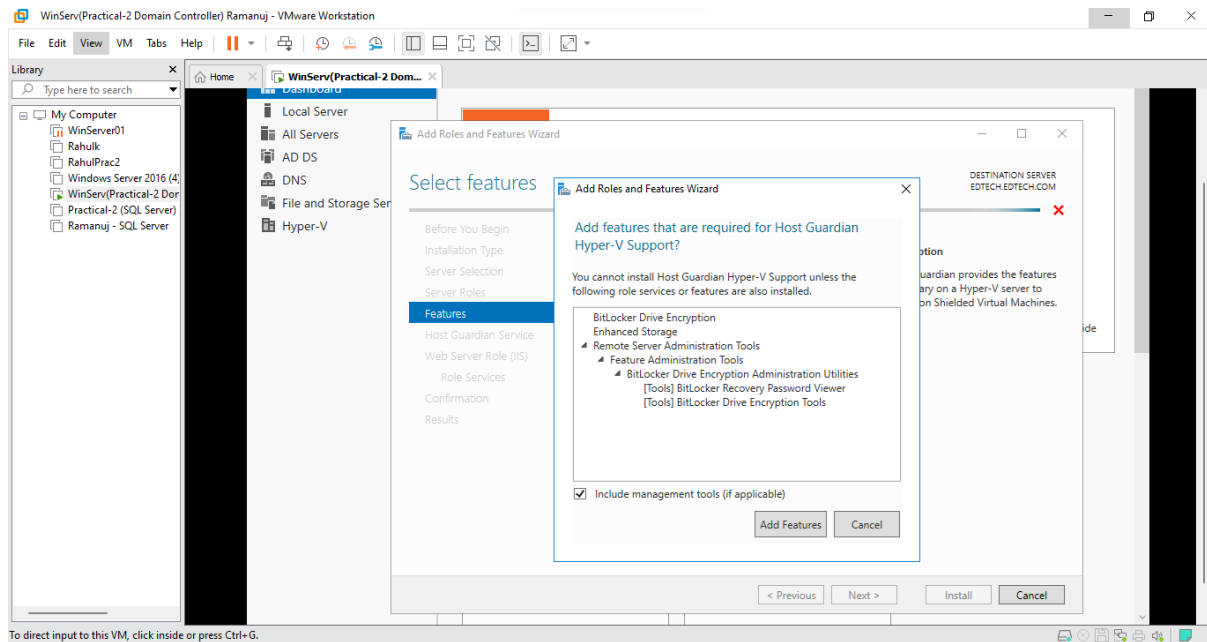
- Select **Host Guardian Service** and Click **Add Features**



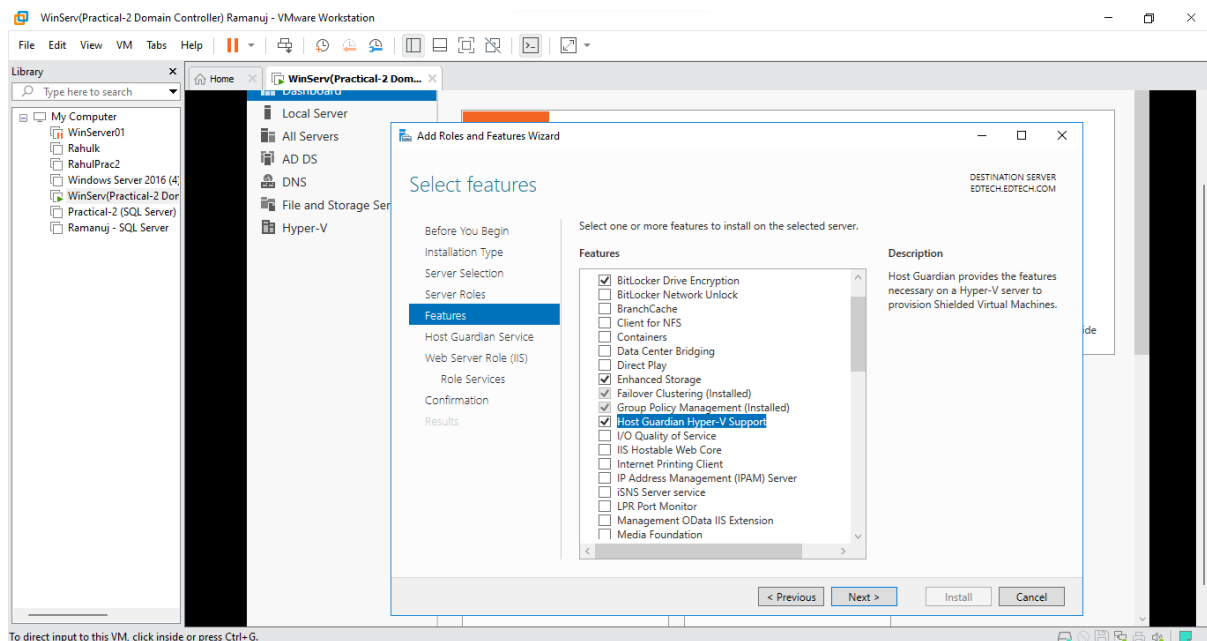
- Click **Next**



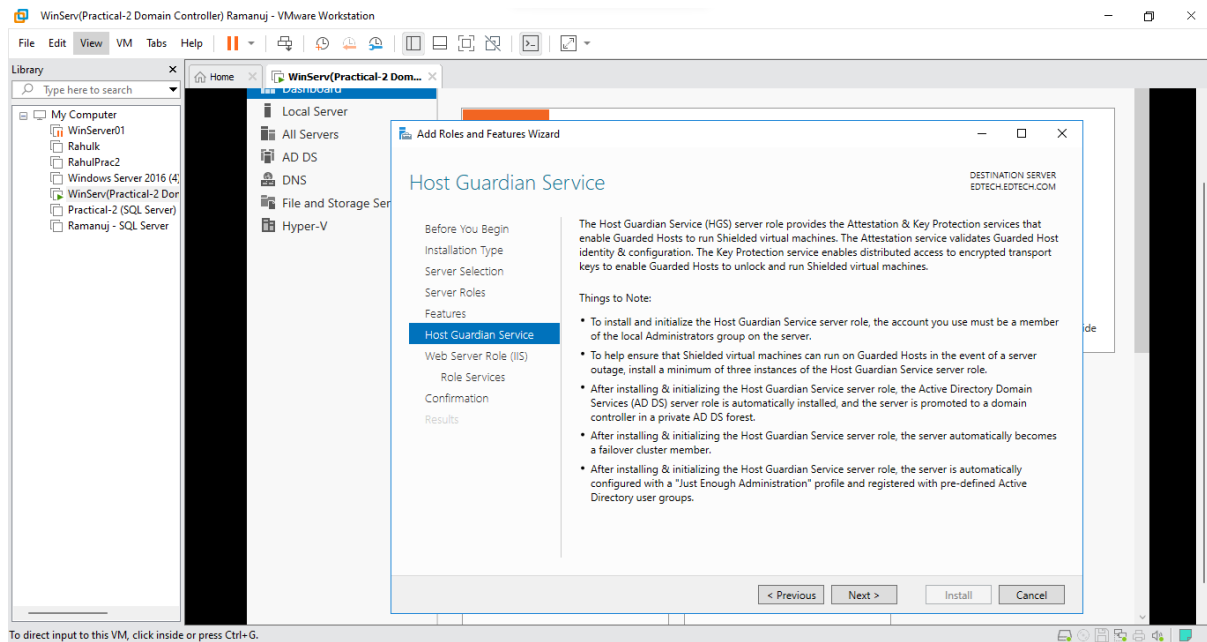
- Select **Host Guardian Hyper-V Support** and Click **Add Features**



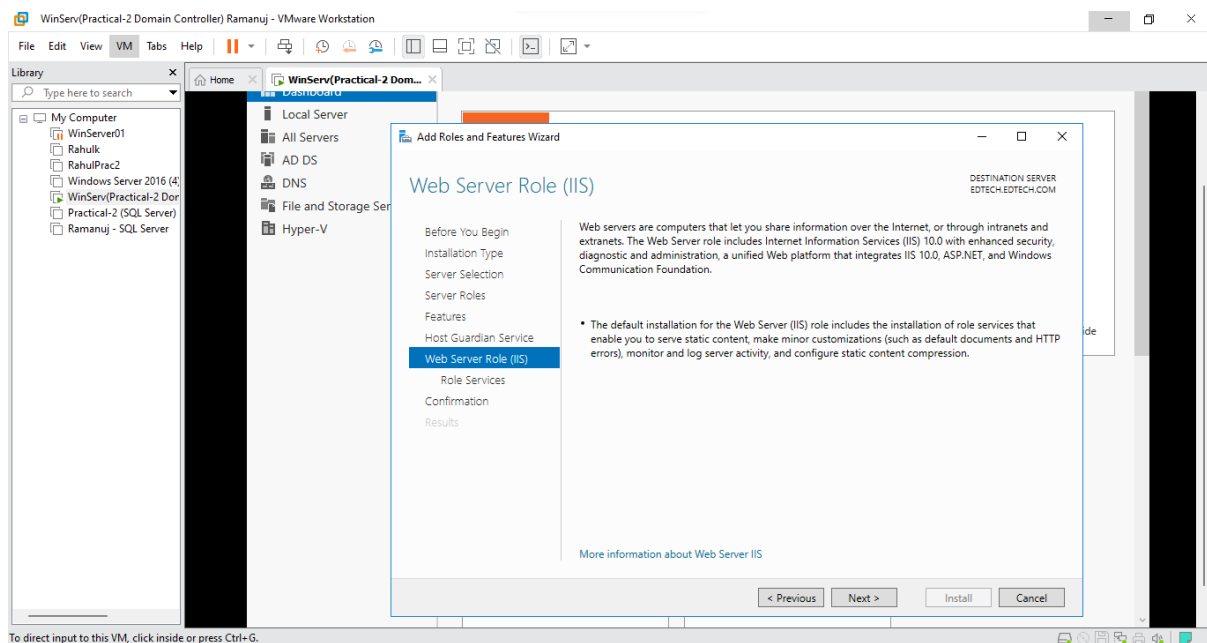
- Click **Next**



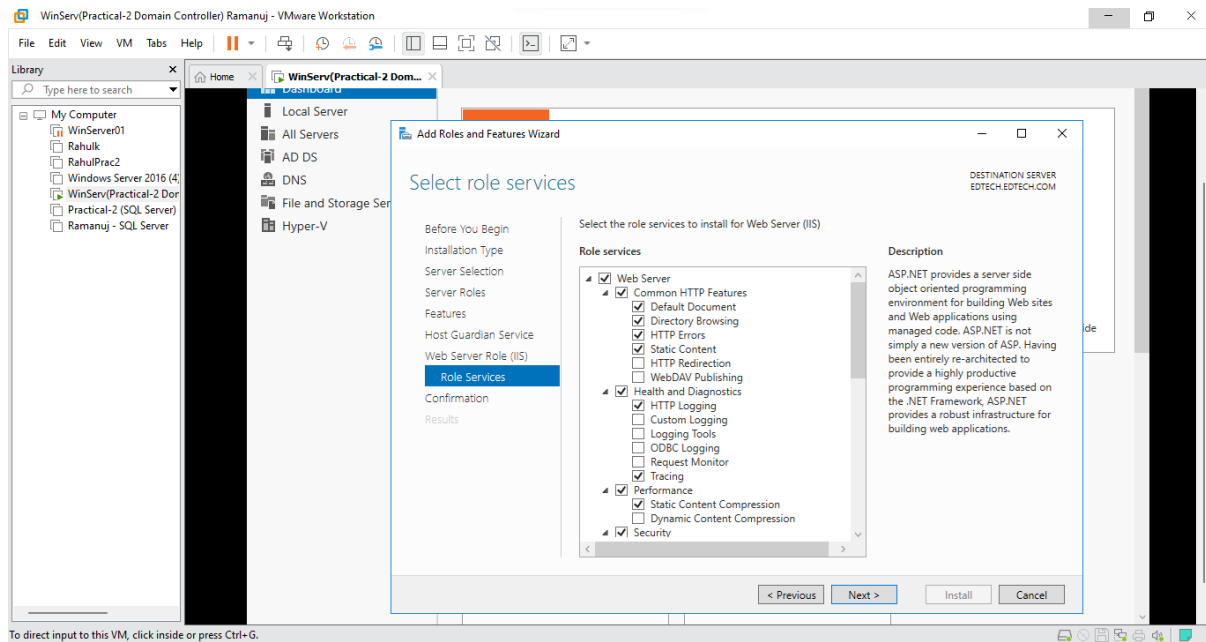
- Click Next



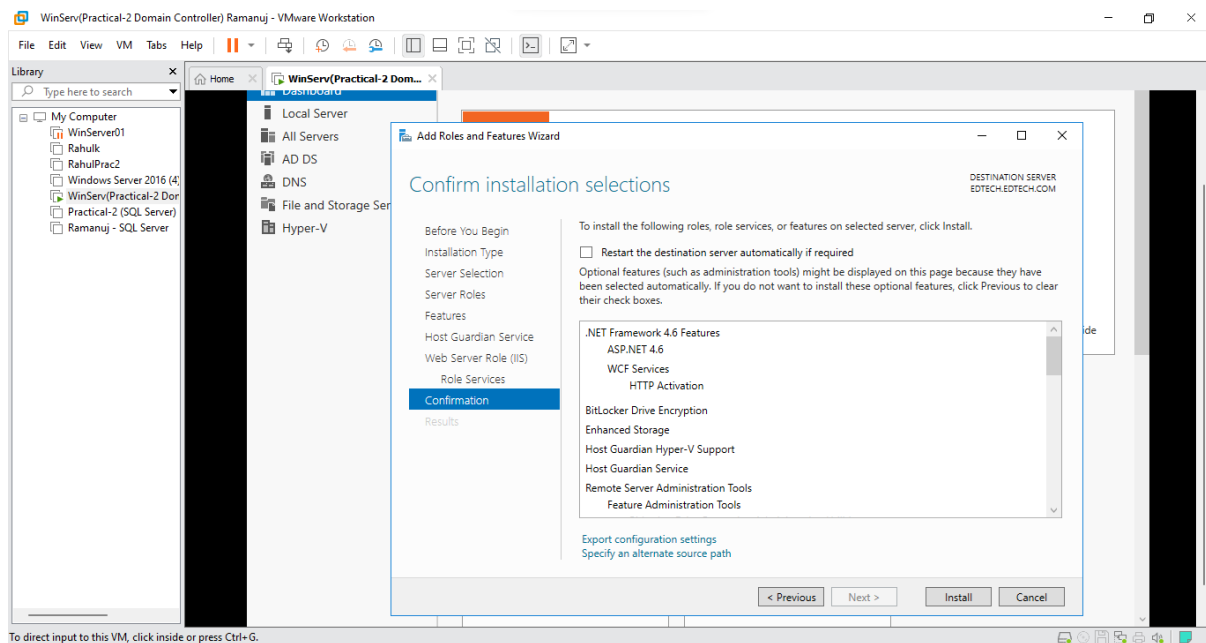
- Click Next



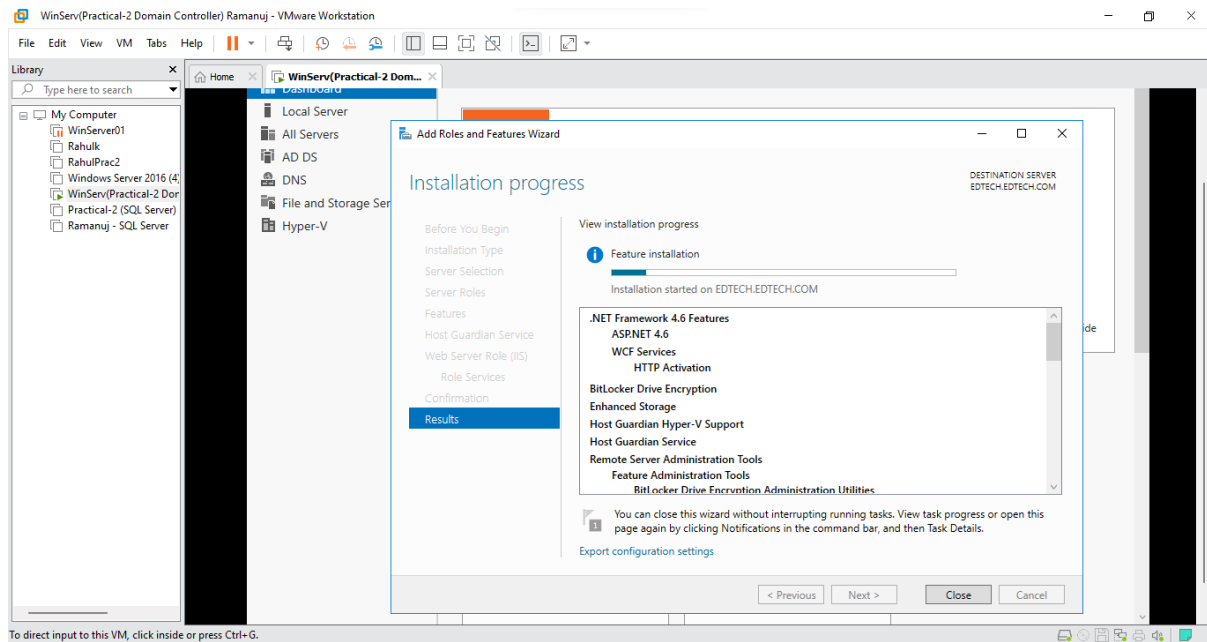
- Keep default values and select **ASP.NET 3.5** and Click **Next**



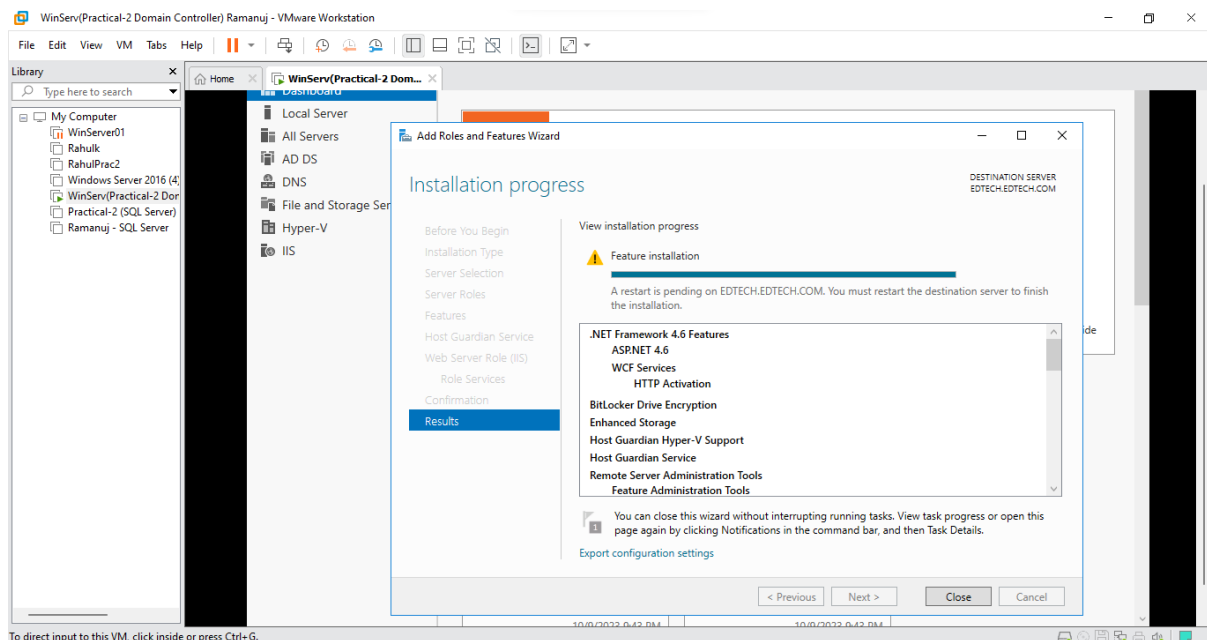
- Click **Install**



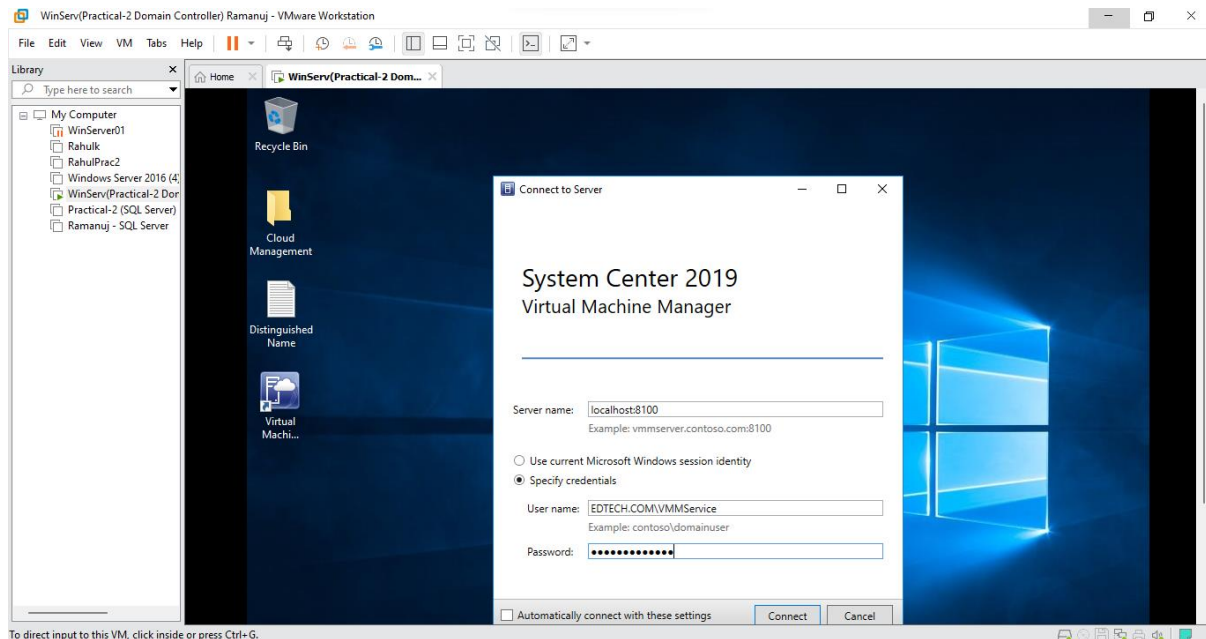
- Features will start installing



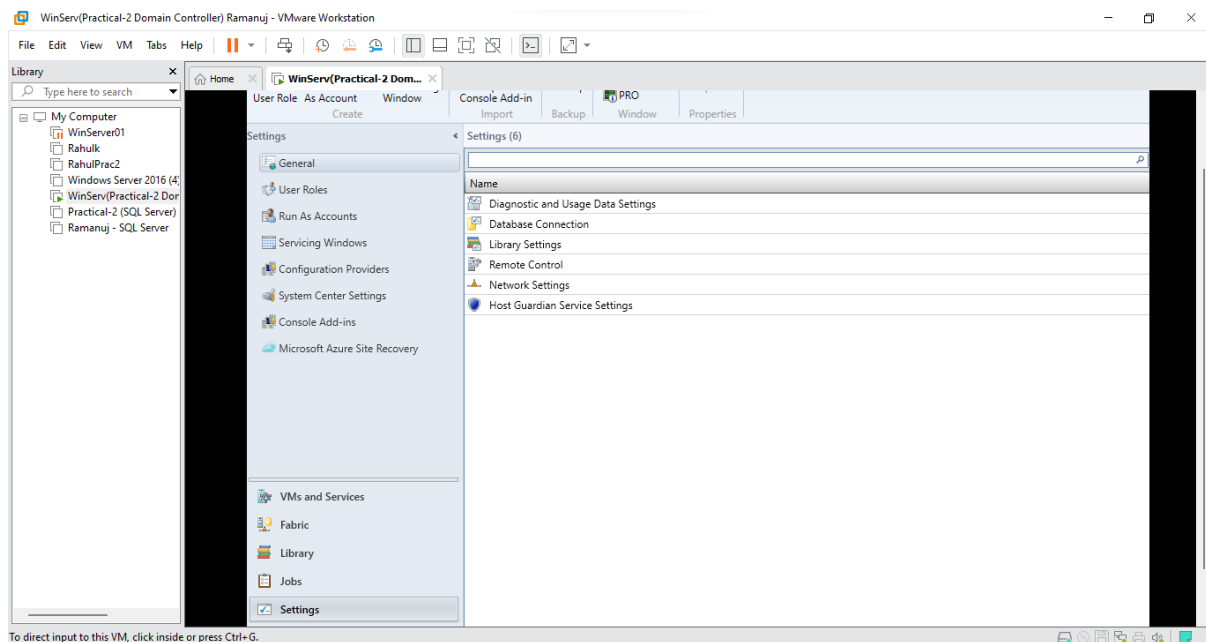
- After installation you will have to restart for the new features to take effect



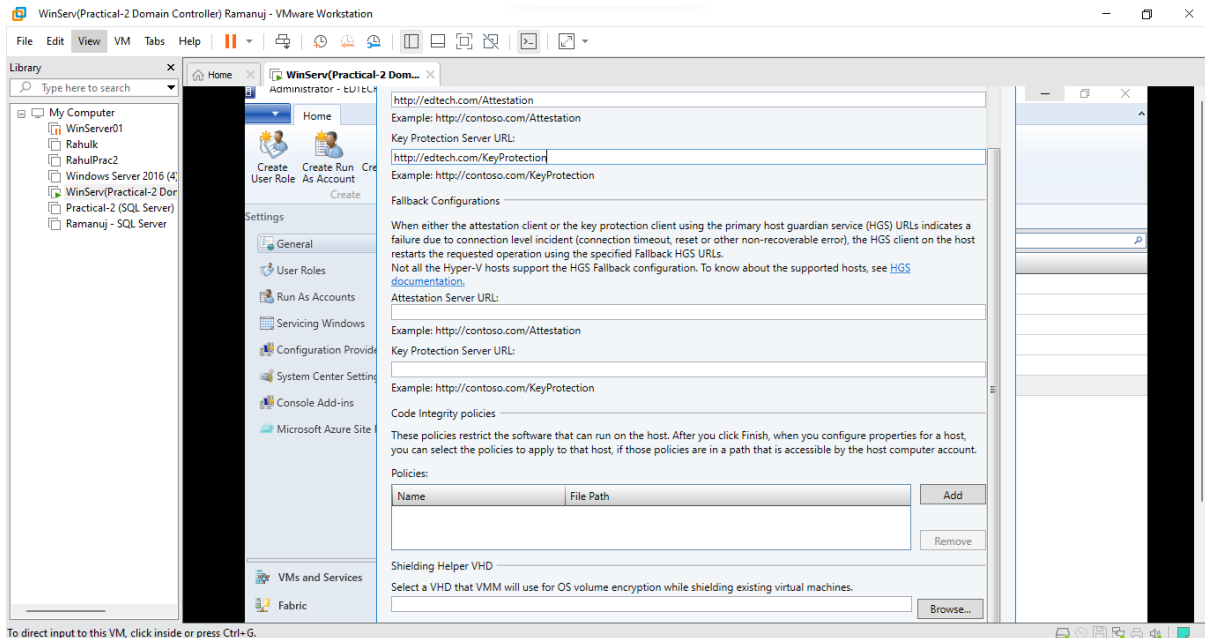
- After restarting open the **Virtual Machine Manager**, Enter your details and Click **Connect**



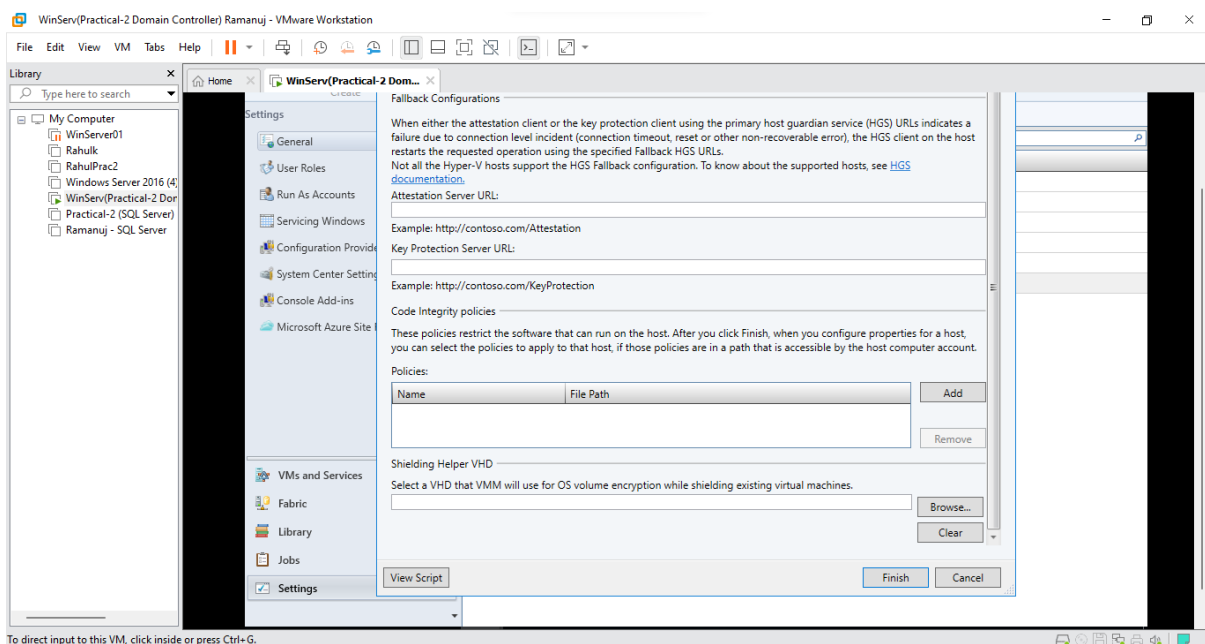
- On the dashboard Click on **Settings** and Select **Host Guardian Service Settings**



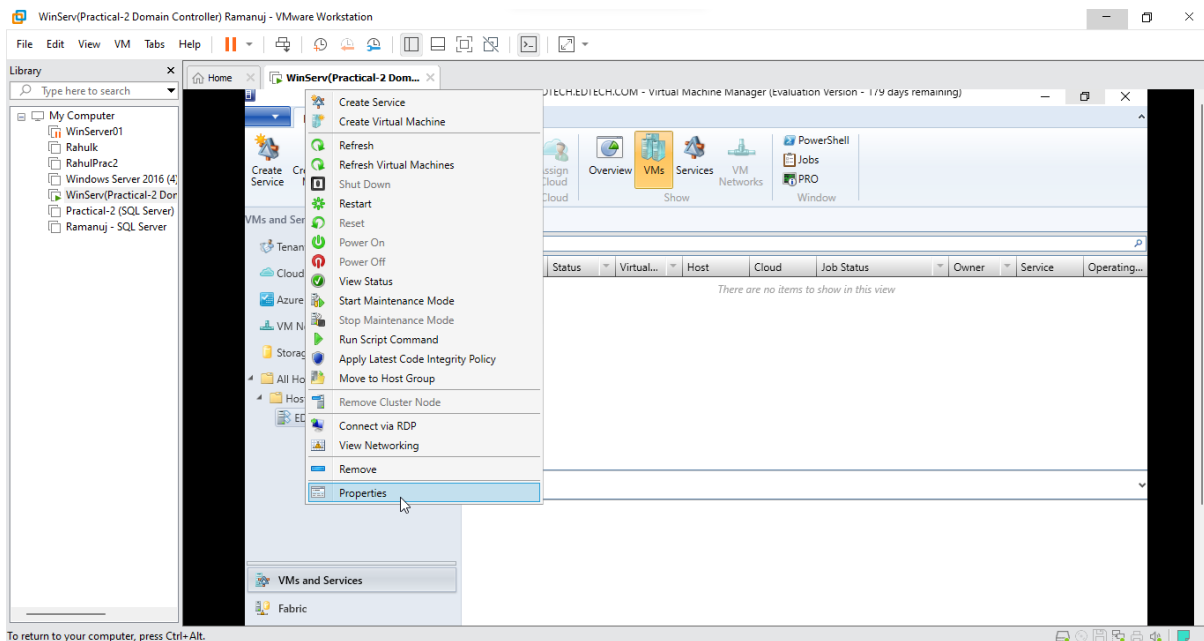
- Within Host Guardian Service Settings Enter your custom domain name followed by Attestation and Repeat the same for KeyProtection.
- (Here it is <http://edtech.com/Attestation> and <http://edtech.com/KeyProtection>)



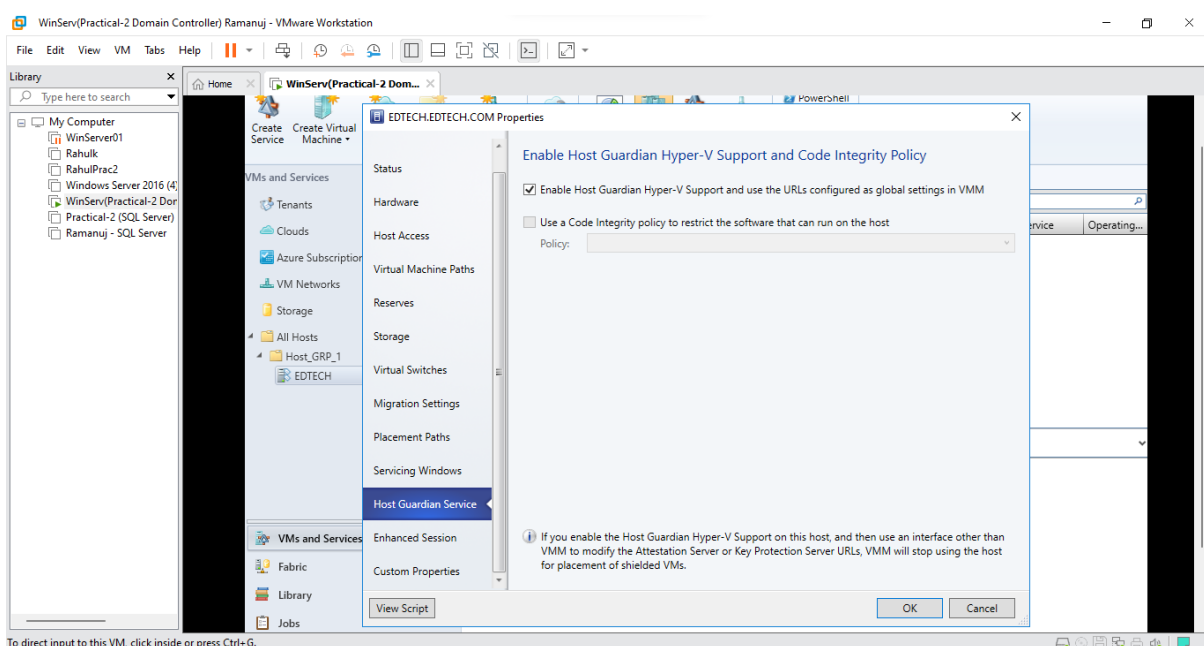
- After entering your details Click on **Finish**



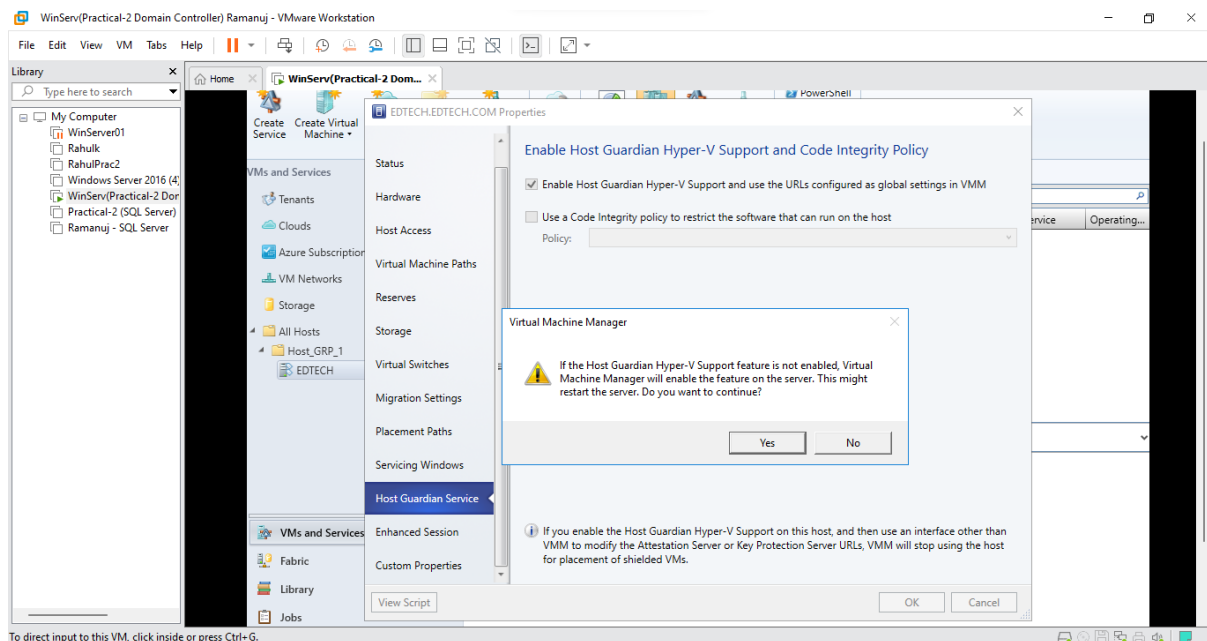
- **Pre-requisite:** Create a Cloud and Create a Server on a Host Group (Here it is **EDTECH**)
- Now within your Host Group Right-Click on **EDTECH** and Select **Properties**



- Within properties Select **Host Guardian Service** and Select **Enable Host Guardian Hyper-V Support** and use the URLs configured as global settings in VMM and Click **Ok**

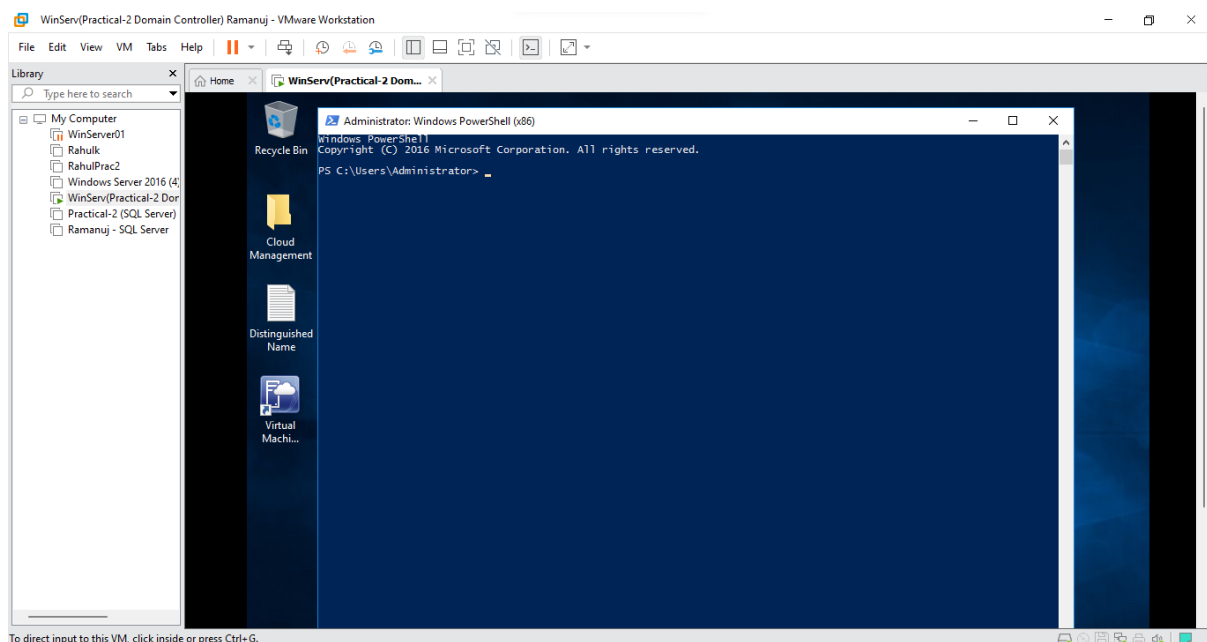


- A Pop-up will come to restart the server for the Host Guardian changes to apply, Click **Yes**



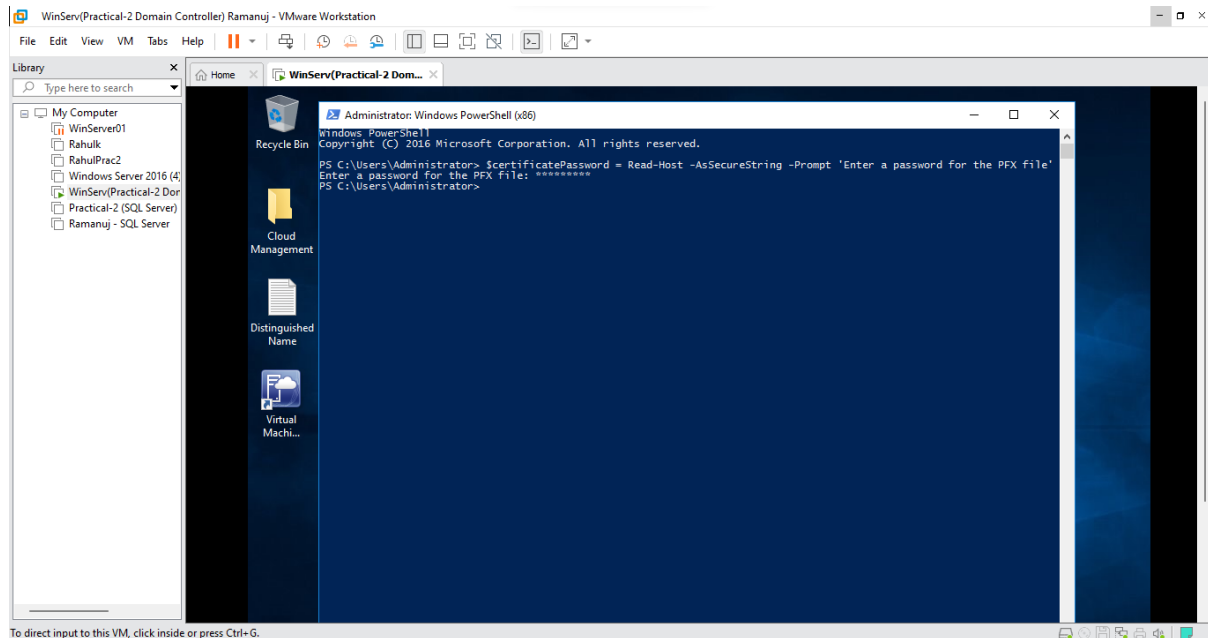
Step 2: Creating the pfx file for the HGS Server

- Open **Windows Powershell** as **Administrator**

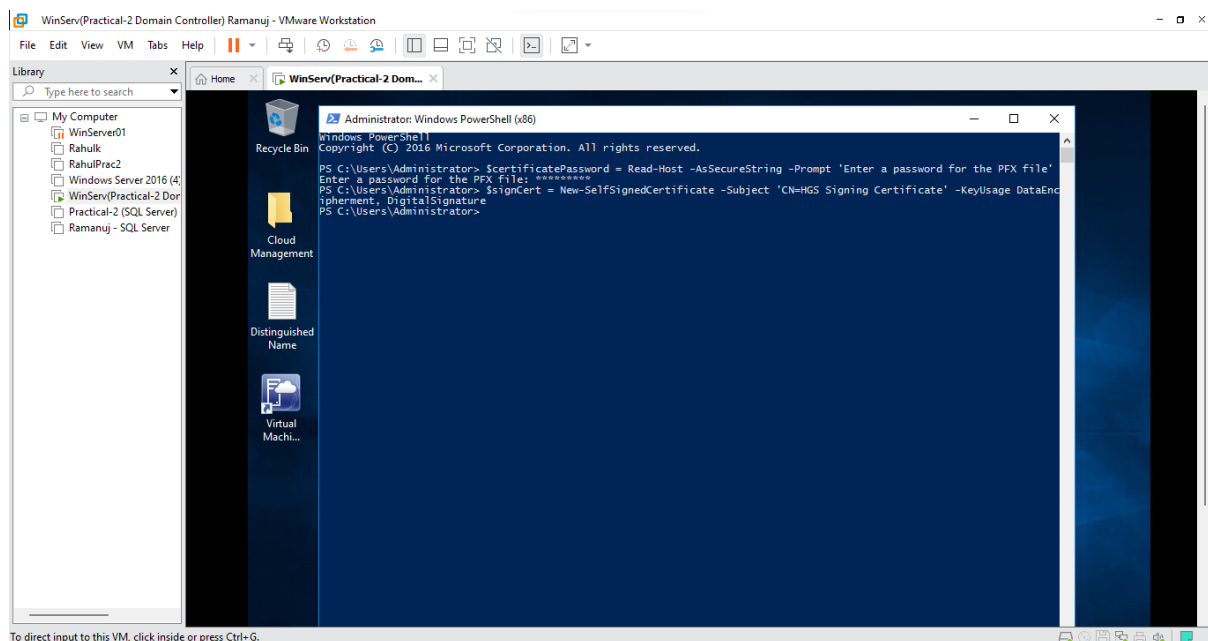


In the Powershell Type the following code:

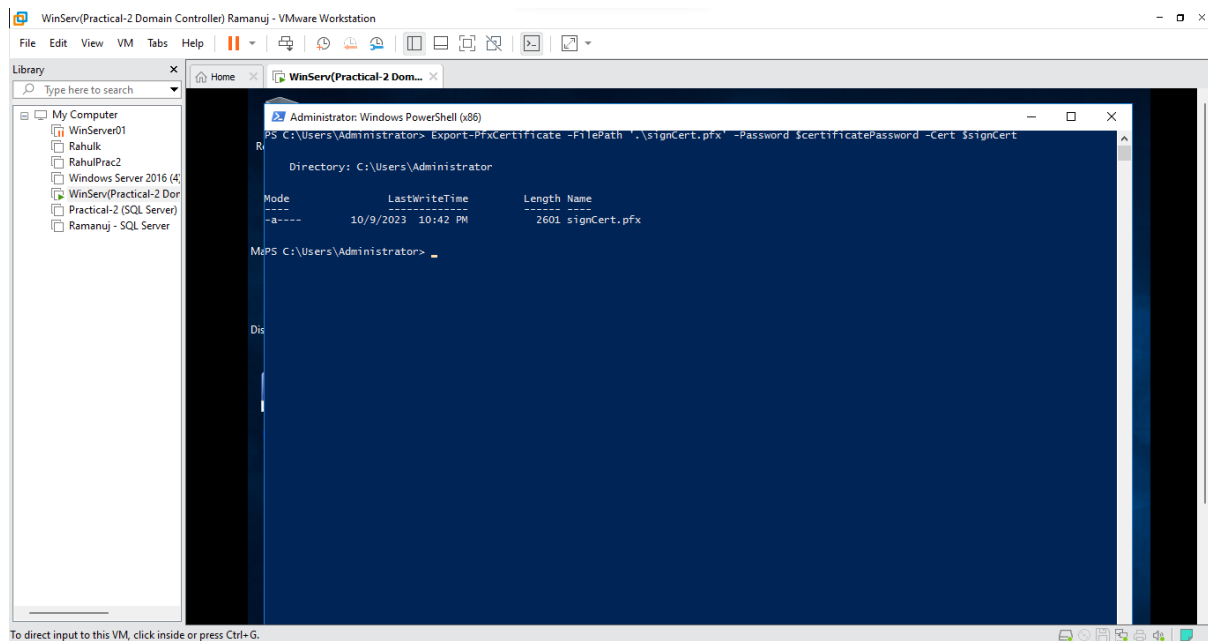
- `$certificatePassword = Read-Host -AsSecureString -Prompt 'Enter a password for the PFX file'`



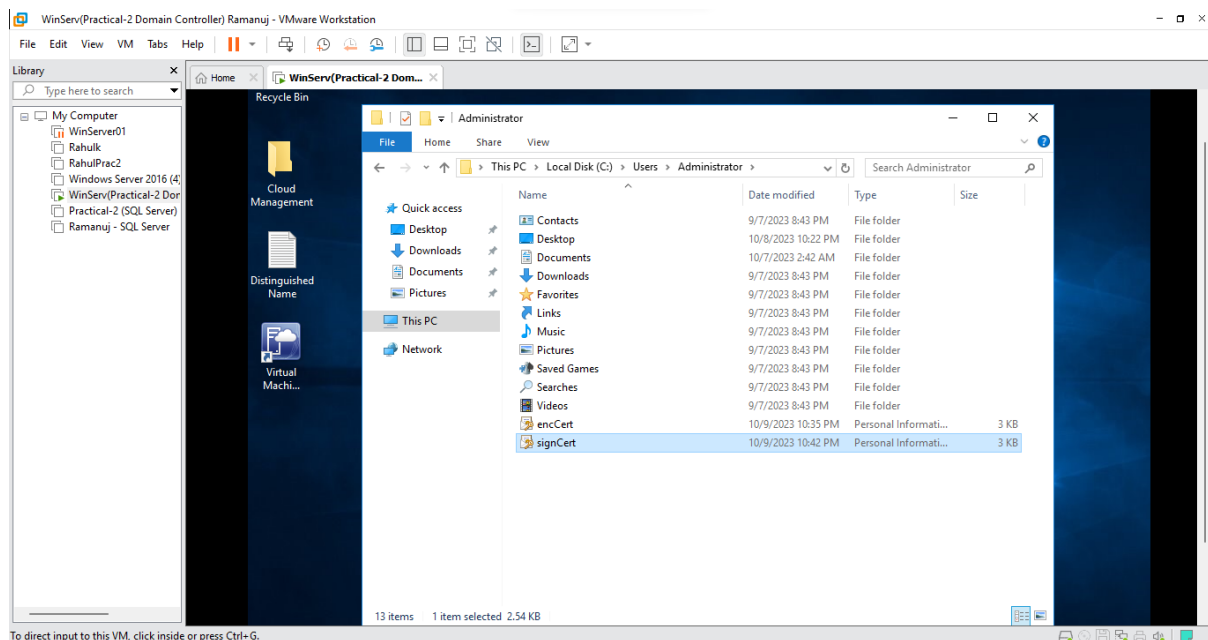
- `$signCert = New-SelfSignedCertificate -Subject 'CN=HGS Signing Certificate' -KeyUsage DataEncipherment, DigitalSignature`



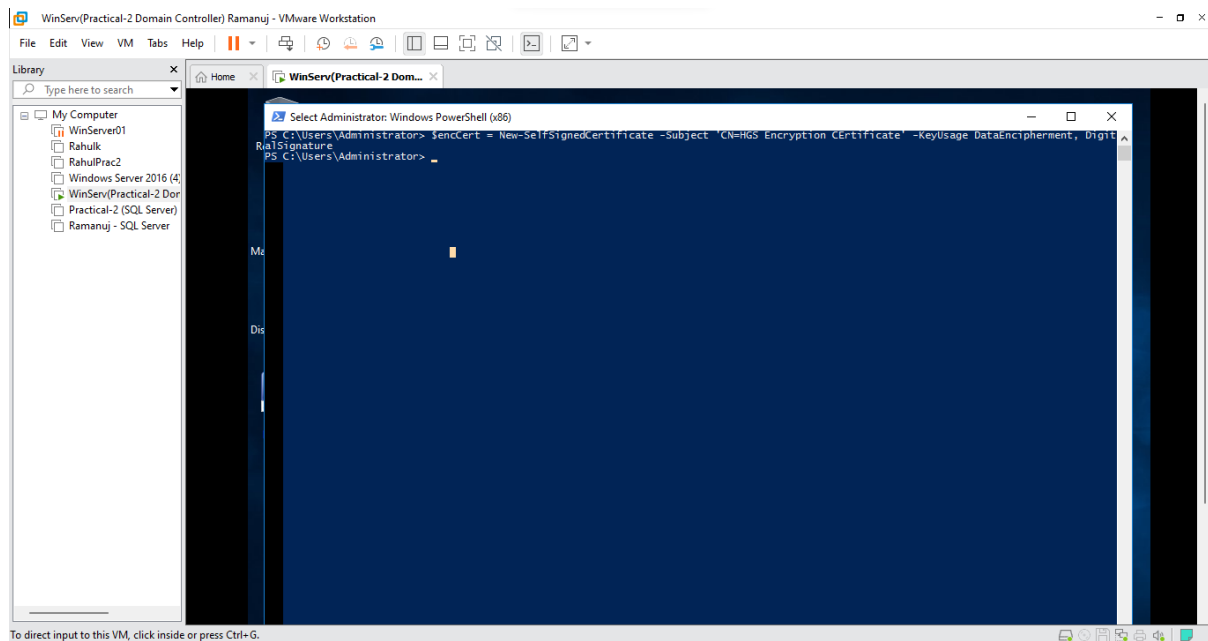
- Export-PfxCertificate –FilePath ‘.\signCert.pfx’ –Password \$certificatePassword –Cert \$signCert



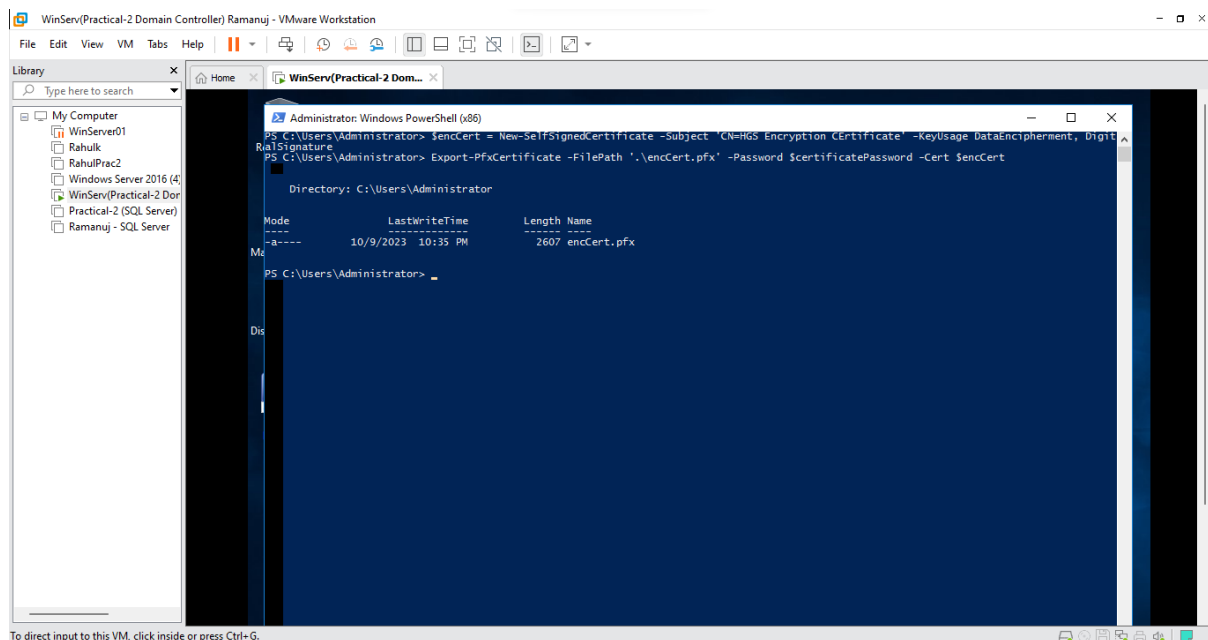
- To confirm the creation of the file Check the **Administrator** folder in **C:\Users**



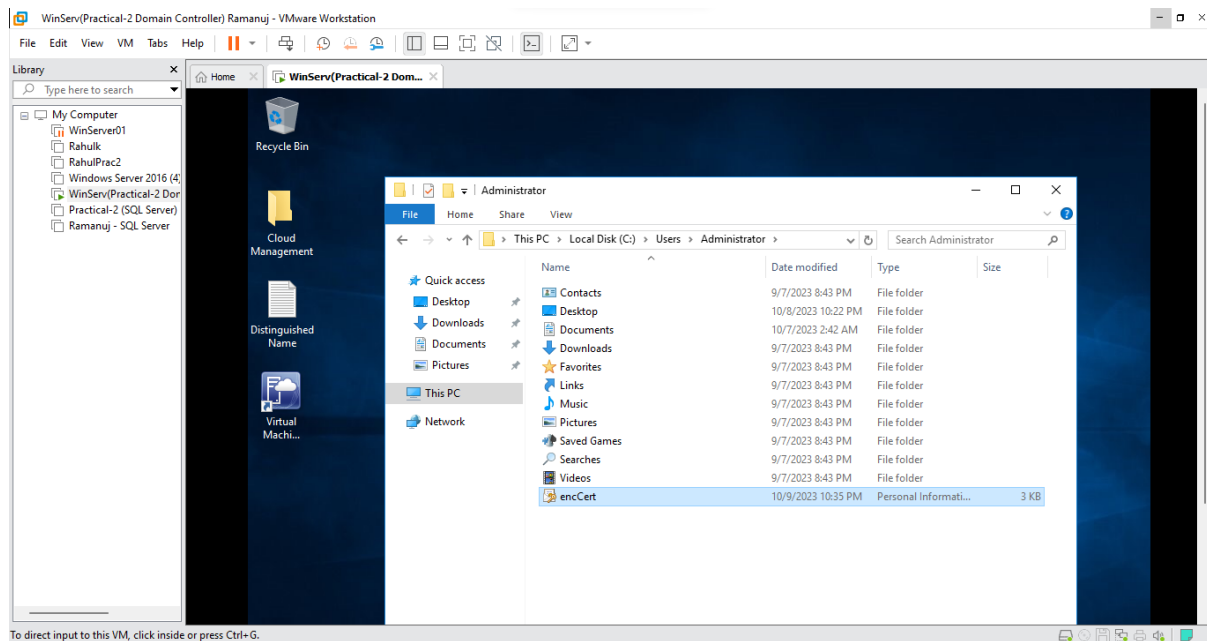
- \$encCert = New-SelfSignedCertificate –Subject ‘CN=HGS Encryption Certificate’ –KeyUsage DataEncipherment, DigitalSignature



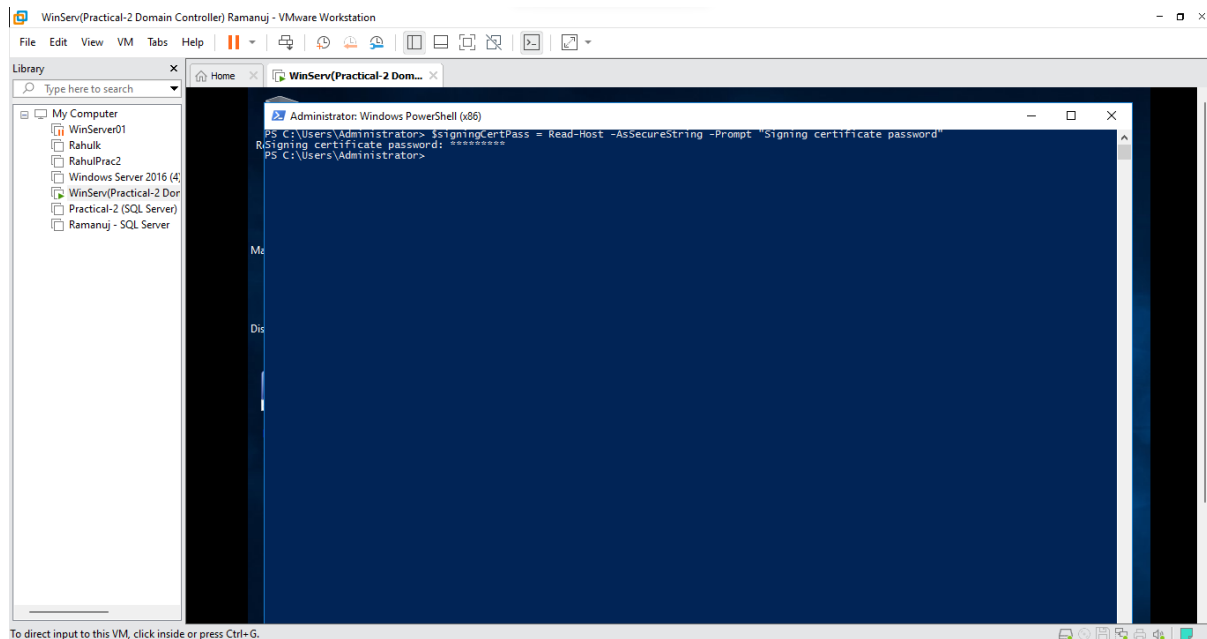
- Export-PfxCertificate –FilePath ‘.\signCert.pfx’ –Password \$certificatePassword –Cert \$signCert



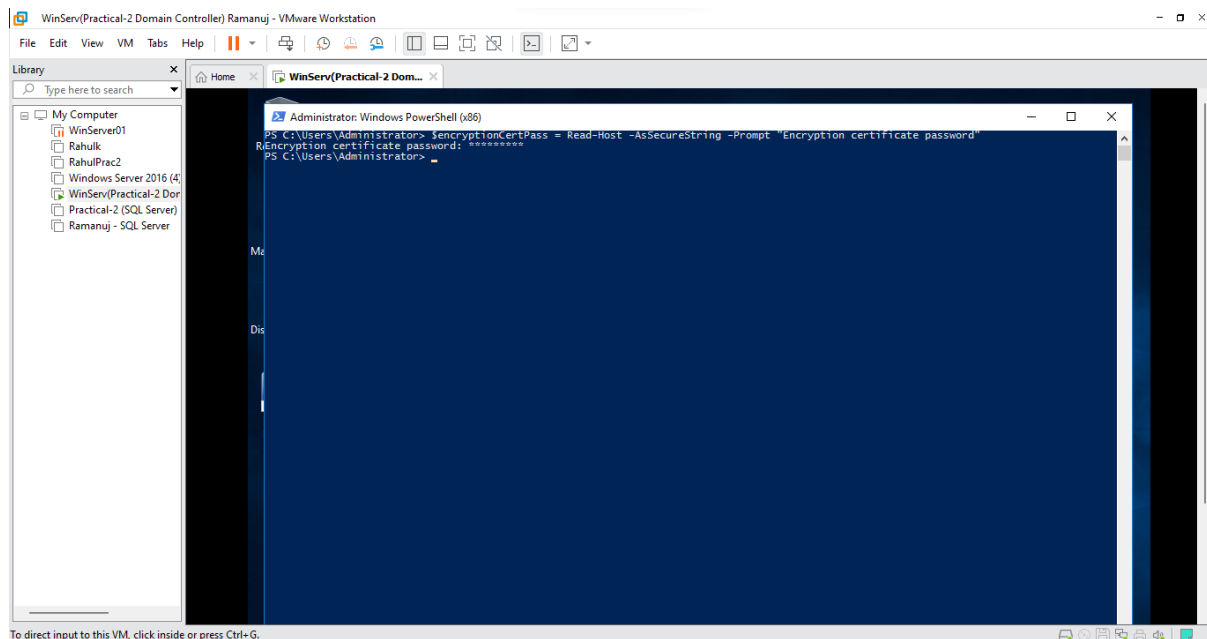
- We confirm the creation of the PFX file by going to the **Administrator** folder in **C:\Users**



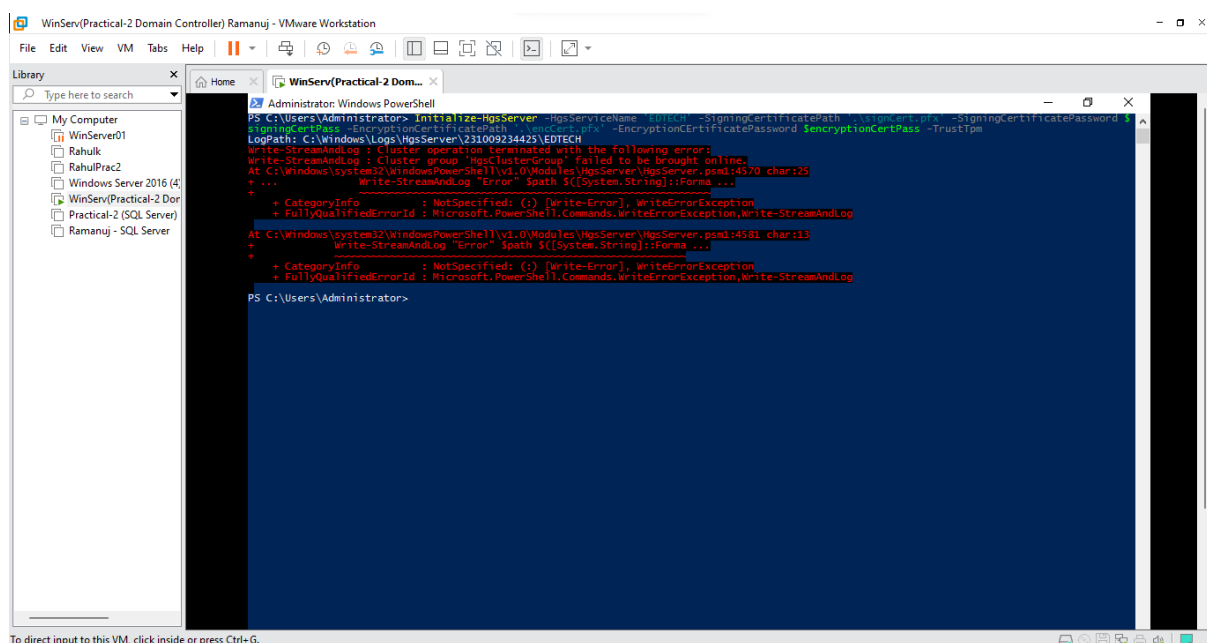
- `$signingCertPass = Read-Host -AsSecureString -Prompt "Signing certificate password"`



- \$encryptionCertPass = Read-Host -AsSecureString -Prompt "Encryption certificate password"



- Initialize-HgsServer -HgsServiceName 'WINSERVER2016' -SigningCertificatePath '.\signCert.pfx' -SigningCertificatePassword \$signingCertPass -EncryptionCertificatePath '.\encCert.pfx' -EncryptionCertificatePassword \$encryptionCertPass -TrustTpm



- To Confirm the creation of the HGS Cluster, Open the **Virtual Machine Manager** and it should show in **Hosts**

